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EXAM PREP

SchweserNotes[™]

Financial Statement Analysis
and Equities

Level I Book 2

KAPLAN[®] SCHWESER

Book 2: Financial Statement Analysis and Equities

SchweserNotes™ 2027

Level I CFA®



SCHWESERNOTES™ 2027 LEVEL I CFA® BOOK 2: FINANCIAL STATEMENT ANALYSIS AND EQUITIES

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CONTENTS

Learning Outcome Statements (LOS)

FINANCIAL STATEMENT ANALYSIS

READING 27

Introduction to Financial Statement Analysis

Module 27.1: Financial Statement Roles

Key Concepts

Answer Key for Module Quizzes

READING 28

Analyzing Income Statements

Module 28.1: Revenue Recognition

Module 28.2: Expense Recognition

Module 28.3: Nonrecurring Items

Module 28.4: Earnings Per Share

Module 28.5: Ratios and Common-Size Income Statements

Key Concepts

Answer Key for Module Quizzes

READING 29

Analyzing Balance Sheets

Module 29.1: Intangible Assets and Marketable Securities

Module 29.2: Common-Size Balance Sheets

Key Concepts

Answer Key for Module Quizzes

READING 30

Analyzing Statements of Cash Flows I

Module 30.1: Cash Flow Introduction and Direct Method CFO

Module 30.2: Indirect Method CFO

Module 30.3: Investing and Financing Cash Flows and IFRS/US GAAP
Differences

Key Concepts
Answer Key for Module Quizzes

READING 31

Analyzing Statements of Cash Flows II

Module 31.1: Analyzing Statements of Cash Flows II
Key Concepts
Answer Key for Module Quizzes

READING 32

Analysis of Inventories

Introduction
Module 32.1: Inventory Measurement
Module 32.2: Inflation Impact on FIFO and LIFO
Module 32.3: Presentation and Disclosure
Key Concepts
Answer Key for Module Quizzes

READING 33

Analysis of Long-Term Assets

Introduction
Module 33.1: Intangible Long-Lived Assets
Module 33.2: Impairment and Derecognition
Module 33.3: Long-Term Asset Disclosures
Key Concepts
Answer Key for Module Quizzes

READING 34

Topics in Long-Term Liabilities and Equity

Module 34.1: Leases
Module 34.2: Deferred Compensation and Disclosures
Key Concepts
Answer Key for Module Quizzes

READING 35

Analysis of Income Taxes

Introduction

Module 35.1: Differences Between Accounting Profit and Taxable
Income

Module 35.2: Deferred Tax Assets and Liabilities

Module 35.3: Tax Rates and Disclosures

Key Concepts

Answer Key for Module Quizzes

READING 36

Financial Reporting Quality

Module 36.1: Reporting Quality

Module 36.2: Accounting Choices and Estimates

Module 36.3: Warning Signs

Key Concepts

Answer Key for Module Quizzes

READING 37

Financial Analysis Techniques

Introduction

Module 37.1: Introduction to Financial Ratios

Module 37.2: Financial Ratios, Part 1

Module 37.3: Financial Ratios, Part 2

Module 37.4: DuPont Analysis

Module 37.5: Industry-Specific Financial Ratios

Key Concepts

Answer Key for Module Quizzes

READING 38

Introduction to Financial Statement Modeling

Module 38.1: Financial Statement Modeling

Key Concepts

Answer Key for Module Quiz

Topic Quiz: Financial Statement Analysis

EQUITIES

READING 39

Equity Instrument Features

Module 39.1: Public and Private Equity Investments

Key Concepts

Answer Key for Module Quiz

READING 40

Equity Jurisdictions, Classes, and the Voting Process

Module 40.1: Rights of Equity Shareholders

Key Concepts

Answer Key for Module Quiz

READING 41

Equity Issuance and Trading

Module 41.1: Equity Markets and Exchanges

Module 41.2: Equity Indexes and Liquidity Measures

Key Concepts

Answer Key for Module Quizzes

READING 42

Sources of Equity Returns

Module 42.1: Dividends, Stock Splits, and Share Repurchases

Module 42.2: Return Calculations for Equities

Key Concepts

Answer Key for Module Quizzes

READING 43

Introduction to Equity Valuation

Module 43.1: Measures and Models of Equity Value

Key Concepts

Answer Key for Module Quiz

READING 44

Discounted Cash Flow (DCF) and Growth Models

Module 44.1: Cash Flow Metrics

Module 44.2: Valuation Models

Module 44.3: Model Limitations and Preferred Stock Valuation

Key Concepts

Answer Key for Module Quizzes

READING 45

Relative Value Equity Valuation Approaches

Module 45.1: Price Multiples and Equity Valuation

Module 45.2: Price and Enterprise Value Multiples

Module 45.3: Peer Groups and Multiples-Based Valuation

Key Concepts

Answer Key for Module Quizzes

READING 46

Financial Statement Forecasting in Equity Valuation

Module 46.1: Financial Statement Forecast Models

Module 46.2: Models Based on Company Characteristics

Key Concepts

Answer Key for Module Quizzes

READING 47

Industry and Competitive Analysis

Module 47.1: Industry Analysis Framework

Module 47.2: Porter's Five Forces and PESTLE Analysis

Key Concepts

Answer Key for Module Quizzes

READING 48

Company Analysis: Past, Present, and Future

Module 48.1: Corporate Strategy and Industry Position

Module 48.2: Revenue, Profitability, and Capital

Key Concepts

Answer Key for Module Quizzes

READING 49

Equity Analyst Research Reports

Module 49.1: Equity Analyst Research Reports

Key Concepts

Answer Key for Module Quiz

READING 50

The Capital Asset Pricing Model, Market Model, and Other Factor-Based Equity Models

Module 50.1: Estimating the Cost of Equity With the Market Model and CAPM

Module 50.2: Arbitrage Pricing Theory

Key Concepts

Answer Key for Module Quizzes

Topic Quiz: Equities

Formulas

Index

Learning Outcome Statements (LOS)

27. Introduction to Financial Statement Analysis

The candidate should be able to:

- a. describe the steps in the financial statement analysis framework.
- b. describe the roles of financial statement analysis.
- c. describe the importance of regulatory filings, financial statement notes and supplementary information, management's commentary, and audit reports.
- d. describe implications for financial analysis of alternative financial reporting systems and the importance of monitoring developments in financial reporting standards.
- e. describe information sources that analysts use in financial statement analysis besides annual and interim financial reports.

28. Analyzing Income Statements

The candidate should be able to:

- a. describe general principles of revenue recognition, specific revenue recognition applications, and implications of revenue recognition choices for financial analysis.
- b. describe general principles of expense recognition, specific expense recognition applications, implications of expense recognition choices for financial analysis and contrast costs that are capitalized versus those that are expensed in the period in which they are incurred.
- c. describe the financial reporting treatment and analysis of non-recurring items (including discontinued operations, unusual or infrequent items) and changes in accounting policies.
- d. describe how earnings per share is calculated and calculate and interpret a company's basic and diluted earnings per share for companies with simple and complex capital structures including those with antidilutive securities.
- e. evaluate a company's financial performance using common-size income statements and financial ratios based on the income statement.

29. Analyzing Balance Sheets

The candidate should be able to:

- a. explain the financial reporting and disclosures related to intangible assets.
- b. explain the financial reporting and disclosures related to goodwill.
- c. explain the financial reporting and disclosures related to financial instruments.
- d. explain the financial reporting and disclosures related to non-current liabilities.
- e. calculate and interpret common-size balance sheets and related financial ratios.

30. Analyzing Statements of Cash Flows I

The candidate should be able to:

- a. describe how the cash flow statement is linked to the income statement and the balance sheet.
- b. describe the steps in the preparation of direct and indirect cash flow statements, including how cash flows can be computed using income statement and balance sheet data.
- c. demonstrate the conversion of cash flows from the indirect to direct method.
- d. contrast cash flow statements prepared under International Financial Reporting Standards (IFRS) and US generally accepted accounting principles (US GAAP).

31. Analyzing Statements of Cash Flows II

The candidate should be able to:

- a. analyze and interpret both reported and common-size cash flow statements.
- b. calculate and interpret free cash flow to the firm, free cash flow to equity, and performance and coverage cash flow ratios.

32. Analysis of Inventories

The candidate should be able to:

- a. describe the measurement of inventory at the lower of cost and net realisable value and its implications for financial statements and ratios.
- b. calculate and explain how inflation and deflation of inventory costs affect the financial statements and ratios of companies that use different inventory valuation methods.
- c. describe the presentation and disclosures relating to inventories and explain issues that analysts should consider when examining a company's inventory disclosures and other sources of information.

33. Analysis of Long-Term Assets

The candidate should be able to:

- a. compare the financial reporting of the following types of intangible assets: purchased, internally developed, and acquired in a business combination.
- b. explain and evaluate how impairment and derecognition of property, plant, and equipment and intangible assets affect the financial statements and ratios.
- c. analyze and interpret financial statement disclosures regarding property, plant, and equipment and intangible assets.

34. Topics in Long-Term Liabilities and Equity

The candidate should be able to:

- a. explain the financial reporting of leases from the perspectives of lessors and lessees.
- b. explain the financial reporting of defined contribution, defined benefit, and stock-based compensation plans.
- c. describe the financial statement presentation of and disclosures relating to long-term liabilities and share-based compensation.

35. Analysis of Income Taxes

The candidate should be able to:

- a. contrast accounting profit, taxable income, taxes payable, and income tax expense and temporary versus permanent differences between accounting profit and taxable income.
- b. explain how deferred tax liabilities and assets are created and the factors that determine how a company's deferred tax liabilities and assets should be treated for the purposes of financial analysis.
- c. calculate, interpret, and contrast an issuer's effective tax rate, statutory tax rate, and cash tax rate.
- d. analyze disclosures relating to deferred tax items and the effective tax rate reconciliation and explain how information included in these disclosures affects a company's financial statements and financial ratios.

36. Financial Reporting Quality

The candidate should be able to:

- a. compare financial reporting quality with the quality of reported results (including quality of earnings, cash flow, and balance sheet items).
- b. describe a spectrum for assessing financial reporting quality.
- c. explain the difference between conservative and aggressive accounting.
- d. describe motivations that might cause management to issue financial reports that are not high quality and conditions that are conducive to issuing low-quality, or even fraudulent, financial reports.
- e. describe mechanisms that discipline financial reporting quality and the potential limitations of those mechanisms.
- f. describe presentation choices, including non-GAAP measures, that could be used to influence an analyst's opinion.

- g. describe accounting methods (choices and estimates) that could be used to manage earnings, cash flow, and balance sheet items.
- h. describe accounting warning signs and methods for detecting manipulation of information in financial reports.

37. Financial Analysis Techniques

The candidate should be able to:

- a. describe tools and techniques used in financial analysis, including their uses and limitations.
- b. calculate and interpret activity, liquidity, solvency, and profitability ratios.
- c. describe relationships among ratios and evaluate a company using ratio analysis.
- d. demonstrate the application of DuPont analysis of return on equity and calculate and interpret effects of changes in its components.
- e. describe the uses of industry-specific ratios used in financial analysis.
- f. describe how ratio analysis and other techniques can be used to model and forecast earnings.

38. Introduction to Financial Statement Modeling

The candidate should be able to:

- a. demonstrate the development of a sales-based pro forma company model.
- b. explain how behavioral factors affect analyst forecasts and recommend remedial actions for analyst biases.
- c. explain how the competitive position of a company based on a Porter's five forces analysis affects prices and costs.
- d. explain how to forecast industry and company sales and costs when they are subject to price inflation or deflation.
- e. explain considerations in the choice of an explicit forecast horizon and an analyst's choices in developing projections beyond the short-term forecast horizon.

39. Equity Instrument Features

The candidate should be able to:

- a. describe basic features and types of equity instruments.
- b. contrast features of publicly listed versus private equity securities.

40. Equity Jurisdictions, Classes, and the Voting Process

The candidate should be able to:

- a. describe differences in economic and voting rights and other characteristics among different jurisdictions and equity classes.
- b. describe the voting process and roles of the issuer's management, board, proxy advisors, asset managers, and asset owners in the process.

41. Equity Issuance and Trading

The candidate should be able to:

- a. describe primary and secondary public equity markets and their functions.
- b. compare exchange, off-exchange, and over-the-counter equities trading.
- c. describe liquidity measures for a publicly listed security and calculate float and average daily volume.
- d. describe types of equity indexes.

42. Sources of Equity Returns

The candidate should be able to:

- a. describe features and uses of dividends, share repurchases, stock splits, and reverse stock splits.
- b. describe dividend payment chronology.
- c. calculate price and total return for equity securities.

43. Introduction to Equity Valuation

The candidate should be able to:

- a. contrast price and value for an equity security.
- b. contrast the use of book value of equity, market capitalization, and enterprise value as indicators of intrinsic value.
- c. describe major categories of equity valuation models and explain their advantages and disadvantages.

44. Discounted Cash Flow (DCF) and Growth Models

The candidate should be able to:

- a. contrast the inputs used in dividend discount, free cash flow to equity (FCFE), free cash flow to the firm (FCFF), and residual income models, and explain the process for using present value models to value equity.
- b. calculate and interpret the intrinsic value of an equity security based on present value models that assume constant cash flow growth or multistage cash flow growth.
- c. explain the shortcomings of constant and multistage cash flow growth assumptions in present value models.
- d. calculate the intrinsic value of a non-callable, non-convertible preferred stock and describe how contingency features affect intrinsic value.

45. Relative Value Equity Valuation Approaches

The candidate should be able to:

- a. explain the process for using multiples to value equity based on the method of comparables and the method of forecasted fundamentals.
- b. calculate and interpret price and enterprise value multiples, and describe their appropriate uses.
- c. determine a potential peer group for multiples-based equity valuation based on a company's industry classification or specific drivers of expected return.
- d. describe the use of multiples based on past, current, and projected future values in estimating equity value.

46. Financial Statement Forecasting in Equity Valuation

The candidate should be able to:

- a. explain the rationale, construction, and uses of financial statement forecast models in equity valuation.
- b. evaluate the construction of an equity valuation model based on characteristics of the subject company.
- c. estimate and interpret the value of an equity security based on the outputs of a financial statement forecast model.

47. Industry and Competitive Analysis

The candidate should be able to:

- a. describe the process of industry and competitive analysis.
- b. describe industry classification methods and compare methods by which companies can be grouped.
- c. determine an industry's size, growth characteristics, profitability, and market share trends.
- d. analyze an industry's structure and external influences using Porter's Five Forces and the PESTLE framework.

48. Company Analysis: Past, Present, and Future

The candidate should be able to:

- a. explain a company's competitive strategy and position within an industry.
- b. evaluate a company's revenue drivers, operating profitability, and pricing power.

- c. evaluate a company's working capital, capital investments, and capital structure.

49. Equity Analyst Research Reports

The candidate should be able to:

- a. describe the elements that should be covered in a detailed company research report.
- b. compare sell-side and buy-side equity analyst reports and their conclusions.
- c. explain why value estimates from the same modeling approach may differ based on analyst assumptions regarding revenue, profitability, investment, and financing.

50. The Capital Asset Pricing Model, Market Model, and Other Factor-Based Equity Models

The candidate should be able to:

- a. explain the use of the CAPM and the market model to estimate a company's required return on equity.
- b. describe the arbitrage pricing theory and the importance of multi-factor models used in equity investing.

READING 27

INTRODUCTION TO FINANCIAL STATEMENT ANALYSIS

MODULE 27.1: FINANCIAL STATEMENT ROLES

LOS 27.a: Describe the steps in the financial statement analysis framework.

The **financial statement analysis framework framework**¹ sets out a generic set of steps for analysts to apply to a multitude of objectives when analyzing debt issues, equity securities, and corporate actions. The framework consists of six steps:

- Step 1: State the objective and context.* Determine what questions the analysis seeks to answer, the form in which this information needs to be presented, and what resources and how much time are available to perform the analysis.
- Step 2: Gather data.* Acquire the company's financial statements and other relevant data on its industry and the economy. Ask questions of the company's management, suppliers, and customers, and visit company sites.
- Step 3: Process the data.* Make any appropriate adjustments to the financial statements. Calculate ratios and perform statistical analysis. Prepare exhibits such as graphs and common-size balance sheets.
- Step 4: Analyze and interpret the data.* Use the data to answer the questions stated in the first step. Decide what conclusions or recommendations the information supports.
- Step 5: Report the conclusions or recommendations.* Prepare a report and communicate it to its intended audience. Be sure the report and its dissemination comply with the Code and Standards that relate to investment analysis and recommendations.
- Step 6: Update the analysis.* Repeat these steps periodically, and change the conclusions or recommendations when necessary.

LOS 27.b: Describe the roles of financial statement analysis.

Financial reporting refers to the way companies show their financial performance to investors, creditors, and other interested parties by preparing and presenting financial statements.

The role of **financial statement analysis** is to use the information in a company's financial statements, along with other relevant information, to make economic decisions. Examples of such decisions include whether to invest in the company's securities or recommend them to investors, whether to extend trade or bank credit to the company, and assigning credit ratings to a company's debt. Analysts use financial statement data to evaluate a company's past performance and current financial position to form opinions about the company's ability to earn profits and generate cash flow in the future. As part of this process, analysts will identify risk factors that affect the company's future profitability and position.



PROFESSOR'S NOTE

This reading deals with financial analysis for external users. Management also performs financial analysis in making everyday decisions. However, management may rely on internal financial information that is likely maintained in a different format and unavailable to external users.

LOS 27.c: Describe the importance of regulatory filings, financial statement notes and supplementary information, management's commentary, and audit reports.

Standard-setting bodies are professional organizations of accountants and auditors that establish financial reporting standards. **Regulatory authorities** are government agencies that have the legal authority to enforce compliance with financial reporting standards.

The two primary standard-setting bodies are the **Financial Accounting Standards Board (FASB)** and the **International Accounting Standards Board (IASB)**. In the United States, the FASB sets forth the US Generally Accepted Accounting Principles (US GAAP). Outside the United States, the IASB establishes the International Financial Reporting Standards (IFRS). Other national standard-setting bodies exist as well. Some of the older IASB standards are referred to as the International Accounting Standards (IAS).

Regulatory authorities, such as the **Securities and Exchange Commission (SEC)** in the United States and the Financial Conduct Authority in the United Kingdom, are established by national governments.

Most national authorities belong to the **International Organization of Securities Commissions (IOSCO)**. Together, the members of IOSCO regulate more than 95% of the world's financial markets. IOSCO is not a regulatory body, but its members work together to improve cross-border cooperation and make national regulations and enforcement more uniform around the world. The IOSCO Objectives and Principles of Securities Regulation are based on three main objectives:

Protecting investors

Ensuring markets are fair, efficient, and transparent

Reducing systemic risk

IOSCO requires issuers to provide full, accurate, and timely disclosure of financial results, risk, and other information used in the decision-making process. It also requires accounting standards that are used to prepare financial statements to be of a high standard and internationally accepted.

The SEC's requirements for financial reporting by US companies are shown in Figure 27.1 as an example of reporting requirements. The SEC has the responsibility of enforcing the Sarbanes-Oxley Act of 2002. The act prohibits a company's external auditor from providing certain additional paid services to the company, to avoid the conflict of interest involved, and to promote auditor independence. The act requires a company's executive management to certify that the financial statements are presented fairly and to include a statement about the effectiveness of the company's internal controls of financial reporting. Additionally, the external auditor must provide a statement confirming the effectiveness of the company's internal controls.

Proxy statements are issued to shareholders when there are matters that require a shareholder vote. These statements, which are also filed with the SEC, are a good source of information about the election of (and qualifications of) board members, compensation, management qualifications, and the issuance of stock options.

In the European Union (EU), each member state has its own securities regulations, but all countries in the EU are required to report using IFRS. The European Commission also has established the European Securities Commission, which advises the European Commission on securities regulation issues, and the European Securities and Market Authority (ESMA), which coordinates regulation within the EU.

Financial statement notes (footnotes) include disclosures that provide further details about the information summarized in the financial statements. Footnotes allow users to improve their assessments of the amount, timing, and uncertainty of the estimates reported in the financial statements. Footnotes do the following:

- | They discuss the basis of presentation such as the fiscal period covered by the statements, whether IFRS or US GAAP is adhered to, and the inclusion of consolidated entities.
- | They provide information about accounting methods, assumptions, and estimates used by management.
- | They provide additional information on information included in the primary financial statements and items such as business acquisitions or disposals, legal actions, employee benefit plans, contingencies and commitments, significant customers, related party transactions, position and performance of segments of the firm, and significant post balance sheet events.
- | They are audited along with the primary financial statements.

Figure 27.1: SEC Required Filings

Form S-1. This is the registration statement filed before the sale of new securities to the public. The statement includes disclosures about the securities offered, audited financial statements (plus interim accounts, if the statement is filed more than three months after year-end), risk assessment, underwriter identification, and the estimated amount and use of the offering proceeds.

Form 10-K. This is the required annual filing that includes information about the business, risks, and its management, audited financial statements and disclosures, and disclosures about legal matters involving the firm. Information required in Form 10-K is similar to that which a firm typically provides in its annual report to shareholders. However, a firm's annual report is not a substitute for the required 10-K filing. Equivalent SEC forms for foreign issuers in the US markets are Form 40-F for Canadian companies and Form 20-F for other foreign issuers.

Form 10-Q. US firms are required to file this form quarterly, with updated interim financial statements (unlike Form 10-K, these statements do not have to be audited) and disclosures about certain events such as significant legal proceedings or changes in accounting policy. Non-US companies are typically required to file the equivalent Form 6-K semiannually.

Form DEF-14A. When a company prepares a proxy statement for its shareholders before the annual meeting or other shareholder vote, it also files the statement with the SEC as Form DEF-14A.

Form 8-K. Companies must file this form to disclose material events including significant asset acquisitions and disposals, changes in management or corporate governance, or matters related to its accountants, its financial statements, or the markets in which its securities trade.

Form 144. A company can issue securities to certain qualified buyers without registering the securities with the SEC, but it must notify the SEC that it intends to do so.

Forms 3, 4, and 5 involve the beneficial ownership of securities by a company's officers and directors. Analysts can use these filings to learn about purchases and sales of company securities by corporate insiders.

Both US GAAP and IFRS require companies to report segment data, but the required disclosure items are only a subset of the required disclosures for the company as a whole. Nonetheless, an analyst can prepare a more detailed analysis and forecast by examining the performance of business or geographic segments separately. Segment profit margins, asset utilization (turnover), and return on assets can be very useful in gaining a clear picture of a firm's overall operations. For forecasting, growth rates of segment revenues and profits can be used to estimate future sales and profits and to determine the changes in company characteristics over time.

A **business segment (operating segment)** is a portion of a larger company that accounts for more than 10% of the company's revenues, assets, or income. An operating segment should be distinguishable from the company's other lines of

business in terms of the risk and return characteristics of the segment. Segments reported should account for a minimum of 75% of the firm's external sales.

The following must be disclosed for each segment within the financial statement notes:

- | Revenue (external and between segments)
- | A measure of profit or loss
- | A measure of assets and liabilities
- | Interest (revenue and expense)
- | Acquisitions of PP&E and intangibles
- | Depreciation and amortization
- | Other noncash expenses
- | Income tax expense
- | Share of equity-accounted investments results

Geographic segments are also identified when they meet the size criterion given previously and the geographic unit has a business environment that is different from that of other segments or the remainder of the company's business.

For example, in its 2016 annual report, Boeing described its business segments as follows:²

- | We are organized based on the products and services we offer. We operate in five principal segments:
 - Commercial Airplanes
 - Our Defense, Space & Security (BDS) business comprises three segments:
 - Boeing Military Aircraft (BMA)
 - Network & Space Systems (N&SS)
 - Global Services & Support (GS&S)
 - Boeing Capital (BCC)

Management commentary (also known as management report, operating and financial review, or **management discussion and analysis [MD&A]**) is one of the most useful sections of an annual report. IFRS guidance recommends that management commentary address the nature of the business, management's

objectives, the company's past performance, the performance measures used, and the company's key relationships, resources, and risks. Analysts must be aware that some parts of management commentary may be unaudited.

For publicly held firms in the United States, the SEC requires management commentary to discuss trends and identify significant events and uncertainties that affect the firm's liquidity, capital resources, and results of operations. Management must also discuss the following:

- | Effects of inflation and changing prices, if material
- | Impact of off-balance-sheet obligations and contractual obligations, such as purchase commitments
- | Accounting policies that require significant judgment by management
- | Forward-looking expenditures and divestitures

An **audit** is an independent review of an entity's financial statements. Public accountants conduct audits and examine the financial reports and supporting records. The objective of an audit is to enable the auditor to provide an opinion on the fairness and reliability of the financial statements.

The independent certified public accounting firm employed by the board of directors is responsible for seeing that the financial statements conform to the applicable accounting standards. The auditor examines the company's accounting and internal control systems, confirms assets and liabilities, and generally tries to determine that there are no material errors in the financial statements. The auditor's report is an important source of information.

The **standard auditor's opinion** contains three parts and states the following:

Whereas the financial statements are prepared by management and are its responsibility, the auditor has performed an independent review.

Generally accepted auditing standards were followed, thus providing *reasonable assurance* that the financial statements contain no material errors.

The auditor is satisfied that the statements were prepared in accordance with accepted accounting principles and that the principles chosen and estimates made are reasonable. The auditor's report must also contain additional explanation when accounting methods have not been used consistently between periods.

An **unqualified opinion** (also known as an unmodified opinion or clean opinion) indicates that the auditor believes the statements are free from material omissions and errors. If the statements make any exceptions to the accounting principles, the auditor may issue a **qualified opinion** and explain these exceptions in the audit report. The auditor can issue an **adverse opinion** if the statements are not presented fairly or are materially nonconforming with accounting standards. If the auditor is unable to express an opinion (e.g., in the case of a scope limitation), a **disclaimer of opinion** is issued. Any opinion other than unqualified is sometimes referred to as a modified opinion.

The auditor's opinion will also contain an explanatory paragraph when a material loss is probable, but the amount cannot be reasonably estimated. These uncertainties may relate to the *going concern assumption* (the assumption that the firm will continue to operate for the foreseeable future), to the valuation or realization of asset values, or to litigation. This type of disclosure may be a signal of serious problems and may call for close examination by the analyst.

Internal controls are the processes by which the company ensures that it presents accurate financial statements. Internal controls are the responsibility of management. For publicly traded firms in the United States, the auditor must express an opinion on the firm's internal controls. The auditor can provide this opinion separately, or as the fourth element of the standard opinion.

An audit report must also contain a section communicating *key audit matters* (international reports) or *critical audit matters* (US), which highlights accounting choices that are of greatest significance to users of financial statements. These would include accounting choices that require significant management judgments and estimates, how significant transactions during a period were accounted for, or choices that the auditor finds especially challenging or subjective—and which, therefore, have a significant likelihood of being misstated.

LOS 27.d: Describe implications for financial analysis of alternative financial reporting systems and the importance of monitoring developments in financial reporting standards.

While the IASB and FASB work together to harmonize changes to accounting standards, some significant differences between IFRS and US GAAP still exist.

Some major differences are outlined in Figure 27.2, but additional differences will be encountered in subsequent modules.

Figure 27.2: Significant Differences Between IFRS and US GAAP

Basis for Comparison	US GAAP	IFRS
Developed by	FASB	IASB
Based on	Rules	Principles
Inventory valuation	FIFO, LIFO, and weighted average	LIFO prohibited
Product development costs	Expensed	May be capitalized
Interest paid	CFO	CFO or CFF
Reversal of inventory write-downs	Prohibited	Allowed

The existence of differences between the two sets of standards require the analyst to exercise caution when making comparisons between firms operating in different jurisdictions.

As financial reporting standards continue to evolve, analysts need to monitor how these developments will affect the financial statements they use. An analyst should be aware of new products and innovations in the financial markets that generate new types of transactions. These might not fall neatly into the existing financial reporting standards. Analysts can use the financial reporting framework as a guide for evaluating what effect new products or transactions might have on financial statements.

To keep up to date on the evolving standards, an analyst can monitor professional journals and other sources, such as the IASB (www.ifrs.org) and FASB (www.fasb.org) websites. CFA Institute produces position papers on financial reporting issues through the CFA Institute Centre for Financial Market Integrity.

Finally, analysts must monitor company disclosures for significant accounting standards and estimates.

LOS 27.e: Describe information sources that analysts use in financial statement analysis besides annual and interim financial reports.

As well as regulated information provided by issuers in filings and financial statements, an analyst can also use additional information sources:

┆ Issuer sources:

- *Earnings calls*. Targeted at investors, analysts, and members of the media, earnings calls include presentations by the company's management and the opportunity for question-and-answer sessions. Firms often provide **earnings guidance** before they release their financial statements. After an earnings announcement, senior management may hold a conference call to answer questions and provide information to complement their regulatory filings.
- Ad hoc presentations and events that are similar in format to an earnings call
- Press releases focusing on major events relevant to the company
- Communications with management, investor relations, and company personnel

Analysts should note that these additional sources of information provided by issuers are unlikely to have been audited.

┆ Public third-party sources:

- Free industry reports, whitepapers, and trade journals
- Government agency-produced economic and industry statistics
- Generalized and industry-specific media sources
- Social media

┆ Proprietary third-party sources:

- Analyst reports
- Reports from data platforms such as Bloomberg, Wind, and FactSet
- Reports from industry-specific agencies and consultancies

┆ Proprietary primary research:

- Studies commissioned by the analyst
- Hands-on experience with the company's products or services
- Data and advice of technical specialists employed by the analyst

An analyst should also review pertinent information on economic conditions and the company's industry and compare the company to its competitors. The necessary information can be acquired from trade journals, statistical reporting services, and government agencies.



MODULE QUIZ 27.1

1. Which of the following statements *least accurately* describes a role of financial statement analysis?
 - A. Use the information in financial statements to make economic decisions.
 - B. Provide reasonable assurance that the financial statements are free of material errors.
 - C. Evaluate an entity's financial position and past performance to form opinions about its future ability to earn profits and generate cash flow.
2. Information about accounting estimates, assumptions, and methods chosen for reporting is *most likely* found in:
 - A. the auditor's opinion.
 - B. financial statement notes.
 - C. management discussion and analysis.
3. If an auditor finds that a company's financial statements have made a specific exception to applicable accounting principles, she is *most likely* to issue a:
 - A. dissenting opinion.
 - B. cautionary note.
 - C. qualified opinion.
4. Information about elections of members to a company's board of directors is *most likely* found in:
 - A. a 10-Q filing.
 - B. a proxy statement.
 - C. footnotes to the financial statements.
5. Which of these steps is *least likely* to be a part of the financial statement analysis framework?
 - A. State the purpose and context of the analysis.
 - B. Determine whether the company's securities are suitable for the client.
 - C. Adjust the financial statement data and compare the company to its industry peers.
6. Which of the following is *least likely* to be a part of segment disclosure?
 - A. Segment sales.
 - B. Segment cost of goods sold.
 - C. Segment earnings.
7. Which of the following sources of information is *most likely* to be classified as a proprietary third-party source?
 - A. Research reports prepared by analysts.
 - B. Trade journals.
 - C. Statistics produced by government agencies.

KEY CONCEPTS

LOS 27.a

The framework for financial analysis has six steps:

State the objective of the analysis.

Gather data.

Process the data.

Analyze and interpret the data.

Report the conclusions or recommendations.

Update the analysis.

LOS 27.b

The role of financial reporting is to provide various users with useful information about a company's performance and financial position.

The role of financial statement analysis is to use the data from financial statements to support economic decisions.

LOS 27.c

Standard-setting bodies are private sector organizations that establish financial reporting standards. The two primary standard-setting bodies are the International Accounting Standards Board (IASB) and, in the United States, the Financial Accounting Standards Board (FASB).

Regulatory authorities are government agencies that enforce compliance with financial reporting standards. Regulatory authorities include the Securities and Exchange Commission (SEC) in the United States and the Financial Conduct Authority in the United Kingdom. Many national regulatory authorities belong to the International Organization of Securities Commissions (IOSCO).

Important information about accounting methods, estimates, and assumptions is disclosed in the footnotes to the financial statements and supplementary schedules. These disclosures also contain information about segment results, commitments and contingencies, legal proceedings, acquisitions or divestitures, issuance of stock options, and details of employee benefit plans.

Management commentary (management discussion and analysis) contains an overview of the company and important information about business trends, future capital needs, liquidity, significant events, and significant choices of accounting methods requiring management judgment.

The objective of audits of financial statements is to provide an opinion on the statements' fairness and reliability.

The auditor's opinion gives evidence of an independent review of the financial statements that verifies that appropriate accounting principles were used, that standard auditing procedures were used to establish reasonable assurance that the statements contain no material errors, and that management's report on the company's internal controls has been reviewed.

An auditor can issue an unqualified (clean) opinion if the statements are free from material omissions and errors, a qualified opinion that notes any exceptions to accounting principles, an adverse opinion if the statements are not presented fairly in the auditor's opinion, or a disclaimer of opinion if the auditor is unable to express an opinion.

A company's management is responsible for maintaining an effective internal control system to ensure the accuracy of its financial statements.

LOS 27.d

Reporting standards are designed to ensure that different firms' statements are comparable to one another and to narrow the range of reasonable estimates on which financial statements are based. This aids users of the financial statements who rely on them for information about the company's activities, profitability, and creditworthiness. An analyst needs to be aware of differences between IFRS and US GAAP when comparing companies in different jurisdictions.

An analyst should be aware of evolving financial reporting standards and new products and innovations that generate new types of transactions.

LOS 27.e

Along with the annual financial statements, important information sources for an analyst include a company's quarterly and semiannual reports, proxy statements, press releases, and earnings guidance, as well as information on the industry and peer companies from external sources.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 27.1

- . **B** This statement describes the role of an auditor, rather than the role of an analyst. The other responses describe the role of financial statement analysis. (LOS 27.b)

- . **B** Information about accounting methods and estimates is contained in the footnotes to the financial statements. (LOS 27.c)
- . **C** An auditor will issue a qualified opinion if the financial statements make any exceptions to applicable accounting standards and will explain the effect of these exceptions in the auditor's report. (LOS 27.c)
- . **B** Proxy statements contain information related to matters that come before shareholders for a vote, such as elections of board members. (LOS 27.c)
- . **B** Determining the suitability of an investment for a client is not one of the six steps in the financial statement analysis framework. The analyst would only perform this function if he also had an advisory relationship with the client. Stating the objective and processing the data are two of the six steps in the framework. The others are gathering the data, analyzing the data, updating the analysis, and reporting the conclusions. (LOS 27.a)
- . **B** Firms are not required to provide detailed financial statements for segments that would include line items such as cost of goods sold, but the following should be disclosed in the segment data:
 - Revenue (external and between segments)
 - A measure of profit or loss
 - A measure of assets and liabilities
 - Interest (revenue and expense)
 - Acquisitions of PP&E and intangibles
 - Depreciation and amortization
 - Other noncash expenses
 - Income tax expense
 - Share of equity-accounted investments results
 (LOS 27.c)
- . **A** Research reports provided by analysts are rarely in the public domain. Trade journals and government statistics are available publicly. (LOS 27.e)

¹ Hennie van Greuning and Sonja Brajovic Bratanovic, *Analyzing and Managing Banking Risk: Framework for Assessing Corporate Governance and Financial Risk*, International Bank for Reconstruction and

Development, April 2003, p. 300.

² Boeing, *The Boeing Company 2016 Annual Report* (USA: 2017),
https://s2.q4cdn.com/661678649/files/doc_financials/annual/2016/2016-Annual-Report.pdf.

READING 28

ANALYZING INCOME STATEMENTS

MODULE 28.1: REVENUE RECOGNITION

LOS 28.a: Describe general principles of revenue recognition, specific revenue recognition applications, and implications of revenue recognition choices for financial analysis.

In a sale of goods where the goods or services are exchanged for cash and returns are not allowed, the recognition of revenue is straightforward: it is recognized at the time of the exchange. The recognition of revenue is not, however, dependent on receiving cash payment. If a sale of goods is made on credit, revenue can be recognized at the time of sale—and an asset, **accounts receivable**, is created on the balance sheet. As a general rule, revenue is recognized in the period in which it is earned, which may not necessarily be the same as the period in which cash is collected from the customer. Revenue is reported net of any returns and allowances in the income statement (e.g., estimated warranty provisions and customer discounts).

If payment for the goods is received before the transfer of the goods or services, a liability, **unearned revenue**, is created when the cash is received (offsetting the increase in the asset *cash*). Revenue is recognized as the goods are transferred to the buyer. As an example, consider a magazine subscription. When the subscription is purchased, an unearned revenue liability is created, and as magazine issues are delivered, revenue is recorded and the liability is decreased.

Converged standards under IFRS and US GAAP take a principles-based approach to revenue recognition issues. The central principle is that a firm should recognize revenue when it has transferred a good or service to a customer. This is consistent with the familiar accrual accounting principle that revenue should be recognized when earned.

The converged standards identify a five-step process¹ for recognizing revenue:

- Step 1:* Identify the contract(s) with a customer.
- Step 2:* Identify the separate or distinct performance obligations in the contract.
- Step 3:* Determine the transaction price.
- Step 4:* Allocate the transaction price to the performance obligations in the contract.
- Step 5:* Recognize revenue when (or as) the entity satisfies a performance obligation.

The standard defines a **contract** as an agreement between two or more parties that specifies their obligations and rights. Collectability must be probable for a contract to exist, but *probable* is defined differently under IFRS and US GAAP, so an identical activity could still be accounted for differently by IFRS and US GAAP reporting firms.

A **performance obligation** is a promise to deliver a distinct good or service. A *distinct* good or service is one that meets the following criteria:

- | The customer can benefit from the good or service on its own or combined with other resources that are readily available.
- | The promise to transfer the good or service can be identified separately from any other promises.

A **transaction price** is the amount a firm expects to receive from a customer in exchange for transferring a good or service to the customer. A transaction price is usually a fixed amount, but it can also be variable (e.g., if it includes a bonus for early delivery).

A firm should recognize revenue only when it is highly probable that it will not have to reverse it. For example, a firm may need to recognize a liability for a refund obligation (and an offsetting asset for the right to returned goods) if revenue from a sale cannot be estimated reliably.

A firm recognizes revenue when the performance obligation is satisfied by transferring the control of the good or service from seller to buyer. Indicators that the customer has obtained control include physical possession by the customer, acceptance of the good or service by the customer, the customer taking on risk and benefits of ownership, the customer holding legal title, and the seller having a right of payment.

For long-term contracts, revenue is recognized based on a firm's progress toward completing a performance obligation over a period of time. Progress toward completion can be measured from the input side (e.g., using the percentage of completion costs incurred as of the statement date). Progress can also be measured from the output side, using engineering milestones or the percentage of total output delivered to date.

A performance obligation is satisfied over a period of time if any of the following three criteria are met:

The customer receives and benefits from the good or service over time as the supplier meets the obligations of the contract (e.g., service and maintenance contracts).

The supplier enhances an existing asset or creates a new asset that the customer controls over the period in which the asset is created or enhanced.

The asset has no alternative use for the supplier, and the supplier has the right to enforce payment for work completed to date (e.g., constructing equipment specific to the needs of a single customer).

The costs to secure a long-term contract, such as sales commissions, must be capitalized; that is, the expense for these costs is spread over the life of the contract.

The following summarizes some examples from IFRS 15 of appropriate revenue recognition under various circumstances.

EXAMPLE: Revenue recognition

1. Performance obligation and progress toward completion (long-term contracts)

A contractor agrees to build a warehouse for a price of \$10 million and estimates the total costs of construction at \$8 million. Although there are several *identifiable components* of the building (site preparation, foundation, electrical components, roof, etc.), these components are not *separate deliverables*, and the performance obligation is the completed building.

During the first year of construction, the builder incurs \$4 million of costs, 50% of the estimated total costs of completion. Based on this expenditure and a belief that the percentage of costs incurred represents an appropriate measure of progress toward completing the performance obligation, the

builder recognizes \$5 million (50% of the transaction price of \$10 million) as revenue for the year.

During the second year of construction, the contractor incurred an additional \$2 million in costs.

The percentage of total costs incurred over the first two years is now $(\$4 \text{ million} + \$2 \text{ million}) / \$8 \text{ million} = 75\%$. The total revenue to be recognized to date is $0.75 \times \$10 \text{ million} = \7.5 million . Because \$5 million of revenue had been recognized in Year 1, \$2.5 million $(= \$7.5 \text{ million} - \$5 \text{ million})$ of revenue will be recognized in Year 2.

This treatment is consistent with the percentage of completion method previously in use, although the new standards do not call it that.

2. Acting as an agent

Consider a travel agent who arranges a first-class ticket for a customer flying to Singapore. The ticket price is \$10,000, made by nonrefundable payment at purchase, and the travel agent receives a \$1,000 commission on the sale. Because the travel agent is not responsible for providing the flight and bears no inventory or credit risk, they are *acting as an agent*. Because they are an agent, rather than a *principal*, they should report revenue equal to their commission of \$1,000, the net amount of the sale. If they were a principal in the transaction, they would report revenue of \$10,000, the gross amount of the sale, and an expense of \$9,000 for the ticket. Note that while gross profit is the same (\$1,000) regardless of selling as a principal or agent, the gross profit margin is very different. If treated as a principal, the margin would be $\$1,000 / \$10,000 = 10\%$, but as an agent, the margin would be $\$1,000 / \$1,000 = 100\%$.

3. Franchising and licensing

Consider a fast food company that both operates restaurants and grants franchisees rights to operate restaurants using its brand name under license, and supplies franchisees with some products used in daily operations. As well as charging a license fee, the company also receives a royalty fee of 2% of the franchisee's turnover. Accounting standards require revenue to be split into categories that have similar characteristics (nature, amounts, timings, and risk factors).

The fast food company would disaggregate revenue into these categories:

- Revenue from company-owned restaurants
- Franchise royalties and fees
- Revenue from supplies to franchises (equipment and food materials)

Franchise fees often grant the franchisee the right to operate over numerous periods and would initially be treated as deferred revenue. Subsequently, it would be amortized to revenue in the income statement over the life of the franchise contract period.

Royalty fees would be included periodically in the franchisor's accounts when they become contractually payable by the franchisee under the terms of the contract.

4. Service versus license

Consider a software supplier that allows customers to purchase a license and install the software on their own machines, or subscribe to a cloud-based solution with access over the internet.

If customers purchase a license and install the software on their own systems, IFRS allows for two treatments:

- a. The software supplier will report revenue over the life of the contract.

Criteria:

- The software supplier will continue to update and enhance the software over the term of the license.
- Customers will be exposed to potential benefits or negative impacts from updates and enhancements.
- Updates and enhancements do not result in a transfer of goods or services.

- b. The software supplier will report revenue at the outset of the contract.

Criteria:

- The license grants the customer the right to use the software as it exists at the start of the contract ("sold as is").
- A separate contract exists for enhancements and updates to the software. The revenue for support services will be recognized when provided (typically over the life of the contract).

If customers access the software without taking physical possession of the software (i.e., cloud-based access), the contract is for a service, and revenue

should be recognized over the life of the contract.

5. Bill-and-hold agreements

Bill-and-hold agreements are a type of sales agreement that involves the customer paying for goods ahead of shipping. Typically, when a customer pays ahead of delivery, the revenue is treated as deferred, but revenue may be recorded before shipping if the supplier can demonstrate that its performance obligations are complete and the customer has control over the good. IFRS criteria include the following: the customer asked for the arrangement, the goods are identified as belonging to the customer, the goods are complete and ready for transfer to the customer, and the goods cannot be redirected to another customer.

Required disclosures under the converged standards include the following:

- ▮ Contracts with customers by category
- ▮ Assets and liabilities related to contracts, including balances and changes
- ▮ Outstanding performance obligations and the transaction prices allocated to them
- ▮ Management judgments used to determine the amount and timing of revenue recognition, including any changes to those judgments



MODULE QUIZ 28.1

1. The first step in the revenue recognition process is to:
 - A. determine the price.
 - B. identify the contract.
 - C. identify the obligations.
2. A contractor agrees to build a bridge for a total price of \$10 million. The project is expected to take four years to complete, at a total cost of \$6.5 million. After Year 1, costs of \$2.5 million have been incurred, and a further \$1 million in costs are incurred in Year 2. The client pays \$2 million in each of the first two years. The amount of revenue the contractor should recognize in Year 2 is *closest* to:
 - A. \$1.54 million.
 - B. \$2.00 million.
 - C. \$5.38 million.
3. A realtor sells one of its client's houses for \$1.4 million, earning a commission of 3%. The costs to the realtor associated with the sale are \$15,000. What is the realtor's gross profit for this transaction?
 - A. \$27,000.
 - B. \$42,000.

C. \$1,385,000.

MODULE 28.2: EXPENSE RECOGNITION

LOS 28.b: Describe general principles of expense recognition, specific expense recognition applications, implications of expense recognition choices for financial analysis and contrast costs that are capitalized versus those that are expensed in the period in which they are incurred.

Expenses are subtracted from revenue to calculate net income. According to the IASB, expenses are decreases in economic benefits during the accounting period in the form of outflows or depletions of assets, or incurrence of liabilities that result in decreases in equity other than those relating to distributions to equity participants.²

If the financial statements were prepared on a cash basis, neither revenue recognition nor expense recognition would be an issue. The firm would simply recognize cash received as revenue and cash payments as expense.

Under the accrual method of accounting, expense recognition is in the period in which the economic benefits of the expenditure are consumed. Three different methods are used to achieve this: the matching principle, expensing as incurred, and capitalization.

Using the **matching principle**, expenses to generate revenue are recognized in the same period as the revenue. Inventory provides a good example. Assume that inventory is purchased during the fourth quarter of one year and sold during the first quarter of the following year. Using the matching principle, both the revenue and the expense (cost of goods sold) are recognized in the first quarter of the second year, when the inventory is sold—not the period in which the inventory was purchased. Another example are goods that are sold with a warranty period. The estimated cost of repairing or replacing faulty goods under the warranty must be estimated and deducted from revenue at the time of sale, rather than when the actual costs are incurred.

Capitalization is an application of the matching principle whereby costs are initially capitalized as assets on the balance sheet and then expensed, using depreciation or amortization, to the income statement over the asset's life as its benefits are consumed. Property, plant, and equipment (PP&E) and intangible assets with finite lives are examples of this.

Not all expenses can be directly tied to revenue generation. These costs are known as **period costs**. Period costs, such as administrative costs, are expensed in the period incurred. For example, if a firm has occupied a leased premise, one year's worth of rent should be expensed to the income statement regardless of whether the rent has been paid.

An accounting policy that recognizes expenses later rather than sooner is seen as aggressive, while a policy that recognizes expenses earlier is conservative. Expensing is, therefore, more conservative than capitalization.

EXAMPLE: Inventory and the matching principle

Imagine a trading company that simply buys and resells goods. If the firm had 100 units of sales during the accounting period, we would expect to see cost of goods sold reflect 100 units of cost if we apply the matching principle. If the firm had 20 units of beginning inventory at the start of the year and purchased a further 90 units, the total number of units available for sale would be 110. The matching concept, therefore, requires the company to remove 10 units that are unsold from the income statement and store them in the balance sheet as a current asset (ending inventory). These 10 units will become the next period's beginning inventory.

Matching Cost of Goods Sold to the Number of Units Sold

	Units	Units
Sales		100
Beginning inventory	20	
Purchases	<u>90</u>	
Available for sale	110	
Ending inventory	<u>(10)</u>	Transfer to B/S
Cost of goods sold		100 Remains in I/S

Assume the firm had the following transactions in the period:

Purchase Made During the Period		Total
Purchase 1	20 units @ \$22 each	\$440
Purchase 2	30 units @ \$25 each	\$750
Purchase 3	30 units @ \$28 each	\$840
Purchase 4	10 units @ \$30 each	<u>\$300</u>
Total		\$2,330
Sales Made During the Period		
Sales	100 units @ \$35	\$3,500

In addition, assume the beginning inventory of 20 units had an associated cost of \$400.

The company has identified that of its 10 units of ending inventory, 8 units relate to Purchase 4, and 2 units relate to Purchase 3.

Ending inventory at cost	Total
2 units @ \$28 each	\$56
8 units @ \$30 each	<u>\$240</u>
Total	\$296

Income Statement	\$	\$
Revenue		3,500
Beginning inventory	400	
Purchases	<u>2,330</u>	
Available for sale	2,730	
Ending inventory	<u>(296)</u>	
Cost of goods sold		<u>2,434</u>
Gross profit		1,066

This approach ensures that the company matches the 100 units of sales with 100 units of cost. The \$296 of ending inventory will be a current asset in the company's balance sheet.



PROFESSOR'S NOTE

This example used the specific identification method to compute the cost of ending inventory. This method is typically used if a firm can identify exactly which items were sold and which items remain in inventory (e.g., an auto dealer records each vehicle sold or in inventory by its identification number).

If the specific cost of each item remaining in ending inventory at period-end cannot be identified, the company will rely on one of three cost flow methodologies to assign cost. The three methods are first-in, first-out (FIFO); last-in, first-out (LIFO); and the weighted average cost method. We will discuss these methods in more detail in our reading on Analysis of Inventories.

Capitalization vs. Expensing

When a firm makes an expenditure, it can either capitalize the cost as an asset on the balance sheet or expense the cost in the income statement in the period incurred. As a general rule, an expenditure that is expected to provide a future economic benefit over multiple accounting periods is capitalized; however, if the future economic benefit is unlikely or highly uncertain, the expenditure is expensed in the period incurred.

An expenditure that is capitalized is initially recorded as an asset on the balance sheet at cost, which is typically its fair value at acquisition plus any costs necessary to prepare the asset for use. Except for land and intangible assets with indefinite lives (such as acquisition goodwill), the cost is then allocated to the income statement over the life of the asset as **depreciation** expense (for tangible assets), **depletion** (for natural resources), or **amortization** expense (for intangible assets with finite lives). Depreciation, depletion, and amortization reduce the carrying value (net book value) of the asset in the balance sheet, and the expense reduces net income in the income statement. Alternatively, if an expenditure is immediately expensed, current-period pretax income is reduced by the amount of the expenditure.

Once an asset is capitalized, subsequent related expenditures that provide more future economic benefits (e.g., rebuilding the asset) are also capitalized. Subsequent expenditures that merely sustain the usefulness of the asset (e.g., regular maintenance) are expensed when incurred.

EXAMPLE: Capitalizing vs. expensing

Northwood Corp. purchased new equipment to be used in its manufacturing plant. The cost of the equipment was \$250,000, including \$5,000 freight and \$12,000 of taxes. In addition to the equipment cost, Northwood paid \$10,000 to install the equipment and \$7,500 to train its employees to use the equipment. Over the asset's life, Northwood paid \$35,000 for repairs and maintenance. At the end of five years, Northwood extended the life of the asset by rebuilding the equipment's motors at a cost of \$85,000.

What amounts should be capitalized on Northwood's balance sheet, and what amounts should be expensed in the period incurred?

Answer:

Northwood should capitalize all costs that provide future economic benefits, including the costs that are necessary to get the asset ready for use. Rebuilding

the equipment's motors extended its life; thus, it increased its future benefits.

Capitalized Costs	
Purchase price	\$250,000 (including freight and taxes)
Installation costs	10,000
Rebuilt motors	<u>85,000</u>
	\$345,000

Costs that do not provide future economic benefits are expensed in the period incurred. The initial training costs are not necessary to get the asset ready for use. Rather, the training costs are necessary to get the employees ready to use the asset. Thus, the training costs are immediately expensed. Repair and maintenance costs are operating expenditures that do not extend the life of the equipment.

Costs Expensed When Incurred	
Initial training costs	\$7,500
Repairs and maintenance	<u>35,000</u>
	\$42,500

EXAMPLE: Impact of capitalization on financial statements

Chair Ltd. was incorporated at the start of the year with the issuance of £40,000 of shares paid in full. The company immediately purchased new equipment to be used in the production process. The machinery cost £12,000 with an estimated useful economic life of four years and no salvage value. The equipment is depreciated using straight-line methodology in the accounts, and we will assume the accounting depreciation is tax deductible. Chair has no other assets and liabilities except cash and PP&E. Assume Chair has revenue of £30,000 per year and an operating profit margin of 40% before considering the impact of the equipment purchase. Chair is subject to a tax rate of 30% and pays no dividends.

Compare and contrast the impact on the financial statement if the cost of equipment is capitalized versus expensed.

Answer:

$$\text{straight-line depreciation of equipment} = \frac{\text{£12,000} - 0}{4 \text{ years}} = \text{£3,000 p.a.}$$

Income Statement: Capitalization of Equipment Cost

Year	1	2	3	4
	£	£	£	£
Revenue	30,000	30,000	30,000	30,000
Operating profit @ 40%	12,000	12,000	12,000	12,000
Depreciation	<u>(3,000)</u>	<u>(3,000)</u>	<u>(3,000)</u>	<u>(3,000)</u>
Income before tax	9,000	9,000	9,000	9,000
Tax @ 30%	<u>(2,700)</u>	<u>(2,700)</u>	<u>(2,700)</u>	<u>(2,700)</u>
Net income	6,300	6,300	6,300	6,300

Income Statement: Expensing of Equipment Cost

Year	1	2	3	4
	£	£	£	£
Revenue	30,000	30,000	30,000	30,000
Operating profit @ 40%	12,000	12,000	12,000	12,000
Equipment expense	<u>(12,000)</u>	<u>0</u>	<u>0</u>	<u>0</u>
Income before tax	0	12,000	12,000	12,000
Tax @ 30%	<u>0</u>	<u>(3,600)</u>	<u>(3,600)</u>	<u>(3,600)</u>
Net income	0	8,400	8,400	8,400

Capitalization spreads the expense of the equipment in the income statement over the life of the asset. Expensing results in the full cost of the equipment passing through the income statement in the first year, but none in subsequent years; in turn, this causes net income to be lower in the first year, but higher in the subsequent years. Capitalization results in less volatility of earnings when compared to expensing.

Balance Sheet: Capitalization of Equipment Cost

End of year	1	2	3	4
	£	£	£	£
Cash	37,300	46,600	55,900	65,200
PP&E (net)	<u>9,000</u>	<u>6,000</u>	<u>3,000</u>	<u>0</u>
Total assets	<u>46,300</u>	<u>52,600</u>	<u>58,900</u>	<u>65,200</u>
Share capital & additional paid-in capital (APIC)	40,000	40,000	40,000	40,000
Retained earnings	<u>6,300</u>	<u>12,600</u>	<u>18,900</u>	<u>25,200</u>
Total equity	<u>46,300</u>	<u>52,600</u>	<u>58,900</u>	<u>65,200</u>

Balance Sheet: Expensing of Equipment Cost

End of year	1	2	3	4
	£	£	£	£
Cash	40,000	48,400	56,800	65,200
PP&E (net)	<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>
Total assets	<u>40,000</u>	<u>48,400</u>	<u>56,800</u>	<u>65,200</u>
Share capital & APIC	40,000	40,000	40,000	40,000
Retained earnings	<u>0</u>	<u>8,400</u>	<u>16,800</u>	<u>25,200</u>
Total equity	<u>40,000</u>	<u>48,400</u>	<u>56,800</u>	<u>65,200</u>

In the earlier years, total assets and equity are higher if the firm capitalizes costs. This results from including the carrying value of the asset (its cost less accumulated depreciation) in the balance sheet. Equity is higher in Year 1 under capitalization due to higher net income, and therefore, higher retained earnings.

These differences narrow over the asset's life until they are identical at the end. This convergence is caused by two factors:

1. The carrying value of the asset decreases as it is depreciated under the capitalization approach.
2. Although net income is lower in Year 1 if the cost is expensed, it is higher in all the subsequent years.

Cash Flow Statement: Capitalization of Equipment Cost

Year	1	2	3	4
	£	£	£	£
Operating cash flow	9,300	9,300	9,300	9,300
Investing cash flow	(12,000)	0	0	0
Financing cash flow	<u>40,000</u>	<u>0</u>	<u>0</u>	<u>0</u>
Change in cash	<u>37,300</u>	<u>9,300</u>	<u>9,300</u>	<u>9,300</u>

We can state operating cash flow as net income plus noncash charges less working capital investment. The only noncash charge in this example is depreciation. Working capital investment is assumed to be zero in this example.

Opening cash	0	37,300	46,600	55,900
Change in cash	<u>37,300</u>	<u>9,300</u>	<u>9,300</u>	<u>9,300</u>
Closing cash	<u>37,300</u>	<u>46,600</u>	<u>55,900</u>	<u>65,200</u>

Cash Flow Statement: Expensing of Equipment Cost

Year	1	2	3	4
	£	£	£	£
Operating cash flow	0	8,400	8,400	8,400
Investing cash flow	0	0	0	0
Financing cash flow	<u>40,000</u>	<u>0</u>	<u>0</u>	<u>0</u>
Change in cash	40,000	8,400	8,400	8,400
Opening cash	0	40,000	48,400	56,800
Change in cash	<u>40,000</u>	<u>8,400</u>	<u>8,400</u>	<u>8,400</u>
Closing cash	<u>40,000</u>	<u>48,400</u>	<u>56,800</u>	<u>65,200</u>

Capitalization requires the investment in equipment to be treated as a cash outflow from investing activities, while expensing treats the cost of equipment as a cash outflow from operating activities. The reason total cash flow differs under the two approaches is that the full tax benefit of the cost is recognized in Year 1 with expensing, while it is spread over the life of the asset under capitalization. At the end of the asset's life, total cash is identical for both approaches.

Financial statement ratios: Capitalization vs. expensing

Ratio	Year 1	Year 2	Year 3	Year 4
Total asset turnover: capitalization	0.7	0.6	0.5	0.5
Total asset turnover: expensing	0.8	0.7	0.6	0.5
Net profit margin: capitalization	21%	21%	21%	21%
Net profit margin: expensing	0%	28%	28%	28%
Return on equity: capitalization	14%	13%	11%	10%
Return on equity: expensing	0%	19%	16%	14%

Total asset turnover (sales / average total assets) is lower if costs are capitalized due to higher balance sheet assets. As the asset's carrying value declines over time and retained earnings converge under the two approaches, the difference in total asset turnover between the two approaches declines.

Net profit margin (net income / sales) is higher in Year 1 if the costs are capitalized because the income statement contains depreciation of £3,000 compared to the full cost of £12,000 if costs are expensed. In subsequent years, expensing will give higher margins as depreciation continues to pass through the income statement when using capitalization, while no cost passes through the income statement after Year 1 if the cost was expensed.

Return on equity (net income / average stockholders' equity) is higher in Year 1 if the cost is capitalized, but lower in subsequent years when compared to expensing. Again, this is due to the impact on net income of depreciation under

capitalization versus the one-off charge to the income statement when expensing.

The preceding example demonstrates how the decision to capitalize or expense a cost reduces comparability among companies. Capitalization increases net income, ROE, and CFO in Year 1 at the expense of decreased values in subsequent years. Analysts should take care to identify different accounting treatments of significant expenditures when comparing companies or industries. Figure 28.1 summarizes the impacts of capitalizing versus expensing.

Figure 28.1: Financial Statement Effects

	Capitalizing	Expensing
Assets & equity	Higher	Lower
Net income (first year)	Higher	Lower
Net income (other years)	Lower	Higher
Income variability	Lower	Higher
ROA & ROE (first year)	Higher	Lower
ROA & ROE (other years)	Lower	Higher
Debt ratio & debt-to-equity	Lower	Higher
CFO	Higher	Lower
CFI	Lower	Higher

The example considered a one-off decision to capitalize or expense. If a company continues to capitalize rather than expense cost in future periods, the profit-enhancing effects of capitalization will continue, provided the capitalized costs in each period exceed depreciation expense.

Capitalized Interest

When a firm constructs an asset for its own use or, in limited circumstances, for resale, the interest that accrues during the construction period is capitalized as a part of the asset's cost. The reasons for capitalizing interest are to accurately measure the cost of the asset and to better match the cost with the revenues generated by the constructed asset. The treatment of construction interest is similar under US GAAP and IFRS.

Capitalized interest is not reported in the income statement as interest expense. Once construction interest is capitalized, the interest cost is allocated to the

income statement through depreciation expense (if the asset is held for use) or COGS (if the asset is held for sale).

Capitalizing interest results in it being reported in the cash flow statement as an outflow from investing activities. This contrasts with the usual treatment of interest paid, which is reported as an outflow from operating activities under US GAAP and can be an operating or financing outflow under IFRS.

For an analyst, both capitalized and expensed interest should be used when calculating interest coverage ratios to get a clearer picture of the company's solvency. Analysts should also adjust income by adding back any depreciation of capitalized interest.

EXAMPLE: Capitalization of interest

Willock AG is a German company specializing in large infrastructure projects. Due to the highly specialized nature of some equipment, Willock constructs equipment that it will use in projects. In the current period, Willock reports EBIT of €160 million and an interest expense of €80 million. Footnote disclosures reveal €20 million of interest has been capitalized in assets in the course of construction in the year, and that depreciation resulting from interest capitalized on assets constructed in prior years that are now in use was €10 million.

Calculate the interest coverage ratio before and after adjusting for capitalized interest.

Answer:

$$\text{interest coverage} = \frac{\text{EBIT}}{\text{interest expense}}$$

$$\text{before adjustment} = \frac{\text{€160 million}}{\text{€80 million}} = 2.0$$

$$\text{after adjustment} = \frac{\text{€160 million} + \text{€10 million}}{\text{€80 million} + \text{€20 million}} = 1.7$$

Additional depreciation in the income statement due to the capitalization of financing costs during the construction phase is reversed for assets that are now in use, increasing EBIT. Interest capitalized in the current period for equipment in the course of construction is added to interest expense.

EXAMPLE: Cash flow treatment of capitalized interest

Continuing the previous example, Willock reported CFO of €70 million and CFI of –€50 million. Willock includes interest paid within its computation of CFO, with the exception of capitalized interest on the construction of equipment. Assuming all interest has been paid by year-end and ignoring any tax implications, what was the impact of interest capitalization on CFO and CFI?

Answer:

€20 million interest was capitalized. Had this been expensed instead, CFI would have been €20 million higher, or –€30 million, and CFO would have been €20 million lower, or €50 million. No adjustment for depreciation is required as it is a noncash charge.

Research and Development Costs

With some exceptions, costs to create intangible assets are expensed as incurred. Important exceptions are research and development costs (under IFRS) and software development costs.

Under IFRS, **research costs**, which are costs aimed at the discovery of new scientific or technical knowledge and understanding, are expensed as incurred. However, **development costs** may be capitalized. Development costs are incurred to translate research findings into a plan or design of a new product or process. To recognize an intangible asset in development, a firm must show that it can complete the asset and intends to use or sell the completed asset, among other criteria.

Under US GAAP, both research and development costs are generally expensed as incurred. However, the costs of creating software for sale to others are treated in a manner similar to the treatment of research and development costs under IFRS. Costs incurred to develop software for sale to others are expensed as incurred until the product's technological feasibility has been established, after which the costs of developing a salable product are capitalized.

To make the financial statements comparable for a company that capitalizes development costs with one that expenses such costs, the income statement should be adjusted to include development costs as an expense, and any current amortization of development cost capitalized in the past should be removed. In the balance sheet, capitalized development costs should be removed, resulting in lower assets and equity. Adjusting the cash flow statement will require

capitalized costs to be removed from CFI and included in CFO, which will decrease CFO.

Bad Debt Expense and Warranty Expense Recognition

If a firm sells goods or services on credit or provides a warranty to the customer, the matching principle requires the firm to estimate bad debt expense or warranty expense. To do so, the firm recognizes the expense in the period of the sale, rather than a later period.

Implications for Financial Analysis

Like revenue recognition, expense recognition requires numerous estimates. Because estimates are involved, it is possible for firms to delay or accelerate the recognition of expenses. Delayed expense recognition increases current net income and is, therefore, more aggressive.

Analysts must consider the underlying reasons for a change in an expense estimate. If a firm decreases its bad debt expense, was it because its collection experience improved, or was it to manipulate net income?

Analysts should also compare a firm's estimates to those of other firms in its industry. If a firm's warranty expense is significantly less than that of a peer firm, is that a result of higher quality products, or is the firm's expense recognition more aggressive than that of the peer firm?

Firms disclose their accounting policies and significant estimates in the financial statement footnotes and in the management discussion and analysis (MD&A) section of the annual report.



MODULE QUIZ 28.2

1. If a company purchases an asset with future economic benefits that are highly uncertain, the company should:
 - A. expense the purchase.
 - B. use straight-line depreciation.
 - C. use an accelerated depreciation method.
2. Red Company immediately expenses its development costs, while Black Company capitalizes its development costs. All else equal, Red Company will:
 - A. show smoother reported earnings than Black Company.
 - B. report higher operating cash flow than Black Company.
 - C. report higher asset turnover than Black Company.

MODULE 28.3: NONRECURRING ITEMS

LOS 28.c: Describe the financial reporting treatment and analysis of non-recurring items (including discontinued operations, unusual or infrequent items) and changes in accounting policies.

The definition of **unusual or infrequent items** is obvious—these events are either unusual in nature or infrequent in occurrence and are material (significant enough to affect the opinions of financial statement users). Examples of items that could be considered unusual or infrequent include the following:

- | Gains or losses from the sale of assets or part of a business, if these activities are not a firm's ordinary operations
- | Impairments, write-offs, and write-downs
- | Restructuring costs

Unusual or infrequent items are included in income from continuing operations and are reported before tax.

Analysts should review unusual or infrequent items to determine whether they truly should be excluded when forecasting a firm's earnings. Some companies appear to be accident-prone and have "unusual or infrequent" losses every year or every few years.

A **discontinued operation** is one that management has decided to dispose of, but either has not yet done so, or has disposed of in the current year after the operation had generated income or losses. To be accounted for as a discontinued operation, the business must be physically and operationally distinct from the rest of the firm, in terms of assets, operations, and investing and financing activities.

The date when the company develops a formal plan for disposing of an operation is referred to as the *measurement date*, and the time between the measurement period and the actual disposal date is referred to as the *phaseout period*. Any income or loss from discontinued operations is reported separately in the income statement, net of tax, after income from continuing operations. Any past income statements presented must be restated, separating the income or loss from the discontinued operations. On the measurement date, the company will accrue any estimated loss during the phaseout period, and any estimated loss on the sale of

the business. Any expected gain on the disposal cannot be reported until after the sale is completed.

Analysis of discontinued operations is straightforward: They do not affect net income from continuing operations. For this reason, analysts should exclude discontinued operations when forecasting future earnings. However, the actual event of discontinuing a business segment or selling assets may provide information about the future cash flows of the firm.

Changes in Accounting Policies and Estimates

Accounting changes include changes in accounting policies, changes in accounting estimates, and prior-period adjustments. Such changes may require either **retrospective application** or **prospective application**. With retrospective application, any prior-period financial statements presented in a firm's current financial statements must be restated, applying the new policy to those statements as well as future statements. Retrospective application enhances the comparability of the financial statements over time. With prospective application, prior statements are not restated, and the new policies are applied only to future financial statements.

Standard-setting bodies, at times, issue a **change in accounting policy**. Sometimes, a firm may change which accounting policy it applies, for example, by changing its inventory costing method or capitalizing rather than expensing specific purchases. Unless it is impractical, changes in accounting policies require retrospective application.

In the recent change to revenue recognition standards, firms were given the option of *modified retrospective application*. This application does not require restatement of prior-period statements; however, beginning values of affected accounts are adjusted for the cumulative effects of the change.

Generally, a **change in accounting estimate** is the result of a change in management's judgment, usually due to new information. For example, management may change the estimated useful life of an asset because new information indicates that the asset has a longer or shorter life than originally expected. Changes in accounting estimates are applied prospectively and do not require the restatement of prior financial statements.

Accounting estimate changes typically do not affect cash flow. An analyst should review changes in accounting estimates to determine their impact on

future operating results.

Sometimes, a change from an incorrect accounting method to one that is acceptable under GAAP or IFRS is required. A correction of an accounting error made in previous financial statements is reported as a **prior-period adjustment** and requires retrospective application. Prior-period results are restated. Disclosure of the nature of any significant prior-period adjustment and its effect on net income is also required.

Prior-period adjustments usually involve errors or new accounting standards and do not typically affect cash flow. Analysts should review adjustments carefully because errors may indicate weaknesses in the firm's internal controls.

Changes in Scope and Exchange Rates

Accounting standards do not require firms to disclose the impact on their financial statements of changes in scope or exchange rates, but analysts must be alert to their effects. In this context, "changes in scope" refer to how acquiring another company affects the size of the combined entity. Mergers and acquisitions can dramatically reduce comparability of company financial statements before and after the acquisition date. If a company conducts overseas trade or owns overseas subsidiaries, fluctuating exchange rates can affect its financial statements because overseas sales and purchases and the income statements of overseas subsidiaries need to be converted to the company's reporting currency.



PROFESSOR'S NOTE

The Level II CFA curriculum addresses accounting and analysis of business combinations and foreign currency translation.



MODULE QUIZ 28.3

1. Changing an accounting estimate:
 - A. is reported prospectively.
 - B. requires restatement of all prior-period statements presented in the current financial statements.
 - C. is reported by adjusting the beginning balance of retained earnings for the cumulative effect of the change.
2. Which of the following transactions would *most likely* be reported below income from continuing operations, net of tax?
 - A. Gain or loss from the sale of equipment used in a firm's manufacturing operation.
 - B. A change from the accelerated method of depreciation to the straight-line method.

- C. The operating income of a physically and operationally distinct division that is currently for sale, but not yet sold.
3. Which of the following statements about nonrecurring items is *least accurate*?
- A. Discontinued operations are reported net of taxes, at the bottom of the income statement before net income.
 - B. Unusual or infrequent items are reported before taxes, above net income from continuing operations.
 - C. A change in accounting principle is reported in the income statement net of taxes, before net income.

MODULE 28.4: EARNINGS PER SHARE

LOS 28.d: Describe how earnings per share is calculated and calculate and interpret a company's basic and diluted earnings per share for companies with simple and complex capital structures including those with antidilutive securities.

Earnings per share (EPS) is one of the most commonly used corporate profitability performance measures for publicly traded firms (nonpublic companies are not required to report EPS data). EPS is reported only for shares of common stock (also known as ordinary stock).

A company may have either a simple or complex capital structure:

- ▮ A **simple capital structure** is one that contains *no* potentially dilutive securities. A simple capital structure contains only common stock, nonconvertible debt, and nonconvertible preferred stock.
- ▮ A **complex capital structure** contains *potentially dilutive securities* such as employee stock options, warrants, or convertible securities.

All firms with complex capital structures must report both *basic* and *diluted* EPS. Firms with simple capital structures report only basic EPS.

Basic EPS

The basic EPS calculation does not consider the effects of any dilutive securities in the computation of EPS:

$$\text{basic EPS} = \frac{\text{net income} - \text{preferred dividends}}{\text{weighted average number of common shares outstanding}}$$

The current year's preferred dividends are subtracted from net income because EPS refers to the per-share earnings *available to common shareholders*. Net

income minus preferred dividends is the income available to common stockholders. Common stock dividends are *not* subtracted from net income because they are a part of the net income available to common shareholders.

The **weighted average number of common shares** is the number of shares outstanding during the year, weighted by the portion of the year they were outstanding.

Effect of Stock Dividends and Stock Splits

A **stock dividend** is the distribution of additional shares to each shareholder in an amount proportional to their current number of shares. If a 10% stock dividend is paid, the holder of 100 shares of stock would receive 10 additional shares.

A **stock split** refers to the division of each “old” share into a specific number of “new” (post-split) shares. The holder of 100 shares will have 200 shares after a 2-for-1 split, or 150 shares after a 3-for-2 split.

The important thing to remember is that each shareholder’s proportional ownership in the company is unchanged by either of these events. Each shareholder has more shares but the same percentage of the total shares outstanding. For calculating EPS, we apply stock dividends and splits retroactively to the beginning of the year, to all shares before the date of the corporate event. Prior years’ weighted average number of common shares is also adjusted as if the stock split or stock dividend had occurred in the prior period, to prevent EPS from appearing to decline when these actions are largely cosmetic.



PROFESSOR’S NOTE

For our purposes here, a stock dividend and a stock split are two ways of doing the same thing. For example, a 50% stock dividend and a 3-for-2 stock split both result in three “new” shares for every two “old” shares.

The following are key things to know about calculating weighted average shares outstanding:

- | The weighting system is days outstanding divided by the number of days in a year, but on the exam, the monthly approximation method will probably be used.
- | Shares issued enter into the computation from the date of issuance.

- Reacquired shares are excluded from the computation from the date of reacquisition.
- Shares sold or issued in a purchase of assets are included from the date of issuance.
- A stock split or stock dividend is applied to all shares outstanding before the split or dividend and to the beginning-of-period weighted average shares. A stock split or stock dividend adjustment is not applied to any shares issued or repurchased after the split or dividend date.

EXAMPLE: Weighted average shares outstanding

Johnson Company has 10,000 shares outstanding at the beginning of the year. On April 1, Johnson issues 4,000 new shares. On July 1, Johnson distributes a 10% stock dividend. On September 1, Johnson repurchases 3,000 shares. Calculate Johnson's weighted average number of shares outstanding for the year, for its reporting of basic earnings per share.

Answer:

Shares outstanding are weighted by the portion of the year that the shares were outstanding. Any shares that were outstanding before the 10% stock dividend must be adjusted for it. Transactions that occur after the stock dividend do not need to be adjusted.



PROFESSOR'S NOTE

Think of the shares before the stock dividend as “old” shares and shares after the stock dividend as “new” shares that each represent ownership of a smaller portion of the company (in this example, 10/11ths of that of an old [pre-stock dividend] share). The weighted average number of shares for the year will be in new shares.

Shares outstanding on January 1: $10,000 \times 1.10 \times 12/12$ of the year = 11,000

Shares issued April 1: $4,000 \times 1.10 \times 9/12$ of the year = 3,300

Shares repurchased September 1: $-3,000 \times 4/12$ of the year = -1,000

Weighted average shares outstanding = 13,300

EXAMPLE: Basic earnings per share

Johnson Company has net income of \$10,000, paid \$1,000 cash dividends to its preferred shareholders, and paid \$1,750 cash dividends to its common

shareholders. Calculate Johnson's basic EPS using the weighted average number of shares from the previous example.

Answer:

$$\begin{aligned}\text{basic EPS} &= \frac{\text{net income} - \text{preferred dividends}}{\text{weighted average shares of common}} \\ &= \frac{\$10,000 - \$1,000}{13,300} = \$0.68\end{aligned}$$



PROFESSOR'S NOTE

Remember, the payment of a cash dividend on common shares is not considered in the calculation of EPS.

Diluted EPS

Before calculating diluted EPS, it is necessary to understand the following terms:

- | **Dilutive securities** are stock options, warrants, convertible debt, or convertible preferred stock that would *decrease* EPS if exercised or converted to common stock.
- | **Antidilutive securities** are stock options, warrants, convertible debt, or convertible preferred stock that would *increase* EPS if exercised or converted to common stock.

We apply the “if converted” method to examine the impact of potentially dilutive securities. This looks at what EPS would have been if the securities were converted at the start of the accounting period, regardless of whether they can actually be converted during the period.

The numerator of the basic EPS equation contains income available to common shareholders (net income less preferred dividends). In the case of diluted EPS, if there are dilutive securities, the numerator must be adjusted as follows:

- | If convertible preferred stock is dilutive (meaning EPS will decrease if it is converted to common stock), the convertible preferred dividends must be added to earnings available to common shareholders.
- | If convertible bonds are dilutive, then the bonds' after-tax interest expense is not considered an interest expense for diluted EPS. Hence, interest expense multiplied by (1 – the tax rate) must be added back to the numerator.



PROFESSOR'S NOTE

Interest paid on bonds is typically tax deductible for the firm. If convertible bonds are converted to stock, the firm saves the interest cost but loses the tax deduction. Thus, only the after-tax interest savings are added back to income available to common shareholders.

The basic EPS denominator is the weighted average number of shares. When the firm has dilutive securities outstanding, the denominator is adjusted for the equivalent number of common shares that would be created by the conversion of all dilutive securities outstanding (convertible bonds, convertible preferred shares, warrants, and options), with each one considered separately to determine if it is dilutive.

If a dilutive security was issued during the year, the increase in the weighted average number of shares for diluted EPS is based on only the portion of the year the dilutive security was outstanding.

Dilutive stock options or warrants increase the number of common shares outstanding in the denominator for diluted EPS. There is no adjustment to the numerator.

The **diluted EPS equation** is:

$$\text{diluted EPS} = \frac{\text{adjusted income available for common shares}}{\text{weighted-average common and potential common shares outstanding}}$$

where:

$$\begin{aligned} \text{adjusted income available for common shares} = & \text{net income} - \text{preferred dividends} \\ & + \text{dividends on convertible preferred stock} \\ & + \text{after-tax interest on convertible debt} \end{aligned}$$

Therefore, diluted EPS is:

$$\text{diluted EPS} = \frac{\left[\text{net income} - \text{preferred dividends} \right] + \left[\text{convertible preferred dividends} \right] + \left(\frac{\text{convertible debt interest}}{\text{interest}} \right) (1 - t)}{\left(\frac{\text{weighted average shares}}{\text{shares}} \right) + \left(\frac{\text{shares from conversion of conv. pfd. shares}}{\text{conv. pfd. shares}} \right) + \left(\frac{\text{shares from conversion of conv. debt}}{\text{conv. debt}} \right) + \left(\frac{\text{shares issuable from stock options}}{\text{stock options}} \right)}$$

The effect of conversion to common shares is included in the calculation of diluted EPS for a given security only if it is, in fact, dilutive. If a firm has more than one potentially dilutive security outstanding, each potentially dilutive security must be examined separately to determine if it is actually dilutive (i.e., would reduce EPS if converted to common stock).

EXAMPLE: EPS with convertible preferred stock

During 20X6, ZZZ Corp. reported net income of \$4.35 million and had 2 million shares of common stock outstanding for the entire year. ZZZ's 7%, \$5 million par value preferred stock is convertible into common stock at a conversion rate of 1.1 shares for every \$10 of par value. Calculate basic and diluted EPS.

Answer:

Step 1: Calculate 20X6 basic EPS:

$$\text{basic EPS} = \frac{\$4,350,000 - (0.07) (\$5,000,000)}{2,000,000} = \$2.00$$

Step 2: Calculate diluted EPS:

- Compute the increase in common stock outstanding if the preferred stock is converted to common stock at the beginning of 20X6: $(\$5,000,000 / \$10) \times 1.1 = 550,000$ shares.
- If the convertible preferred shares were converted to common stock, there would be no preferred dividends paid. Therefore, you should add back the convertible preferred dividends that had previously been subtracted from net income in the numerator.
- Compute diluted EPS as if the convertible preferred stock were converted into common stock:

$$\begin{aligned} \text{diluted EPS} &= \frac{\text{net. inc.} - \text{pref. div.} + \text{convert. pref. dividends}}{\text{wt. avg. shares} + \text{convert. pref. common shares}} \\ \text{diluted EPS} &= \frac{\$4,350,000}{2,000,000 + 550,000} = \$1.71 \end{aligned}$$

- Check to see if diluted EPS is less than basic EPS ($\$1.71 < \2.00). If the answer is yes, the preferred stock is dilutive and must be included in diluted EPS as computed previously. If the answer is no, the preferred stock is antidilutive, and conversion effects are not included in diluted EPS.

A quick way to check whether convertible preferred stock is dilutive is to divide the preferred dividend by the number of shares that will be created if the preferred stock is converted. For ZZZ:

$$\frac{\$5,000,000 \times 0.07}{550,000} = \$0.64$$

Because this is less than basic EPS, the convertible preferred is dilutive.

EXAMPLE: EPS with convertible debt

During 20X6, YYY Corp. had earnings available to common shareholders of \$2.5 million and had 1 million shares of common stock outstanding for the entire year, for basic EPS of \$2.50. During 20X5, YYY issued 2,000, \$1,000 par, 5% bonds for \$2 million (issued at par). Each of these bonds is convertible to 120 shares of common stock. The tax rate is 30%. Calculate the 20X6 diluted EPS.

Answer:

Compute the increase in common stock outstanding if the convertible debt is converted to common stock at the beginning of 20X6:

$$\text{shares issuable for debt conversion} = (2,000)(120) = 240,000 \text{ shares}$$

If the convertible debt is considered converted to common stock at the beginning of 20X6, then there would be no interest expense related to the convertible debt. Therefore, it is necessary to increase YYY's after-tax net income for the after-tax effect of the decrease in interest expense:

$$\text{increase in income} = [(2,000)(\$1,000)(0.05)] (1 - 0.30) = \$70,000$$

Compute diluted EPS as if the convertible debt were common stock:

$$\text{diluted EPS} = \frac{\text{net. inc.} - \text{pref. div.} + \text{convert. int.} (1 - t)}{\text{wt. avg. shares} + \text{convertible debt shares}}$$

$$\text{diluted EPS} = \frac{\$2,500,000 + \$70,000}{1,000,000 + 240,000} = \$2.07$$

Check to make sure that *diluted EPS is less than basic EPS* (\$2.07 < \$2.50). If diluted EPS is more than the basic EPS, the convertible bonds are *antidilutive* and should not be treated as common stock in computing diluted EPS.

A quick way to determine whether the convertible debt is dilutive is to calculate its per share impact:

$$\frac{\text{convertible debt interest} (1 - t)}{\text{convertible debt shares}}$$

If this per share amount is greater than basic EPS, the convertible debt is antidilutive, and the effects of conversion should not be included when calculating diluted EPS.

If this per share amount is less than basic EPS, the convertible debt is dilutive, and the effects of conversion should be included in the calculation of diluted EPS.

For YYY:

$$\frac{\$70,000}{240,000} = \$0.29$$

The company's basic EPS is \$2.50, so the convertible debt is dilutive, and the effects of conversion should be included in the calculation of diluted EPS.

Stock options and warrants are dilutive only when their exercise prices are less than the average market price of the stock over the year. If the options or warrants are dilutive, use the **treasury stock method** to calculate the number of shares used in the denominator:

- ▮ The treasury stock method assumes that the funds received by the company from the exercise of the options would be used to hypothetically purchase shares of the company's common stock in the market at the average market price.
- ▮ The net increase in the number of shares outstanding (the adjustment to the denominator) is the number of shares created by exercising the options less the number of shares hypothetically repurchased with the proceeds of exercise.

EXAMPLE: EPS with stock options

During 20X6, XXX Corp. reported earnings available to common shareholders of \$1.2 million and had 500,000 shares of common stock outstanding for the entire year, for basic EPS of \$2.40. XXX has 100,000 stock options outstanding the entire year. Each option allows its holder to purchase one share of common stock at \$15 per share. The average market price of XXX's common stock during 20X6 is \$20 per share. Calculate diluted EPS.

Answer:

Number of common shares created if the options are exercised = 100,000

Cash inflow if the options are exercised = \$15 per share × 100,000 shares = \$1,500,000

Number of shares that can be purchased with these funds = \$1,500,000 / \$20 = 75,000 shares

$$\text{diluted EPS} = \frac{\$1,200,000}{500,000 + 25,000} = \$2.29$$

A quick way to calculate the net increase in common shares from the potential exercise of stock options or warrants when the exercise price is less than the

average market price is:

$$\left[\frac{\text{AMP} - \text{EP}}{\text{AMP}} \right] \times N$$

where:

AMP = average market price over the year

EP = exercise price of the options or warrants

N = number of common shares that the options and warrants can be converted into

For XXX:

$$\frac{\$20 - \$15}{\$20} \times 100,000 \text{ shares} = 25,000 \text{ shares}$$



MODULE QUIZ 28.4

1. Hall Corp. had 100,000 shares of common stock outstanding at the beginning of the year. Hall issued 30,000 shares of common stock on May 1. On July 1, the company issued a 10% stock dividend. On September 1, Hall issued 1,000, 10% bonds, each convertible into 21 shares of common stock. What is the weighted average number of shares to be used in computing diluted EPS, assuming the convertible bonds are dilutive?
 - A. 132,000.
 - B. 139,000.
 - C. 146,000.
2. An analyst has gathered the following information about a company:
 - 300,000 shares outstanding
 - 100,000 warrants exercisable at \$50 per share
 - Average share price: \$55
 - Year-end share price: \$60How many shares should be used in computing diluted EPS?
 - A. 9,091.
 - B. 90,909.
 - C. 309,091.
3. An analyst has gathered the following information about a company:
 - 100,000 common shares outstanding from the beginning of the year
 - Earnings: \$125,000
 - 1,000, 7%, \$1,000 par bonds convertible into 25 shares each, outstanding as of the beginning of the year
 - Tax rate: 40%The company's diluted EPS is *closest* to:
 - A. \$1.22.
 - B. \$1.25.

- C. \$1.34.
4. An analyst has gathered the following information about a company:
- 50,000 common shares outstanding from the beginning of the year
 - Warrants outstanding all year on 50,000 shares, exercisable at \$20 per share
 - Stock is selling at year-end for \$25
 - Average price of the company's stock for the year was \$15
- How many shares should be used in calculating the company's diluted EPS?
- A. 16,667.
B. 50,000.
C. 66,667.

MODULE 28.5: RATIOS AND COMMON-SIZE INCOME STATEMENTS

LOS 28.e: Evaluate a company's financial performance using common-size income statements and financial ratios based on the income statement.

Common-Size Income Statements

A vertical **common-size income statement** expresses each category of the income statement as a percentage of revenue. The common-size format standardizes the income statement by eliminating the effects of size. This allows for comparison of income statement items over time (time-series analysis) and across firms (cross-sectional analysis). For example, the following are year-end income statements of industry competitors North Company and South Company:

	North Co.	South Co.
Revenue	\$75,000,000	\$3,500,000
Cost of goods sold	<u>52,500,000</u>	<u>700,000</u>
Gross profit	\$22,500,000	\$2,800,000
Administrative expense	11,250,000	525,000
Research expense	<u>3,750,000</u>	<u>700,000</u>
Operating profit	\$7,500,000	\$1,575,000

Notice that North is significantly larger and more profitable than South when measured in absolute dollars. North's gross profit is \$22,500,000, as compared to South's gross profit of \$2,800,000. Similarly, North's operating profit of \$7,500,000 is significantly greater than South's operating profit of \$1,575,000.

Once we convert the income statements to common-size format, we can see that South is the more profitable firm on a relative basis. South's gross profit of 80% and operating profit of 45% are significantly greater than North's gross profit of 30% and operating profit of 10%.

	North Co.	South Co.
Revenue	100%	100%
Cost of goods sold	<u>70%</u>	<u>20%</u>
Gross profit	30%	80%
Administrative expense	15%	15%
Research expense	<u>5%</u>	<u>20%</u>
Operating profit	10%	45%

Common-size analysis can also be used to examine a firm's strategy. South's higher gross profit margin may be the result of technologically superior products. Notice that South spends more on research than North on a relative basis. This may allow South to charge a higher price for its products.

In most cases, expressing expenses as a percentage of revenue is appropriate. One exception is income tax expense. Tax expense is more meaningful when expressed as a percentage of pretax income. The result is known as the **effective tax rate**.

Income Statement Ratios

Margin ratios can be used to measure a firm's profitability quickly. **Gross profit margin** is the ratio of gross profit (revenue minus cost of goods sold) to revenue (sales):

$$\text{gross profit margin} = \frac{\text{gross profit}}{\text{revenue}}$$

Gross profit margin can be increased by raising prices or reducing production costs. A firm might be able to increase prices if its products can be differentiated from other firms' products as a result of factors such as brand names, quality, technology, or patent protection. This was illustrated in the previous example whereby South's gross profit margin was higher than North's.

Another popular margin ratio is **net profit margin**. Net profit margin is the ratio of net income to revenue:

$$\text{net profit margin} = \frac{\text{net income}}{\text{revenue}}$$

Net profit margin measures the profit generated after considering all expenses. Like gross profit margin, net profit margin should be compared over time and with the firm's industry peers.

Any subtotal found in the income statement can be expressed as a percentage of revenue. For example, operating profit divided by revenue is known as **operating profit margin**. Pretax accounting profit divided by revenue is known as **pretax margin**.



MODULE QUIZ 28.5

1. A vertical common-size income statement expresses each category of the income statement as a percentage of:
 - A. assets.
 - B. gross profit.
 - C. revenue.
2. Which of the following would *most likely* result in higher gross profit margin, assuming no fixed costs?
 - A. A 10% increase in the number of units sold.
 - B. A 5% decrease in production cost per unit.
 - C. A 7% decrease in administrative expenses.

KEY CONCEPTS

LOS 28.a

Revenue is recognized when earned, and expenses are recognized when incurred.

Accounting standards identify a five-step process for recognizing revenue:

- Step 1:* Identify the contract(s) with a customer.
- Step 2:* Identify the performance obligations in the contract.
- Step 3:* Determine the transaction price.
- Step 4:* Allocate the transaction price to the performance obligations in the contract.
- Step 5:* Recognize revenue when (or as) the entity satisfies a performance obligation.

Information that can influence the choice of revenue recognition method includes progress toward completion of a performance obligation, variable

considerations and their likelihood of being earned, revisions to contracts, and whether the firm is acting as a principal or an agent in a transaction.

LOS 28.b

The matching principle requires that firms match revenues recognized in a period with the expenses required to generate them. One application of the matching principle is seen in accounting for inventory, with cost of goods sold as the cost of units sold from inventory that are included in current-period revenue. Other costs, such as depreciation of fixed assets or administrative overhead, are period costs, and they are taken without regard to revenues generated during the period.

With capitalization, the asset value is put on the balance sheet, and the cost is expensed through the income statement over the asset's useful life through either depreciation or amortization. Compared to expensing, capitalization results in the following:

- | Lower expense and higher net income in period of acquisition, and higher expense (depreciation or amortization) and lower net income in each of the remaining years of the asset's life
- | Higher assets and equity
- | Lower CFI and higher CFO because the cost of a capitalized asset is classified as an investing cash outflow
- | Higher ROE and ROA in the initial period, and lower ROE and ROA in subsequent periods because net income is lower and both assets and equity are higher
- | Lower debt-to-assets and debt-to-equity ratios because assets and equity are higher

Users of financial data should analyze the reasons for any changes in estimates of expenses and compare these estimates with those of peer companies.

LOS 28.c

Results of discontinued operations are reported below income from continuing operations, net of tax, from the date the decision to dispose of the operations is made. These results are segregated because they likely are nonrecurring and do not affect future net income.

Unusual or infrequent items are reported before tax and above income from continuing operations. An analyst should determine how “unusual” or “infrequent” these items really are for the company when estimating future earnings or firm value.

Changes in accounting standards, changes in accounting methods applied, and corrections of accounting errors require retrospective restatement of all prior-period financial statements included in the current statement. A change in an accounting estimate, however, is applied prospectively (to subsequent periods) with no restatement of prior-period results.

LOS 28.d

$$\text{basic EPS} = \frac{\text{net income} - \text{preferred dividends}}{\text{weighted average number of common shares outstanding}}$$

When a company has potentially dilutive securities, it must report diluted EPS. For any convertible preferred stock, convertible debt, warrants, or stock options that are dilutive, the calculation of diluted EPS is as follows:

$$\text{diluted EPS} = \frac{\left[\text{net income} - \text{preferred dividends} \right] + \left[\begin{array}{c} \text{convertible} \\ \text{preferred} \\ \text{dividends} \end{array} \right] + \left(\begin{array}{c} \text{convertible} \\ \text{debt} \\ \text{interest} \end{array} \right) (1 - t)}{\left(\begin{array}{c} \text{weighted} \\ \text{average} \\ \text{shares} \end{array} \right) + \left(\begin{array}{c} \text{shares from} \\ \text{conversion of} \\ \text{conv. pfd. shares} \end{array} \right) + \left(\begin{array}{c} \text{shares from} \\ \text{conversion of} \\ \text{conv. debt} \end{array} \right) + \left(\begin{array}{c} \text{shares} \\ \text{issuable from} \\ \text{stock options} \end{array} \right)}$$

A dilutive security is one that, if converted to its common stock equivalent, would decrease EPS. An antidilutive security is one that would not reduce EPS if converted to its common stock equivalent.

LOS 28.e

A vertical common-size income statement expresses each item as a percentage of revenue. The common-size format standardizes the income statement by eliminating the effects of size. Common-size income statements are useful for trend analysis and for comparisons with peer firms.

Two popular profitability ratios are gross profit margin (gross profit / revenue) and net profit margin (net income / revenue). A firm can often achieve higher profit margins by differentiating its products from the competition.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 28.1

- . **B** The five steps in revenue recognition are as follows:

Step 1: Identify the contract or contracts with the customer.
Step 2: Identify the performance obligations in the contract(s).
Step 3: Determine a transaction price.
Step 4: Allocate the transaction price to the performance obligations.
Step 5: Recognize revenue when (or as) the performance obligations have been satisfied.

(LOS 28.a)

- . **A** In Year 1, the contractor has incurred 38.5% (\$2.5 million / \$6.5 million) of expected total costs and should recognize 38.5% of total revenue (\$10 million $\times$ 38.5% = \$3.85 million). By the end of Year 2, a total of \$3.5 million costs have been incurred, 53.8% of expected total costs. Cumulatively, by the end of Year 2, total revenue of 53.8% $\times$ \$10 million = \$5.38 million should have been recognized, leaving \$1.53 million (\$5.38 million – \$3.85 million) to be recognized in Year 2. The amount paid by the client does not affect revenue. (LOS 28.a)
- . **A** The realtor is acting as an agent, and so should recognize only the commission as revenue. Revenue should therefore be: (\$1.4 million $\times$ 3%) = \$42,000. Gross profit will be \$42,000 – \$15,000 = \$27,000. (LOS 28.a)

Module Quiz 28.2

- . **A** If the future economic benefits of a purchase are highly uncertain, a company should expense the purchase in the period it is incurred. (LOS 28.b)
- . **C** As compared to a firm that capitalizes its expenditures, a firm that immediately expenses expenditures will report lower assets. Thus, asset turnover (revenue / average assets) will be higher for the expensing firm (lower denominator). (LOS 28.b)

Module Quiz 28.3

- . **A** A change in an accounting estimate is reported prospectively. No restatement of prior-period statements is necessary. (LOS 28.c)
- . **C** A physically and operationally distinct division that is currently for sale is

treated as a discontinued operation. The income from the division is reported net of tax below income from continuing operations. Gains and losses on sales of operating assets, as well as depreciation expense, are reported pretax, above income from continuing operations. (LOS 28.c)

- . **C** A change in accounting principle requires retrospective application; that is, all prior-period financial statements currently presented are restated to reflect the change. (LOS 28.c)

Module Quiz 28.4

- . **B** The new stock is weighted by 8 / 12. The bonds are weighted by 4 / 12 and are not affected by the stock dividend.

$$\text{basic shares} = \{[100,000 \times (12 / 12)] + [30,000 \times (8 / 12)]\} \times 1.10 = 132,000$$

$$\text{diluted shares} = 132,000 + [21,000 \times (4 / 12)] = 139,000$$

(LOS 28.d)

- . **C** Because the exercise price of the warrants is less than the average share price, the warrants are dilutive. Use the Treasury stock method to determine the denominator impact:

$$\frac{\$55 - \$50}{\$55} \times 100,000 \text{ shares} = 9,091 \text{ shares}$$

Thus, the denominator will increase by 9,091 shares to 309,091 shares. The question asks for the total, not just the impact of the warrants. (LOS 28.d)

- . **B** First, calculate basic EPS:

$$\frac{\$125,000}{100,000} = \$1.25$$

Next, check if the convertible bonds are dilutive:

$$\text{numerator impact} = (1,000 \times 1,000 \times 0.07) \times (1 - 0.4) = \$42,000$$

$$\text{denominator impact} = (1,000 \times 25) = 25,000 \text{ shares}$$

$$\text{per share impact} = \frac{\$42,000}{25,000 \text{ shares}} = \$1.68$$

Because \$1.68 is greater than the basic EPS of \$1.25, the bonds are antidilutive. Thus, diluted EPS = basic EPS = \$1.25. (LOS 28.d)

- . **B** The warrants, in this case, are antidilutive. The average price per share of \$15 is less than the exercise price of \$20. The year-end price per share is not relevant. The denominator consists of only the common stock for basic EPS. (LOS 28.d)

Module Quiz 28.5

- . **C** Each category of the income statement is expressed as a percentage of revenue (sales). (LOS 28.e)
- . **B** A 5% decrease in per unit production cost will increase gross profit by reducing the cost of goods sold. Assuming no fixed costs, gross profit margin will remain the same if sale quantities increase. Administrative expenses are not included in gross profit margin. (LOS 28.e)

¹ IFRS 15, *Revenue From Contracts With Customers*.

² IASB *Framework for the Preparation and Presentation of Financial Statements*, paragraph 4.25(b).

READING 29

ANALYZING BALANCE SHEETS

MODULE 29.1: INTANGIBLE ASSETS AND MARKETABLE SECURITIES

LOS 29.a: Explain the financial reporting and disclosures related to intangible assets.

Intangible assets are nonmonetary assets that lack physical substance. Securities are not considered intangible assets. Intangible assets are either identifiable or unidentifiable. **Identifiable intangible assets** can be acquired separately or are the result of rights or privileges conveyed to their owner. Examples of identifiable intangibles are patents, trademarks, and copyrights. **Unidentifiable intangible assets** cannot be acquired separately and may have an unlimited life. The best example of an unidentifiable intangible asset is goodwill.

Under International Financial Reporting Standards (IFRS), identifiable intangibles that are *purchased* can be reported on the balance sheet using the cost model or the revaluation model, although the revaluation model can only be used if an active market for the intangible asset exists. Both models are basically the same as the measurement models used for property, plant, and equipment. Under US GAAP, only the cost model is allowed.

Except for certain legal costs, intangible assets that are *created internally*, such as research and development costs, are expensed as incurred under US GAAP. Under IFRS, a firm must identify the research stage (discovery of new scientific or technical knowledge) and the development stage (using research results to plan or design products). The firm must expense costs incurred during the research stage but can capitalize costs incurred during the development stage. Criteria a project must meet for the firm to capitalize development costs include the following: the project is technically feasible, the resources exist to complete

the project, a market exists for the product, and the company has the intention and resources to complete the project and sell the product.

Finite-lived intangible assets are amortized over their useful lives and tested for impairment in the same way as PP&E. The firm must review its amortization method and useful life estimates at least annually. Intangible assets with indefinite lives are not amortized, but they are tested for impairment at least annually.

Under IFRS and US GAAP, all of the following should be expensed as incurred:

- | Start-up and training costs
- | Administrative overhead
- | Advertising and promotion costs
- | Relocation and reorganization costs
- | Termination costs

Some analysts choose to eliminate intangible assets when they evaluate balance sheets. However, analysts should consider the value to the firm of each intangible asset before making any adjustments.

EXAMPLE: Measuring intangible assets

The R&D department of Lowe S.A. worked on two projects during the year:

- Project 1 aims to develop hydrogen fuel cells for motor vehicles. The company has not yet developed a working prototype propulsion unit, but it believes if successful, in the long run it could revolutionize the motor industry by producing environmentally clean vehicles.
- Project 2 is to develop a new type of catalytic converter, which would remove more particulates from exhaust fumes than those currently available. The company has developed a working prototype and is now working on a commercial version of the product. Lowe believes the demand from auto manufacturers would be high and has the resources to develop and launch the product.

R&D Department Period Costs

	Project 1	Project 2
	€m	€m
Materials	150	120
Direct labor	80	60
Production overhead	40	30
Administrative overhead	30	30

What costs will be capitalized, and what costs should be expensed, if Lowe reports under IFRS?

Answer:

Project 1: This project is not yet technically feasible, so is still in the research phase. Under IFRS and US GAAP, Lowe should expense all costs.

Project 2: The existence of a working prototype suggests the project is technically feasible and that the project is in the development phase. The company has the resources and intention to complete the project and believes there would be high customer demand when launched. Under IFRS, the development costs should be capitalized while US GAAP would require expensing.

Costs to be capitalized = materials + direct labor + production overhead (120 + 60 + 30) = €210 million.

Administrative costs are expensed as incurred.

LOS 29.b: Explain the financial reporting and disclosures related to goodwill.

Balance sheet **goodwill** results from acquiring another business. Goodwill is the amount by which the purchase price is greater than the fair value of the acquired company's identifiable net assets (assets minus liabilities).

Acquirers are often willing to pay more than the fair value of a target's identifiable net assets because the target may have assets that are not reported on its balance sheet. For example, the target's reputation and customer loyalty certainly have value, but that value is not quantifiable. Also, the target may have research and development assets that remain off the balance sheet because of accounting standards. Finally, part of the acquisition price may reflect perceived synergies from the business combination. For example, the acquirer may be able to eliminate duplicate facilities and reduce payroll after the acquisition.



PROFESSOR'S NOTE

Occasionally, the purchase price of an acquisition is less than fair value of the identifiable net assets. In this case, the difference is immediately recognized as a gain in the acquirer's income statement.

Goodwill is only created in a purchase acquisition. Internally generated goodwill is expensed as incurred.

Because it is an intangible asset with indefinite life, goodwill is not amortized but must be tested for impairment at least annually. If goodwill is impaired, the company decreases its value and recognizes a loss in the income statement. The impairment loss does not affect cash flow. Goodwill impairment suggests an acquired business is now worth less than the price the company paid to acquire it, because the company has reduced its estimate of the future excess returns the acquired business is expected to generate.

Acquiring firms might try to take advantage of the fact that goodwill is not amortized, manipulating net income upward by allocating more of an acquisition price to goodwill and less to the identifiable assets, especially when fair value is subjective. Lower-valued identifiable assets result in less future depreciation and amortization expense, and therefore higher net income.

Accounting goodwill should not be confused with **economic goodwill**. Economic goodwill derives from the expected future performance of the firm, while accounting goodwill is the result of past acquisitions.

Some analysts believe goodwill represents the present value of excess returns that the acquired company is expected to contribute, and so it is a legitimate asset. Other analysts believe goodwill is not a genuine asset because it cannot be sold separately and might result from simply overpaying for an acquired company. Examining impairments of goodwill can be a way to judge how successful companies' past acquisitions have been.

To improve comparability when computing ratios, analysts should eliminate goodwill from balance sheets and goodwill impairment charges from income statements. Also, analysts should evaluate future acquisitions in terms of the price paid relative to the earning power of the acquired assets.

LOS 29.c: Explain the financial reporting and disclosures related to financial instruments.

Financial instruments are contracts that give rise to both a financial asset of one entity and a financial liability or equity instrument of another entity.¹ Financial instruments can be found on the asset side and the liability side of the balance sheet. Financial instruments held as assets include investment securities (stocks and bonds), derivatives, loans, and receivables.

Accounting standards require some financial instruments to be measured at historical cost, some at amortized cost, and some at fair value. Financial assets measured at cost include unquoted equity investments (for which fair value cannot be reliably measured) and loans to and notes receivable from other entities.

Under US GAAP, debt securities acquired with the intent to hold them until they mature are classified as **held-to-maturity securities** and measured at amortized cost. Amortized cost is equal to the original issue price minus any principal payments, plus any amortized discount or minus any amortized premium, minus any impairment losses. Subsequent changes in market value are ignored.

Financial assets measured at fair value, also known as **mark-to-market accounting**, include trading securities, available-for-sale securities, and derivatives.

Trading securities (also known as held-for-trading securities) are debt securities acquired with the intent to sell them in the near term. Trading securities are reported on the balance sheet at fair value, with unrealized gains and losses (changes in market value before the securities are sold) recognized in the income statement. All equity securities holdings with quoted market prices (except those that give a company significant influence over a firm) are treated this way. Unrealized gains and losses are also known as holding period gains and losses. **Derivative instruments** are treated the same as trading securities.



PROFESSOR'S NOTE

Accounting for equity investments when the share owner has significant influence or control over a firm is addressed at Level II.

Available-for-sale securities are debt securities that are not expected to be held to maturity or traded in the near term. Like trading securities, available-for-sale securities are reported on the balance sheet at fair value. However, any unrealized gains and losses are not recognized in the income statement, but are reported in other comprehensive income as a part of shareholders' equity.

For all financial securities, dividend and interest income and realized gains and losses (actual gains or losses when the securities are sold) are recognized in the income statement.

Figure 29.1 summarizes the different classifications and measurement bases of financial assets under US GAAP.

Figure 29.1: Financial Asset Measurement Bases—US GAAP

Historical Cost	Amortized Cost	Fair Value
Unlisted equity investments	Held-to-maturity securities	Trading securities
Loans and notes receivable		Available-for-sale securities
		Derivatives

EXAMPLE: Classification of investment securities

Triple D Corporation, a US GAAP reporting firm, purchased a 6% bond, at par, for \$1 million at the beginning of the year. Interest rates have recently increased, and the market value of the bond declined \$20,000. Determine the bond's effect on Triple D's financial statements under each classification of securities.

Answer:

If the bond is classified as a *held-to-maturity* security, the bond is reported on the balance sheet at \$1,000,000. Interest income of \$60,000 [$\$1,000,000 \times 6\%$] is reported in the income statement.

If the bond is classified as a *trading* security, the bond is reported on the balance sheet at \$980,000. The \$20,000 unrealized loss and \$60,000 of interest income are both recognized in the income statement.

If the bond is classified as an *available-for-sale* security, the bond is reported on the balance sheet at \$980,000. Interest income of \$60,000 is recognized in the income statement. The \$20,000 unrealized loss is reported as part of other comprehensive income.

IFRS Treatment of Marketable Securities

Under IFRS, the three classifications of investment securities are as follows:

Securities measured at amortized cost (corresponds to the treatment of held-to-maturity securities under US GAAP): IFRS requires that cash flows are solely interest and principal and that the business model is to hold the security to maturity.

Securities measured at fair value through other comprehensive income (corresponds to the treatment of available-for-sale securities under US GAAP)

Securities measured at fair value through profit and loss (corresponds to the treatment of trading securities under US GAAP)

While the three different treatments are essentially the same as those used under US GAAP, there are significant differences in how securities are classified under IFRS and US GAAP. Similarities and differences are as follows:

- | Under both IFRS and US GAAP, loans, notes receivable, debt securities that a firm intends to hold until maturity, and unlisted securities for which fair value cannot be reliably determined, are all *measured at (amortized) historical cost*.
- | Under IFRS, debt securities for which a firm intends to collect interest payments but also to sell the securities are *measured at fair value through other comprehensive income*. This is similar to the treatment of available-for-sale securities under US GAAP.
- | Under IFRS, firms may make an irrevocable choice at the time of purchase to account for equity securities as *measured at fair value through other comprehensive income*. Equity securities cannot be classified as available-for-sale under US GAAP.
- | Under IFRS, financial assets that do not fit either of the other two classifications are *measured at fair value through profit and loss* (unrealized gains and losses reported on the income statement).
- | Under IFRS, firms can make an irrevocable choice to carry any financial asset at *fair value through profit and loss*. This choice is not available under US GAAP.

Figure 29.2 summarizes the different classifications of financial assets under IFRS.

Figure 29.2: Financial Asset Classifications—IFRS

Measured at Amortized Cost	Measured at Fair Value Through Other Comprehensive Income	Measured at Fair Value Through Profit and Loss
■ Debt securities acquired with the intent to hold them to maturity ■ Loans receivable ■ Notes receivable ■ Unlisted equity securities if fair value cannot be determined reliably	■ Debt securities acquired with intent to collect interest payments but sell before maturity ■ Equity securities only if this treatment is chosen at time of purchase	■ Debt securities acquired with intent to sell in near term ■ Equity securities (unless fair value through OCI is chosen at time of purchase) ■ Derivatives ■ Any security not assigned to the other two categories ■ Any security for which this treatment is chosen at time of purchase

LOS 29.d: Explain the financial reporting and disclosures related to non-current liabilities.

Long-term financial liabilities include bank loans, notes payable, bonds payable, and some derivatives. If the financial liabilities are not issued at face value, the liabilities are usually reported on the balance sheet at amortized cost. If the issuance value differs from face value, any premium or discount is amortized through interest expense over the life of the liability. Amortized cost at any point in the liability's life is equal to the issue price minus any principal payments, plus any amortized discount or minus any amortized premium. Amortizing premiums and discounts causes the balance sheet liability to move toward face value at maturity of the liability.

In some cases, financial liabilities are reported at fair value. Examples include held-for-trading liabilities such as a short position in a stock (which may be classified as a short-term liability), derivative liabilities, and nonderivative liabilities with exposures hedged by derivatives.

Deferred tax liabilities are income taxes payable in future periods as a result of timing differences between financial accounting and tax accounting. Deferred tax liabilities are created when the amount of income tax expense recognized in the income statement is greater than taxes payable using tax accounting. This can occur when expenses or losses are tax deductible before they are recognized in the income statement. A common example is when a firm uses an accelerated

depreciation method for tax purposes and the straight-line method for financial reporting. Deferred tax liabilities are also created when revenues or gains are recognized in the income statement before they are taxable. For example, a firm often recognizes the earnings of a subsidiary before any distributions (dividends) are made. Eventually, deferred tax liabilities will reverse when the taxes are paid.



PROFESSOR'S NOTE

We explain deferred tax items in more detail in our reading on Analysis of Income Taxes.



MODULE QUIZ 29.1

1. For a company reporting under IFRS, product development costs:
 - A. must always be capitalized.
 - B. may be capitalized.
 - C. must be expensed.
2. The SF Corporation has created employee goodwill by reorganizing its retirement benefit package. An independent management consultant estimated the value of the goodwill at \$2 million. In addition, SF recently purchased a patent that was developed by a competitor. The patent has an estimated useful life of five years. Should SF report the goodwill and patent on its balance sheet?

	<u>Goodwill</u>	<u>Patent</u>
A. Yes		No
B. No		Yes
C. No		No
3. At the beginning of the year, the Parent Company purchased all 500,000 shares of Sub, Inc. for \$15 per share. Just before the acquisition date, Sub's balance sheet reported net assets of \$6 million. Parent determined the fair value of Sub's property and equipment was \$1 million higher than reported by Sub. What amount of goodwill should Parent report as a result of its acquisition of Sub?
 - A. \$0.
 - B. \$500,000.
 - C. \$1,500,000.
4. At the beginning of the year, Company P purchased \$80,000 face value of Company S corporate bonds for \$77,000. Company P intends to hold these bonds for several years but sell them before they mature. At the end of the year, the market value of the bonds was \$75,000. What amount should Company P report on its balance sheet at year-end for the investment in Company S bonds?
 - A. \$75,000.
 - B. \$77,000.
 - C. \$80,000.

MODULE 29.2: COMMON-SIZE BALANCE SHEETS

LOS 29.e: Calculate and interpret common-size balance sheets and related financial ratios.

A vertical **common-size balance sheet** expresses each item of the balance sheet as a percentage of total assets. The common-size format standardizes the balance sheet by eliminating the effects of size. This allows for comparison over time (time-series analysis) and across firms (cross-sectional analysis).

EXAMPLE: Common-size balance sheets

The following are the balance sheets of industry competitors East Company (East) and West Company (West).

	East	West
Cash	\$2,300	\$1,500
Accounts receivable	3,700	1,100
Inventory	<u>5,500</u>	<u>900</u>
Current assets	11,500	3,500
Plant and equipment	32,500	11,750
Goodwill	<u>1,750</u>	<u>0</u>
Total assets	\$45,750	\$15,250
Current liabilities	\$10,100	\$1,000
Long-term debt	<u>26,500</u>	<u>5,100</u>
Total liabilities	36,600	6,100
Equity	<u>9,150</u>	<u>9,150</u>
Total liabilities and equity	\$45,750	\$15,250

Convert each balance sheet to a vertical common-size balance sheet and interpret the results.

Answer:

East is obviously the larger company. By converting the balance sheets to common-size format, we can eliminate the size effect.

	East	West
Cash	5%	10%
Accounts receivable	8%	7%
Inventory	<u>12%</u>	<u>6%</u>
Current assets	25%	23%
Plant and equipment	71%	77%
Goodwill	<u>4%</u>	<u>0%</u>
Total assets	100%	100%
Current liabilities	22%	7%
Long-term debt	<u>58%</u>	<u>33%</u>
Total liabilities	80%	40%
Equity	<u>20%</u>	<u>60%</u>
Total liabilities and equity	100%	100%

East's investment in current assets of 25% of total assets is slightly higher than West's current assets of 23%. However, East's current liabilities of 22% of total assets are significantly higher than West's current liabilities of 7%. Thus, East is less liquid and may have more difficulty paying its current obligations when due. However, West's superior working capital position may not be an efficient use of resources. The investment returns on working capital are usually lower than the returns on long-term assets.

A closer look at current assets reveals that East reports less cash as a percentage of assets than West. In fact, East does not have enough cash to satisfy its current liabilities without selling more inventory and collecting receivables. East's inventories of 12% of total assets are higher than West's inventories of 6%. Carrying higher inventories may be an indication of inventory obsolescence. Further analysis of inventory is necessary.

Not only are East's current liabilities higher than West's, but East's long-term debt of 58% of total assets is much greater than West's long-term debt of 33%. Thus, East may have trouble satisfying its long-term obligations because its capital structure consists of more debt.

Common-size analysis can also be used to examine a firm's strategies. East appears to be growing through acquisitions because it is reporting goodwill. West is growing internally because no goodwill is reported. It could be that East is financing the acquisitions with debt.

The percentages on a common-size balance sheet are examples of **balance sheet ratios**, which compare one balance sheet item to another balance sheet item (total assets in this case). Balance sheet ratios, along with common-size analysis, can be used to evaluate a firm's liquidity and solvency. The results can be compared over time and across firms.

Liquidity ratios measure the firm's ability to satisfy its short-term obligations as they come due. Liquidity ratios include the current ratio, the quick ratio, and the cash ratio:

$$\text{current ratio} = \frac{\text{current assets}}{\text{current liabilities}}$$

$$\text{quick ratio} = \frac{\text{cash} + \text{marketable securities} + \text{receivables}}{\text{current liabilities}}$$

$$\text{cash ratio} = \frac{\text{cash} + \text{marketable securities}}{\text{current liabilities}}$$

Although all three ratios measure a firm's ability to pay current liabilities, they should be considered collectively. For example, assume Firm A has a higher current ratio but a lower quick ratio than Firm B. This must result from Firm A having higher inventory than Firm B, because the quick ratio (also known as the acid-test ratio) excludes inventory from current assets. Similar analysis can be performed by comparing the quick ratio to the cash ratio, which excludes inventory and receivables.

Solvency ratios measure a firm's ability to satisfy its long-term obligations. Solvency ratios include the long-term debt-to-equity ratio, the total debt-to-equity ratio, the debt ratio, and the financial leverage ratio:

$$\text{long-term debt-to-equity ratio} = \frac{\text{long-term debt}}{\text{total equity}}$$

$$\text{total debt-to-equity ratio} = \frac{\text{total debt}}{\text{total equity}}$$

$$\text{debt ratio} = \frac{\text{total debt}}{\text{total assets}}$$

$$\text{financial leverage ratio} = \frac{\text{total assets}}{\text{total equity}}$$

All four ratios measure solvency, but they should be considered collectively. For example, Firm A might have a higher long-term debt-to-equity ratio but a lower total debt-to-equity ratio as compared to Firm B. This would indicate that Firm B is using more short-term debt to finance itself.

When calculating solvency ratios, debt is typically considered to be any interest-bearing obligation. The financial leverage ratio captures the impact of all interest-bearing and non-interest-bearing obligations.

Analysts must understand the limitations of balance sheet ratio analysis:

- | Comparisons with peer firms are limited by differences in accounting standards and estimates.
- | Ratios might not be comparable for firms that operate in different industries.
- | Interpretation of ratios requires significant judgment.
- | Balance sheet data are only measured at a single point in time.



MODULE QUIZ 29.2

1. A vertical common-size balance sheet expresses each category of the balance sheet as a percentage of:
 - A. assets.
 - B. equity.
 - C. revenue.
2. Which of the following ratios are used to measure a firm's liquidity and solvency?

<u>Liquidity</u>	<u>Solvency</u>
A. Current ratio	Quick ratio
B. Debt-to-equity ratio	Financial leverage ratio
C. Cash ratio	Total debt ratio

KEY CONCEPTS

LOS 29.a

Intangible assets created internally are expensed as incurred. Purchased intangibles with finite lives are treated in a way similar to tangible assets. Purchased intangibles with indefinite lives are not amortized but must be tested for impairment periodically.

Under IFRS, research costs are expensed as incurred and development costs are capitalized if certain criteria are satisfied. Both research and development costs are expensed under US GAAP.

LOS 29.b

Goodwill is the excess of purchase price over the fair value of identifiable net assets in a business acquisition. Goodwill is not amortized, but must be tested for impairment at least annually.

LOS 29.c

Under IFRS, debt securities acquired with the intent to hold them to maturity are measured at amortized cost. Debt securities acquired with the intent to collect interest payments but sell before maturity are measured at fair value through other comprehensive income. Debt securities acquired with the intent to sell them in the near term, as well as equity securities and derivatives, are measured at fair value through profit and loss.

IFRS permits firms to elect, irrevocably at the time of purchase, to measure equity securities at fair value through other comprehensive income, or any security at fair value through profit and loss.

Under US GAAP, held-to-maturity securities are reported at amortized cost. Trading securities, available-for-sale securities, and derivatives are reported at fair value. For trading securities and derivatives, unrealized gains and losses are recognized in the income statement. Unrealized gains and losses for available-for-sale securities are reported in equity (other comprehensive income). Equity securities cannot be classified as available-for-sale.

LOS 29.d

Financial liabilities that are not issued at face value are reported at amortized cost. The liability will move toward par value at maturity as the premium or discount (relative to par) is amortized. Held-for-trading liabilities and derivative liabilities are reported at fair value.

Deferred tax liabilities result from temporary timing differences between a firm's tax reporting and its financial reporting.

LOS 29.e

A vertical common-size balance sheet expresses each item of the balance sheet as a percentage of total assets. The common-size format standardizes the balance sheet by eliminating the effects of size. This allows for comparison over time (time-series analysis) and across firms (cross-sectional analysis).

Balance sheet ratios, along with common-size analysis, can be used to evaluate a firm's liquidity and solvency. Liquidity ratios measure the firm's ability to satisfy its short-term obligations as they come due. Liquidity ratios include the current ratio, the quick ratio, and the cash ratio.

Solvency ratios measure the firm's ability to satisfy its long-term obligations. Solvency ratios include the long-term debt-to-equity ratio, the total debt-to-

equity ratio, the debt ratio, and the financial leverage ratio.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 29.1

- . **B** Product development costs may be capitalized under IFRS if certain criteria are met (e.g., the company has identified a customer base, has sufficient resources to complete the product, and intends to sell the product). US GAAP requires all product development expenditure to be expensed. (LOS 29.a)
- . **B** Goodwill developed internally is expensed as incurred. The purchased patent is reported on the balance sheet. (LOS 29.a)
- . **B** The purchase price of \$7,500,000 ($\$15 \text{ per share} \times 500,000 \text{ shares}$) – fair value of net assets of \$7,000,000 ($\$6,000,000 \text{ book value} + \$1,000,000 \text{ increase in property and equipment}$) = goodwill of \$500,000. (LOS 29.a)
- . **A** Debt securities acquired with the intent to sell before maturity are reported on the balance sheet at their fair values. (LOS 29.b)

Module Quiz 29.2

- . **A** Each category of the balance sheet is expressed as a percentage of total assets. (LOS 29.e)
- . **C** The current ratio, quick ratio, and cash ratio measure liquidity. The debt-to-equity, the total debt ratio, and the financial leverage ratio measure solvency. (LOS 29.e)

¹ IAS 32, *Financial Instruments: Presentation*, 32,11.

READING 30

ANALYZING STATEMENTS OF CASH FLOWS I

MODULE 30.1: CASH FLOW INTRODUCTION AND DIRECT METHOD CFO

LOS 30.a: Describe how the cash flow statement is linked to the income statement and the balance sheet.

The **cash flow statement** provides information for a reporting period beyond that available from the income statement, which is based on accrual, rather than cash, accounting. Analysts use cash flow statements to understand:

- | A company's cash receipts and cash payments during an accounting period
- | A company's operating, investing, and financing activities
- | The impact of accrual accounting events on cash flows
- | A company's quality of earnings

An analyst can use the statement of cash flows to assess a firm's liquidity, solvency, and financial flexibility, including:

- | Whether regular operations generate enough cash to sustain the business
- | Whether the firm generates enough cash to pay off existing debts as they mature
- | Whether the firm is likely to need additional financing
- | Whether the firm can meet unexpected obligations
- | Whether the firm can take advantage of new business opportunities as they arise

The cash flow statement reconciles the beginning and ending balances of cash in the balance sheet over an accounting period. The change in cash is a result of the firm's operating, investing, and financing activities, as follows:

$$\begin{array}{rcl}
 & \text{Operating cash flow} & \\
 + / - & \text{Investing cash flow} & \\
 + / - & \text{Financing cash flow} & \\
 \hline
 = & \text{Change in cash balance} & \\
 + & \text{Beginning cash balance} & \\
 \hline
 = & \text{Ending cash balance} &
 \end{array}$$

Earnings are considered to be of high quality when operating cash flows, which can also be called cash flow from operations (CFO), are close to or higher than reported earnings. If earnings are consistently higher than CFO, their quality is lower because the accruals-based earnings are not backed by cash creation from operating activities.

Like the statement of cash flows, the income statement is a “flow” statement (sometimes referred to as a dynamic statement), as it shows the company’s performance between two balance sheet dates. However, due to the accruals concept, net income is not the same as cash generated by the company. In addition, many financing and investing cash flows do not affect the income statement at the time of the transactions.

With respect to the balance sheet, operating activities relate (with a few exceptions) to the firm’s current assets and current liabilities. Investing activities typically relate to the firm’s noncurrent assets, and financing activities typically relate to the firm’s noncurrent liabilities and equity.

Transactions for which the timing of revenue or expense recognition differs from the receipt or payment of cash are reflected in changes in balance sheet accounts. For example, when revenues (sales) exceed cash collections, the firm has sold items on credit, and accounts receivable (an asset) increase. The opposite occurs if customers repay more on their outstanding accounts than the firm extends in new credit: cash collections exceed revenues, and accounts receivable decrease. Similarly, when purchases from suppliers exceed cash payments, accounts payable (a liability) increase. When cash payments exceed purchases, accounts payable decrease.

It is helpful to understand how transactions affect each balance sheet account. For example, accounts receivable are increased by sales on credit and decreased by cash collections. We can summarize this relationship as follows:

Beginning accounts receivable	Opening balance sheet
+ Sales	Income statement
– <u>Cash collections</u>	Cash flow statement
= Ending accounts receivable	Closing balance sheet

Knowing three of the four variables, we can solve for the fourth. For example, if beginning accounts receivable are €10,000, ending accounts receivable are €15,000, and sales are €68,000, then cash collections must equal €63,000.

By rearranging this relationship, we can say that cash = sales – ending accounts receivable + beginning accounts receivable, or cash = sales – change in accounts receivable. Cash = €68,000 – (€15,000 – €10,000) = €63,000.

Revenue recognition standards affect when a sale is recorded in the income statement, but not the timing of cash flows from the customer. If a firm receives cash from a customer in advance of the sale being recorded in the income statement (unearned revenue), the cash received will appear in the cash flow statement at the time of payment, not when the sale is recorded. For example, if a company provided a service agreement to a customer, the cash received is reflected in the cash flow statement when the customer pays for the contract (normally in advance), but revenue recognized in the income statement is spread over the life of the contract.

EXAMPLE: Balance sheet and income statement impacts on cash flow

Loftus Communications Limited provides equipment sales and installation of telecommunication equipment. Loftus also provides maintenance for systems installed under service contracts. Revenue from maintenance contracts is deferred and recognized over the life of the contract.

Selected Financial Statement Data: Loftus Communications Limited

Financial year	20X2	20X1
	£m	£m
Income statement extract		
Revenues	2,000,000	1,800,000
Balance sheet extract		
Accounts receivable	900,000	500,000
Unearned revenue (deferred)	1,000,000	300,000

Calculate the cash received from customers in 20X2.

Answer:

	20X2
	£m
Revenue	2,000,000
Increase in accounts receivable	(400,000)
Increase in unearned revenue	<u>700,000</u>
Cash received from customers	<u>2,300,000</u>

The increase in accounts receivable represents credit extended to customers, which is a use of cash from the firm's perspective. The increase in unearned revenue represents services that customers have paid for in advance, a source of cash.

Understanding these interrelationships is not only useful in preparing the cash flow statement, but is also helpful in uncovering accounting shenanigans, as we will see in our reading on Financial Reporting Quality.

LOS 30.b: Describe the steps in the preparation of direct and indirect cash flow statements, including how cash flows can be computed using income statement and balance sheet data.

**PROFESSOR'S NOTE**

Throughout the discussion of the direct and indirect methods, remember the following points:

- The terms *cash flow from operations (CFO)*, *cash flow from operating activities*, and *operating cash flows (OCF)* are used interchangeably in the Level I CFA curriculum.
- Two methods can be used in the accounts to present CFO. CFO is presented differently under the direct and indirect methods, but the result is the same under both methods.
- Cash flow from investing (CFI) and cash flow from financing (CFF) are each presented the same way regardless of which method a firm chooses to present CFO.
- An increase in an asset account is a use of cash, and a decrease in an asset account is a source of cash.

- An increase in a liability account is a source of cash, and a decrease in a liability is a use of cash.
- Sources of cash are positive numbers (cash inflows), and uses of cash are negative numbers (cash outflows).

The Direct Method for Cash Flow From Operating Activities

The direct method of presenting a firm's statement of cash flows shows only cash payments and cash receipts over the period. The sum of these inflows and outflows is CFO. The direct method gives an analyst more information than the indirect method (which we will explain later in this reading). The analyst can see the actual amounts that went to each use of cash and that were received from each source of cash. This information can help the analyst to better understand the firm's performance over time and to forecast future cash flows.

The following are common components that appear on a statement of cash flows presented using the direct method:

- | Cash collected from customers (typically the main component of CFO)
- | Cash used in the production of goods and services (cash inputs)
- | Cash operating expenses, such as salaries
- | Cash paid for interest
- | Cash paid for taxes

These are steps for the direct method:

Step 1: Start at the top of the income statement with revenue.

Step 2: Examine the balance sheet for any assets or liabilities (typically current) relating to the income statement item.

Step 3: Compute the change in the balance sheet asset or liability.

Step 4: Adjust the income statement for the change in the balance sheet amount using the following rules:

- Subtract an increase in an asset (a use of cash), or add a decrease in an asset (a source of cash).
- Add an increase in a liability (a source of cash), or subtract a decrease in a liability (a use of cash).

For these rules to work consistently, we must *treat expense items as negative numbers* (e.g., cost of goods sold, wages and salaries) before we adjust them.

Step 5: After adjusting the income statement item for the change in the balance sheet asset or liability, move to the next item in the income statement.

Step 6: Ignore any noncash charges (e.g., depreciation, gains and losses on asset disposal). A noncash charge is any amount in the income statement that is due to accounting treatment rather than actual cash flows.

Step 7: Once all income statement items have been adjusted for accruals, total the amount to get CFO.

EXAMPLE: CFO using the direct method

Use the following balance sheet and income statement to prepare operating cash flows using the direct method. The company produces its financial statements under US GAAP.

Income Statement for 20X7

	\$
Sales	104,000
Expenses	
Cost of goods sold	40,000
Wages	5,000
Depreciation	7,000
Interest	<u>1,000</u>
Total expenses	<u>53,000</u>
Income from continuing operations	51,000
Gain from sale of land	10,000
Loss on disposals of PP&E	<u>2,000</u>
Pretax income	59,000
Provision for income taxes	<u>20,000</u>
Net income	39,000
Common dividends declared	<u>8,500</u>

Balance Sheets for 20X7 and 20X6

Assets	20X7 \$	20X6 \$
Current assets		
Cash	53,000	11,500
Accounts receivable	10,000	9,000
Inventory	<u>5,000</u>	<u>7,000</u>
Total current assets	68,000	27,500
Noncurrent assets		
Land	35,000	40,000
Gross plant and equipment	69,000	60,000
Less: accumulated depreciation	<u>(12,000)</u>	<u>(9,000)</u>
Net plant and equipment	57,000	51,000
Goodwill	<u>10,000</u>	<u>10,000</u>
Total assets	170,000	128,500
Liabilities		
Current liabilities		
Accounts payable	9,000	5,000
Wages payable	4,500	8,000
Interest payable	3,500	3,000
Unearned revenue	6,000	2,000
Taxes payable	5,000	4,000
Dividends payable	<u>6,000</u>	<u>1,000</u>
Total current liabilities	34,000	23,000
Noncurrent liabilities		
Bonds payable	15,000	10,000
Deferred tax liability	<u>20,000</u>	<u>15,000</u>
Total liabilities	69,000	48,000
Stockholders' equity		
Common stock	15,000	20,000
Additional paid in capital	<u>25,000</u>	<u>30,000</u>
Contributed capital	40,000	50,000
Retained earnings	<u>61,000</u>	<u>30,500</u>
Total equity	<u>101,000</u>	<u>80,500</u>
Total liabilities and stockholders' equity	170,000	128,500

Answer:

Start at the top of the income statement and adjust each line for the change in balance sheet asset and liabilities that arise due to the accruals process.

Cash collected from customers	\$
Sales	104,000
Increase in accounts receivable	(1,000)
Increase in unearned revenue liability	<u>4,000</u>
Cash collected	107,000

To calculate cash paid to suppliers, adjust cost of goods sold for the change in inventory (the result is purchases for the period) and for the change in accounts payable.

Cash paid to suppliers	\$
Cost of goods sold	(40,000)
Decrease in inventory	<u>2,000</u>
Purchases	(38,000)
Increase in accounts payable	<u>4,000</u>
Cash paid to suppliers	(34,000)

All expenses, including cost of goods sold, must be treated as negative values if we wish to apply the rules for sources and uses of cash. Expenses in the income statement may be shown as either negative or positive values in the financial statements at the company's discretion. Users of the accounts are expected to know that revenue and gains increase net income, and that expenses and losses decrease it.

Wages paid	\$
Wage expense	(5,000)
Decrease in wages payable	<u>(3,500)</u>
Cash paid to employees	(8,500)

The next line in the income statement is depreciation. When using the direct method we ignore depreciation because it is not a cash flow.

Interest paid	\$
Interest expense	(1,000)
Increase in interest payable	<u>500</u>
Cash interest paid	(500)

We also ignore the gain from sale of land and the loss on disposal of PP&E because these items relate to investing activities (CFI) and not operating activities (CFO). Later in this reading we will explain how to determine the related cash flows when we calculate CFI.

The last item in the income statement before net income is the tax expense. Here we must adjust not only for changes in the current liability for taxes payable, but also for any changes in deferred tax assets and deferred tax liabilities, which are typically noncurrent items. We will explain these further in our reading on Analysis of Income Taxes.

Cash paid to tax authorities	\$
Tax expense (provision for income taxes)	(20,000)
Increase in taxes payable liability	1,000
Increase in deferred tax liability	<u>5,000</u>
Cash paid for taxes	(14,000)

Having arrived at the bottom of the income statement, we can sum the cash flows to compute CFO:

20X7 CFO

	\$	\$
Cash collected from customers		107,000
Cash paid to suppliers	34,000	
Wages paid to employees	8,500	
Cash interest paid	500	
Cash paid to tax authorities	<u>14,000</u>	
Total cash operating expenses		<u>(57,000)</u>
Operating cash flows		50,000

MODULE 30.2: INDIRECT METHOD CFO

Using the indirect method of presenting CFO, we begin with net income and adjust it for differences between accounting items and actual cash inflows and outflows.

Non-cash-based items in the income statement can be described as either noncash charges or working capital investment. For example, depreciation is a noncash charge as it is deducted in calculating net income, but it requires no cash outlay. Therefore, we must add depreciation (and amortization) back to net income for the period.

Another adjustment to net income on an indirect statement of cash flows is to subtract gains and add back losses on the disposal of assets. Proceeds from the sale of fixed assets are an investing cash flow. Because gains and losses relate to

CFI activities, we need to remove them from net income to calculate CFO under the indirect method.



PROFESSOR'S NOTE

Candidates are often confused by why noncash charges and gains and losses are ignored when using the direct method, but adjusted for under the indirect method. The key to understanding this is to realize that these items have been included in arriving at net income, which is the starting point of our computation when we use the indirect method. That is, net income includes some non-cash and non-operating items that we have to unwind to arrive at CFO.

Under the indirect method, we also need to adjust net income for any change in balance sheet accounts, just as we do with the direct method. If, for example, accounts receivable went up during the period, we know that sales during the period were greater than the cash collected from customers. We need to reduce net income to reflect the fact that sales, rather than cash collected, were used in calculating net income.

The net change in a company's total operating assets and liabilities is known as its **investment in working capital**. We can think of CFO as net income adjusted for noncash charges and the investment in working capital:

$$\text{CFO} = \text{NI} + \text{NCC} - \text{WC}_{\text{INV}}$$

Noncash charges are gains and losses that have passed through the income statement but are not cash flows. While we call them “charges,” in practice they can either increase or decrease net income. Gains and losses on asset disposals are classic examples. Figure 30.1 lists some of the typical noncash charges in company income statements.

Figure 30.1: Typical Noncash Charges, Gains, and Losses

Add Back	Depreciation, depletion, and amortization
	Loss on asset disposal
	Asset impairments and write-downs
	Losses on early retirement of debt
	Amortization of bond discounts (under the amortized cost method)
	Increases in deferred tax liabilities, decreases of deferred tax assets
	Losses of equity accounted associates
Subtract	Gains on asset disposals
	Gains on early retirement of debt
	Reversals of impairment and write-downs
	Amortization of bond premiums (under the amortized cost method)
	Decreases in deferred tax liabilities, increases in deferred tax assets

Working capital investment represents the investment in noncash working capital. This means we need to look at the change in current assets and liabilities that relate to the operating revenues and expenses and ignore any assets or liabilities that do not relate to operating items.



PROFESSOR'S NOTE

The definition of *working capital* is not the same in all parts of the Level I CFA curriculum. In our module covering ratios, we will see it defined as total current assets minus total current liabilities. Here and in Corporate Finance, it is defined as operating assets minus operating liabilities; this is often referred to as noncash working capital.

In practice, when looking at the current assets, we ignore cash and any short-term investments (except trading securities, for which cash flows are treated as CFO). Dividends payable and any short-term interest-bearing debt instruments are also excluded, as they affect cash flow from financing (CFF), not CFO.

Adjust for changes in working capital accounts as follows:

Add back	Decreases in current operating assets
	Increases in current operating liabilities
Subtract	Increases in current operating assets
	Decreases in current operating liabilities

The steps in calculating CFO under the indirect method can be summarized as follows:

Step 1: Begin with net income.

Step 2: Add back all noncash charges to income (such as depreciation and amortization) and subtract all noncash components of revenue. Subtract gains or add losses that resulted from financing or investing cash flows (such as gains from sale of land).

Step 3: Adjust for working capital by adding or subtracting changes to balance sheet operating accounts as follows:

- Subtract increases in operating asset accounts (uses of cash), and add decreases (sources of cash).
- Add increases in operating liability accounts (sources of cash), and subtract decreases (uses of cash).

EXAMPLE: Statement of cash flows using the indirect method

Use the balance sheet and income statement presented in the previous direct method example to prepare operating cash flows under the indirect method.

Answer:

Step 1: Start with net income of \$39,000.

Step 2: Add back noncash charges:

Depreciation = \$7,000

Change in deferred tax liability = \$5,000

Loss on disposal of PP&E = \$2,000

Deduct noncash gains:

Gain from sale of land = \$10,000

Step 3: Subtract increases in receivables and inventories and add increases in payables.

	\$
Net income	39,000
Noncash charges	
Depreciation	7,000
Increase in deferred tax liability	5,000
Loss on disposal of PP&E	2,000
Gain from sale of land	<u>(10,000)</u>
Subtotal	43,000
Investment in working capital	
Increase in receivables	(1,000)
Decrease in inventories	2,000
Increase in accounts payable	4,000
Decrease in wages payable	(3,500)
Increase in interest payable	500
Increase in unearned revenue liability	4,000
Increase in taxes payable	<u>1,000</u>
Operating cash flows	50,000

Alternatively, we can look at noncash working capital in aggregate rather than line by line.

	20X7	20X6
	\$	\$
Total current assets	68,000	27,500
Less cash	<u>(53,000)</u>	<u>(11,500)</u>
Operating assets	15,000	16,000
 Total current liabilities	34,000	23,000
Less dividends payable	<u>(6,000)</u>	<u>(1,000)</u>
Operating liabilities	28,000	22,000
 Non-cash working capital	(13,000)	(6,000)
Investment in working capital	<u>(\$7,000)</u>	

We can then calculate CFO as $NI + NCC - WC_{INV}$:

$$CFO = 39,000 + 4,000 - (-7,000) = \$50,000$$

Our examples demonstrate that whether CFO is calculated using the direct or indirect method, we arrive at the same figure.

Both IFRS and US GAAP encourage the use of a statement of cash flows in the direct format. Under US GAAP, a statement of cash flows under the direct method must include footnote disclosure of the indirect method. Most companies, however, report cash flows using the indirect method, which requires no additional disclosure. Later in this reading we will illustrate the method an analyst can use to create a statement of cash flows in the direct method format when the company reports using the indirect method.



MODULE QUIZ 30.1, 30.2

1. The Continental Corporation reported sales revenue of \$150,000 for the current year. If accounts receivable decreased \$10,000 during the year and accounts payable increased \$4,000 during the year, cash collections were:
 - A. \$154,000.
 - B. \$160,000.
 - C. \$164,000.

Use the following data, prepared under US GAAP, to answer Questions 2 through 4.

Income statement selected data:

	20X7
	\$
Revenues	2,000,000
Cost of goods sold	1,400,000
Tax expense	200,000

Balance sheet selected data:

	20X7	20X6
	\$	\$
Long-lived assets		
Deferred tax	30,000	20,000
Current assets		
Accounts receivable	200,000	150,000
Inventory	260,000	280,000
Current liabilities		
Accounts payable	120,000	150,000
Unearned revenue	220,000	150,000
Tax payable	150,000	240,000
Long-term liabilities		
Deferred tax	150,000	100,000

2. What was cash collected from customers in 20X7?
 - A. \$1,950,000.

- B. \$1,980,000.
 - C. \$2,020,000.
3. What was cash tax paid in 20X7?
- A. \$160,000.
 - B. \$250,000.
 - C. \$290,000.
4. What was cash paid to suppliers in 20X7?
- A. \$1,350,000.
 - B. \$1,410,000.
 - C. \$1,450,000.
5. Using the following information, what is the firm's cash flow from operations?

Net income	\$120
Decrease in accounts receivable	20
Depreciation	25
Increase in inventory	10
Increase in accounts payable	7
Decrease in wages payable	5
Increase in deferred tax liabilities	15
Profit from the sale of land	2

- A. \$158.
 - B. \$170.
 - C. \$174.
6. Net income for Monique, Inc., for the year ended December 31, 20X7 was \$78,000. Its accounts receivable balance at December 31, 20X7, was \$121,000, and this balance was \$69,000 at December 31, 20X6. The accounts payable balance at December 31, 20X7, was \$72,000, and it was \$43,000 at December 31, 20X6. Depreciation for 20X7 was \$12,000, and there was an unrealized gain of \$15,000 included in 20X7 income from the change in value of trading securities. Which of the following amounts represents Monique's cash flow from operations for 20X7?
- A. \$52,000.
 - B. \$67,000.
 - C. \$82,000.
7. From an analyst's perspective, an advantage of the indirect method for presenting operating cash flow is that the indirect method:
- A. shows operating cash received and paid.
 - B. provides more information than the direct method.
 - C. shows the difference between net income and operating cash flow.

MODULE 30.3: INVESTING AND FINANCING CASH FLOWS AND IFRS/US GAAP

DIFFERENCES

Cash flow from investing activities (CFI) consists of the cash inflows and outflows that result from acquiring or disposing of long-term assets and certain investments.

Cash flow from financing activities (CFF) consists of the cash inflows and outflows that result from transactions affecting a firm's capital structure, such as borrowing, repaying debt, and issuing or redeeming equity securities.

Examples of each cash flow classification, in accordance with US GAAP, are presented in Figure 30.2.

Figure 30.2: US GAAP Cash Flow Classifications

Operating Activities	
<i>Inflows</i>	<i>Outflows</i>
Cash collected from customers	Cash paid to employees and suppliers
Interest and dividends received	Cash paid for other expenses
Sale proceeds from trading securities	Acquisition of trading securities
	Interest paid on debt or leases
	Taxes paid
Investing Activities	
<i>Inflows</i>	<i>Outflows</i>
Sale proceeds from PP&E and intangibles	Acquisition of PP&E and intangibles
Sale proceeds from debt and equity investments	Acquisition of debt and equity investments
Principal received from loans made to others	Loans made to others
Financing Activities	
<i>Inflows</i>	<i>Outflows</i>
Principal amounts of debt issued	Principal paid on debt or leases
Proceeds from issuing stock	Payments to reacquire stock
	Dividends paid to shareholders

Debt and equity investments (other than trading securities) and loans made to others are reported as investing cash outflows. However, under US GAAP the income from these investments (interest and dividends received) is reported as operating cash inflows. Principal amounts borrowed are reported as financing cash inflows, but interest paid is reported as an operating cash outflow. Finally, dividends paid to the firm's shareholders are financing cash outflows. Later in

this reading we will see that IFRS offers companies more discretion than US GAAP with regard to how they may classify these cash flows.



PROFESSOR'S NOTE

Don't confuse dividends received and dividends paid. Under US GAAP, dividends received are operating cash flows, and dividends paid are financing cash flows.

EXAMPLE: Computing CFI

Returning to the financial statements we used in our CFO examples, we will use the following information to compute CFI:

	20X7	20X6
Balance sheet extract	\$	\$
Land	35,000	40,000
Gross PP&E	69,000	60,000
Accumulated depreciation	<u>(12,000)</u>	<u>(9,000)</u>
Net PP&E	57,000	51,000

Income statement extract	
Depreciation	7,000
Gain on sale of land	10,000
Loss on disposal of PP&E	2,000

A footnote disclosure reveals that the company purchased PP&E for \$25,000 during 20X7.

Answer:

CFI will comprise additions to PP&E and the disposal proceeds from the sale of PP&E and land.

The footnote disclosure shows acquisitions of PP&E were \$25,000; however, gross PP&E has only increased by \$9,000. This, coupled with the disposal loss in the income statement, indicates that the company must have disposed of PP&E during 20X7.

Step 1: Compute cost of disposed PP&E	\$
Beginning gross PP&E	60,000
Acquisitions gross cost	25,000
Disposals gross cost	<u>(X)</u>
Ending gross PP&E	69,000

The value (X) we need to calculate is the cost of the assets that have been disposed of; that is, the gross book value the company recorded when it first acquired these assets. By rearranging the reconciliation, the gross cost of the disposed asset is computed as follows:

beginning PP&E + acquisitions – ending gross PP&E = disposals gross cost

disposals gross cost = \$60,000 + \$25,000 – \$69,000 = \$16,000

Next, we can see that accumulated depreciation has increased by \$3,000 when the 20X7 depreciation expense is \$7,000. The difference is also due to the asset disposal, because when a company disposes of an asset, it removes the accumulated depreciation on that asset from the total.

Step 2: Compute accumulated depreciation on disposed PP&E	\$
Beginning accumulated depreciation	9,000
Depreciation expense	7,000
Accumulated depreciation on disposed PP&E	<u>(X)</u>
Ending accumulated depreciation	12,000

Accumulated depreciation on disposed PP&E = beginning accumulated depreciation + depreciation expense – ending accumulated depreciation
 = \$9,000 + \$7,000 – \$12,000 = \$4,000.

Step 3: Compute carrying value of PP&E disposal	\$
Gross cost	16,000
Accumulated depreciation	<u>(4,000)</u>
Carrying value before disposal	12,000



PROFESSOR'S NOTE

Steps 1–3 can be combined for a quicker result:

Shortcut approach	\$
Beginning carrying value	51,000
Depreciation expense	(7,000)
Additions to PP&E	25,000
Carrying value of assets disposed	<u>(X)</u>
Ending carrying value	57,000

Carrying value of assets disposed = beginning carrying value – depreciation expense + additions to PP&E – ending carrying value

$$= \$51,000 - \$7,000 + \$25,000 - \$57,000 = \$12,000$$

When an asset is disposed of, the carrying value is removed from the balance sheet and netted against proceeds received from the sale, and any difference is reported as an accounting gain or loss in the income statement. A gain results if the proceeds from the sale exceed the carrying value, and a loss results if the proceeds are less than the carrying value. The only elements of the disposal that are cash flows are the disposal proceeds.

<i>Step 4: Compute disposal proceeds</i>	<i>\$</i>
Disposal proceeds	X
Carrying value removed from balance sheet	<u>(12,000)</u>
Disposal gain/(loss) in income statement	<u>(2,000)</u>

The disposal proceeds on the sale of PP&E must have been \$10,000.

We carry out similar calculations for the disposal of land. This will be simpler because land is not depreciated, so the carrying value is its gross cost. Because the footnotes do not mention any acquisitions of land, the carrying value of the land disposed of is the change in carrying value reported on the balance sheet.

$$\text{Carrying value of disposed land} = \$40,000 - \$35,000 = \$5,000$$

<i>Compute disposal proceeds</i>	<i>\$</i>
Disposal proceeds	X
Carrying value removed from balance sheet	<u>(5,000)</u>
Disposal gain/(loss) in income statement	<u>10,000</u>

The disposal proceeds on the sale of land must have been \$15,000.

Finally we can combine these results to determine cash flow from investing:

$$\text{CFI} = -\text{cash paid for PP\&E acquisitions} + \text{disposal proceeds}$$

$$\text{CFI} = -\$25,000 + \$10,000 + \$15,000 = \$0$$

The cost of new assets acquired was, by coincidence, perfectly matched by the proceeds from asset disposals.

EXAMPLE: Computing cash flow from financing (CFF)

Returning to the financial statements from our CFO examples, we will need the following information to compute CFF:

	20X7	20X6
Balance sheet extract	\$	\$
Current liabilities		
Dividends payable	6,000	1,000
Noncurrent liabilities		
Bonds payable	15,000	10,000
Stockholders' equity		
Common stock	15,000	20,000
Additional paid-in capital	<u>25,000</u>	<u>30,000</u>
Contributed capital	40,000	50,000
Retained earnings	61,000	30,500
Other financial statement data		
Net income	39,000	
Dividend declared	8,500	

A footnote disclosure reveals that the bonds outstanding had been issued at face value (par).

Answer:

We can begin by determining cash flows from issuing or repaying bonds:

	\$
Beginning bonds payable	10,000
Net principal flows	<u>X/(X)</u>
Ending bonds payable	15,000

net principal flow = ending bonds payable – beginning bonds payable =
\$15,000 – \$10,000 = \$5,000 (inflow)

Next we can determine cash flows from issuing or redeeming equity shares.

Contributed capital is the sum of common stock at par and additional paid-in capital, and reflects the price at which the company issued shares.

Reconciliation of equity contributed capital	\$
Beginning contributed capital	50,000
Net proceeds from repurchases and issuance	<u>X/(X)</u>
Ending contributed capital	40,000

net proceeds from buyback and issuance = ending contributed capital – beginning contributed capital = \$50,000 – \$40,000 = –\$10,000 (use of cash, or a net share repurchase of \$10,000)

Next we determine cash dividends paid, which can be a two-step process:

Step 1: Calculate dividend declared if it is not given.

In this example, the dividend declared of \$8,500 was given. If it had not been, we could compute it based on the change in retained earnings:

Reconciliation of retained earnings	\$
Beginning retained earnings	30,500
Net income	39,000
Dividend declared	<u>(X)</u>
Ending retained earnings	61,000

dividends declared = beginning retained earnings + net income – ending retained earnings

$$= \$30,500 + \$39,000 - \$61,000 = \$8,500$$

Step 2: Adjust the dividend declared for changes in the dividends payable liability. Simply because it is declared does not mean it has been paid. While the dividend declared does not pass through the income statement, it does reduce retained earnings; therefore, we treat it as a negative value. We then apply the increase/decrease rules for balance sheet liabilities:

Dividends paid	\$
Dividend declared	(8,500)
Increase in dividend payable liability	<u>5,000</u>
Cash dividend paid	(3,500)

Computation of CFF

	\$
Net principal on bonds	5,000
Net proceeds from share repurchase and issuance	(10,000)
Cash dividends	<u>(3,500)</u>
Cash flow from financing	(8,500)

Having computed CFO, CFI, and CFF, we can complete the cash flow statement.

Total cash flow

	\$
Cash flow from operations	50,000
Cash flow from investing	0
Cash flow from financing	(8,500)
Total cash flow	41,500
20X6 balance sheet cash	11,500
20X7 balance sheet cash	53,000

The total cash flow of \$41,500 is equal to the increase in cash. The difference between beginning cash and ending cash should be used as a check figure to ensure that the total cash flow calculation is correct.



PROFESSOR'S NOTE

The Level I CFA curriculum makes a few simplifications in the calculation of CFF.

For bonds issued at a premiums or discounts, the difference relative to par is amortized over the life of the bond. Amortization of premiums and discounts are not cash flows:

$$\text{coupon} + \text{amortized discount} - \text{amortized premium} = \text{interest expense}$$

With *amortization of a discount bond*, amortization will increase the interest expense and carrying value.

With *amortization of a premium bond*, amortization will decrease the interest expense and carrying value.

The curriculum mentions the noncash element affecting interest expense (see Figure 30.1), but does not mention the impact on the balance sheet carrying value.

The Level I curriculum sidesteps such complications by assuming that bonds are issued at par value in computations of CFF, which means there are no premiums or discounts to amortize.

A second simplification in the Level I curriculum is to assume that stock buybacks are transacted at the same price that the shares were initially issued for. In practice, this is unlikely, and if the buyback price differs from issue price, adjustments to retained earnings are made.

direct method.

The only difference between the indirect and direct methods of presentation is in the cash flow from operations (CFO) section. CFO under the direct method can be computed using a combination of the income statement and a statement of cash flows prepared under the indirect method.



PROFESSOR'S NOTE

The Level I CFA curriculum describes a three-step method for converting from indirect cash flow statements to direct cash flow statements. Provided that you understood the direct and indirect methods presented earlier, you will find little that you do not already know (although it is presented slightly differently).

Here is the three-step process:

Step 1: Aggregate all revenues and gains and all expenses and losses.

Step 2: Remove all noncash charges and disaggregate the remaining items.

Step 3: Convert from accruals to cash flows by adjusting for the change in working capital.

EXAMPLE: Conversion from indirect to direct CFO

Using the same data as the direct and indirect examples, the three stages are as follows:

	\$
<i>Step 1:</i> Total revenues and gains	114,000
Total expenses and losses	<u>75,000</u>
Net income	39,000

<i>Step 2:</i> Revenues less noncash charges	104,000
(\$114,000 – \$10,000)	
\$10,000 = gain on asset disposal	
Expenses less noncash charges	61,000
(\$75,000 – \$7,000 – \$2,000 – \$5,000)	
\$7,000 = depreciation	
\$2,000 = loss on disposal of PP&E	
\$5,000 = change in deferred tax liability	
Cost of goods sold	40,000
Wages	5,000
Interest	1,000
Tax payable	<u>15,000</u>
(Tax provision – increase in DTL)	
Total	61,000
<i>Step 3:</i> Cash collected from customers	107,000
\$104,000 – \$1,000 + \$4,000	
Cash paid to suppliers	(34,000)
–\$40,000 + \$2,000 + \$4,000	
Cash paid to employees	(8,500)
–\$5,000 – \$3,500	
Cash interest paid	(500)
–\$1,000 + \$500	
Cash paid to tax authorities*	<u>(14,000)</u>
–\$15,000 + \$1,000	
Net cash flow from operating activities	50,000

* Note that the computation of cash paid looks slightly different from the direct method example. This is because the change in deferred tax is treated as a noncash charge and adjusted for in Step 2.

LOS 30.d: Contrast cash flow statements prepared under International Financial Reporting Standards (IFRS) and US generally accepted accounting principles (US GAAP).

Recall that under US GAAP, dividends paid to the firm's shareholders are reported as financing activities, while interest paid is reported in operating

activities. Interest received and dividends received from investments are also reported as operating activities.

International Financial Reporting Standards (IFRS) allow more flexibility in the classification of cash flows. Under IFRS, interest and dividends received may be classified as either operating *or* investing cash inflows. Dividends paid to the company's shareholders and interest paid on the company's debt may be classified as either operating *or* financing cash outflows.

Another important difference relates to income taxes paid. Under US GAAP, all taxes paid are reported as operating cash outflows—even taxes related to investing and financing transactions. Under IFRS, income taxes are reported as operating cash outflows unless they are associated with an investing or financing transaction.

For example, consider a company that sells land that was held for investment for \$1 million. Income taxes on the sale total \$160,000. Under US GAAP, the firm reports an inflow of cash from investing activities of \$1 million and an outflow of cash from operating activities of \$160,000. Under IFRS, the firm can report a net inflow of \$840,000 from investing activities.

The differences between US GAAP and IFRS are summarized in Figure 30.3.

Figure 30.3: Differences Between US GAAP and IFRS

	US GAAP	IFRS
Interest received	CFO	CFO or CFI
Interest paid	CFO	CFO or CFF
Dividends received	CFO	CFO or CFI
Dividends paid	CFF	CFO or CFF
Bank overdraft	Treated as balance sheet debt	Treated as balance sheet cash
Taxes	CFO	May be split between CFO, CFI, and CFF according to the nature of the transaction that caused tax to become payable
Presentation of CFO	Direct preferred, but indirect allowed; a reconciliation of net income to CFO must be disclosed if using direct method	Direct preferred, but indirect allowed



MODULE QUIZ 30.3

Assuming US GAAP, use the following data to answer Questions 1 and 2.

Net income	\$45
Depreciation	75
Taxes paid	25
Interest paid	5
Dividends paid	10
Cash received from sale of company building	40
Issuance of preferred stock	35
Repurchase of common stock	30
Purchase of machinery	20
Issuance of bonds	50
Debt retired through issuance of common stock	45
Paid off long-term bank borrowings	15
Profit on sale of building	20

- Cash flow from investing activities is:
 - \$30.
 - \$20.
 - \$50.
- Cash flow from financing activities is:
 - \$30.
 - \$55.
 - \$75.
- Which of the following items is *least likely* considered a cash flow from financing activity under US GAAP?
 - Receipt of cash from the sale of bonds.
 - Payment of cash for dividends.
 - Payment of interest on debt.
- Which of the following would be *least likely* to cause a change in investing cash flow?
 - The sale of a division of the company.
 - The purchase of new machinery.
 - An increase in depreciation expense.
- Issuing bonds is classified as:
 - an investing activity.
 - a financing activity.
 - having no cash flow impact.
- The sale of land is classified as a(n):
 - operating activity.
 - investing activity.

- C. financing activity.
- 7. Which balance sheet items are *most likely* to be linked to cash flows from financing?
 - A. Long-lived assets.
 - B. Current assets and liabilities.
 - C. Long-term liabilities and equity.
- 8. Under IFRS, interest expense may be classified as:
 - A. either operating cash flow or financing cash flow.
 - B. operating cash flow only.
 - C. financing cash flow only.
- 9. Under US GAAP, dividends received from investments are classified as:
 - A. operating cash flow.
 - B. investing cash flow.
 - C. financing cash flow.

KEY CONCEPTS

LOS 30.a

Cash flow from operations is not the same as earnings because of the accruals process. To calculate CFO, balance sheet operating assets and liabilities are used to adjust income statement revenues and expenses to cash flows.

Cash flows can be computed as the income statement figure – increase in related operating assets + decreases in related operating assets + increases in related operating liabilities – decrease in related operating liabilities.

Operating activities typically relate to the firm's current assets and current liabilities. Investing activities typically relate to noncurrent assets. Financing activities typically relate to noncurrent liabilities and equity.

LOS 30.b

Under the direct method of presenting CFO, each line item of the accruals-based income statement is adjusted to get cash receipts or cash payments. The main advantage of the direct method is that it presents clearly the firm's operating cash receipts and payments.

Under the indirect method of presenting CFO, net income is adjusted for transactions that affect net income but do not affect operating cash flow, such as depreciation and gains or losses on asset sales, and for changes in balance sheet items. The main advantage of the indirect method is that it focuses on the

differences between net income and operating cash flow and gives the user of the accounts an indication of earnings quality.

CFI is calculated by determining the changes in asset accounts that result from investing activities. The cash flow from selling an asset is its book value plus any gain on the sale (or minus any loss on the sale).

CFF is the sum of net cash flows from creditors (new borrowings minus principal repaid) and net cash flows from shareholders (new equity issued minus share repurchases minus cash dividends paid).

LOS 30.c

An indirect cash flow statement can be converted to a direct cash flow statement by adjusting each income statement account for changes in associated balance sheet accounts and by eliminating noncash and nonoperating items (i.e., applying the direct method).

LOS 30.d

Differences in cash flow classifications between US GAAP and IFRS:

	US GAAP	IFRS
Interest received	CFO	CFO or CFI
Interest paid	CFO	CFO or CFF
Dividends received	CFO	CFO or CFI
Dividends paid	CFF	CFO or CFF
Bank overdraft	Treated as balance sheet debt	Treated as balance sheet cash
Taxes	CFO	May be split between CFO, CFI, and CFF according to the nature of the transaction that caused tax to become payable
Presentation of CFO	Direct preferred but indirect allowed; a reconciliation of net income to CFO must be disclosed if using the direct method	Direct preferred but indirect allowed

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 30.1, 30.2

- . **B** \$150,000 sales + \$10,000 decrease in accounts receivable = \$160,000 cash collections. The change in accounts payable does not affect cash collections. Accounts payable result from a firm's purchases from its suppliers. (Module 30.1, LOS 30.a, LOS 30.b)
- . **C** Revenue must be adjusted for the change in accounts receivable asset and unearned revenue liability:

	\$
Revenue	2,000,000
Increase in accounts receivable	(50,000)
Increase in unearned revenue	<u>70,000</u>
Cash from customers	<u>2,020,000</u>

(Module 30.1, LOS 30.a, LOS 30.b)

- . **B** The income statement tax expense must first be adjusted to remove the impact of any changes to deferred tax assets and liabilities to arrive at tax payable. Tax payable represents the tax owed to the tax authorities on this period's earnings. The tax-payable figure then must be adjusted for changes in tax-payable liabilities to arrive at the cash tax paid.

	\$
Tax expense	(200,000)
Increase in deferred tax liability	50,000
Increase in deferred tax asset	<u>(10,000)</u>
Tax payable	<u>(160,000)</u>
Decrease in tax payable liability	<u>(90,000)</u>
Cash taxes paid	<u>(250,000)</u>

(Module 30.1, LOS 30.a, LOS 30.b)

- . **B** The cost of goods sold needs to be adjusted for both change in inventory and change in accounts payable.

	\$
Cost of goods sold	(1,400,000)
Decrease in inventory	<u>20,000</u>
Purchases	(1,380,000)
Decrease in accounts payable	(30,000)
Cash paid to suppliers	<u>(1,410,000)</u>

(Module 30.1, LOS 30.a, LOS 30.b)

- . **B** Net income – profits from sale of land + depreciation + decrease in receivables – increase in inventories + increase in accounts payable – decrease in wages payable + increase in deferred tax liabilities = $120 - 2 + 25 + 20 - 10 + 7 - 5 + 15 = \170 . Note that the profit on the sale of land should be subtracted from net income because this transaction is classified as investing, not operating. (Module 30.2, LOS 30.b)

. **A**

Net income	\$78,000
Depreciation	12,000
Unrealized gain	(15,000)
Increase in accounts receivable	(52,000)
Increase in accounts payable	<u>29,000</u>
Cash flow from operations	\$52,000

(Module 30.2, LOS 30.b)

- . **C** The indirect method reconciles the difference between net income and CFO. The direct method shows operating cash received and paid—and, therefore, provides more information on its face than the indirect method. (Module 30.2, LOS 30.b)

Module Quiz 30.3

- . **B** Cash from sale of building – purchase of machinery = $40 - 20 = \$20$. (Module 30.3, LOS 30.b)
- . **A** Sale of preferred stock + issuance of bonds – principal payments on bank borrowings – repurchase of common stock – dividends paid = $35 + 50 - 15 - 30 - 10 = \30 . Note that we did not include \$45 of debt retired through issuance of common stock because this was a noncash transaction. Knowing how to handle noncash transactions is important. (Module 30.1, LOS 30.b)
- . **C** The payment of interest on debt is an *operating* cash flow under US GAAP. (Module 30.3, LOS 30.d)
- . **C** Depreciation does not represent a cash flow. To the extent that it affects the firm's taxes, an increase in depreciation changes operating cash flows, but not investing cash flows. (Module 30.3, LOS 30.b)
- . **B** Issuing bonds is classified as a financing activity. (Module 30.3, LOS 30.b)

- . **B** The sale of land is classified as an investing activity. (Module 30.3, LOS 30.b)
- . **C** Financing cash flows are linked primarily to changes in long-term liabilities and equity. Changes in current assets and liabilities tend to be linked to operating cash flows. Changes in long-lived assets are typically linked to investing cash flows. (Module 30.3, LOS 30.b)
- . **A** Under IFRS, interest expense can be classified as either an operating cash flow or financing cash flow. (Module 30.3, LOS 30.d)
- . **A** Dividends received from investments are classified as operating cash flow under US GAAP. (Module 30.3, LOS 30.d)

READING 31

ANALYZING STATEMENTS OF CASH FLOWS II

MODULE 31.1: ANALYZING STATEMENTS OF CASH FLOWS II

LOS 31.a: Analyze and interpret both reported and common-size cash flow statements.

The cash flow statement provides information to assess the firm's liquidity, solvency, and financial flexibility. An analyst can use the statement of cash flows to determine the following:

- | Whether regular operations generate enough cash to sustain the business
- | Whether enough cash is generated to pay off existing debts as they mature
- | Whether the firm is likely to need additional financing
- | Whether unexpected obligations can be met
- | Whether the firm can take advantage of new business opportunities as they arise

Cash flow analysis begins with evaluating the firm's sources and uses of cash from operating, investing, and financing activities. Sources and uses of cash change as the firm moves through its life cycle. For example, when a firm is in the early stages of growth, it may experience negative operating cash flow as it uses cash to increase inventory and receivables. Early-stage firms usually finance this negative operating cash flow externally by issuing debt or equity securities. These sources of financing are not sustainable. Eventually, the firm must begin generating positive operating cash flow or external investors will stop providing capital. Over the long term, successful firms must generate operating cash flows that exceed capital expenditures and provide a return to debt and equity holders.

An analyst should identify how a firm is generating its operating cash flow. The firm's earnings-related activities should provide its operating cash flow.

However, a firm can also generate positive operating cash flow in the near term by decreasing noncash working capital, such as by liquidating inventory and receivables or increasing payables. Decreasing noncash working capital is not sustainable because inventories and receivables cannot fall below zero and creditors will not extend credit indefinitely unless payments are made when due.

Operating cash flow provides a check on the quality of a firm's earnings. A stable relationship of operating cash flow and net income is an indication of quality earnings. Earnings that significantly exceed operating cash flow may be an indication of aggressive (or even improper) accounting choices such as recognizing revenues too soon or delaying the recognition of expenses. An analyst should also evaluate the variability of net income and operating cash flow as an indicator of a firm's risk.

An analyst should also examine a firm's sources and uses of cash from investing activities. Increasing capital expenditures, a use of cash, is usually an indication of growth. Conversely, a firm may reduce capital expenditures or even sell capital assets to save or generate cash. An analyst should inquire into the reasons for this because it may result in higher cash outflows in the future when the firm needs to replace older fixed assets. Investing cash flows may also result from acquisitions or investments in securities.

The financing activities section of the cash flow statement reveals information about whether the firm is generating cash flow by issuing debt or equity. It also indicates whether the firm is using cash to repay debt, reacquire stock, or pay dividends. For example, an analyst would certainly want to know if a firm issued debt and used the proceeds to reacquire stock or pay dividends to shareholders.

Common-Size Cash Flow Statements

As with the income statement and balance sheet, an analyst can use common-size analysis to interpret a cash flow statement.

The cash flow statement can be converted to common-size format by expressing each line item as a percentage of revenue. Alternatively, each inflow of cash can be expressed as a percentage of total cash inflows, and each outflow of cash can be expressed as a percentage of total cash outflows.

A revenue-based common-size cash flow statement is useful for identifying trends and forecasting future cash flow. Because each line item of the cash flow statement is stated in terms of revenue, once future revenue is forecast, cash flows can be estimated for those items that are tied to revenue.

EXAMPLE: Common-size cash flow statement analysis

Triple Y Corporation's common-size cash flow statement is shown in the following table. Each item has been stated as a percentage of revenue. Explain the decrease in Triple Y's total cash flow as a percentage of revenues.

Triple Y Corporation

Cash Flow Statement (Percentage of Revenues)			
Year	20X9	20X8	20X7
Net income	13.4%	13.4%	13.5%
Depreciation	4.0%	3.9%	3.9%
Accounts receivable	−0.6%	−0.6%	−0.5%
Inventory	−10.3%	−9.2%	−8.8%
Prepaid expenses	0.2%	−0.2%	0.1%
Accrued liabilities	5.5%	5.5%	5.6%
Operating cash flow	12.2%	12.8%	13.8%
Cash from sale of fixed assets	0.7%	0.7%	0.7%
Purchase of plant and equipment	−12.3%	−12.0%	−11.7%
Investing cash flow	−11.6%	−11.3%	−11.0%
Sale of bonds	2.6%	2.5%	2.6%
Cash dividends	−2.1%	−2.1%	−2.1%
Financing cash flow	0.5%	0.4%	0.5%
Total cash flow	1.1%	1.9%	3.3%

Answer:

Operating cash flow has decreased as a percentage of revenues. This appears to be due largely to accumulating inventories. Investing activities, specifically purchases of plant and equipment, have also required an increasing percentage of the firm's cash flow. These observations are consistent with a growing firm, but an analyst should inquire whether the increase in inventories was intended or unintended.

LOS 31.b: Calculate and interpret free cash flow to the firm, free cash flow to equity, and performance and coverage cash flow ratios.

Free cash flow is a measure of cash available for discretionary use, after a firm has covered its capital expenditures. This fundamental cash flow measure is often used for valuation. Analysts have several ways to measure free cash flow. Two of the more common measures are free cash flow to the firm and free cash flow to equity.

Free Cash Flow to the Firm

Free cash flow to the firm (FCFF) is cash available to all investors, both equity owners and debtholders. FCFF can be calculated by starting with either net income or operating cash flow. FCFF is calculated from net income as follows:

$$\text{FCFF} = \text{NI} + \text{NCC} + [\text{Int} \times (1 - \text{tax rate})] - \text{FC}_{\text{INV}} - \text{WC}_{\text{INV}}$$

where:

NI = net income

NCC = noncash charges (depreciation and amortization)

Int = cash interest paid

FC_{INV} = fixed capital investment (net capital expenditures)

WC_{INV} = working capital investment



PROFESSOR'S NOTE

Fixed capital investment is cash spent on fixed assets minus cash received from selling fixed assets. We cannot assume it is the same as CFI, which may also include cash flows from investments in securities and repaid principal from loans made.

Cash interest paid, net of tax, is added back to net income. This is because FCFF is the cash flow available to stockholders and debtholders. Because interest is paid to (and therefore “available to”) the debtholders, it must be included in FCFF.

Notice that three components of the equation are just computing operating cash flows under the indirect method:

$$\text{CFO} = \text{NI} + \text{NCC} - \text{WC}_{\text{INV}}$$

For this reason, FCFF can also be calculated from operating cash flow as follows:

$$\text{FCFF} = \text{CFO} + [\text{Int} \times (1 - \text{tax rate})] - \text{FC}_{\text{INV}}$$

where:

CFO = cash flow from operations

Int = cash interest paid

FC_{INV} = fixed capital investment (net capital expenditures)

For firms that follow IFRS, it is not necessary to adjust for interest that is included as a part of financing activities. Additionally, firms that follow IFRS can report dividends paid as operating activities. In this case, the dividends paid would be added back to CFO. Again, the goal is to calculate the cash flow that is available to the shareholders and debtholders. It is not necessary to adjust dividends for taxes because dividends paid are not tax deductible.

Free Cash Flow to Equity

Free cash flow to equity (FCFE) is the cash flow available for distribution to common shareholders. FCFE can be calculated as follows:

$$\text{FCFE} = \text{CFO} - \text{FC}_{\text{INV}} + \text{net borrowing}$$

where:

CFO = cash flow from operations

FC_{INV} = fixed capital investment (net capital expenditures)

net borrowing = debt issued – debt repaid



PROFESSOR'S NOTE

If net borrowing is negative (debt repaid exceeds debt issued), we would subtract net borrowing in calculating FCFE.

If a firm that follows IFRS has subtracted dividends paid in calculating CFO, dividends must be added back when calculating FCFE.

EXAMPLE: Free cash flow

Using the financial statement extracts presented next, calculate the company's FCFF and FCFE. Assume a tax rate of 40%.

	\$
Net income	39,000
Noncash charges	
Depreciation	7,000
Increase in deferred tax liability	5,000
Loss on disposal of PP&E	2,000
Gain from sale of land	<u>(10,000)</u>
Subtotal	43,000
Investment in working capital	
Increase in receivables	(1,000)
Decrease in inventories	2,000
Increase in accounts payable	4,000
Decrease in wages payable	(3,500)
Increase in interest payable	500
Increase in unearned revenue liability	4,000
Increase in taxes payable	<u>1,000</u>
Operating cash flows	<u>50,000</u>

Interest paid	\$
Interest expense	(1,000)
Increase in interest payable	<u>500</u>
Cash interest paid	<u>(500)</u>

Cash flow from investing (CFI)	\$
Cash paid for PP&E	(25,000)
Disposal proceeds	25,000
Cash flow from investing	<u>0</u>

Cash flow from financing (CFF)	\$
Net principal on bonds	5,000
Net proceeds from share repurchase and issuance	(10,000)
Cash dividends	<u>(3,500)</u>
Cash flow from financing	<u>(8,500)</u>

Answer:

In this example, CFI is equal to FC_{INV} due to the absence of cash flows related to financial investments.

$$\begin{aligned}
 FCFF &= CFO + [\text{cash interest paid} \times (1 - \text{tax rate})] - \text{fixed capital investment} \\
 &= \$50,000 + \$500(1 - 0.4) - \$0 = \$50,300
 \end{aligned}$$

$FCFE = CFO - \text{fixed capital investment} + \text{net borrowing}$

$$= \$50,000 - \$0 + \$5,000 = \$55,000$$

Alternatively, $FCFE = FCFF - [\text{interest paid} \times (1 - \text{tax rate})] + \text{net borrowing}$

$$= \$50,300 - \$500(1 - 0.4) + \$5,000 = \$55,000$$

Other Cash Flow Ratios

Just as with the income statement and balance sheet, the cash flow statement can be analyzed either over time or by comparing it to those of other firms. Cash flow ratios can be categorized as performance ratios and coverage ratios.

Performance Ratios

The **cash flow-to-revenue ratio** measures the amount of operating cash flow generated for each dollar of revenue:

$$\text{cash-flow-to-revenue ratio} = \frac{\text{CFO}}{\text{net revenue}}$$

The **cash return-on-assets ratio** measures the return of operating cash flow attributed to all providers of capital:

$$\text{cash-return-on-assets ratio} = \frac{\text{CFO}}{\text{average total assets}}$$

The **cash return-on-equity ratio** measures the return of operating cash flow attributed to shareholders:

$$\text{cash-return-on-equity ratio} = \frac{\text{CFO}}{\text{average total equity}}$$

The **cash-to-income ratio** measures the ability to generate cash from firm operations:

$$\text{cash-to-income ratio} = \frac{\text{CFO}}{\text{operating income}}$$

Cash flow per share is a variation of basic earnings per share measured by using CFO instead of net income:

$$\text{cash flow per share} = \frac{\text{CFO} - \text{preferred dividends}}{\text{weighted average number of common shares}}$$

Note: If common dividends were classified as operating activities under IFRS, they should be added back to CFO for calculating cash flow per share.

Coverage Ratios

The **debt coverage ratio** measures financial risk and leverage:

$$\text{debt coverage ratio} = \frac{\text{CFO}}{\text{total debt}}$$

The **interest coverage ratio** measures the firm's ability to meet its interest obligations:

$$\text{interest coverage ratio} = \frac{\text{CFO} + \text{interest paid} + \text{taxes paid}}{\text{interest paid}}$$

Note: If interest paid was classified as a financing activity under IFRS, no interest adjustment is necessary.

The **reinvestment ratio** measures the firm's ability to acquire long-term assets with operating cash flow:

$$\text{reinvestment ratio} = \frac{\text{CFO}}{\text{cash paid for long-term assets}}$$

The **debt payment ratio** measures the firm's ability to satisfy long-term debt with operating cash flow:

$$\text{debt payment} = \frac{\text{CFO}}{\text{cash long-term debt repayment}}$$

The **dividend payment ratio** measures the firm's ability to make dividend payments from operating cash flow:

$$\text{dividend payment ratio} = \frac{\text{CFO}}{\text{dividends paid}}$$

The **investing and financing ratio** measures the firm's ability to purchase assets, satisfy debts, and pay dividends:

$$\text{investing and financing ratio} = \frac{\text{CFO}}{\text{cash outflows from investing and financing activities}}$$



MODULE QUIZ 31.1

1. In preparing a common-size cash flow statement, each cash flow is expressed as a percentage of:
 - A. total assets.
 - B. total revenue.
 - C. the change in cash.
2. To calculate free cash flow to the firm based on operating cash flow, an analyst should add interest expense net of tax and subtract:
 - A. noncash charges.
 - B. fixed capital investment.

- C. working capital investment.
- 3. The reinvestment ratio measures a firm's ability to use its operating cash flow to:
 - A. pay dividends.
 - B. invest in working capital.
 - C. acquire long-lived assets.

KEY CONCEPTS

LOS 31.a

An analyst should determine whether a company is generating positive operating cash flow over time that is greater than its capital spending needs and whether the company's accounting policies are causing reported earnings to diverge from operating cash flow.

A common-size cash flow statement shows each item as a percentage of revenue, or shows each cash inflow as a percentage of total inflows and each outflow as a percentage of total outflows.

LOS 31.b

Free cash flow to the firm (FCFF) is the cash available to all investors, both equity owners and debtholders:

- FCFF = net income + noncash charges + [cash interest paid $\times$ (1 – tax rate)] – fixed capital investment – working capital investment
- FCFF = CFO + [cash interest paid $\times$ (1 – tax rate)] – fixed capital investment

Free cash flow to equity (FCFE) is the cash flow that is available for distribution to the common shareholders after all obligations have been paid.

$$\text{FCFE} = \text{CFO} - \text{fixed capital investment} + \text{net borrowing}$$

Cash flow performance ratios (e.g., cash return on equity or on assets) and cash coverage ratios (e.g., debt coverage or cash interest coverage) provide information about the firm's operating performance and financial strength.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 31.1

- . **B** The cash flow statement can be converted to common-size format by expressing each line item as a percentage of revenue. (LOS 31.a)

- . **B** FCFF can be calculated from CFO by adding interest expense net of tax and subtracting fixed capital investment. (LOS 31.b)
- . **C** The reinvestment ratio is CFO/cash paid for long-term assets. (LOS 31.b)

READING 32

ANALYSIS OF INVENTORIES

INTRODUCTION

This reading assumes that you are familiar with the cost flow methodologies first-in, first-out (FIFO), last-in, first-out (LIFO), and average cost. You should also be able to compute cost of sales (cost of goods sold, or COGS), gross profit margin, and ending inventory under each cost flow approach. You should also be aware of how to adjust financial statements from LIFO to FIFO using disclosures relating to the LIFO reserve. Ensure that you are aware of which costs can be capitalized in ending inventory and which costs must be expensed. For candidates who need to review these concepts, we have provided supplementary study material in the “Resource Library” section of our online Learning Management System.

MODULE 32.1: INVENTORY MEASUREMENT

LOS 32.a: Describe the measurement of inventory at the lower of cost and net realisable value and its implications for financial statements and ratios.

Under IFRS, inventory is reported on the balance sheet at the lower of cost or net realizable value. **Net realizable value (NRV)** is equal to the expected sales price less the estimated selling costs and completion costs. If NRV is less than the balance sheet value of inventory, the inventory is “written down” to NRV, and the loss is recognized in the income statement either as a separate line item or by increasing cost of goods sold. If the value of the inventory recovers later, the inventory can be “written up,” and the gain is recognized in the income statement as a separate line item or by reducing COGS by the amount of the recovery. Because inventory is valued at the lower of cost or NRV, inventory cannot be written up by more than it was previously written down.



PROFESSOR'S NOTE

The write-down (or subsequent write-up) of inventory is usually accomplished through the use of a valuation allowance account, which is a contra-asset account similar to accumulated depreciation. By using a valuation allowance account, the firm is able to separate the original cost of inventory from the carrying value of the inventory.

Under US GAAP, companies that use inventory cost methods other than LIFO or the retail method report inventories at the lower of cost or NRV. For companies using LIFO or the retail method, inventory is reported on the balance sheet at the **lower of cost or market**. Market is usually equal to replacement cost, but it cannot be greater than NRV or less than NRV minus a normal profit margin. If replacement cost exceeds NRV, then market is NRV. If replacement cost is less than NRV minus a normal profit margin, then market is NRV minus a normal profit margin.



PROFESSOR'S NOTE

Think of lower of cost or market, where *market* cannot be outside a range of values. The range is from NRV minus a normal profit margin, to NRV. So, the size of the range is the normal profit margin. *Net* means the sales price less selling and completion costs.

The retail method is not explained in the Level I curriculum. Companies that resell merchandise can use this method. The methodology involves computing the cost of goods available for sale (beginning inventory + purchases). The cost of goods sold is estimated by taking sales and deducting the normal profit margin. Finally, ending inventory is computed as cost of goods sold less goods available for sale.

If the market value of inventory decreases below its cost, the inventory is written down to market on the balance sheet. The decrease in value is recognized in the income statement by increasing COGS for relatively small changes in value, or by recording the loss from the inventory write-down separately for a relatively large change in value. The market value becomes the new cost basis.

If there is a subsequent recovery in value, no write-up is allowed under US GAAP. This applies to companies using lower of cost or NRV as well as those using lower of cost or market.

EXAMPLE: Inventory write-down

Zoom, Inc. (Zoom), sells digital cameras. Per-unit cost information pertaining to Zoom's inventory is as follows:

Original cost	\$210
Estimated selling price	\$225
Estimated selling costs	\$22
Net realizable value	\$203
Replacement cost	\$197
Normal profit margin	\$12

What are the per-unit carrying values of Zoom's inventory using (1) lower of cost or NRV and (2) lower of cost or market?

Answer:

Using the lower of cost or NRV, because the original cost of \$210 exceeds net realizable value ($\$225 - \$22 = \$203$), the inventory is written down to the net realizable value of \$203. The \$7 decrease in value ($\$203 \text{ NRV} - \$210 \text{ original cost}$) is reported in the income statement.

Using the lower of cost or market, market is equal to the replacement cost of \$197 because NRV of \$203 is greater than replacement cost, and NRV minus a normal profit margin ($\$203 - \$12 = \$191$) is less than replacement cost. Because original cost exceeds market (replacement cost), the inventory is written down to \$197, and a \$13 loss ($\$197 \text{ replacement cost} - \$210 \text{ original cost}$) is reported in the income statement.

EXAMPLE: Inventory write-up

Assume that in the year after the write-down in the previous example, NRV and replacement cost both increase by \$10. What is the impact of the recovery under IFRS, and under US GAAP if lower of cost or market is used?

Answer:

Under IFRS, Zoom will write up inventory to \$210 per unit and recognize a \$7 gain in its income statement. The write-up (gain) is limited to the original write-down of \$7. The carrying value cannot exceed original cost.

Under US GAAP, no write-up is allowed. The per-unit carrying value will remain at \$197. Zoom will simply recognize higher profit when the inventory is sold.

LIFO ending inventory is based on older, lower costs (assuming inflation) than under FIFO. Because cost is the basis for determining whether an impairment has occurred, LIFO firms are less likely to recognize inventory write-downs than firms using FIFO or weighted average cost.

Analysts must understand how an inventory write-down or write-up affects a firm's ratios. For example, a write-down may significantly affect inventory turnover in current and future periods. Thus, comparability of ratios across periods may be an issue.

In certain industries, reporting inventory above historical cost is permitted under IFRS and US GAAP. This exception applies primarily to producers and dealers of commodities, such as agricultural and forest products, mineral ores, and precious metals. Under this exception, inventory is reported at NRV, and any unrealized gains and losses from changing market prices are recognized in the income statement. If an active market exists for the commodity, the quoted market price is used to value the inventory. Otherwise, recent market transactions are used.

A write-down of inventory to NRV affects the financial statements and ratios in several ways. Assuming the write-down is reported as part of the cost of sales, these effects in the period of the write-down include the following:

- | Because inventory is a current asset, an inventory write-down decreases both current and total assets.
- | The current ratio (CA/CL) decreases. The quick ratio is unaffected because inventories are not included in its numerator.
- | Inventory turnover (COGS/average inventory) increases, which decreases days' inventory on hand and the cash conversion cycle.
- | The decrease in total assets increases total asset turnover and increases the debt-to-assets ratio.
- | Equity decreases, increasing the debt-to-equity ratio.
- | The increase in COGS decreases gross margin, operating margin, and net margin.
- | The percentage decrease in net income can be expected to be greater than the percentage decrease in assets or equity. As a result, both return on assets (ROA) and return on equity (ROE) decrease.

For periods that follow a write-down of inventory to NRV, COGS may be decreased by lower inventory carrying values, which will increase profitability. Together with the decreases in assets and equity from the write-down, an increase in net income from decreased COGS will increase reported ROA and ROE in subsequent periods.

MODULE 32.2: INFLATION IMPACT ON FIFO AND LIFO

LOS 32.b: Calculate and explain how inflation and deflation of inventory costs affect the financial statements and ratios of companies that use different inventory valuation methods.

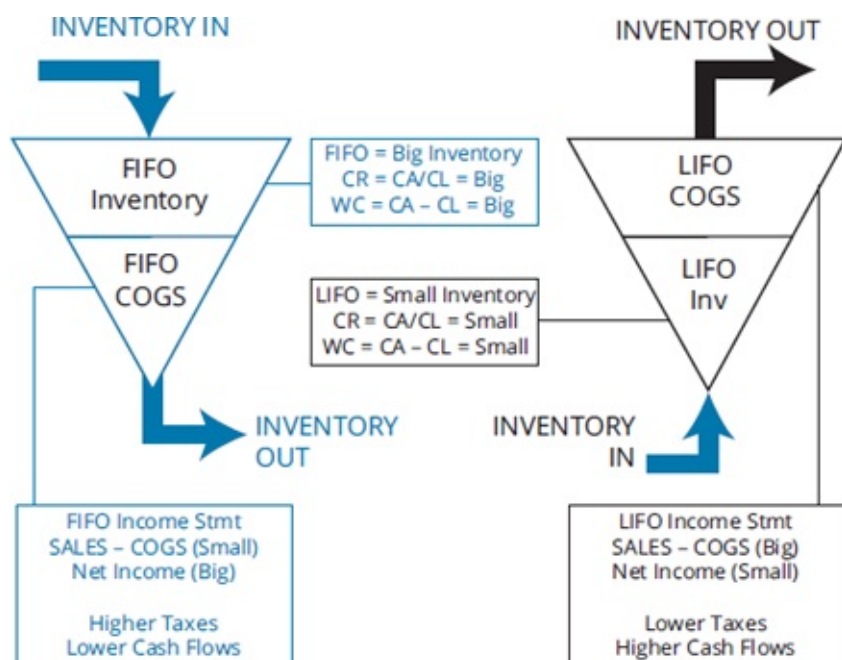
During inflationary periods and with stable or increasing inventory quantities, LIFO COGS is higher than FIFO COGS. This is because the last units purchased have a higher cost than the first units purchased. Under LIFO, the more costly last units purchased are assumed to be the first units sold. Of course, higher COGS under LIFO will result in lower gross profit and net income compared to FIFO.

Using similar logic, we can see that LIFO ending inventory is lower than FIFO ending inventory because under LIFO, ending inventory is valued using older, lower costs.

During deflationary periods, assuming stable or increasing inventory quantities, the cost flow effects of using LIFO and FIFO will be reversed; that is, LIFO COGS will be lower, and LIFO ending inventory will be higher. This makes sense because the most recent lower-cost purchases are assumed to be sold first under LIFO, and the units in ending inventory are assumed to be the earliest purchases with higher costs.

Consider the diagram in Figure 32.1 to help visualize the FIFO-LIFO difference during periods of rising prices and growing inventory levels.

Figure 32.1: LIFO and FIFO Diagram—Rising Prices and Growing Inventory Balances



Remember, it's not the older or newer physical inventory units that are reported in the income statement and balance sheet; rather, it is the *costs* that are assigned to the units sold and to the units remaining in inventory.



PROFESSOR'S NOTE

Be able to describe the effects of LIFO and FIFO, assuming inflation, in your sleep. When prices are falling, the effects are simply reversed. When you are finished with this review, take the time to look at these graphs and relationships again to solidify the concepts in your mind.

During periods of stable prices, the LIFO, FIFO, and average cost methods will yield the same results for inventory, COGS, and gross profit. During periods of trending prices (up or down), different cost flow methods may result in significant differences in these items.



PROFESSOR'S NOTE

The presumption in this section is that inventory quantities are stable or increasing.

When prices are rising or falling, FIFO provides the most useful measure of ending inventory. Recall that FIFO inventory is made up of the most recent purchases. These purchase costs can be viewed as a better approximation of current cost, and thus a better approximation of economic value. LIFO inventory,

by contrast, is based on older costs that may differ significantly from current economic value.

Changing prices can also produce significant differences between COGS under LIFO and FIFO. Recall that LIFO COGS is based on the most recent purchases. As a result, when prices are rising, LIFO COGS will be higher than FIFO COGS. When prices are falling, LIFO COGS will be lower than FIFO COGS. Because LIFO COGS is based on the most recent purchases, LIFO COGS produces a better approximation of current cost in the income statement.

When prices are changing, the average cost method will produce values of COGS and ending inventory between those of FIFO and LIFO.

Because COGS is subtracted from revenue in calculating gross profit, gross profit is also affected by the choice of cost flow method. Assuming inflation, higher COGS under LIFO will result in lower gross profit. In fact, all profitability measures (gross profit, operating profit, income before taxes, and net income) will be affected by the choice of cost flow method.

Figure 32.2: Effects of Inventory Valuation Methods

	FIFO	LIFO
Cost of sales	Lower	Higher
Ending inventory	Higher	Lower
Gross profit	Higher	Lower
Note: This assumes increasing prices and stable or increasing inventory levels.		

Effects on Ratios

Again, assuming increasing prices and stable or rising inventory levels, we can trace the ratio effects of the adjustments of LIFO values to FIFO values.

Profitability. Compared to FIFO, LIFO produces higher COGS in the income statement and results in lower earnings. Any profitability measure that includes COGS will be higher under FIFO. For example, FIFO COGS will result in higher gross margins, operating margins, and net profit margins as compared to LIFO.

Liquidity. Compared to FIFO, LIFO results in a lower inventory value on the balance sheet. Because inventory (a current asset) is higher under FIFO, the current ratio, a popular measure of liquidity, is also higher under FIFO. Working capital is higher under FIFO as well, because current assets are higher.

Activity. Inventory turnover ($\text{COGS} / \text{average inventory}$) is higher for firms that use LIFO compared to firms that use FIFO. Under LIFO, COGS is valued at more recent, higher costs (higher numerator), while inventory is valued at older, lower costs (lower denominator). Adjusting to FIFO values will result in lower turnover and higher days of inventory on hand ($365 / \text{inventory turnover}$).

Solvency. Adjusting to FIFO results in higher total assets because inventory is higher. Higher total assets under FIFO result in higher stockholders' equity (assets – liabilities). Because total assets and stockholders' equity are higher under FIFO, the debt ratio and the debt-to-equity ratio are lower under FIFO compared to LIFO.

LIFO Liquidation

A **LIFO liquidation** occurs when a LIFO firm's inventory quantities decline. Older, lower costs are included in COGS compared to a situation in which inventory quantities are not declining. LIFO liquidation results in higher profit margins and higher income taxes compared to what they would be if inventory quantities were not declining. The extra profit reported with a LIFO liquidation inflates operating margins by recognizing historical inflationary gains from increasing inventory prices as income in the current period. Increases in profit margins from LIFO liquidation are not sustainable, however, because a firm cannot continue forever to sell existing inventory without replenishment.

Management could use a LIFO liquidation (draw down inventory) to artificially inflate current period earnings. Inventory declines can also be caused by events outside the management's control, such as strikes or materials shortages at a key supplier that make inventory reduction involuntary, or a decline in expected customer orders that results in a voluntary reduction in inventory to suit market conditions.

Analysts must look to the LIFO reserve disclosures in the footnotes to see if the LIFO reserve has decreased over the period, which would indicate the possibility of a LIFO liquidation that requires adjustment of profit margins if its impact has been significant.

EXAMPLE: Comparing FIFO and LIFO with a LIFO liquidation

Willock Corporation is a retailer incorporated at the start of Year 1. Sales prices and purchase costs have been inflating at 5% per annum. Using the data

provided on sales and purchase units and prices compute Willocks gross profit and gross profit margin using FIFO and LIFO cost flow methodology. (Assume all purchases each year are made at the beginning of the year for the indicated prices.)

Data: Units	Year 1	Year 2	Year 3
Sales	10,000	12,000	16,000
Beginning Inventory	0	4,000	7,000
Purchases	<u>14,000</u>	15,000	<u>10,000</u>
Available for sale	14,000	19,000	17,000
Ending inventory	<u>(4,000)</u>	(7,000)	<u>(1,000)</u>
Units sold	10,000	12,000	16,000

Sales and purchase prices	Year 1	Year 2	Year 3
Sales price	\$100	\$105	\$110
Purchase price	\$80	\$84	\$88

Answer:

FIFO Gross Profit	Year 1	Year 2	Year 3
	\$	\$	\$
Sales	1,000,000	1,260,000	1,760,000
Beginning inventory	0	320,000	588,000
Purchases	<u>1,120,000</u>	<u>1,260,000</u>	<u>880,000</u>
Available for sale	1,120,000	1,580,000	1,468,000
Ending inventory	<u>-320,000</u>	<u>-588,000</u>	<u>-88,000</u>
Cost of goods sold	800,000	992,000	1,380,000
Gross profit	\$200,000	\$268,000	\$380,000

LIFO Gross Profit	Year 1	Year 2	Year 3
	\$	\$	\$
Sales	1,000,000	1,260,000	1,760,000
Beginning inventory	0	320,000	572,000
Purchases	<u>1,120,000</u>	<u>1,260,000</u>	<u>880,000</u>
Available for sale	1,120,000	1,580,000	1,452,000
Ending inventory	<u>-320,000</u>	<u>-572,000</u>	<u>-80,000</u>
Cost of goods sold	800,000	1,008,000	1,372,000
Gross profit	\$200,000	\$252,000	\$388,000

In Year 1, LIFO and FIFO gross profit is the same. This is because there was no beginning inventory and we assumed the purchases in the year were all at

the same price. If purchases had been made at different prices during Year 1, FIFO and LIFO results would differ.

In Year 2, FIFO gives a higher gross margin because ending inventory is valued at the most recent purchase price of \$84, compared to LIFO which values 4,000 units of ending inventory at the older purchase price of \$80 and the remaining 3,000 units at \$84. This result should not be a surprise as both purchase prices and ending inventory quantities increased.

In Year 3, LIFO gives a higher gross margin even though purchase prices are inflating. This unusual result is caused by a LIFO liquidation. Inventory levels decreased during the year because the number of units purchased was less than the number of units sold.

Key Ratios	Year 1	Year 2	Year 3
FIFO gross margin	20.00%	21.27%	21.59%
LIFO gross margin	20.00%	20.00%	22.05%

The impact of the LIFO liquidation can be detected in the company's gross profit margins. In Year 2, because of both rising purchase prices and increasing inventory levels, FIFO reports a higher gross margin (gross profit/sales). In Year 3 the LIFO liquidation releases older purchase prices previously stored in ending inventory through cost of goods sold, resulting in a lower cost of goods sold figure and a higher gross profit margin.



MODULE QUIZ 32.1, 32.2

1. Kamp, Inc., sells specialized bicycle shoes. At year-end, due to a sudden increase in manufacturing costs, the replacement cost per pair of shoes is \$55. The original cost is \$43, and the current selling price is \$50. The normal profit margin is 10% of the selling price, and the selling costs are \$3 per pair. According to US GAAP, which of the following amounts should each pair of shoes be reported on Kamp's year-end balance sheet?
 - A. \$42.
 - B. \$43.
 - C. \$47.
2. Poulter Products reports under IFRS and wrote its inventory value down from cost of \$400,000 to net realizable value of \$380,000. The *most likely* financial statement effect of this change is a(n):
 - A. increase in cost of sales.
 - B. decrease in depreciation charges.
 - C. loss reported as other comprehensive income.
3. Which of the following statements relating to US GAAP inventory valuation is *most accurate*?

- A. Companies using FIFO should report inventory in the balance sheet at the lower of cost or market value.
 - B. Inventory can be written up, but not by more than it was previously written down.
 - C. When establishing market value, net realizable value should be used as the upper limit if replacement cost exceeds net realizable value.
4. Which of the following statements relating to the effects of an inventory write-down is *least accurate*?
- A. Asset turnover will increase.
 - B. The quick ratio will decrease.
 - C. The debt-to-equity ratio will increase.
5. Under which inventory cost flow assumption does inventory on the balance sheet *best* approximate its current cost?
- A. First-in, first-out.
 - B. Weighted average cost.
 - C. Last-in, first-out.
6. In periods of rising prices and stable inventory quantities, which of the following *best* describes the effect on gross profit of using LIFO as compared to using FIFO?
- A. Lower.
 - B. Higher.
 - C. The same.
7. In an inflationary environment, a LIFO liquidation will *most likely* result in an increase in:
- A. inventory.
 - B. accounts payable.
 - C. operating profit margin.

MODULE 32.3: PRESENTATION AND DISCLOSURE

LOS 32.c: Describe the presentation and disclosures relating to inventories and explain issues that analysts should consider when examining a company's inventory disclosures and other sources of information.

Inventory disclosures, usually found in the financial statement footnotes, are useful for evaluating the firm's inventory management. The disclosures are also useful for adjusting a firm's financial statements to make them comparable with other firms in the industry.

Required inventory disclosures are similar under US GAAP and IFRS and include the following:

- ▮ Cost flow method (LIFO, FIFO, etc.) used

- | Total carrying value of inventory, with carrying value by classification (raw materials, work in progress, and finished goods), if appropriate
- | Carrying value of inventories reported at fair value less selling costs
- | The cost of inventory recognized as an expense (COGS) during the period
- | Amount of inventory write-downs (valuation allowances) during the period
- | Reversals of inventory write-downs during the period, including a discussion of the circumstances of reversal (IFRS only because US GAAP does not allow reversals)
- | Carrying value of inventories pledged as collateral

Inventory Ratios

Merchandising firms, such as wholesalers and retailers, purchase inventory that is ready for sale. In this case, inventory is reported in one account on the balance sheet. Manufacturing firms normally report inventory using three separate accounts: raw materials, work in progress, and finished goods. Analysts can use these disclosures, along with other sources of information such as Management's Discussion and Analysis, economic data specific to the industry, industry trade publications, and other sections of the firm's financial reports, as a signal of a firm's future revenues and earnings.

For example, an increase in raw materials or work in progress inventory may indicate that the firm expects increase in demand. Higher demand should result in higher revenues and earnings. Conversely, an increase in finished goods inventory, while raw materials and work in progress are decreasing, may indicate decreasing demand and potential inventory write-downs in the future.

Analysts should also examine the relationship between sales and finished goods. Finished goods inventory growing faster than sales may indicate declining demand and excessive or potentially obsolete inventory. Obsolete inventory will result in lower earnings in the future when the inventory is written down. In addition, too much inventory is costly, as the firm may incur storage costs, insurance premiums, and inventory taxes. Holding too much inventory uses cash that might be deployed more efficiently somewhere else.

The inventory turnover ratio measures how quickly a firm is selling its inventory. Generally, high inventory turnover (low days of inventory on hand) is desirable. However, inventory turnover can be too high. A firm with an inventory turnover

ratio that is too high may not be carrying enough inventory to satisfy customers' needs, which can cause the firm to lose sales. High inventory turnover may also indicate that inventory write-downs have occurred. Write-downs are usually the result of poor inventory management.

To further assess the explanation for high inventory turnover, we can look at inventory turnover relative to sales growth within the firm and industry. High turnover with slower growth may be a sign of inadequate inventory quantities. Alternatively, sales growth at or above the industry average supports the conclusion that high inventory turnover reflects greater efficiency.

EXAMPLE: Inventory ratio analysis

Hyzer Corporation is a US manufacturer of products for use in the sport of disc golf. Hyzer presents the following financial statements data:

Hyzer Corporation Balance Sheets	20X5 \$000	20X6 \$000	20X7 \$000
Assets			
Cash	950	1,250	2,675
Trade receivables	520	1,520	3,020
Inventories	<u>500</u>	<u>900</u>	<u>300</u>
Total current assets	1,970	3,670	5,995
Gross plant and equipment	5,600	5,400	5,400
Accumulated depreciation	<u>2,400</u>	<u>2,980</u>	<u>3,650</u>
Net plant and equipment	<u>3,200</u>	<u>2,420</u>	<u>1,750</u>
Total assets	5,170	6,090	7,745
Liabilities and equity			
Trade payables	320	220	720
Accrued expenses	75	175	275
Interest payable	45	195	215
Tax payable	80	156	295
Short-term debt	<u>200</u>	<u>120</u>	<u>0</u>
Current liabilities	720	866	1,505
Long-term debt	1,800	1,850	2,150

Hyzer Corporation Balance Sheets	20X5 \$000	20X6 \$000	20X7 \$000
Common stock	1,000	1,100	1,050
Additional paid-in-capital	500	550	525
Retained earnings	<u>1,150</u>	<u>1,724</u>	<u>2,515</u>
Stockholders' equity	<u>2,650</u>	<u>3,374</u>	<u>4,090</u>
Total liabilities and equity	5,170	6,090	7,745

	20X6 \$000	20X7 \$000
Revenue	5,500	7,500
Cost of sales	<u>2,600</u>	<u>4,100</u>
Gross profit	2,900	3,400
Operating expenses	<u>1,900</u>	<u>2,150</u>
Operating profit	1,000	1,250
Disposal (loss)/gain	(30)	80
Interest expense	<u>150</u>	<u>200</u>
Earnings before tax	820	1,130
Taxes	<u>246</u>	<u>339</u>
Net income	574	791

20X5 sales = \$5,300,000

A review of the footnote disclosure reveals that Hyzer uses FIFO to establish the cost of raw materials. Hyzer records balance sheet inventory values at the lower of cost and NRV.

The following footnote disclosure is given for inventory balances:

Inventory net value	20X5 \$000	20X6 \$000	20X7 \$000
Raw materials	120	207	68
Valuation allowance	<u>-20</u>	<u>-27</u>	<u>-2</u>
Net carrying value	100	180	66
Work in progress	50	95	31
Valuation allowance	<u>0</u>	<u>-5</u>	<u>-1</u>
Net carrying value	50	90	30
Finished goods	403	706	221
Valuation allowance	<u>-53</u>	<u>-76</u>	<u>-17</u>
Net carrying value	350	630	204
Total net carrying value	500	900	300

Use the data in the financial statements to calculate and comment on the following ratios for 20X6 and 20X7:

- Inventory turnover and days of inventory on hand
- Gross profit margin and sales growth
- Current and quick ratios

Answer:

Key ratios summary	20X6	20X7
Inventory turnover	3.7	6.8
Days of inventory on hand	98	53
Gross profit margin	53%	45%
Sales growth	4%	36%
Current ratio	4.24	3.98
Quick ratio	3.20	3.78

Inventory turnover (cost of sales / average inventory):

$$20X6: \$2,600,000 / [(\$500,000 + \$900,000) / 2] = 3.71$$

$$20X7: \$4,100,000 / [(\$900,000 + \$300,000) / 2] = 6.83$$

Days of inventory on hand (365 / inventory turnover):

$$20X6: 365 / 3.71 = 98 \text{ days}$$

$$20X7: 365 / 6.83 = 53 \text{ days}$$

Inventory turnover has increased dramatically in 20X7, and as a result, days of inventory on hand has dropped. At first glance, this could be assumed to be due to improvements in working capital management. Inventory balances at year-end have dropped dramatically from \$900,000 in 20X6 to \$300,000 in 20X7, which is a significant decline—and therefore, is unlikely to be explained purely by improvements in inventory management.

The decrease in inventory balances could be misinterpreted as indicating that the company expects demand for its products to decline significantly. The inventory disclosure shows that raw materials, work in progress, and finished goods have declined substantially in 20X7. Typically, if customer demand declined significantly, we would expect raw materials and work in progress to decline relative to finished goods. This indicates that raw materials were 20% of inventory in 20X6 (\$180,000 / \$900,000) and 22% in 20X7 (\$66,000 / \$300,000), while work in progress was 10% in 20X6 (\$90,000 / \$900,000) and remained at 10% in 20X7 (\$30,000 / \$300,000). Raw materials and work in

progress have not declined proportionately relative to finished goods. This indicates that lower future demand is not expected.

Another clue for the analyst is that the valuation allowance for finished goods has decreased as a proportion of the cost of finished goods, rather than increased.

$$20X6: \$76,000 / \$706,000 = 10.7\%$$

$$20X7: \$17,000 / \$221,000 = 7.7\%$$

If product obsolescence due to expected future decreases in sales was a significant factor, the valuation allowance for finished goods should have increased proportionally. Remember that the valuation allowance indicates that NRV has declined below cost.

These factors combined indicate to the analyst that slowing customer demand and obsolescence is not the reason for the dramatic decline in 20X7 ending inventory.

Looking at sales growth, the analyst can see a significant increase in 20X7:

$$20X6: (\$5,500,000 / \$5,300,000) - 1 = 3.77\%$$

$$20X7: (\$7,500,000 / \$5,500,000) - 1 = 36.36\%$$

It would seem unlikely (but not impossible) that a company seeing such increases in sales would be facing a dramatic decline in the next period. This is another indicator that the declining inventory balances are not caused by obsolescence or falling demand.

Gross profit margins, however, did decline in 20X7, suggesting that the cost of raw materials, manufacturing costs, or both increased:

$$20X6: \$2,900,000 / \$5,500,000 = 52.73\%$$

$$20X7: \$3,400,000 / \$7,500,000 = 45.33\%$$

Positive sales growth, coupled with declining margins, shows that the cost of producing disc golf products has increased. This, coupled with the decline in inventory balances in 20X7, could indicate that the company is facing supply-side shocks. Declining margins reinforce the suspicion that the significant drop in inventory levels is unlikely to be solely explained by efficiency improvements. The sales growth of 36% in 20X7 should also make the analyst question why inventory balances have declined so substantially.

The analyst should seek out further explanation for the decline. Potential sources are as follows:

- Management discussion and analysis
- Significant events disclosed in the accounts
- Conferences and communication with the senior management
- Media and industry reports and journals
- The accounts of companies in the same industry

Looking at Hyzer's liquidity, the current ratio declined between 20X6 and 20X7 while the quick ratio improved. The major difference between the two measures is the inclusion of inventory in the current ratio.

Current ratio: (current assets / current liabilities)

$$20X6: \$3,670,000 / \$866,000 = 4.24$$

$$20X7: \$5,995,000 / \$1,505,000 = 3.98$$

Quick ratio: (cash + market securities + receivables) / current liabilities

$$20X6: \$2,770,000 / \$866,000 = 3.20$$

$$20X7: \$5,695,000 / \$1,505,000 = 3.78$$

Some analysts prefer the quick ratio because it removes the impact of inventory on liquidity. Inventory is one of the most illiquid of current assets, as there can be a significant delay (days of inventory on hand + days sales outstanding) between holding inventory and receiving cash. Using this measure, Hyzer's liquidity shows a moderate improvement.



MODULE QUIZ 32.3

1. Which of the following inventory disclosures would *least likely* be found in the footnotes of a firm following IFRS?
 - A. The amount of loss reversals, from previously written-down inventory, recognized during the period.
 - B. The carrying value of inventories that collateralize a short-term loan.
 - C. The separate carrying values of raw materials, work in progress, and finished goods computed under the LIFO cost flow method.
2. Paul Neimer calculates the following horizontal common-size inventory data for Redpine Manufacturing, Inc.:

	Year 1	Year 2	Year 3	Year 4
Sales	1.00	1.10	1.18	1.25
Inventories:				
Raw materials	1.00	1.09	1.07	1.04
Work in progress	1.00	1.11	1.15	1.17
Finished goods	1.00	1.10	1.21	1.33

Based on these data, Neimer should *most likely* conclude that Redpine:

- A. has an increasing inventory turnover ratio.
 - B. anticipates declining demand for its products.
 - C. might be losing sales due to inadequate inventory.
3. Which of the following is *most likely* for a firm with high inventory turnover and lower sales growth than the industry average? The firm:
- A. is managing its inventory effectively.
 - B. may have obsolete inventory that requires a write-down.
 - C. may be losing sales by not carrying enough inventory.
4. During a period of increasing prices, compared to reporting under LIFO, a firm that reports using FIFO for inventory will have a:
- A. lower gross margin.
 - B. higher current ratio.
 - C. higher asset turnover.

KEY CONCEPTS

LOS 32.a

Under IFRS, inventories are valued at the lower of cost or net realizable value (NRV). Inventory write-ups are allowed, but only to the extent that a previous write-down to NRV was recorded.

Under US GAAP, inventories are valued at the lower of cost or NRV for companies using cost methods other than LIFO or the retail method. For companies using LIFO or the retail method, inventories are valued at the lower of cost or market. Market is usually equal to replacement cost, but it cannot exceed NRV or be less than NRV minus a normal profit margin. No subsequent write-up is allowed for any company reporting under US GAAP.

A write-down of inventory value from cost to NRV will do the following:

- | Decrease inventory, assets, and equity. A contra account (valuation allowance) is used to reduce the carrying value of inventory and most often increase cost of sales. If the write-down is materially large, it will be disclosed on the face of the income statement as its own line item.
- | Increase asset turnover, the debt-to-equity ratio, and the debt-to-assets ratio.
- | Decrease net income and the net profit margin, as well as ROA and ROE for a typical firm.

LOS 32.b

In periods where inventory levels and purchase costs are rising, the following occur:

- | FIFO produces the highest ending inventory value and the lowest cost of sales.
- | Average cost will produce ending inventory and cost of sales between LIFO and FIFO.
- | LIFO produces the lowest ending inventory and the highest cost of sales.
- | FIFO results in higher profitability and liquidity ratios, and lower inventory turnover and solvency ratios.
- | If inventory levels are increasing but costs are decreasing, these results are reversed.

If LIFO is used and inventory levels decline, a LIFO liquidation has occurred, which will distort profit margins in that period.

Analysts may have to adjust financial statements to compare firms that use different inventory cost methods.

LOS 32.c

Required inventory disclosures areas are as follows:

- | The cost flow method (LIFO, FIFO, etc.) used
- | Total carrying value of inventory and carrying value by classification (raw materials, work in progress, and finished goods), if appropriate
- | Carrying value of inventories reported at fair value less selling costs
- | The cost of inventory recognized as an expense (COGS) during the period
- | Amount of inventory write-downs during the period
- | Reversals of inventory write-downs during the period (IFRS only because US GAAP does not allow reversals)
- | Carrying value of inventories pledged as collateral

An analyst should examine inventory disclosures to determine the following:

- | Whether the finished goods category is growing while raw materials and goods in process are declining, which may indicate decreasing demand and potential future inventory write-downs
- | Whether raw materials and goods in process are increasing, which may indicate increasing future demand and higher earnings

- | Whether increases in finished goods are greater than increases in sales, which may indicate decreasing demand or inventory obsolescence and potential future inventory write-downs

Inventory turnover, days of inventory on hand, and gross profit margin can be used to evaluate the quality of a firm's inventory management:

- | Inventory turnover that is too low (high days of inventory on hand) may be an indication of slow-moving or obsolete inventory.
- | High inventory turnover, together with low sales growth relative to the industry, may indicate inadequate inventory levels and lost sales because customer orders could not be fulfilled.
- | High inventory turnover, together with high sales growth relative to the industry average, suggest that high inventory turnover reflects greater efficiency rather than inadequate inventory.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 32.1, 32.2

- . **B** Market is equal to the replacement cost as long as replacement cost is within a specific range. The upper bound is net realizable value (NRV), which is equal to the selling price (\$50) less selling costs (\$3) for an NRV of \$47. The lower bound is NRV (\$47) less normal profit margin (10% of selling price = \$5) for a net amount of \$42. Since replacement cost is greater than NRV (\$47), market equals NRV (\$47). Additionally, we have to use the lower of cost (\$43) or market (\$47) principle, so the shoes should be recorded at cost of \$43. (Module 32.1, LOS 32.a)
- . **A** The write-down in inventory value from cost to NRV is reported on the income statement either as an addition to cost of sales or as a separate line item, not as other comprehensive income. Depreciation will not be affected as inventory is not depreciated. (Module 32.1, LOS 32.a)
- . **C** When establishing market value, replacement cost should be used subject to upper and lower limits. The upper limit is net realizable value, and the lower limit is net realizable value minus the normal profit margin. The lower of cost or market value is only applicable to companies using either the LIFO or retail sales method. Companies using specific identification,

FIFO, or average cost should use the lower of cost or NRV. Inventory write-downs under US GAAP cannot be reversed. (Module 32.1, LOS 32.a)

- . **B** The quick ratio $[(\text{cash} + \text{receivables} + \text{marketable securities}) / \text{current liabilities}]$ will not be affected because the numerator does not include inventory. Inventory write-downs will decrease the carrying value of inventory in the balance sheet, reducing current and total assets. The write-down will decrease income for the period. Asset turnover $(\text{revenue} / \text{average total assets})$ will increase as revenue (sales) is not affected, but balance sheet assets decline. The debt-to-equity ratio $(\text{interest-bearing liabilities} / \text{stockholders' equity})$ will increase as the write-down reduces net income and therefore retained earnings and stockholders' equity. (Module 32.1, LOS 32.a)
- . **A** Under FIFO, ending inventory is made up of the most recent purchases, thereby providing a closer approximation of current cost. (Module 32.2, LOS 32.b)
- . **A** Compared to FIFO, COGS calculated under LIFO will be higher because the most recent, higher-cost units are assumed to be the first units sold. Higher COGS under LIFO will result in lower gross profit $(\text{revenue} - \text{COGS})$. (Module 32.2, LOS 32.b)
- . **C** In a LIFO liquidation, older and lower costs are included in cost of sales. Thus, cost of sales per unit decreases and profit margins increase. (Module 32.2, LOS 32.b)

Module Quiz 32.3

- . **C** While the separate carrying values of raw materials, work in progress, and finished goods are required disclosure for some firms, LIFO is not permitted under IFRS. (LOS 32.c)
- . **B** Redpine's finished goods inventory is growing faster than sales, while work in progress inventory is growing more slowly than sales and raw materials inventory is decreasing. These data are consistent with Redpine reducing production in response to decreasing demand. Inventory turnover ratios cannot be calculated directly from the common-size data given, but finished goods inventory increasing faster than sales suggests inventory turnover is likely decreasing. (LOS 32.c)

- . **C** High inventory turnover coupled with low sales growth relative to the industry may be an indication of inadequate inventory levels. In this case, the firm may be losing sales by not carrying enough inventory. (LOS 32.c)
- . **B** Compared to using LIFO, using FIFO would produce lower COGS, higher gross operating income, and higher ending inventory, so current assets and the current ratio would be higher. Consequently, gross margin would be higher and asset turnover would be lower under the FIFO inventory method. (LOS 32.c)

READING 33

ANALYSIS OF LONG-TERM ASSETS

INTRODUCTION

Before studying this reading, ensure you have reviewed capitalization versus expensing costs in our reading on Analyzing Income Statements and understand the implications of these approaches on the financial statements. You should also understand the various types of long-term assets, PP&E, and intangibles (from our reading on Understanding Balance Sheets and the supplementary study materials in our online Resource Library). Ensure you have understood the two models of recording assets in the balance sheet. Under US GAAP, the cost method is required. IFRS allows companies to choose either the cost method or revaluation model for each class of asset. You should also ensure you understand depreciation and amortization methods (Resource Library).

MODULE 33.1: INTANGIBLE LONG-LIVED ASSETS

LOS 33.a: Compare the financial reporting of the following types of intangible assets: purchased, internally developed, and acquired in a business combination.

Intangible assets are long-term assets that lack physical substance, such as patents, brand names, copyrights, and franchises. Some intangible assets have finite lives, while others have indefinite lives.

The cost of a finite-lived intangible asset is amortized over its useful life. Indefinite-lived intangible assets are not amortized, but they are tested for impairment at least annually. If impaired, the reduction in value is recognized in the income statement as a loss in the period in which the impairment is recognized.

Intangible assets are also considered either identifiable or unidentifiable. Under IFRS, an **identifiable intangible asset** must be the following:

- | Capable of being separated from the firm or arise from a contractual or legal right
- | Controlled by the firm
- | Expected to provide future economic benefits

In addition, the future economic benefits must be probable, and the asset's cost must be reliably measurable.

An **unidentifiable intangible asset** is one that cannot be purchased separately and may have an indefinite life. The most common example of an unidentifiable intangible asset is goodwill. Goodwill is the excess of purchase price over the fair value of the identifiable assets (net of liabilities) acquired in a business combination.

Not all intangible assets are reported on the balance sheet. Accounting for an intangible asset depends on whether the asset was created internally, purchased externally, or obtained as part of a business combination.

Intangible Assets Created Internally

With some exceptions, costs to create intangible assets are expensed as incurred. Important exceptions are research and development costs (under IFRS) and software development costs.

Under IFRS, **research costs**, which are costs aimed at the discovery of new scientific or technical knowledge and understanding, are expensed as incurred. However, **development costs** may be capitalized. Development costs are incurred to translate research findings into a plan or design of a new product or process. To recognize an intangible asset in development, a firm must show that it can complete the asset and intends to use or sell the completed asset, among other criteria.

Under US GAAP, both research and development costs are generally expensed as incurred. However, the costs of creating software for sale to others are treated in a manner similar to the treatment of research and development costs under IFRS. Costs incurred to develop software for sale to others are expensed as incurred until the product's technological feasibility has been established, after which the costs of developing a salable product are capitalized. Costs incurred

developing software for internal use must be expensed until it is probable that the firm will complete the project and use the software as intended. Technological feasibility is more difficult to demonstrate than the probability of completing and using the software.

Purchased Intangible Assets

Like tangible assets, an intangible asset purchased from another party is initially recorded on the balance sheet at cost, typically its fair value at acquisition.

If the intangible asset is purchased as part of a group, the total purchase price is allocated to each asset on the basis of its fair value. For analytical purposes, an analyst is usually more interested in the type of asset acquired rather than the value assigned on the balance sheet. For example, recently acquired franchise rights may provide insight into the firm's future operating performance. In this case, the allocation of cost is not as important.

The financial statement effects of capitalizing intangible assets are the same as the effects of capitalizing other expenditures. Capitalizing results in higher net income in the first year and lower net income in the subsequent years. Similarly, assets, equity, and operating cash flow are all higher when expenditures are capitalized.

A company that generates its intangible assets internally will have lower balance sheet assets than a company that purchases its intangibles. Internally generated intangibles are typically expensed, while purchased intangibles are capitalized. This can have a considerable impact on financial statements and ratios.

Intangible Assets Obtained in a Business Combination

The **acquisition method** is used to account for business combinations. Under the acquisition method, the purchase price is allocated to the identifiable assets and liabilities of the acquired firm on the basis of fair value. This will involve establishing the fair value of any intangibles that the acquired company had internally developed and including them in the balance sheet at the date of acquisition. To capitalize these intangible assets, which had previously expensed as incurred, the asset must be an identifiable intangible asset. Any remaining amount of the purchase price is recorded as **goodwill** (see Figure 33.1). Goodwill is said to be an unidentifiable asset that cannot be separated from the business itself.

Figure 33.1: Goodwill

Purchase price	X
Fair market value of net assets acquired	(X)
Fair market value of identifiable intangibles previously not recognized	<u>(X)</u>
Goodwill	X

Only goodwill created in a business combination is capitalized on the balance sheet. The costs of any internally generated “goodwill” are expensed in the period incurred.



MODULE QUIZ 33.1

1. The cost of an intangible asset is *most likely* to be amortized if the asset has a(n):
 - A. finite life and was purchased.
 - B. finite life and was created internally.
 - C. indefinite life and was acquired in a business combination.
2. StefoJo, PLC, is acquiring a controlling interest in Dykes Limited. Dykes has internally developed intangibles that have previously been expensed. Which of the following comments is *most accurate*?
 - A. StefoJo should capitalize any identifiable internally generated intangibles.
 - B. Goodwill generated resulting from Dykes' reputation and customer loyalty should be capitalized.
 - C. Capitalizing intangible assets Dykes had previously expensed will increase the goodwill recorded at the acquisition date.
3. Ferdinand, Inc., is developing software for internal use within its business operations. Assuming Ferdinand follows US GAAP, which of the following statements is *least accurate*?
 - A. The software development costs can only be capitalized once the software is technically feasible.
 - B. Capitalized software development will increase earnings in the year of capitalization but reduce earnings in subsequent periods.
 - C. The software development costs should be capitalized once it is probable that the software will be completed and used as intended.

MODULE 33.2: IMPAIRMENT AND DERECOGNITION

LOS 33.b: Explain and evaluate how impairment and derecognition of property, plant, and equipment and intangible assets affect the financial statements and ratios.

Depreciation and amortization represent the spreading of an asset's cost to match

the benefits earned over an asset's life, and they cause the balance sheet carrying value to decline. An **impairment** is an unanticipated decline in an asset's value, causing it to fall below the carrying value. **Derecognition** occurs when an asset is disposed of or retired.

Impairments

Both IFRS and US GAAP require firms to write down impaired assets by recognizing a loss in the income statement, but the standards are applied slightly differently. If the impairment loss is material in size and nature, it will be shown as an unusual or infrequent item.



PROFESSOR'S NOTE

The following discussion applies to both tangible and intangible long-lived assets with finite lives that are held for use.

Impairments Under IFRS

Under IFRS, the firm must annually assess whether events or circumstances indicate that an impairment of an asset's value has occurred. For example, there may have been a significant decline in the market value of the asset, or a significant change in the asset's physical condition. If so, the asset's value must be tested for impairment.

An asset is impaired when its carrying value (original cost less accumulated depreciation) exceeds the **recoverable amount**. The recoverable amount is the greater of its fair value less any selling costs and its **value in use**. The value in use is the present value of its future cash flow stream from continued use and disposal. Value in use is a highly subjective figure requiring estimation of future cash flows, disposal proceeds, and the selection of an appropriate discount rate.

If impaired, the asset's value must be written down on the balance sheet to the recoverable amount. An impairment loss, equal to the excess of carrying value over the recoverable amount, is recognized in the income statement.

Under IFRS, an impairment loss on an identifiable long-lived asset can be reversed if the asset's value recovers in the future. However, the loss reversal is limited to the original impairment loss.

Impairments Under US GAAP

Under US GAAP, an asset is tested for impairment only when events and circumstances indicate the firm may not be able to recover the carrying value through future use.

Determining an impairment and calculating the loss potentially involve two steps. In the first step, the asset is tested for impairment by applying a **recoverability test**. If the asset is impaired, the second step involves measuring the loss.

Regarding *recoverability*, an asset is considered impaired if the carrying value (original cost less accumulated depreciation) is greater than the asset's future *undiscounted* cash flow stream. Because the recoverability test is based on estimates of future undiscounted cash flows, tests for impairment involve considerable management discretion. Like value in use under IFRS, estimating the future undiscounted cash flows from an asset's use and disposal is highly subjective.

Regarding *loss measurement*, if impaired, the asset's value is written down to fair value on the balance sheet—and a loss, equal to the excess of carrying value over the fair value of the asset (or the *discounted* value of its future cash flows, if the fair value is not known), is recognized in the income statement. The discounted value of future cash flows is equivalent of value in use under IFRS.

Under US GAAP, loss recoveries are typically not permitted.



PROFESSOR'S NOTE

The difference between testing for impairment and measuring the impairment loss can be confusing. In testing for impairment under US GAAP, undiscounted cash flows are used. Once impairment has been detected, the loss is based on fair value, or the discounted expected future cash flows. Using undiscounted cash flows to test for impairment keeps PP&E assets from becoming “impaired” by increases in the discount rate when interest rates increase. In measuring the impairment loss, US GAAP uses fair value, not fair value less selling costs.

EXAMPLE: Asset impairment

Information related to equipment owned by Brownfield Company follows:

Original cost	\$900,000
Accumulated depreciation to date	\$100,000
Expected future cash flows	\$795,000
Fair value	\$790,000
Value in use	\$785,000
Selling costs	\$30,000

Assuming Brownfield will continue to use the equipment, test the asset for impairment under both IFRS and US GAAP and discuss the results.

Answer:

The carrying value of the equipment is \$900,000 original cost – \$100,000 accumulated depreciation = \$800,000, and the recoverable amount under IFRS is \$785,000 (greater of \$785,000 value in use and \$760,000 fair value less selling costs). Under IFRS, the asset is written down on the balance sheet to the \$785,000 recoverable amount, and a \$15,000 loss (\$800,000 carrying value – \$785,000 recoverable amount) is recognized in the income statement.

Under US GAAP, the asset is impaired because the \$795,000 expected future cash flows is less than the \$800,000 carrying value. The asset is written down on the balance sheet to its \$790,000 fair value, and a \$10,000 loss (\$800,000 carrying value – \$790,000 fair value) is recognized on the income statement.

Impairment reduces an asset's carrying value on the balance sheet. An impairment charge is recognized as a loss in the income statement, reducing assets and equity (retained earnings). In the year of impairment, return on assets (ROA) and return on equity (ROE) will decrease because the impairment charge reduces net income.

In subsequent periods, net income will be higher than it would have been without the impairment charge because depreciation or amortization will be lower (the asset has a lower depreciable value). Both ROA and ROE will increase in periods after the impairment charge because both equity and assets will fall as a result of the impairment charge. Asset turnover will increase in the period in which the impairment charge is taken, and in subsequent periods.

Asset impairment has no impact on cash flow because the impairment does not reduce taxable income; it is an unrealized loss until the asset is disposed of.

Analysis of Impairments

An impairment loss essentially indicates that the firm has not recognized sufficient depreciation or amortization expense, and has overstated earnings as a result.

The judgment required in determining asset impairments gives management considerable discretion about the timing and amounts of impairment charges. Consequently, impairment decisions present an opportunity for management to manipulate earnings. Waiting to recognize an impairment loss until a period of relatively high earnings would tend to smooth earnings.

Alternatively, existing managements may take more impairment charges in periods when earnings will be poor due to external (macroeconomic or industry) factors. New managements may also choose to take more or greater impairment charges when they take over. In either case, the resulting low earnings might not be perceived as the “fault” of management, and lower values for assets and equity give a boost to ROA and ROE going forward.

Intangible Assets With Indefinite Lives

Intangible assets with indefinite lives are not amortized; rather, they are tested for impairment at least annually. An impairment loss is recognized when the carrying amount exceeds fair value.

Long-Lived Assets Held for Sale

If a firm intends to sell an asset, it is probable that the asset will be sold, and the asset is available for immediate sale, then it must be reclassified from *held for use* to *held for sale*. When a firm reclassifies an asset as held for sale, the asset is no longer depreciated or amortized. The held-for-sale asset is impaired if its carrying value exceeds its fair value less selling costs. If impaired, the asset is written down to net realizable value, and the loss is recognized in the income statement.

For long-lived assets held for sale, the loss can be reversed under IFRS and US GAAP if the value of the asset recovers in the future. However, the loss reversal is limited to the original impairment loss. Thus, the carrying value of the asset after reversal cannot exceed the carrying value before the impairment was recognized.

Derecognition

Eventually, long-term assets are removed from the balance sheet. Derecognition occurs when assets are sold, exchanged, or abandoned.

When a long-term asset is sold, the asset is removed from the balance sheet and the difference between the sale proceeds and the carrying value of the asset is reported as a gain or loss in the income statement, as seen in Figure 33.2. The carrying value is equal to original cost minus accumulated depreciation and any impairment charges.

Figure 33.2: Derecognition of Long-Term Assets

Disposal proceeds	X	Cash flow from investing
Carrying value before disposal	(X)	Removed from balance sheet
Disposal gain (loss)	X/(X)	Accounting gain or (loss) in income statement

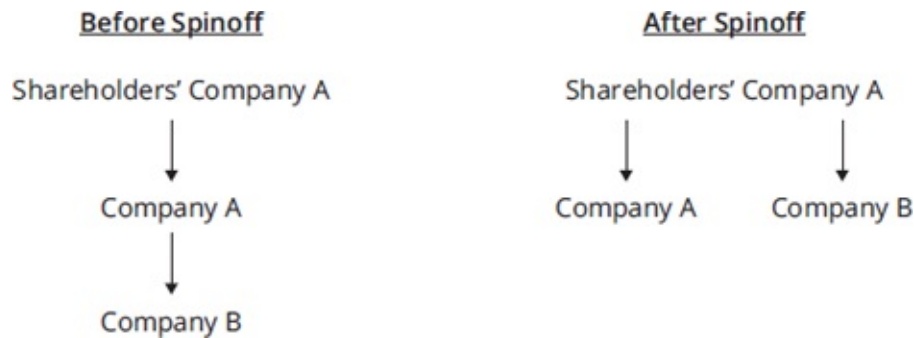
The gain or loss is reported in the income statement with other gains and losses, or reported separately as an unusual or infrequent item if material. Also, if the firm presents its cash flow statement using the indirect method, the gain or loss is removed from net income to compute cash flow from operations because the proceeds from selling a long-lived asset are an investing cash inflow.

If a long-lived asset is abandoned, the treatment is similar to a sale, except there are no proceeds. In this case, the carrying value of the asset is removed from the balance sheet, and a loss of that amount is recognized in the income statement.

If a long-lived asset is exchanged for another asset, a gain or loss is computed by comparing the carrying value of the old asset with fair value of the old asset (or the fair value of the new asset, if that value is clearly more evident). The carrying value of the old asset is removed from the balance sheet, and the new asset is recorded at its fair value. If no reliable measure of fair value exists, the new asset is valued at the carrying value of the old asset.

A **spinoff** is the transfer of assets that constitute an entire division or subsidiary into a new legal entity, upon which the shares of the spinnee are subsequently distributed to its shareholders, and the shareholders do not surrender any stock of the spinnor.¹ See Figure 33.3 as follows.

Figure 33.3: Spinoff



Before the spinoff, Company A (the spinnor) owns Company B (the spinnee), and shareholders just have shares in Company A. After the spinoff, the shareholders have shares in both Company A and Company B. Company B is no longer included in the consolidated accounts of Company A.

Once the spinoff becomes probable, the balance sheet assets and liabilities of the spinnee will be transferred from assets held for use to assets held for sale in the spinnor's accounts. Assets held for sale may be referred to as assets held for distribution, in this context. No profit or loss is recorded on the disposal in the income statement.



MODULE QUIZ 33.2

1. According to US GAAP, an asset is impaired when:
 - A. the firm cannot fully recover the carrying amount of the asset through operations.
 - B. accumulated depreciation plus salvage value exceeds acquisition cost.
 - C. the present value of future cash flows from an asset exceeds its carrying value.
2. Using the following data, what is the income statement difference between an impairment recorded under US GAAP and an impairment recorded under IFRS?

Original cost	\$250,000
Accumulated depreciation to date	\$150,000
Expected future cash flows	\$105,000
Fair value	\$95,000
Value in use	\$90,000
Selling costs	\$10,000

- A. \$0.
 - B. \$5,000.
 - C. \$10,000.
3. In the year after an impairment charge on a finite-lived identifiable intangible asset, compared to not taking the charge, net income is *most likely* to be:
 - A. lower.
 - B. higher.
 - C. unaffected.

4. A firm recently recognized a \$15,000 loss on the sale of machinery used in its manufacturing operation. The original cost of the machinery was \$100,000 and the accumulated depreciation at the date of sale was \$60,000. What amount did the firm receive from the sale?
 - A. \$25,000.
 - B. \$45,000.
 - C. \$85,000.
5. Other things equal, which of the following actions related to property, plant, and equipment will *most likely* decrease a firm's return on assets (ROA) in future periods?
 - A. Impairment.
 - B. Derecognition.
 - C. Upward revaluation.
6. Sinclair S.r.l. has recently exchanged an old asset for a new asset. Which of the following comments is *least accurate*?
 - A. The difference between the fair value and the carrying value of the disposed asset will be recognized in the income statement as a gain or loss.
 - B. The new asset will be recorded in the balance sheet at the fair value of the assets disposed of.
 - C. If the fair value of the asset received and the fair value of the asset disposed of cannot be established, a disposal loss will be recorded in the income statement equal to the carrying value of the disposed asset.

MODULE 33.3: LONG-TERM ASSET DISCLOSURES

LOS 33.c: Analyze and interpret financial statement disclosures regarding property, plant, and equipment and intangible assets.

IFRS Disclosures

Under IFRS, the firm must disclose the following for each class of property, plant, and equipment (PP&E):

- | Basis for measurement (usually historical cost)
- | Depreciation method
- | Depreciation expense in the period
- | Useful lives or depreciation rate
- | Gross carrying value and accumulated depreciation at the start and end of the period

- | Reconciliation of carrying amounts from the beginning of the period to the end of the period

The firm must also disclose the following:

- | Title restrictions and assets pledged as collateral
- | Agreements to acquire PP&E in the future

If the revaluation (fair value) model is used, the firm must disclose the following:

- | The revaluation date
- | How fair value was determined
- | Carrying value using the historical cost model
- | Revaluation surplus in other comprehensive income (OCI)

Under IFRS, the disclosure requirements for intangible assets are similar to those for PP&E, except that the firm must disclose whether the useful lives are finite or indefinite.

For impaired assets, the firm must disclose the following:

- | Amounts of impairment losses and reversals by asset class
- | Where the losses and loss reversals are recognized in the income statement
- | Circumstances that caused the impairment loss or reversal

US GAAP Disclosures

Under US GAAP, the PP&E disclosures include the following:

- | Depreciation expense by period
- | Balances of major classes of assets by nature and function, such as land, improvements, buildings, machinery, and furniture
- | Accumulated depreciation by major classes, or in total
- | General description of depreciation methods used

Under US GAAP, the disclosure requirements for intangible assets are similar to those for PP&E. In addition, the firm must provide an estimate of amortization expense for the next five years.

For impaired assets, the firm must disclose the following:

- | A description of the impaired asset

- | Circumstances that caused the impairment
- | How fair value was determined
- | The amount of loss
- | Where the loss is recognized in the income statement

Under IFRS, the depreciation and amortization expenses may be shown on the face of the income statement if producing an income statement using the nature of the expense approach. If the company prepares the income statement using the function of the expense method, it will not be shown on the face; instead, it will be included in the cost of sales and sales general and admin. If the indirect method is used to produce cash flow from operations, depreciation and amortization will appear as noncash charges. For companies adopting the direct method, depreciation will not appear in the computation of operating cash flows; however, US GAAP requires the indirect method reconciliation of net income to operating cash flows to be disclosed in the footnotes. Cash received on asset disposals and spent on asset acquisitions are presented as cash flows from investing activities.

Analyzing Long-Lived Asset Disclosures

Fixed-asset turnover shows revenue per dollar of fixed assets and indicates how efficiently a firm is using its long-term assets to generate sales. A higher ratio is interpreted as greater efficiency.

$$\text{fixed asset turnover} = \text{revenue} / \text{average fixed assets}$$

Financial statement disclosures provide an analyst considerable information about a company's fixed assets and depreciation (amortization) methods. An analyst can use the data to estimate the average age of the firm's assets. The average age is useful for two reasons:

Older, less-efficient assets may make a firm less competitive.

The average age of assets helps an analyst to estimate the timing of major capital expenditures and a firm's future financing requirements.

The level of detail provided in footnote disclosures regarding fixed assets and depreciation varies across firms. Because assets are often grouped by their useful lives, the following methods of estimating the average age, economic life, and remaining useful life of a firm's assets do not produce precise values, but they can highlight issues for further investigation. The following computations

assume zero salvage values and straight-line depreciation, and as a result, they are less insightful for asset classes subject to accelerated depreciation methods and substantial salvage values.

Three useful calculations (in years) for an analyst are as follows.

Average Age

$$\text{average age} = \frac{\text{accumulated depreciation}}{\text{annual depreciation expense}}$$

This calculation is more accurate for a firm that uses straight-line depreciation. The calculation can be significantly affected by the mix of assets.

Total Useful Life

$$\text{total useful life} = \frac{\text{historical cost (gross cost)}}{\text{annual depreciation expense}}$$

Historical cost is gross PP&E before deducting accumulated depreciation.

Remaining Useful Life

$$\text{remaining useful life} = \frac{\text{ending net PP\&E}}{\text{annual depreciation expense}}$$

Net PP&E is equal to original cost (gross PP&E) minus accumulated depreciation.



PROFESSOR'S NOTE

The remaining useful life can also be approximated by subtracting the average age from the total useful life.

EXAMPLE: Calculating average age and total useful life

At the end of 20X8, a company has gross PP&E of \$3 million and accumulated depreciation of \$1 million. During the year, the depreciation expense was \$500,000.

What is the average age, total useful life, and remaining useful life of the company's PP&E?

Answer:

$$\begin{aligned}\text{average age} &= \frac{\text{accumulated depreciation}}{\text{depreciation expense}} = \frac{\$1,000,000}{\$500,000} = 2 \text{ years} \\ \text{total useful life} &= \frac{\text{historical cost}}{\text{depreciation expense}} = \frac{\$3,000,000}{\$500,000} = 6 \text{ years} \\ \text{remaining useful life} &= \frac{\text{ending net PP\&E}}{\text{depreciation expense}} = \frac{\$2,000,000}{\$500,000} = 4 \text{ years}\end{aligned}$$

Another popular metric is the ratio of annual capital expenditures to depreciation expense. This ratio provides information about whether the firm is maintaining its production capacity by replacing its PP&E at the same rate as its assets are being depreciated.



MODULE QUIZ 33.3

- Which of the following disclosures would *least likely* be found in the financial statement footnotes of a firm?
 - Accumulated depreciation.
 - Carrying values by asset class.
 - Average age of assets.
- Metallurgy, Inc., reported depreciation expense of \$15 million for the most recent year. Beginning-of-year gross PP&E and accumulated depreciation were \$287 million and \$77 million, respectively. If end-of year gross PP&E and accumulated depreciation were \$300 million and \$80 million, the estimated remaining useful life of PP&E is *closest* to:
 - 10 years.
 - 15 years.
 - 20 years.
- Ruby, Inc., reported depreciation expense of \$90 million for the most recent year. Beginning-of-year gross PP&E and accumulated depreciation were \$300 million and \$120 million, respectively. If end-of year gross PP&E and accumulated depreciation were \$350 million and \$200 million, the estimated average age of PP&E is *closest* to:
 - 1.3 years.
 - 1.6 years.
 - 2.2 years.
- For impairments, US GAAP *least likely* requires firms to disclose:
 - amounts of impairment reversals by asset class.
 - how fair value was determined.
 - circumstances that led to the impairment.

KEY CONCEPTS

LOS 33.a

The cost of a purchased finite-lived intangible asset is amortized over its useful life. Indefinite-lived intangible assets are not amortized, but they are tested for impairment at least annually. The cost of internally developed intangible assets is expensed.

Under IFRS, research costs are expensed, but development costs may be capitalized. Under US GAAP, both research and development costs are expensed as incurred, except in the case of software.

The acquisition method is used to account for assets acquired in a business combination. The purchase price is allocated to the fair value of identifiable assets of the acquired firm less its liabilities. Any excess of the purchase price above the fair value of the acquired firm's net assets is recorded as goodwill, an unidentifiable intangible asset that cannot be separated from the business itself.

Compared to expensing an asset's cost, capitalization results in the following:

- | Lower expense and higher net income in period of acquisition, and higher expense (depreciation or amortization) and lower net income in each of the remaining years of the asset's life
- | Higher assets and equity
- | Lower CFI and higher CFO because the cost of a capitalized asset is classified as an investing cash outflow
- | Higher ROE and ROA in the initial period, and lower ROE and ROA in subsequent periods because net income is lower, and both assets and equity are higher
- | Lower debt-to-assets and debt-to-equity ratios because assets and equity are higher

LOS 33.b

Impairment charges decrease net income, assets, and equity, which results in lower ROA and ROE and higher debt-to-equity and debt-to-assets ratios for a typical firm.

Under IFRS, an asset is impaired when its carrying value exceeds the recoverable amount. The recoverable amount is the greater of fair value less selling costs and the value in use (present value of expected cash flows). If impaired, the asset is written down to the recoverable amount. Loss recoveries are permitted, but not above historical cost.

Under US GAAP, an asset is impaired if its carrying value is greater than the asset's undiscounted future cash flows. If impaired, the asset is written down to its fair value. Subsequent recoveries are not allowed for assets held for use.

Asset impairments result in losses in the income statement. Impairments have no impact on cash flow, as they have no tax or other cash flow effects until disposal of the asset.

Derecognition of assets can result in either a gain or loss on the income statement. A loss will reduce net income and assets, while a gain will increase net income and assets.

When a long-lived asset is *sold*, the difference between the sale proceeds and the carrying (book) value of the asset is reported as a gain or loss in the income statement.

When a long-lived asset is *abandoned*, the carrying value is removed from the balance sheet, and a loss is recognized in that amount.

If a long-lived asset is *exchanged* for another asset, a gain or loss is computed by comparing the carrying value of the old asset with fair value of the old asset (or fair value of the new asset, if more clearly evident).

LOS 33.c

There are differences in the disclosure requirements for long-lived assets under IFRS and US GAAP. However, firms are generally required to disclose the following:

- | Carrying values for each class of asset
- | Accumulated depreciation and amortization
- | Title restrictions and assets pledged as collateral
- | For impaired assets, the loss amount and the circumstances that caused the loss
- | For revalued assets (IFRS only), the revaluation date, how fair value was determined, and the carrying value using the historical cost model

Analysts can use disclosures of the historical cost, accumulated depreciation (amortization), and annual depreciation (amortization) expense to estimate average age of assets, total useful life of assets, and remaining useful life of assets.

$$\text{average age} = \frac{\text{accumulated depreciation}}{\text{annual depreciation expense}}$$

$$\text{total useful life} = \frac{\text{historical cost}}{\text{annual depreciation expense}}$$

$$\text{remaining useful life} = \frac{\text{ending net PP\&E}}{\text{annual depreciation expense}}$$

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 33.1

- . **A** The cost of an intangible asset is amortized if the asset has a finite life and was purchased or acquired in a business combination. Development costs for internally generated intangible assets may be capitalized under IFRS, but research costs are expensed as incurred. (LOS 33.a)
- . **A** Any identifiable internally generated intangible assets that the targeted company had previously expensed must be included in the balance sheet assets acquired. Internally generated goodwill is not an identifiable intangible asset and would not be included in the fair market value of assets acquired. The capitalization of previously unrecognized identifiable intangibles will increase the fair market value of assets acquired and reduce the amount of the purchase price allocated to goodwill. (LOS 33.a)
- . **A** Development costs for software intended for *external sale* may be capitalized once the software is considered technically feasible. Development costs for software intended for *use within the business* may be capitalized once it is probable that the software will be completed and used as intended. Capitalization stores costs in the balance sheet rather than recognizing them in the income statement at the time it is incurred, resulting in higher earnings in the year of capitalization. Future earnings will be lower if costs are capitalized, as the intangible asset is amortized over the period it generates benefits. (LOS 33.a)

Module Quiz 33.2

- . **A** An asset is impaired when the firm cannot recover the carrying value. Under US GAAP, recoverability is tested based on undiscounted future cash flows. (LOS 33.b)

- . **C** The carrying value of the asset is equal to $\$250,000 - \$150,000 = \$100,000$. Under US GAAP, a two-step process is used. The recoverability test is used to identify if an impairment has occurred. An impairment has occurred if the carrying value is higher than the estimated undiscounted cash flows from the asset's use and disposal. In this case, $\$105,000$ is greater than the $\$100,000$ carrying value, so the asset is not impaired under US GAAP. Under IFRS, carrying value is compared to the recoverable amount. The recoverable amount is the higher of fair value less selling costs and value in use. In this case, value in use would be used as $\$90,000$ is greater than $\$85,000$. IFRS would record an impairment of $\$10,000$, equal to carrying value less value in use. (LOS 33.b)
- . **B** Because a finite-lived identifiable intangible asset would be amortized, amortization expense in the year after the reduction from the impairment charge would be lower (the carrying value of the asset would most likely be lower), increasing net income. (LOS 33.b)
- . **A** Gain or loss is equal to the sale proceeds minus the carrying value (cost minus accumulated depreciation) at the time of sale. Given the loss of $\$15,000$ and carrying value of $\$40,000$ ($\$100,000 - \$60,000$), we can solve for the proceeds of $\$25,000$ ($-\$15,000 + \$40,000$). (LOS 33.b)
- . **C** An upward revaluation will increase the book value of assets and increase depreciation expense in future periods (decreasing net income), both of which reduce ROA. Impairment would have the opposite effects, reducing the carrying value of assets and decreasing future depreciation. Derecognizing an asset may increase, decrease, or not affect ROA in future periods. (LOS 33.b)
- . **C** The gain or loss on an asset's exchange is calculated as the difference between the fair value of the asset that has been disposed of and the carrying value of the disposed asset. The new asset should be recorded in the balance sheet at the fair value of the asset disposed of. If the fair value of the disposed asset is difficult to estimate, then the gain or loss on disposal is computed as the fair value of the asset acquired in the exchange less the carrying value of the asset disposed of. The new asset would be recorded at its fair value in the balance sheet. If neither the fair value of the asset disposed of nor acquired can be established, the new asset will be

recorded at the carrying value of the disposed asset. In this situation, there will be no gain or loss recorded on disposal. (LOS 33.b)

Module Quiz 33.3

- . **C** The average age is not a required disclosure. However, it can be calculated given other disclosures. (LOS 33.c)
- . **B** The remaining useful life can be estimated as ending net PP&E value divided by annual depreciation, so $(300 - 80) / 15 = 14.66$ years. (LOS 33.c)
- . **C** The average age can be estimated as ending accumulated depreciation value divided by annual depreciation, so $200 / 90 = 2.22$ years. (LOS 33.c)
- . **A** US GAAP requires the disclosure of impairment losses, but not reversals. US GAAP typically does not allow the reversal of impairments, with the exception of assets held for sale. (LOS 33.c)

¹ Definition per US GAAP, ASC 505-60.

READING 34

TOPICS IN LONG-TERM LIABILITIES AND EQUITY

MODULE 34.1: LEASES

LOS 34.a: Explain the financial reporting of leases from the perspectives of lessors and lessees.

Instead of purchasing an asset, a firm may choose to lease the asset. With a lease, a firm (the **lessee**) essentially purchases the right to use an asset from another firm (the **lessor**) for a specified period, which can range from a month to many years. The lessee makes periodic payments to the lessor for the use of the asset. Thus, a lease can be considered an alternative to financing the purchase of an asset.

To be a lease, a contract must meet the following three requirements:

It must refer to a specific asset.

It must give the lessee effectively all the asset's economic benefits during the term of the lease.

It must give the lessee the right to determine how to use the asset during the term of the lease.

The advantages of leasing rather than purchasing an asset may include the following:

- | *Less initial cash outflow.* Typically, a lease requires only a small down payment, if any.
- | *Less costly financing.* Because a lease is effectively secured by the leased asset if the lessee defaults, the interest rate implicit in a lease contract may be less than the interest rate would be on a loan to purchase the asset.

- | *Less risk of obsolescence.* At the end of a lease, the lessee often returns the leased asset to the lessor, and therefore, does not bear the risk of an unexpected decline in the asset's end-of-lease value. Given that the lessor bears the risk of obsolescence, this increases the lessor's risk and is reflected in a higher implicit interest rate within the lease. Some leases include guaranteed residual income clauses, whereby the lessee guarantees a minimum value for the leased assets at the end of the lease term. In this case, the risk of obsolescence remains with the lessee.

Under IFRS and US GAAP, any lease in which both the benefits and the risks of ownership are substantially transferred to the lessee is classified as a **finance lease**. If either the benefits or the risks of ownership are not substantially transferred to the lessee, a lease is classified as an **operating lease**. Any given lease will be classified the same way by the lessee and the lessor.

Financial reporting standards require a lease to be classified as a finance lease if it meets *any* of the following five conditions:

Ownership of the leased asset transfers to the lessee.

The lessee has an option to buy the asset and is expected to exercise it.

The lease is for most of the asset's useful life.

The present value of the lease payments is greater than or equal to the asset's fair value.

The lessor has no other use for the asset (i.e., the asset is of a specialized nature only suitable for use by the lessee).

Leases that are not classified as finance leases are classified as operating leases.

Lessee Accounting

IFRS requires the lessee to record a **right-of-use asset (ROU asset)** and a **lease liability** (both equal to the present value of the lease payments) on the balance sheet. This treatment is required for all leases, except those that are short term (up to 12 months) or are of low value (up to USD 5,000). This treatment creates a debt instrument (the lease liability) and an ROU asset, resulting in a balance sheet that is comparable to the lessee issuing debt and using the proceeds to buy an asset.

The ROU asset is intangible rather than PP&E. The ROU asset will be amortized over the term of the lease, with the amortization amount each period recorded on

the income statement. Each lease payment is split between interest and principal repayment. The lease liability decreases each period by the principal portion of each lease payment. While the lease asset and the liability both begin with the same value and reach zero at the end of the lease, they will have different values during the life of the lease, as the following example illustrates.



PROFESSOR'S NOTE

All the examples in the Level I CFA curriculum assume that payments are made in arrears (the end of each period) rather than at the start. In practice, most leases require payment at the start of each period. We follow the curriculum in our examples.

EXAMPLE: Lessee accounting for a finance lease

The Affordable Company (Affordable) leases a machine for its own use for four years with annual payments of \$10,000. At the end of the lease, which is also the end of the machine's useful life, Affordable will return the machine to the lessor. The interest rate implicit in the lease is 5%. Assuming that the ROU asset is amortized on a straight-line basis over the term of the lease, calculate the impact of the lease on Affordable's financial statements for each of the four years.

Answer:

The lease is classified as a finance lease because the lease is in effect for the asset's useful life.

The present value of the lease payments is as follows:

$$N = 4; I/Y = 5; PMT = -10,000; FV = 0; CPT \rightarrow PV = 35,460$$

This amount will be recognized on the balance sheet as an ROU asset and as a lease liability.

The ROU asset will be amortized straight-line over the four years, decreasing each year by $\$35,460 / 4 = \$8,865$. This amount will be recognized each year on the income statement as an amortization expense.

The lease liability will be treated as if it were an amortizing loan.

Year	(1) Beginning Liability \$	(2) Interest Expense = (1) × 5% \$	(3) Lease Payment \$	(4) Principal Repayment = (3) – (2) \$	Ending Lease Liability = (1) – (4) \$	Book Value of ROU Asset \$
0					35,460	35,460
1	35,460	1,773	10,000	8,227	27,233	26,595
2	27,233	1,362	10,000	8,638	18,595	17,730
3	18,595	930	10,000	9,070	9,525	8,865
4	9,525	475	10,000	9,525	0	0

The interest expense will be recognized each year on the income statement, separately from the amortization expense for the ROU asset. On the balance sheet, the ROU asset value decreases by \$8,865 each year, and the lease liability is reduced by the principal repayment from Column 4. Note that the book value of the ROU asset is less than the book value of the lease liability during the life of the lease. This is because the principal repayment in the early years of the lease is less than the straight-line amortization of the ROU asset. In the later years, the principal repayment is greater than the straight-line amortization so that at the end of the lease, both the asset and the liability reach zero.

On the cash flow statement, the repayment of principal will be classified as a cash outflow from financing. Under IFRS, the interest portion of each payment may be classified as either an operating or a financing cash outflow. Under US GAAP, the interest portion is classified as an operating cash outflow.

Under US GAAP, other than these differences in cash flow classification, a finance lease (that is not short term) is reported, just as we have described for leases under IFRS.

For an *operating lease* (that is not short term) under US GAAP, a lease liability is also recorded and amortized as under IFRS. However, the ROU asset is not amortized straight-line. Instead, it is amortized by the same amount each period as the decrease in the lease liability, so that the asset and the liability are equal in each period of the lease. On the income statement, interest expense and amortization of the ROU asset are not reported separately as they are for a finance lease; they are combined and reported as lease expense. On the cash flow statement, the full lease payment is classified as an operating cash outflow.

EXAMPLE: Lessee accounting for an operating lease under US GAAP

Using the same data from the previous example, calculate the impact of the lease on Affordable's financial statements if Affordable reports under US GAAP and the lease is classified as an operating lease.

Answer:

The lease liability will be treated the same way as it is for a finance lease, but the book value of the ROU asset will remain equal to the book value of the lease liability.

Year	(1) Beginning Liability \$	(2) Interest = (1) × 5% \$	(3) Lease Payment \$	(4) Principal Repayment = (3) – (2) \$	Ending Lease Liability = (1) – (4) \$	Book Value of ROU Asset \$
0					35,460	35,460
1	35,460	1,773	10,000	8,227	27,233	27,233
2	27,233	1,362	10,000	8,638	18,595	18,595
3	18,595	930	10,000	9,070	9,525	9,525
4	9,525	475	10,000	9,525	0	0

On the balance sheet, the book value of the ROU asset is amortized by the same amount each period, as the lease liability (i.e., the amortization of the asset) is equal to the principal repayment.

On the income statement, the lease expense will equal interest plus amortization of the ROU asset, which will equal the lease payment.

Amortization is equal to the principal repayment each period, and the principal repayment equals the lease payment minus interest. As a result, amortization plus interest each period is equal to the lease payment of \$10,000.

On the cash flow statement, the entire \$10,000 cash outflow is classified as cash from operations.

Comparison of finance and operating leases: US GAAP

Balance Sheet				
Year	ROU Asset		Lease Liability	
	Finance Lease	Operating Lease	Finance Lease	Operating Lease
	\$	\$	\$	\$
0	35,460	35,460	35,460	35,460
1	26,595	27,233	27,233	27,233
2	17,730	18,595	18,595	18,595
3	8,865	9,525	9,525	9,525
4	0	0	0	0

Income Statement				
Year	Finance Lease			Lease Expense, Operating Lease
	Interest	Amortization	Total	
	\$	\$	\$	
0				
1	1,773	8,865	10,638	10,000
2	1,362	8,865	10,227	10,000
3	930	8,865	9,795	10,000
4	475	8,865	9,340	10,000

Note the following when comparing finance lease accounting and operating lease accounting:

- The lease liability is the same each year under both methods. The ROU asset is not the same, except at initiation of the lease and at the end of the lease. Operating leases report higher ROU assets.
- The ROU asset matches the lease liability in each year for the operating lease. The ROU asset only matches the lease liability at the start and end of the lease's life for the finance lease.
- Finance lease methodology reports both interest and amortization expenses in the income statement. Operating lease methodology reports a lease expense equal to the payment, which is not split into interest and amortization components.

- In the early years of a lease's life, the income statement expense is greater in total for a finance lease compared to an operating lease. In the later years, income statement expense is less for a finance lease. In total, the same amount of expense goes through the income statement under both methodologies.

The impact on the financial statements is summarized in Figure 34.1.

Figure 34.1: Impact on Financial Statements

	Finance Lease (IFRS and US GAAP)	Operating Lease (US GAAP)
Balance sheet assets (ROU asset)	Lower	Higher
Balance sheet liabilities	Same	Same
Income statement earnings (early years)	Lower	Higher
Income statement earnings (later years)	Higher	Lower
EBIT	Higher	Lower
Interest expense	Higher	Lower
Operating cash flows (CFO)	Higher	Lower
Financing cash flows (CFF)	Lower	Higher

For short-term (<12 months) or low-value leases under IFRS, and for short-term leases under US GAAP, no lease asset or liability is reported on the balance sheet. Each period, the lease payment is reported as rental expense on the income statement, spread straight-line over the life of the lease. A summary of the approaches is shown in Figure 34.2.

Figure 34.2: Lessee Accounting

IFRS		
<u>Operating lease</u>	<u>Finance lease</u>	<u>Short term or <\$5,000</u>
Both treated as finance leases for accounting		Expensed straight-line
US GAAP		
<u>Operating lease</u>	<u>Finance lease</u>	<u>Short term</u>
Different methodology		Expensed straight-line

Lessor Accounting

Under both IFRS and US GAAP, there are two lease classifications for lessors, finance leases and operating leases, just as for lessees. At the initiation of a finance lease, the lessor removes the leased asset from its balance sheet and adds

a **lease receivable** asset, equal to the value of the expected lease payments. If this value is different from the asset's book value, the lessor will recognize a profit or a loss. Over the term of the lease, the lessor will use the effective interest method (the same method we have just seen for lessees) to amortize the lease receivable and will report the interest portion of the lease payments as income. This interest income is included in the lessor's revenue for the period if the firm is a manufacturer or dealer of the leased asset. On the cash flow statement, the entire cash inflow is classified as cash from operations.

If manufacturing or dealing in the leased equipment is the main business operation of the firm, the sales proceeds will be in the revenue line and the carrying value of the asset will be a cost of sale. In other words, the treatment is like selling inventory. Both IFRS and US GAAP refer to this treatment as a **sales-type lease**.

If the lessor is a financing company, rather than a manufacturer of the leased equipment, a gain or loss is not recognized at initiation. Instead, the gain or loss is deferred, and it is recognized over the life of the lease as interest income (for a gain) or an expense (for a loss). IFRS and US GAAP refer to these leases as **direct financing leases**.

EXAMPLE: Lessor accounting for a finance lease

The Expensive Company is the lessor to Affordable Company from our lessee accounting example. Expensive is a manufacturer of the equipment being leased.

From the previous lessee examples, the lease had annual payments of \$10,000 over a four-year period. At the end of the lease, which is also the end of the machine's useful life, Affordable will return the machine to the lessor. The interest rate implicit in the lease is 5%. Expensive believes the asset will have a residual value of \$2,000 at the end of the lease period. The carrying value of the asset, recorded as inventory in the lessor's balance sheet, was \$30,000.

Calculate the impact of the lease on Expensive's financial statements for each of the four years if the lease is treated as a finance lease.

Answer:

Profit or loss on derecognition of asset:

The present value of the lease payments is computed at the 5% rate implicit in the lease. In our example, the PV of lease payments = \$35,460 (the same as in

the lessee example). This is treated as revenue at the beginning of the lease.

$$\text{The present value of the expected residual value} = \frac{\$2,000}{(1.05)^4} = \$1,645.$$

The cost of sales is equal to the asset's carrying value less the present value of the residual value: $\$30,000 - \$1,645 = \$28,355$.

	\$
Revenue = present value of lease payments	35,460
Cost of sale = carrying value – present value of residual value	<u>(28,355)</u>
Gross profit or (loss)	7,105

Balance sheet lease receivable asset and interest income:

The sum of the present value of lease payments and the salvage value is referred to the **net investment in the lease** and will be the initial value of the lease receivable asset in the balance sheet.

$$\text{net investment in the lease} = \$35,460 + \$1,645 = \$37,105$$

The net investment in the lease is equal to the asset's fair value. This is because the interest rate implicit in the lease is the internal rate of return that makes the present value of the lease payments received and the residual value equal the asset's fair value.

Year	(1) Beginning Lease Receivable Asset (Net Investment in the Lease)	(2) Interest Income = (1) × 5%	(3) Lease Payment Received	(4) Principal Repayment = (3) – (2)	Ending Lease Receivable Asset = (1) – (4)
	\$	\$	\$	\$	\$
0					37,105
1	37,105	1,855	10,000	8,145	28,960
2	28,960	1,448	10,000	8,552	20,408
3	20,408	1,020	10,000	8,980	11,428
4	11,428	572	10,000	9,428	2,000

At the end of the lease period, the lease receivable asset is equal to the expected residual value of the asset. If the lessor disposes of the asset for any amount that differs from the residual value, a further gain or loss will be reported in the income statement.

For an operating lease, the lessor does not remove the leased asset from its balance sheet. The lessor will continue to record the depreciation expense over the life of the asset. On the income statement, the lessor reports the lease payments as income, while depreciation and other costs associated with leasing the asset are reported as expenses. As with a finance lease, the entire cash inflow is classified as cash from operations.

EXAMPLE: Lessor accounting for an operating lease

Using the same data, calculate the impact on Expensive's financial statements for each of the four years if the lease is treated as an operating lease. Assume straight-line depreciation with a salvage value of \$2,000.

Answer:

The asset remains in the lessor's balance sheet, typically within plant, property, and equipment. The asset will continue to be depreciated over the life of the lease. No lease receivable asset is created. Payments from the lessee are treated as rental income (lease revenue) on a straight-line basis in the income statement.

$$\text{depreciation expense} = \frac{(\$30,000 - \$2,000)}{4 \text{ years}} = \$7,000$$

Year	Net Plant, Property, and Equipment \$	Depreciation Expense \$	Lease Revenue \$	Net Income Statement Impact \$
0	30,000			
1	23,000	7,000	10,000	3,000
2	16,000	7,000	10,000	3,000
3	9,000	7,000	10,000	3,000
4	2,000	7,000	10,000	3,000

Comparing the financial statements for lessors using the finance lease and operating lease approaches shows that the choices have a dramatic impact, with the exception of the cash flow statement.

Lessor: Finance vs. Operating Lease Comparison

Year	Balance Sheet		Income Statement		Cash Flow Statement	
	Finance Lease	Operating Lease	Finance Lease	Operating Lease	Finance Lease	Operating Lease
	Lease Receivable	Net PP&E	Interest Revenue	Net of Lease Revenue and Depreciation	CFO	CFO
	\$	\$	\$	\$	\$	\$
0	37,105	30,000				
1	28,960	23,000	8,960*	3,000	10,000	10,000
2	20,408	16,000	1,448	3,000	10,000	10,000
3	11,428	9,000	1,020	3,000	10,000	10,000
4	2,000	2,000	<u>572</u>	<u>3,000</u>	<u>10,000</u>	<u>10,000</u>
Total			12,000	12,000	40,000	40,000

*Year 1 contains the profit on the derecognition of the asset and Year 1 interest revenue (\$7,105 + \$1,855).

Note that the same total income hits the income statement under both treatments, but the amounts in each individual year vary.



MODULE QUIZ 34.1

- Compared to purchasing a long-lived asset using debt financing, leasing the asset *most likely*:
 - is more costly to the lessee.
 - requires a greater initial cash outflow from the lessee.
 - allows the lessee to avoid the risk of obsolescence.
- Under IFRS, which of the following lease types *least likely* requires a lessee to create a right-of-use (ROU) asset and a lease liability?
 - Low-value leases.
 - Operating leases.
 - Finance leases.
- During the life of a long-term lease under IFRS, the lessee recognizes:
 - interest expense only.
 - amortization expense and interest expense.
 - neither amortization expense nor interest expense.
- Criteria for reporting a lease as a finance lease *least likely* include that the:
 - present value of the lease payments is less than the fair value of the asset.
 - lease term is for substantially most of the asset's useful life.
 - lessee directs the use of the asset and retains the benefits from the asset's use.
- For a lessor, operating leases result in:

- A. interest income recorded in the income statement.
 - B. a profit or loss at the beginning of the lease.
 - C. depreciation on the leased asset.
6. For a lessee with an operating lease, which of the following is *most accurate*?
- A. Both IFRS and US GAAP report interest and amortization in the income statement.
 - B. For both IFRS and US GAAP, the right-of-use (ROU) asset will equal the lease liability over the life of the lease.
 - C. IFRS will typically result in a lower ROU asset than US GAAP.
7. For a lessor, cash flows from a lease are classified as:
- A. operating.
 - B. investing.
 - C. financing.
8. If the lessor in a finance lease is a manufacturer or dealer of leased equipment, the lessor will:
- A. retain the asset in its balance sheet and continue to depreciate it.
 - B. record higher revenues at lease inception when compared to an operating lease.
 - C. record lease revenue in its income statement rather than interest and amortization.

MODULE 34.2: DEFERRED COMPENSATION AND DISCLOSURES

LOS 34.b: Explain the financial reporting of defined contribution, defined benefit, and stock-based compensation plans.

Pension plans and stock-based awards are examples of **deferred compensation**, where employees earn compensation in the current period but do not receive cash flows until later. Typically, the accounting for deferred compensation requires judgment and assumptions by management.

Pension Plans

A **pension** is a form of deferred compensation earned over time through employee service. The most common pension arrangements are defined contribution plans and defined benefit plans.

A **defined contribution plan** is a retirement plan in which the firm contributes a sum each period to the employee's retirement account. The firm's contribution can be based on any number of factors, including years of service, the employee's age, compensation, profitability, or even a percentage of the employee's contribution. In any event, the firm makes no promise to the

employee regarding the future value of the plan assets. The investment decisions are left to the employee, who assumes all of the investment risk.

The financial reporting requirements for defined contribution plans are straightforward. The pension expense is simply equal to the employer's contribution. Once the contribution is paid, there is no future obligation to report on the balance sheet as a liability.

In a **defined benefit plan**, the firm promises to make periodic payments to employees after retirement. The benefit is usually based on the employee's years of service and the employee's compensation at, or near, retirement. For example, an employee might earn a retirement benefit of 2% of her final salary for each year of service. Consequently, an employee with 20 years of service and a final salary of \$100,000 would receive \$40,000 ($\$100,000 \text{ final salary} \times 2\% \times 20 \text{ years of service}$) each year upon retirement until death. Because the employee's future benefit is defined, the employer assumes the investment risk.

A company that offers defined pension benefits typically funds the plan by contributing assets to a separate legal entity, usually a trust. The plan assets are managed to generate the income and principal growth necessary to pay the pension benefits as they come due. The fair value of plan assets is just the current value of the pool of assets at today's date.

Financial reporting for a defined benefit plan is much more complicated than for a defined contribution plan because the employer must estimate the value of the future obligation to its employees. The obligation involves forecasting numerous variables, such as future compensation levels, employee turnover, average retirement age, mortality rates, and an appropriate discount rate. Due to the complexity, firms employ the services of external actuaries to estimate the values. The liability side of the pension plan is the present value of the expected payments to the employees from retirement to death discounted back to today's date.

For a defined benefit plan, the **net pension asset** or **net pension liability**, referred to as **funded status**, is a key element for analysis. If the fair value of the plan's assets is greater than the estimated pension obligation, the plan is said to be *overfunded*, and the sponsoring firm records a net pension asset on its balance sheet. If the fair value of the plan's assets is less than the estimated pension obligation, the plan is *underfunded*, and the firm records a net pension liability.

For plans relating to post-retirement health care benefits, the liability represents

the present value of expected health care premiums post retirement. These plans are typically not prefunded with a pool of assets; therefore, they will always be a liability on the balance sheet.

The change in the net pension asset or liability is recognized on the firm's financial statements each year. Some components are included in net income, while others are recorded as changes to other comprehensive income (OCI). The total economic cost of running the plan is the same under US GAAP and IFRS; however, the split between the income statement and OCI differs.



PROFESSOR'S NOTE

Accounting for defined benefit pension plans is addressed in more detail at Level II.

Accounting for Defined Benefit Plans Under IFRS

The change in funded status comprises three elements:

Service cost. This represents the present value of the additional benefits employees are entitled to after retirement because they have worked an additional year. This element also includes “past service costs,” which represent changes to the benefits earned in previous periods resulting from alterations to the plan and other factors.

Net interest expense or income. This is calculated as the net pension asset or liability times the plan's discount rate. If the plan's beginning funded status is a net asset, the company will report interest income; if the plan is underfunded at the start of the year, the company will report interest expense.



PROFESSOR'S NOTE

Due to the interest element, analysts should treat a balance sheet liability as a debt instrument for leverage computations.

Remeasurements. There are two sources of remeasurement: actuarial gains and losses, and the difference between actual and expected returns on plan assets. Actuarial gains and losses are changes in the net pension asset or liability that result from changing actuarial estimates, such as the rate of salary growth, discount rate, employee turnover, retirement age, and mortality rates. The difference between expected and actual return on plan assets results from the expected return being included in the income statement and the actual return being a component of funded status in the balance sheet.

The income statement expense under IFRS only includes service costs and net interest expense or income. Remeasurement gains and losses are taken directly to other comprehensive income within stockholders' equity.

Accounting for Defined Benefit Plans Under US GAAP

The change in a net pension asset or liability has five components under US GAAP. The first three are recognized in the income statement each period, while the last two go to other comprehensive income.

Service costs for the current period.

Interest expense or income.

The expected return on plan assets.

Past service costs.

Actuarial gains and losses.

One of the differences from IFRS pension accounting is that past service costs are recognized in other comprehensive income, rather than in the income statement as part of employee service costs. These costs are amortized over the employees' service period. Actuarial gains and losses are typically treated the same way, but US GAAP allows firms to recognize them in the period incurred.

For manufacturing companies, pension expense is allocated to inventory and cost of goods sold for employees who provide direct labor to production and to salary or administrative expenses for other employees. As a result, pension expense does not appear separately on the income statement for manufacturing companies. An analyst must examine the financial statement notes to find the details of these companies' pension expense.

Share-Based Compensation

Share-based compensation is designed to align the interest of managers and stockholders and reduce agency costs (which we described in the Corporate Finance topic area). Share-based compensation does not require cash outflows from the company, but issuing employees stock will dilute the proportional ownership of existing shareholders and reduce earnings per share.

A criticism of share-based compensation is that an individual employee is unlikely able to directly influence the company's stock price—which, to a certain extent, is driven by the ebbs and flows of the markets. There is also a danger that stock award may make managers too risk-averse, to prevent declines in the value of their holdings. Stock options, on the other hand, may cause managers to take on too much risk, because options have asymmetrical payoffs.

An option has value if the stock price is above the exercise price, but its value cannot fall below zero if the stock price is below the exercise price.

IFRS and US GAAP both require the company to estimate the **fair value** of any stock-based compensation at the **grant date**, and to expense it to the income statement over the **vesting period**. The vesting period (service period) is the time between the grant date and when the employee receives the stock or can first exercise a stock option.

Stock grants. These are shares awarded outright, with restrictions, or contingent on performance. The fair value of the stock grant is the share price on the grant date. If vesting is not immediate, the compensation expense will be recognized between the grant date and the vesting date, which is known at the service period. If vesting is immediate, then the full fair value is recognized as an expense in the income statement on the grant date.

Performance shares. These are stock grants that depend on meeting a set performance target. Typically, the performance targets are not share price related, but instead focus on metrics like return on equity. While this addresses the concern that the individual may have little influence over share price, it may create incentives for managers to manipulate financial statements.

Stock grants that do not vest until certain criteria are met (typically, length of service or performance goals) are often referred to as **restricted stock units**.

Employee stock options are options to invest in the company's stock at a given price (the exercise price) at a future date. Unlike stock grants, which have value as long as the stock price is greater than zero, an option only has value if the stock price is above the exercise price. If an employee stock option is exercised, the company issues new shares in return for the exercise price. Option valuation models, such as the Black-Scholes-Merton or binomial models, are needed to compute the fair value of the option on the grant date. Some of the inputs that the models require are highly subjective—in particular, the assumed volatility of the company's stock price over the life of the option. The fair value is then expensed between the grant date and the vesting date (the date the option can first be exercised).

Stock grants have the following effects on financial statements:

The grant date is when the fair value is established. Fair value is normally the market price on this date.

If vesting is immediate, the full fair value is expensed to the income statement, and both common stock and additional paid-in capital (APIC) are increased by this amount.

If there is a length of time between the grant date and the vesting period (service period), a compensation expense is recognized in each year over the service period in the income statement on a straight-line basis. This appears in equity in an account such as a *share-based compensation reserve* or APIC. Over the entire service period, the total amount of the fair value of the option on the grant date will have been expensed to the income statement.

At the end of the service period, any amount remaining in an equity reserve will be recycled into common stock and APIC.

Stock options have the following effects on financial statements:

The fair value of the option is established using the option valuation methodology adopted.

As with grants, at the grant date, there is no impact on common stock or APIC, and the fair value is spread, straight-line, over the service period as a compensation expense in the income statement and an increase in APIC or a share-based compensation reserve (IFRS 2 does not specify a required equity account).

For both grants and options, the expense in the income statement reduces retained earnings; however, stockholders' equity remains constant by increasing equity by the same amount.

On exercise of the option, cash increases by the exercise price received. Stockholders' equity increases by the same amount, split between common stock at par and APIC, with any amount in an equity reserve also being recycled into APIC.

Stock-based appreciation rights (SARs) generate cash for the holders that is linked to stock performance. The employee receives payments based on the change in value of the company's shares without needing to hold the stock. These have the advantage of aligning employees' and shareholders' interest without creating new shares and diluting existing shareholders. They have payoffs similar to stock options, and so do not introduce bias toward risk-averse behavior. The downside for the company is that they result in cash outflows when the stock performs well. Non-exchange-traded firms may use a version of

this called **phantom stock**, where the cash payments are related to the performance of a hypothetical stock.

LOS 34.c: Describe the financial statement presentation of and disclosures relating to long-term liabilities and share-based compensation.

Lease Disclosures

The objective of lease disclosures is to provide users of financial statements with a basis to assess the effect of leasing activities on the entity's financial position, performance, and cash flows. To achieve that objective, lessees and lessors disclose both qualitative and quantitative information.

As indicated in IFRS 16, here is what lessee disclosures must include:

- | The carrying amount included in the balance sheet for the ROU asset by class of underlying asset
- | Total cash outflows relating to the lease
- | The interest expense included in the income statement resulting from the lease liability
- | Depreciation (amortization) expensed in the period on the ROU asset, by class of underlying asset
- | Expenses relating to variable lease payments not included in lease liabilities



PROFESSOR'S NOTE

Variable lease payments are included in the disclosure for lessees and lessors, but they are not defined by the Level I curriculum. There are two types of variable lease payments. The first is where the rental payments are linked to an index or rate (e.g., a market reference interest rate or the rate of inflation). The current lease liability and ROU asset are based on the current value of the index or rate and then are remeasured when there are changes. The second type relates to variable payments depending on future sales or use of the asset. These items are not included in the lease liability or ROU asset, and instead are expensed when occurred.

- | Additions to ROU assets
- | Maturity analysis of lease liabilities and the split between current and long-term liabilities



PROFESSOR'S NOTE

Next year's principal repayment will be a current liability, and the remaining principal repayments will be reported in long-term liabilities.

- | Income statement expenses relating to low-value and short-term leases
- | Quantitative and qualitative information on the following:
 - The nature of the leasing activities
 - Future cash outflows to which the lessee is exposed to that are not reflected in the lease liability (could include any guarantees of the leased asset's residual value)
 - Restrictions and covenants imposed by the lease
 - Sale and leaseback transactions (where the lessee sells the asset to the lessor and then immediately leases it back)

As indicated in IFRS 16, lessors must disclose the following for finance leases:

- | Selling profit or loss on derecognition of the asset
- | Finance income (interest) recognized in the income statement relating to the net investment in the lease (lease receivable asset)
- | Income relating to variable lease payments not included in the measurement of the net investment in the lease
- | Qualitative and quantitative explanation of significant changes in the net investment in the lease
- | Maturity analysis of lease payments receivable
- | Reconciliation of undiscounted lease payments to the net investment in the lease

As indicated in IFRS 16, lessors must disclose the following for operating leases:

- | Lease income recognized in the income statement, separately disclosing income for variable lease payments that do not depend on an index or rate
- | Maturity analysis of lease payments receivable—at a minimum, must show undiscounted lease payments to be received in each of the next five years and aggregated amounts beyond five years.
- | The underlying asset remains in the lessor's balance sheet and must comply with the disclosure for the following:

- IAS 16 for leases of property, plant and equipment, disaggregated by class
- IAS 36: Impairments

Pension Disclosures

For defined contribution plans, the only disclosure requirement of IAS 19 is separate disclosure of the employer's contribution expensed in the income statement.

For defined benefit plans, IAS 19 sets out the following objectives:

Explain the characteristics and risks of its defined benefit plan. Much of the risk in defined benefit plans relates to the funded status of the plan—and in particular, underfunded plans.

Identify amounts in the financial statements relating to defined benefit plans.

Describe how defined benefit plans affect amounts, timings, and uncertainties relating to future cash flows. (Note that the cash flows relating to defined benefit plans are the employers' contribution.)

While the firm has discretion regarding how to achieve these objectives, some minimum disclosures are required. Disclosure requirements are as follows:

- | The nature of the plan, governance, regulatory framework, and risk exposures
- | Reconciliations of beginning and ending values for funded status, the present value of the defined benefit obligation, and plan assets, showing components of the change (includes changes in the balance sheet net asset or liability that are taken to the income statement expense and to OCI)
- | Sensitivity analysis showing how changes to key actuarial assumptions (discount rate, rate of salary growth, mortality rates, etc.) affect the present value of the defined benefit obligation
- | Composition of plan assets by asset type
- | Expected employer contributions for the next period and beyond
- | The maturity profile of the defined benefit obligation

Share-Based Compensation Disclosures

The objective of these disclosures is to provide users of the accounts with sufficient information to understand the nature and extent of stock-based

compensation arrangements, including their impacts on current and future cash flows. Firms must disclose:

- | The nature of the plan and key details such as grant date, vesting date, and service period, as well as settlement characteristics of employee stock options (physical delivery or cash settlement)
- | How the fair value at the grant date was determined
- | The effect of share-based transactions on earnings and the financial position (i.e., impacts on the income statement and balance sheet)



MODULE QUIZ 34.2

1. A net pension asset or liability can be associated with:
 - A. defined benefit pension plans only.
 - B. defined contribution pension plans only.
 - C. either defined benefit or defined contribution pension plans.
2. Which of the following is *most accurate* concerning defined benefit pension plans under both IFRS and US GAAP?
 - A. The income statement expense is the same.
 - B. The amounts taken to other comprehensive income (OCI) are the same.
 - C. The total periodic cost of the plan is the same.
3. Which of the following is *least likely* a criticism of employee stock options?
 - A. The binary nature of option payoffs may encourage excessive risk taking.
 - B. The fair value requires subjective estimates at the grant date.
 - C. They result in cash outflows for the company on exercise.

KEY CONCEPTS

LOS 34.a

Advantages of leasing rather than purchasing an asset may include a smaller initial cash outflow, lower-cost financing, and less risk of obsolescence.

A lease transfers the benefits and risks of ownership of the asset to the lessee. A finance lease must meet at least one of the following criteria:

Ownership of the leased asset transfers to the lessee.

The lessee has an option to buy the asset and is expected to exercise it.

The lease is for most of the asset's useful life.

The present value of the lease payments is greater than or equal to the asset's fair value.

The lessor has no other use for the asset (i.e., the asset is of a specialized nature only suitable for use by the lessee).

If none are met, the lease is an operating lease.

Lessee Reporting

Under IFRS, for both finance and operating leases, except for short-term leases, a lessee reports a right-of-use (ROU) asset and a lease liability on its balance sheet, both equal to the present value of the promised lease payments. The interest portion of each lease payment is reported as an interest expense, while the principal repayment portion of each payment reduces the lease liability. For short-term or low-value leases, rent expense is reported on the income statement, and no balance sheet entries are required.

Under US GAAP, reporting for a finance lease is the same as under IFRS. For an operating lease, the reporting is similar as under IFRS, except that the entire lease payment is recorded as a lease expense and the amortization of the ROU asset is equal to the principal repayment. For short-term leases, rent expense is reported on the income statement, and no balance sheet entries are required.

Lessor Accounting

A lease that is classified as finance or operating by the lessee is classified the same way by the lessor.

Under both IFRS and US GAAP, with a finance lease, the lessor removes the leased asset from its balance sheet and adds a lease receivable asset. The lessor reports the interest portion of the lease payments as income. For an operating lease, the lessor keeps the leased asset on its balance sheet, reports lease payments as income, and reports depreciation and other costs as expenses.

For a lessor, finance leases can be sales-type leases or direct financing leases. For a dealer or manufacturer of the leased equipment, the sales-type treatment is used. A profit or loss is recorded on the leased asset, as if it was inventory sold at initiation of the lease. For a direct financing lease, any gain or loss on derecognition is deferred and recognized in the income statement over the life of the lease as interest.

LOS 34.b

Defined Benefit Pension Plan Reporting

A firm reports a net pension liability on its balance sheet if the fair value of a defined benefit plan's assets is less than the estimated pension obligation, or a net pension asset if the fair value of the plan's assets is greater than the estimated pension obligation. The change in the net pension asset or liability is reflected in a firm's income statement, and as changes to accumulated other comprehensive income (OCI) each period.

IFRS refers to the estimated pension liability as the present value of defined benefit obligations (PVDBO), while US GAAP uses the term projected benefit obligation (PBO). In practice, the PVDBO and the PBO are the same. Both are equal to the benefits earned to date, to be paid between retirement and death, discounted back to the balance sheet date. The computations are complex and require actuaries to estimate the assumptions and values.

Given consistent actuarial assumptions, the total periodic cost of the defined benefit plan is the same under IFRS and US GAAP. The amounts taken in the income statement and OCI, however, differ.

Service cost is the present value of the additional benefits earned by employees for being a member of the plan for an additional service period. Both IFRS and US GAAP include this in the income statement expense.

The interest cost is the increase in the PVDBO (PBO) due to the passage of time. IFRS nets the interest cost with the expected return on plan assets to show interest income or expense. US GAAP shows these two components separately as they can be computed using different rates. Whether reported net or separately, they are a component of the income statement expense.

Changes to the estimated pension liability as a result of actuarial estimates changing and the difference between actual and expected return on plan assets are taken directly to OCI. IFRS refers to these amounts as remeasurements, while US GAAP refers to them as actuarial gains and losses. US GAAP may potentially amortize these amounts through future income statement expenses, while IFRS does not.

Past service costs are changes to the estimated pension liability as a result of benefits earned in prior periods changing if the plan is amended. IFRS expenses these amounts in the income statement in the year of change. US GAAP takes these amounts to OCI and then amortizes them to the income statement over the remainder of the employees' service life.

Defined Contribution Pension Plan Reporting

A pension expense for a defined contribution pension plan is equal to the employer's contributions. There is no balance sheet asset or liability reported, providing the employer's contribution has been made by year-end.

Stock Grants and Options Reporting

Stock grants and stock options serve to reward employees and align employees and shareholders' interest to reduce agency cost. Advantages to the company include the lack of cash outflows. Disadvantages include the employees' relative lack of influence over the company's stock price and dilution of existing shareholders.

Stock grants can have immediate or delayed vesting. The fair value of a stock grant is established on the grant date and is equal to the stock's fair value. If vesting is immediate, the fair value is expensed to the income statement, and common stock and additional paid-in capital (APIC) are adjusted in the balance sheet, as if the company was issuing shares. If there is a period between the grant and vesting, it is referred to as the service period. The fair value of the stock grant is spread straight-line as a compensation expense in the income statement and increases common stock and APIC in the balance sheet.

Stock options give the employee the right to buy shares in the future at the exercise price. A criticism of employee stock options is that the binary nature of payoffs can lead to excessive risk taking. The fair value of the option must be estimated using subjective assumptions. The fair value is then expensed straight-line to the income statement over the vesting period. The vesting period is from the grant date until the first date the options can be exercised. A compensation expense is recorded in the income statement, and APIC increased in the balance sheet. At the point the options are exercised in the future cash increases by the exercise price received, common stock increases by the par value of share issued and any balance is taken to APIC. If the options expire and are not exercised, no adjustments need to be made.

LOS 34.c

Both IFRS and US GAAP set out the objectives of disclosure for leases, defined benefit pension plans, and stock-based compensation. The objective of the disclosure is to provide the users of the accounts with sufficient information to understand the nature, risks, and extent of leases and compensation plans,

including their impacts on current and expected cash flows. While companies have discretion over the disclosure needed to achieve these objectives, the accounting standards provide guidance.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 34.1

- . **C** Avoiding the risk of obsolescence is one of the advantages of leasing assets instead of purchasing them. At the end of a lease, the lessee often returns the leased asset to the lessor, and therefore does not bear the risk of an unexpected decline in the asset's end-of-lease value. The interest rate implicit in a lease contract may be less than the interest rate on a loan to purchase the asset. The terms of a lease may not require all the covenants typically included in loan agreements or bond indentures. (LOS 34.a)
- . **A** Both operating leases and finance leases report an ROU asset and a lease liability, and both are accounted for identically. Under US GAAP, operating and financing leases also report both an ROU asset and lease liability; however, the accounting for the two types is slightly different. Exceptions exist for low-value assets and leases with durations of less than one year under IFRS (US GAAP has no monetary value criteria). (LOS 34.a)
- . **B** At lease inception, the lessee records a right-of-use (ROU) asset and a lease liability, both equal to the present value of the lease payments. In each period over the life of the lease, the lessee recognizes interest expense for the interest portion of the lease payments and amortization expense on the ROU asset. (LOS 34.a)
- . **A** IFRS and US GAAP treat a lease as a finance lease (for both lessee and lessor) if any of the following criteria are met:
 1. Ownership of the leased asset transfers to the lessee.
 2. The lessee has an option to buy the asset and is expected to exercise it.
 3. The lease is for most of the asset's useful life.
 4. The present value of the lease payments is greater than or equal to the asset's fair value.
 5. The lessor has no other use for the asset (i.e., the asset is of a specialized nature only suitable for use by the lessee).

(LOS 34.a)

- . **C** For a lessor, the asset remains in their balance sheet if treated as an operating lease, and is depreciated over its life. There is no derecognition of the asset, so there is no gain or loss at the outset of the lease. Payments from the lessee are treated as rental income. (LOS 34.a)
- . **C** US GAAP reports an income statement expense that is equal to the lease payment. While conceptually, the amount contains both interest and amortization, they are not shown separately on the income statement. IFRS reports both interest and amortization in the income statement, and the aggregate of the two amounts will not equal the lease payment. Under US GAAP, the ROU asset and the lease liability are equal at all points over the lease's life. Under IFRS, the only points where the ROU asset is the same as the lease liability are at initiation and the end of the lease. IFRS will result in a lower ROU asset than US GAAP. Under IFRS, the ROU asset is amortized over the life of the asset (using either straight-line or accelerated methods). Under US GAAP, the amortization is equal to the principal payment on the liability. Principal payments initially are lower due to the high interest cost, but as the lease ages and the liability decreases, the interest element of each payment will decrease and the principal element will increase. The result is that accumulated amortization under IFRS is greater than US GAAP, resulting in a lower ROU asset. (LOS 34.a)
- . **A** Cash flows from a lease are operating cash inflows for the lessor. (LOS 34.a)
- . **B** Finance leases for a lessor may be classified as either sales-type leases or direct financing leases. If the lessor is a manufacturer or dealer of the leased equipment, the asset is treated as inventory sold at the outset of the lease. Revenue is reported as well as a cost of sales amount. The revenue is the present value of lease receipts. Cost of sales is the carrying value of the asset less the present value of any residual value. The lease is a direct financing lease if the lessor is a financing company (buying an asset on behalf of the lessee and leasing it out). Any gain or loss on derecognition of the asset does not affect the income statement on lease initiation. The gain or loss is deferred and recognized in the income statement, as interest, over the life of the lease. For an operating lease, revenue is recognized when lease payments from the lessee fall due. (LOS 34.a)

Module Quiz 34.2

- . **A** Defined benefit pension plans can be overfunded and result in a net pension asset, or they can be underfunded and result in a net pension liability. Defined contribution plans do not result in balance sheet assets or liabilities because they are neither owned by the sponsoring firm nor obligations of the sponsoring firm. (LOS 34.b)
- . **C** The total cost of the defined benefit pension plan is identical for both IFRS and US GAAP. IFRS and US GAAP differ in the amounts recognized in the income statement and the amounts taken to OCI. (LOS 34.b)
- . **C** On exercise, the company receives the option's exercise price (a cash inflow to the company) and creates new shares. This dilutes existing stockholders' ownership proportions in the company. The fair value of the option at the grant date requires subjective estimation—in particular, the volatility of the company's stock price over the service period. By nature, options have binary payoffs, being either zero if the stock price is below the exercise price or positive if the stock price is above the exercise price. This may lead to excessive risk taking because the option holder gains if the stock price increases, but if the stock price decreases the option holder does not experience losses except in the sense that the options go unexercised. (LOS 34.b)

READING 35

ANALYSIS OF INCOME TAXES

MODULE 35.1: DIFFERENCES BETWEEN ACCOUNTING PROFIT AND TAXABLE INCOME

LOS 35.a: Contrast accounting profit, taxable income, taxes payable, and income tax expense and temporary versus permanent differences between accounting profit and taxable income.

Financial accounting standards (IFRS and US GAAP) are often different than income tax laws and regulations. As a result, the amount of income tax expense recognized in the income statement may differ from the actual taxes owed to the taxing authorities.

Let's begin by defining the terms we will use to distinguish items on a company's tax return (the information it files with its country's tax authorities) from the comparable items on its financial statements.

Tax Return Terminology

- | **Taxable income.** This is income subject to tax based on the tax return.
- | **Taxes payable.** This is the tax liability caused by *taxable income*. This is also known as the current tax expense, but do not confuse this with *income tax expense* (see next).
- | **Income tax paid.** This is the actual cash flow for income taxes, including payments or refunds from other years.
- | **Tax loss carryforward.** This is a current or past loss that can be used to reduce taxable income (thus, taxes payable) in the future. It can result in a deferred tax asset.
- | **Tax base.** This is the net amount of an asset or liability used for tax reporting purposes.

Financial Reporting Terminology

- | **Accounting profit.** This is pretax financial income based on financial accounting standards (also known as *income before tax* and *earnings before tax*).
- | **Income tax expense.** This is the expense recognized in the income statement that includes taxes payable and *changes* in deferred tax liabilities and assets.
- | **Deferred tax liabilities.** These are balance sheet amounts that result from an excess of income tax expense over taxes payable that are expected to result in future cash outflows.
- | **Deferred tax assets.** These are balance sheet amounts that result from an excess of taxes payable over income tax expense that are expected to be recovered from future operations. These can also result from tax loss carryforwards.
- | **Valuation allowance.** This is the reduction of deferred tax assets based on the likelihood the assets will not be realized.
- | **Carrying value.** This is the net balance sheet value of an asset or liability.
- | **Permanent difference.** This is a difference between taxable income (tax return) and pretax income (income statement) that will not reverse in the future.
- | **Temporary difference.** This is a difference between the tax base and the carrying value of an asset or liability that will result in either taxable amounts or deductible amounts in the future. Several examples of how temporary differences arise are presented later in this review.

Differences Between Tax and Financial Reporting

Differences between the treatment of an accounting item for tax reporting and for financial reporting can occur when the following occur:

- | The timing of revenue and expense recognition in the income statement and the tax return differ
- | Certain revenues and expenses are recognized in the income statement, but never on the tax return or vice versa
- | Assets and/or liabilities have different carrying amounts and tax bases
- | Gain or loss recognition in the income statement that differs from the tax return

- | Tax losses from prior periods may offset future taxable income
- | Financial statement adjustments may not affect tax return, or may be recognized in different periods

Differences between the balance sheet carrying value and the tax base of an asset may be permanent or temporary. Temporary timing differences result from the same total amounts passing through the income statement and tax returns over time, but with differences in each individual period. These temporary timing differences result in the creation of deferred tax assets and deferred tax liabilities, which we will discuss next. Permanent timing differences do not result in deferred tax assets or liabilities.

Deferred Tax Liabilities

A **deferred tax liability (DTL)** is created when the income tax expense (income statement) is greater than taxes payable (tax return) due to temporary differences. DTLs occur when

- | revenues (or gains) are recognized in the income statement before they are included on the tax return due to temporary differences, and
- | expenses (or losses) are tax deductible before they are recognized in the income statement.

DTLs are expected to reverse (i.e., they are caused by temporary differences) and result in future cash outflows when the taxes are paid. These may be referred to as “taxable temporary differences” because the firm will pay more tax when they reverse.

A DTL is most often created when an accelerated depreciation method is used on the tax return and straight-line depreciation is used on the income statement.

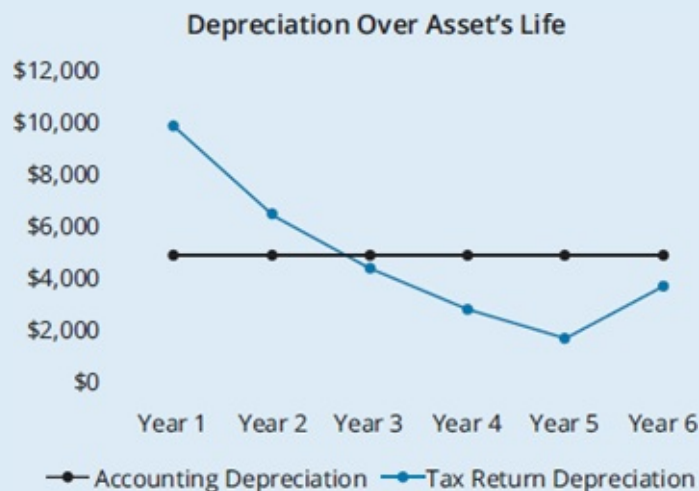
EXAMPLE: Temporary timing differences

Flippy Discs Corporation purchased an injection-molding machine for \$30,000 at the start of the current period. The machine has a useful economic life of six years and no expected residual value. For accounting purposes, the machine will be depreciated straight line; however, a double declining balance will be used by the tax authorities. Calculate depreciation for accounting and for tax, and discuss the timing differences.

Answer:

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Total
Accounting depreciation	\$5,000	\$5,000	\$5,000	\$5,000	\$5,000	\$5,000	\$30,000
Tax return depreciation	\$10,000	\$6,667	\$4,444	\$2,963	\$1,975	\$3,951	\$30,000

The total amount of depreciation over the life of the asset is the same. The accelerated method used in the tax returns results in more depreciation in the early years of the asset's life, but less in the later years relative to the straight-line method. This is a classic example of a taxable temporary difference.



Visually, we can see that the tax return depreciation is greater in the first two years, creating a timing difference, which then begins to reverse from Year 3.

Due to the temporary timing differences, taxable income will initially be lower than earnings before tax in the accounts. The result of this is that Flippy pays lower tax in Years 1 and 2. Once the tax return depreciation drops below the accounting depreciation, taxable income will be greater than earnings before tax, and Flippy will pay more tax. The temporary timing difference is created and then reverses, resulting in less tax due in the early years and more tax due in the later years. We know with certainty that when the timing difference reverses, Flippy will pay more tax, meeting the definition of a liability.

Deferred Tax Assets

A **deferred tax asset (DTA)** is created when taxes payable (tax return) are greater than income tax expense (income statement) due to temporary differences. DTAs occur when

- revenues (or gains) are taxable before they are recognized in the income statement,
- expenses (or losses) are recognized in the income statement before they are tax deductible, and
- tax loss carryforwards are available to reduce future taxable income.

Similar to DTLs, DTAs are expected to reverse through future operations. However, DTAs are expected to provide future tax savings, while DTLs are expected to result in future cash outflows. Timing differences that lead to the creation of DTAs may be referred to as “deductible temporary differences” because the creation or increase of a DTA will reduce the income statement tax expense. A firm that has taxable losses in excess of its taxable income can carry those excess losses forward and use them to reduce taxable income (and taxes) in future periods.

Post-employment benefits, unearned revenue, warranty expenses, and tax loss carryforwards are typical causes of DTAs.

Taxable and Deductible Temporary Differences

Initially, if taxable income (on the tax return) is lower than earnings before tax (on the income statement) because of a temporary timing difference, a DTL will be created. If taxable income is initially greater than earnings before tax, a DTA will be created.

Another way of analyzing temporary differences is to compare the balance sheet carrying values of assets and liabilities to their tax base values (Figure 35.1).

Figure 35.1: Comparison of Carrying Value to Tax Base for Temporary Timing Differences

Balance Sheet	Carrying Value vs. Tax Base	Deferred Tax Asset or Liability
Asset	Carrying value > tax base	Deferred tax liability
Asset	Carrying value < tax base	Deferred tax asset
Liability	Carrying value > tax base	Deferred tax asset
Liability	Carrying value < tax base	Deferred tax liability



PROFESSOR'S NOTE

Our online Resource Library includes supplemental material on how the tax bases of balance sheet assets and liabilities are calculated.

Permanent Differences

A **permanent difference** between taxable income and pretax income is one that will not reverse in the future. Permanent differences do not create DTAs or DTLs. Permanent differences can be caused by revenue that is not taxable, expenses that are not deductible, or tax credits that result in a direct reduction of taxes. For example, in the United States, interest received on municipal bonds is typically not taxable (but appears on the financial statements as pretax income), and the cost of life insurance on key company officers is typically not tax deductible (but appears on the financial statements as a pretax expense).

Permanent differences cause the firm's effective tax rate to differ from the statutory tax rate. We will examine statutory, effective, and cash tax rates later in this reading.

Tax Expense

A company's tax expense is the amount included in the income statement and may be referred to as a tax provision. It includes the amount the company owes the tax authorities (tax payable) based on this year's taxable income, along with any changes in DTAs and DTLs:

$$\text{tax expense} = \text{tax payable} + \Delta\text{DTL} - \Delta\text{DTA}$$

Creation of, or increases in, DTLs increase tax expense. Creation of, or increases in, DTAs reduce tax expense. This is because DTAs are expected to provide future tax savings, while DTLs are expected to result in future cash outflows for tax. Because the firm's actions in the current period either caused deferred tax items to be created or changed their values, the accruals concept requires the changes to be recognized in tax expense in the current period.

Changes in the Enacted Tax Rate

Changes in the company's statutory tax rate will also result in changes to DTLs and DTAs. Both DTLs and DTAs are created at the statutory tax rate in effect at the point of the initial timing difference. If the statutory tax rate changes, existing DTLs and DTAs must be adjusted. Because a DTL or DTA must reverse to zero when the timing differences reverse, they need to reflect the statutory tax rate expected to be in effect when reversal occurs. An increase in the statutory rate will increase both DTAs and DTLs, while a decrease will reduce them. Like

any other change in a DTA or DTL, this will also affect tax expense in the income statement.



MODULE QUIZ 35.1

1. Which of the following tax definitions is *least accurate*?
 - A. Taxable income is income based on the rules of the tax authorities.
 - B. Taxes payable are the amount due to the government.
 - C. Income tax expense equals pretax income times the statutory tax rate.
2. If the tax base of an asset is less than the carrying value of the asset and the difference is not expected to reverse in the future, this will result in:
 - A. the creation of a DTL.
 - B. the creation of a DTA.
 - C. neither the creation of a DTA nor a DTL.
3. Which of the following is *least likely* to result in the creation of a deferred tax asset (DTA)?
 - A. Expenses or losses that are tax deductible before they are recognized in the income statement.
 - B. Revenues or gains that are taxable before they are recognized in the income statement.
 - C. Tax loss carryforwards that are available to reduce future taxable income.
4. Sidewinder Corporation has a year-end accounts receivable carrying value of \$500,000 after including a doubtful debt provision of 5% of year-end receivables. In the tax returns, bad debt is only deducted when it is written off. This treatment will *most likely* result in:
 - A. the creation of a DTL.
 - B. the creation of a DTA.
 - C. neither the creation of a DTA nor a DTL.

5. Roadrunner, Inc., reported the following information in its tax returns and financial statements:

Tax return extract:	\$
Tax payable for the year ended 20X7	500,000
Balance sheet extracts:	
DTA year-end 20X7	300,000
DTA year-end 20X6	350,000
DTL year-end 20X7	500,000
DTL year-end 20X6	400,000

Which of the following is *closest* to the tax expense reported in Roadrunner's 20X7 income statement?

- A. \$500,000.
 - B. \$550,000.
 - C. \$650,000.
6. An increase in the tax rate causes the balance sheet value of a DTA to:
 - A. decrease.

- B. increase.
- C. remain unchanged.

MODULE 35.2: DEFERRED TAX ASSETS AND LIABILITIES

LOS 35.b: Explain how deferred tax liabilities and assets are created and the factors that determine how a company's deferred tax liabilities and assets should be treated for the purposes of financial analysis.

Creation of DTAs and DTLs

As discussed, DTAs and DTLs result from temporary timing differences between earnings before tax and taxable income. For a timing difference to be temporary, the amount that passes through the income statement and the tax return over the life of the asset or liability must be the same, even though the amounts differ that pass through each during individual periods.

EXAMPLE: Deferred tax assets and liabilities

Firebird Corporation is a new subsidiary set up at the start of the year by its parent company, Wraith Incorporated. Firebird's statutory tax rate is 30%, and it has no permanent timing differences. Tax returns and income statements are presented for the first three years of its operation.

Tax return: Firebird Corporation

	Year 1	Year 2	Year 3
	\$000	\$000	\$000
Revenue	100,000	120,000	130,000
Cost of sales	28,500	34,200	37,050
Other expenses	19,950	23,940	25,935
Depreciation	10,000	10,000	10,000
Warranty costs	2,000	5,000	8,000
Interest expense	<u>10,000</u>	<u>12,000</u>	<u>12,500</u>
Taxable income	29,550	34,860	36,515
Tax payable 30%	8,865	10,458	10,955

Income statement: Firebird Corporation

	Year 1	Year 2	Year 3
	\$000	\$000	\$000
Revenue	95,000	114,000	123,500
Cost of sales	<u>28,500</u>	<u>34,200</u>	<u>37,050</u>
Gross profit	66,500	79,800	86,450
Other expenses	19,950	23,940	25,935
Depreciation	8,000	8,000	8,000
Interest expense	<u>10,000</u>	<u>12,000</u>	<u>12,500</u>
Earnings before tax	28,550	35,860	40,015
Tax expense	<u>8,565</u>	<u>10,758</u>	<u>12,005</u>
Net income	19,985	25,102	28,010

Firebird acquired PP&E for \$40 million at the start of Year 1. For financial statement reporting, the PP&E is depreciated straight-line over five years, while for tax reporting, it is depreciated straight-line over four years. Its residual value is assumed to be zero in both cases.

Firebird estimates that warranty expense will be 5% of gross revenues. For this example, we will assume all of the year-end liability values for warranties are likely to pass through future tax returns as the expenditure is incurred.

Analyze the differences between the tax return and the income statement, and calculate any deferred tax items that Firebird will recognize.

Answer:

The first noticeable difference is the revenue figures. Remember that revenue is reported net of any returns and allowances in the income statement (e.g., items such as estimated warranty provisions and customer discounts). The tax return has a line for warranty costs, which indicates that these may be the source of some of the difference. Another significant difference between the two statements are the depreciation lines. Finally, the tax return shows tax payable on taxable income, which differs from the tax expense in the income statement. The latter includes tax payable and changes in DTAs and DTLs.

1. Timing differences related to depreciation

$$\text{tax allowable depreciation} = \frac{\$40,000,000 - \$0}{4 \text{ years}} = \$10,000,000$$

$$\text{accounting depreciation} = \frac{\$40,000,000 - \$0}{5 \text{ years}} = \$8,000,000$$

In total, \$40 million of depreciation will pass through the tax returns and income statements, but different amounts will be reported for each individual

year. Depreciation will be higher in the tax returns than in the income statement for each of the first four years. Firebird will recognize a DTL because the firm pays lower tax initially and higher tax on the reversal of the timing difference. We may also say the asset's carrying value is greater than its tax base, resulting in a DTL. After Year 4, depreciation in the tax return will be zero, while depreciation in the income statement will continue, causing the timing difference to reverse.

End of Period	Year 1	Year 2	Year 3	Year 4	Year 5
	\$000	\$000	\$000	\$000	\$000
Carrying value	32,000	24,000	16,000	8,000	0
Tax base	<u>30,000</u>	<u>20,000</u>	<u>10,000</u>	<u>0</u>	<u>0</u>
Timing difference	2,000	4,000	6,000	8,000	0
DTL @ 30%	600	1,200	1,800	2,400	0
Δ DTL	+600	+600	+600	+600	-2,400

The timing difference grows over the first four years and then reverses in Year 5. Carrying value is equal to the asset's cost less accumulated accounting depreciation. The tax base can be viewed as cost less accumulated tax-allowable depreciation. (Formally, it is defined as the total depreciation that will pass through the tax returns in future periods.)

2. Timing differences related to warranties

A warranty is a commitment to replace or repair a good if it becomes defective during the stated warranty period. Due to the accruals process—and in particular, the matching concept—the cost of meeting future warranty claims must be estimated and accounted for at the time of the sale. The estimated future warranty expense is deducted from revenue, and a warranty liability is created in the balance sheet. Tax authorities, however, are unwilling to allow companies to save tax on *estimated* expenses. As a result, they typically require the expenditure to repair or replace faulty goods to have been incurred before they allow a deduction in the tax returns.

The following table shows the warranty provision, given Firebird's estimate of 5% of sales, and the expenditure actually incurred in each of the three years (from the tax return).

	Year 1	Year 2	Year 3
	\$000	\$000	\$000
Warranty provision	5,000	6,000	6,500
Warranty expenditure	2,000	5,000	8,000

The balance sheet liability will increase by each year's warranty provision, offset by the actual expense incurred:

Carrying Value	Year 1 \$000	Year 2 \$000	Year 3 \$000
Warranty liability (at start of period)	0	3,000	4,000
Warranty provision	5,000	6,000	6,500
Warranty expenditure	<u>(2,000)</u>	<u>(5,000)</u>	<u>(8,000)</u>
Warranty liability (at end of period)	3,000	4,000	2,500

The tax base of a liability is its carrying value less the amount that *will* pass through future tax returns. (There is one exception to this, which is any form of revenue received in advance. These amounts are taxed when received—in other words, before income statement recognition—and the tax base becomes the carrying value of the liability less amounts that *will not* pass through tax returns in the future.)

Recall that we have assumed all of the year-end liability values for warranties are likely to pass through future tax returns as the expenditure is incurred. This will result in a zero value for the tax base in each year.

End of Period	Year 1 \$000	Year 2 \$000	Year 3 \$000
Carrying value	3,000	4,000	2,500
Tax base	<u>0</u>	<u>0</u>	<u>0</u>
Timing difference	3,000	4,000	2,500
DTA @ 30%	900	1,200	750
Δ DTA	+900	+300	−450

3. Income tax expense

Now that we have calculated the deferred tax items, we can illustrate how income tax expense was determined for financial statement reporting.

	Year 1 \$000	Year 2 \$000	Year 3 \$000
Tax payable	8,865	10,458	10,955
+Change in DTL	600	600	600
−Change in DTA	<u>(900)</u>	<u>(300)</u>	<u>450</u>
Tax expense	8,565	10,758	12,005

Realizability of Deferred Tax Assets

Although deferred taxes are created from temporary differences that are expected to reverse in the future, neither DTAs nor DTLs are carried on the balance sheet at their discounted present value. However, DTAs are assessed at each balance sheet date to determine the likelihood that the company will have sufficient future taxable income to recover the tax assets. Without future taxable income, a DTA is worthless.

If there is sufficient doubt that the benefits of DTAs will be realized in future periods, both IFRS and US GAAP require companies to recognize the diminished value of the DTAs. Under IFRS, companies simply reduce a DTA, which increases tax expense in that period. US GAAP employs different mechanics with the same result. Under US GAAP, the full DTA is still carried on the balance sheet, but it must be reduced by a **valuation allowance**. The valuation allowance is a contra account that reduces the net balance sheet value of the DTA. Creating or increasing a valuation allowance increases income tax expense and decreases net income.

It is up to management to defend the valuation of all DTAs. If a company has order backlogs or existing contracts that are expected to generate future taxable income, a valuation allowance might not be necessary. However, if a company has cumulative losses over the past few years or a history of inability to use tax loss carryforwards, then the company likely needs to reduce its DTAs to reflect the likelihood they may never be realized. Because of the subjective nature of estimating future profitability, adjustments to DTAs could be a source of potential earnings management.

Treatment for Analytical Purposes

If DTLs are expected to reverse in the future, an analyst should treat them as liabilities. If, however, an analyst does not expect them to reverse in the foreseeable future (typically because of expected continued growth in capital expenditures), the recommended adjustment is to treat them as equity (decrease the DTL and increase equity by the same amount). The key question is, “When or will the total DTL be reversed in the future?” In practice, the treatment of deferred taxes for analytical purposes varies. An analyst must decide on the appropriate treatment on a case-by-case basis.



1. If the tax base of an asset exceeds the asset's carrying value and a reversal is expected in the future, then:
 - A. a DTA is created.
 - B. a DTL is created.
 - C. neither a DTA nor a DTL is created.
2. Enigma KK uses straight-line depreciation in its accounts, while using accelerated depreciation for tax purposes. For both financial and tax reporting, the company assumes a five-year useful economic life and zero residual value. In the current year, accounting depreciation is ¥10,000,000 and tax allowable depreciation is ¥8,000,000. Which of the following statements relating to the impact of depreciation in the current year is *most accurate*?
 - A. The DTL will increase.
 - B. The tax expense will be greater than tax payable.
 - C. The carrying value of the asset is greater than the tax base.
3. A US GAAP reporting firm reports an increased valuation allowance at the end of the current period. What effect will this have on the firm's income tax expense in the current period?
 - A. Increase.
 - B. Decrease.
 - C. No effect.
4. An analyst is comparing a firm to its competitors. The firm has a DTL that results from accelerated depreciation for tax purposes. The firm is expected to continue increasing its purchases of PP&E in the foreseeable future. How should the liability be treated for analysis purposes?
 - A. It should be treated as equity at its full value.
 - B. It should be treated as a liability at its full value.
 - C. The present value should be treated as a liability, with the remainder being treated as equity.

MODULE 35.3: TAX RATES AND DISCLOSURES

LOS 35.c: Calculate, interpret, and contrast an issuer's effective tax rate, statutory tax rate, and cash tax rate.

Some firms' reported income tax expense differs from the amount that would result from multiplying its earnings before taxes by the statutory income tax rate. Recall that the statutory rate is the tax rate of the jurisdiction where the firm operates. Any differences generally result from the following:

- ▮ Different tax rates in different tax jurisdictions (countries)
- ▮ Permanent tax differences: tax credits, tax-exempt income, nondeductible expenses, and tax differences between capital gains and operating income

- Changes in tax rates and legislation
- Tax holidays in some countries

Typically, temporary timing differences do not cause the statutory and effective rates to differ.

Analysts should consider the following three tax rates:

Statutory tax rate: corporate income tax rate in which the company is domiciled

$$\text{Effective tax rate} = \frac{\text{income tax expense}}{\text{pretax income}}$$

$$\text{Cash tax rate} = \frac{\text{cash taxes paid}}{\text{pretax income}}$$

The footnote disclosure in the financial statements must reconcile the statutory tax rate to the effective tax rate. Understanding the differences between reported income tax expense and the amount based on the statutory income tax rate will enable the analyst to better forecast future earnings and cash flow.

When estimating future earnings and cash flows, the analyst should understand each element of the reconciliation, including its relative impact, how it has changed with time, and how it is likely to change in the future.

When analyzing trends in tax rates, it is important to only include reconciliation items that are continuous in nature rather than those that are sporadic. Items such as different tax rates in different countries, tax-exempt income, and nondeductible expenses tend to be continuous. Other items are almost always sporadic, such as large asset sales and tax holiday savings. Watch for special conditions such as termination dates for a tax holiday or a requirement to pay the accumulated taxes at some point in the future. An analyst should review the disclosures of each financial statement based on the footnotes and management discussion and analysis.

EXAMPLE: Analyzing a tax rate reconciliation

Novelty Distribution Company (NDC) does business in the United States and abroad. NDC's reconciliation between effective and statutory tax rates for three years is provided in the following table. Analyze the changes in effective tax rates over the three years shown.

Statutory US federal income tax rate reconciliation

	20X3	20X4	20X5
Statutory US federal income tax rate	35.0%	35.0%	35.0%
State income taxes, net of related federal income tax benefit	2.1%	2.2%	2.3%
Benefits and taxes related to foreign operations	(6.5%)	(6.3%)	(2.7%)
Tax rate changes	0.0%	0.0%	(2.0%)
Capital gains on sale of assets	0.0%	(3.0%)	0.0%
Special items	(1.6%)	8.7%	2.5%
Other, net	0.8%	0.7%	(1.4%)
Effective income tax rates	29.8%	37.3%	33.7%

Answer:

In some cases an analyst may want to convert the reconciliation from percentages to absolute numbers. However, for this example, the trends can be analyzed simply by using the percentages.

The effective tax rate is upward trending over the three-year period. Contributing to the upward trend is an increase in the state income tax rate and a decrease in benefits related to taxes on foreign income. In 20X4, a loss related to the sale of assets partially offset an increase in taxes created by special items. In 20X3 and 20X5, the special items and the other items also offset each other. The fact that the special items and other items are so volatile over the three-year period suggests that it will be difficult for an analyst to forecast the effective tax rate for NDC for the foreseeable future without additional information. This volatility also reduces comparability with other firms.

LOS 35.d: Analyze disclosures relating to deferred tax items and the effective tax rate reconciliation and explain how information included in these disclosures affects a company's financial statements and financial ratios.

Companies are required to disclose details on the sources of the temporary differences that cause DTAs and DTLs to be reported on the balance sheet. Common examples of temporary differences include the following:

- A DTL results from using *accelerated depreciation for tax purposes* and straight-line depreciation for the financial statements. An analyst should

consider the firm's growth rate and capital spending levels when determining whether the difference will actually reverse.

- | *Impairments* generally result in a DTA because the write-down is recognized immediately in the income statement, but the deduction on the tax return is generally not allowed until the asset is sold or disposed of.
- | *Restructuring* generates a DTA because the costs are recognized for financial reporting when the restructuring is announced, but they are not deducted for tax until actually paid. Restructuring usually results in significant cash outflows (net of the tax savings) in the years after the restructuring costs are reported.
- | In the United States, firms that use LIFO for their financial statements are required to use LIFO for tax purposes, so no temporary differences result. However, in countries where this is not a requirement, temporary differences can result from the *choice of inventory cost flow method*.
- | *Post-employment benefits* and *deferred compensation* are both recognized for financial reporting when earned by the employee, but they are not deducted for tax purposes until actually paid. These can result in a DTA that will be reversed when the benefits or compensation are paid.
- | A deferred tax adjustment is made to stockholders' equity to reflect the future tax impact of unrealized gains or losses on *available-for-sale marketable securities* that are taken directly to equity. No DTL is added to the balance sheet for the future tax liability when gains or losses are realized.

Typically, the following deferred tax information is disclosed:

- | DTLs, DTAs, any valuation allowance, and the net change in the valuation allowance over the period
- | Any unrecognized DTL for undistributed earnings of subsidiaries and joint ventures
- | Current-year tax effect of each type of temporary difference
- | Components of income tax expense
- | Reconciliation of reported income tax expense and the tax expense based on the statutory rate
- | Tax loss carryforwards and credits

EXAMPLE: Analyzing deferred tax item disclosures

WCCO, Inc.'s, income tax expense has consistently been larger than taxes payable over the last three years. WCCO disclosed in the footnotes to its 20X5 financial statements the major items recorded as DTAs and DTLs (in millions of dollars), as shown in the following table.

Deferred tax disclosures

	20X5	20X4	20X3
Employee benefits	\$278	\$310	\$290
International tax loss carryforwards	<u>101</u>	<u>93</u>	<u>115</u>
Subtotal	379	403	405
Valuation allowance	<u>(24)</u>	<u>(57)</u>	<u>(64)</u>
Deferred tax assets	355	346	341
Property, plant, and equipment	452	361	320
Unrealized gains on available-for-sale securities	<u>67</u>	<u>44</u>	<u>23</u>
Deferred tax liabilities	519	405	343
Deferred income taxes	\$164	\$59	\$2

Given this information, explain why income tax expense has exceeded taxes payable over the last three years. Also explain the effect of the change in the valuation allowance on WCCO's earnings for 20X5.

Answer:

The company's DTA balance results from international tax loss carryforwards and employee benefits (most likely, pension and other post-retirement benefits), offset by a valuation allowance. The company's DTL balance results from property, plant, and equipment (most likely from using accelerated depreciation methods for tax and straight-line on the financial statements) and unrealized gains on securities classified as available for sale (because the unrealized gain is not taxable until realized).

Income tax expense is equal to taxes payable plus deferred income tax expense (change in DTLs minus change in DTAs). Because DTLs have been growing faster than DTAs, deferred income tax expense has been positive, resulting in income tax expense being higher than taxes payable.

Management decreased the valuation allowance by \$33 million in 20X5. This resulted in a reduction in deferred income tax expense and an increase in reported earnings for 20X5. Decreasing the valuation allowance implies that management has increased its estimate of future taxable income against which it can use DTAs.



MODULE QUIZ 35.3

1. Which of the following rates is *most likely* to be useful for the analyst for forecasting future earnings?
 - A. The statutory tax rate.
 - B. The effective tax rate.
 - C. The cash tax rate.
2. Valkyrie AG is a German company that has a wholly owned subsidiary in Italy. Valkyrie earns pretax income of €100 million in each country. If the tax rate in Germany is 30% and the tax rate in Italy is 20%, what is the company's effective tax rate?
 - A. 20%.
 - B. 25%.
 - C. 30%.
3. Reactor SA is a French firm with significant global operations:

Earnings Before Tax	Year 1	Year 2	Year 3
	€m	€m	€m
French operations	200	150	210
Overseas operations	<u>50</u>	<u>75</u>	<u>90</u>
Total	250	225	300

Income Taxes (Income Statement)	Year 1	Year 2	Year 3
	€m	€m	€m
Current:			
French operations	56	30	59
Overseas operations	<u>12</u>	<u>20</u>	<u>21</u>
	68	50	80
Deferred:			
French operations	8	(4)	6
Overseas operations	<u>(1)</u>	<u>2</u>	<u>2</u>
	7	2	8
Total	75	48	88

The effective tax rate was highest in:

- A. Year 1.
 - B. Year 2.
 - C. Year 3.
4. Which of the following is *least likely* to explain why the statutory and effective tax rates differ?
 - A. The presence of permanent timing differences.
 - B. Overseas business operations.
 - C. Changes in deferred tax assets (DTAs) and deferred tax liabilities (DTLs).

KEY CONCEPTS

LOS 35.a

Deferred tax terminology is as follows:

- | *Taxable income*. Income subject to tax based on the tax return.
- | *Accounting profit*. Pretax income from the income statement based on financial accounting standards.
- | *Deferred tax assets (DTAs)*. Balance sheet asset value that results when taxes payable (tax return) are greater than income tax expense (income statement) and the difference is expected to reverse in future periods.
- | *Deferred tax liabilities (DTLs)*. Balance sheet liability value that results when income tax expense (income statement) is greater than taxes payable (tax return) and the difference is expected to reverse in future periods.
- | *Valuation allowance*. Reduction of DTAs (contra account) based on the likelihood that the future tax benefits will not be realized.
- | *Taxes payable*. The tax liability from the tax return. Note that this term also refers to a liability that appears on the balance sheet for taxes due but not yet paid.
- | *Income tax expense*. Expense recognized in the income statement that includes taxes payable and changes in DTAs and DTLs.

If taxable income is less than pretax income and the cause of the difference is expected to reverse in future years, a DTL is created. If taxable income is greater than pretax income and the difference is expected to reverse in future years, a DTA is created.

The balance of a DTA or DTL is equal to the difference between the tax base and the carrying value of the asset or liability, multiplied by the tax rate.

Income tax expense and taxes payable are related through the change in the DTA and the change in the DTL:

$$\text{income tax expense} = \text{taxes payable} + \Delta \text{DTL} - \Delta \text{DTA}$$

A permanent difference is a difference between taxable income and pretax income that will not reverse in the future. Permanent differences do not create DTAs or DTLs.

When a firm's enacted tax rate increases (decreases), DTAs and DTLs are both increased (decreased) to reflect the new rate. Changes in these values will also affect income tax expense.

LOS 35.b

A DTL is created when income tax expense (income statement) is higher than taxes payable (tax return). Deferred tax liabilities occur when revenues (or gains) are recognized in the income statement before they are taxable on the tax return, or expenses (or losses) are tax deductible before they are recognized in the income statement.

A DTA is created when taxes payable (tax return) are higher than income tax expense (income statement). DTAs are recorded when revenues (or gains) are taxable before they are recognized in the income statement, when expenses (or losses) are recognized in the income statement before they are tax deductible, or when tax loss carryforwards are available to reduce future taxable income.

The tax base for a depreciable fixed asset is the amount of tax-allowable depreciation that will be deducted in future tax returns.

The tax base of a liability is equal to the liability's carrying value less amounts that will be included in future taxable income. With liabilities relating to income received in advance, the tax base is equal to the carrying value of the liability less any amounts that will not be included in future taxable income.

Asset carrying value > tax base	Deferred tax liability
Asset carrying value < tax base	Deferred tax asset
Liability carrying value > tax base	Deferred tax asset
Liability carrying value < tax base	Deferred tax liability

If it is more likely than not that some or all of a DTA will not be realized (because of insufficient future taxable income to recover the tax asset), then the gross DTA must be reduced by a valuation allowance under US GAAP. Under IFRS a firm reports a smaller DTA if future recoverability is uncertain but does not report a valuation allowance.

DTLs that are not expected to reverse, typically because of expected continued growth in capital expenditures, should be treated for analytical purposes as equity. If DTLs are expected to reverse, they should be treated for analytical purposes as liabilities.

LOS 35.c

Firms are required to reconcile their effective income tax rate with the applicable statutory rate in the country where the business is domiciled. Analyzing trends in individual reconciliation items can aid in understanding past earnings trends and in predicting future effective tax rates. Where adequate data are provided, they can also be helpful in predicting future earnings and cash flows, or for adjusting financial ratios.

Tax rates are as follows:

Statutory tax rate: corporate income tax rate in which the company is domiciled

$$\text{Effective tax rate} = \frac{\text{income tax expense}}{\text{pretax income}}$$

$$\text{Cash tax rate} = \frac{\text{cash taxes paid}}{\text{pretax income}}$$

LOS 35.d

Typically, the following deferred tax information is disclosed in the footnotes:

- | DTLs, DTAs, any valuation allowance, and the net change in the valuation allowance over the period
- | Any unrecognized DTL for undistributed earnings of subsidiaries and joint ventures
- | Current-year tax effect of each type of temporary difference
- | Components of income tax expense
- | Reconciliation of reported income tax expense and the tax expense based on the statutory rate
- | Tax loss carryforwards and credits

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 35.1

- . C Pretax income and income tax expense are not always linked because of temporary and permanent differences. (LOS 35.a)
- . C Timing differences that do not reverse, referred to as permanent differences, do not result in the creation of DTAs or DTLs. If the timing difference was expected to reverse, a DTL would be created because carrying value > tax base. (LOS 35.a)

- . **A** Expenses that are tax deductible before they are recognized in the income statement result in deferred tax liabilities. For example, accelerated depreciation in the tax returns versus straight-line depreciation in the accounts results in more depreciation being recognized in the tax returns, in early periods, and results in carrying value exceeding the tax base. If revenues or gains are taxable before they are recognized in the income statement, DTAs are created. Unearned revenue is a good example of this. Tax loss carryforwards result from negative taxable income. Rather than triggering the tax authorities to pay tax to the firm, they are deferred and can be offset against future taxable income. As a result, the firm will pay less tax in the future, and a DTA is created. (LOS 35.a)
- . **B** The doubtful debt provision reduces the balance sheet carrying value, but not the tax base of receivables. If the tax base of the asset is greater than the carrying value, this results in a DTA. (LOS 35.a)
- . **C** The income statement tax expense = tax payable + Δ DTL – Δ DTA.

Increases in DTLs increase the tax expense, and increases in DTAs reduce the tax expense. The opposite is true of decreases.

Change in DTA: $\$300,000 - \$350,000 = -\$50,000$

Change in DTL: $\$500,000 - \$400,000 = \$100,000$

Tax expense = $\$500,000 + \$100,000 - (-\$50,000) = \$650,000$

(LOS 35.a)

- . **B** An increase in the tax rate will increase any DTAs and DTLs. (LOS 35.a)

Module Quiz 35.2

- . **A** If the tax base of an asset exceeds the carrying value, a DTA is created. Taxable income will be lower in the future when the reversal occurs. (LOS 35.b)
- . **C** The carrying value is higher than the tax base of the asset due to the use of accelerated depreciation in the tax returns versus straight-line in the accounts. As this year's tax allowable depreciation is now lower than this year's accounting depreciation, the DTL will begin to decline. Decreases in the DTL reduce the tax expense. The difference between carrying value and tax base will decrease as the timing difference reverses until, at the end of the asset's life, they both equal zero. (LOS 35.b)

- . **A** Recognizing a greater valuation allowance reduces the net value of a DTA, which increases income tax expense in the current period. (LOS 35.b)
- . **A** The DTL is not expected to reverse in the foreseeable future because the firm is expected to continue to increase its investment in depreciable assets, and accelerated depreciation for tax on the newly acquired assets delays the reversal of the DTL. The liability should be treated as equity at its full value. (LOS 35.b)

Module Quiz 35.3

- . **B** The effective rate is most useful for forecasting future earnings; however, it may need to be adjusted to remove the impact of transitory components. The statutory rate is the tax rate where the company is domiciled for reporting purposes. The cash tax rate is most applicable when forecasting future tax payments and cash flows. (LOS 35.c)

- . **B** Tax in Germany = €100 million × 30% = €30 million

$$\text{Tax in Italy} = \text{€100 million} \times 20\% = \text{€20 million}$$

$$\text{Effective tax rate} = \frac{30 + 20}{100 + 100} = \frac{50}{200} = 25\%$$

(LOS 35.c)

- . **A** The effective tax rate is measured as the tax expense divided by earnings before tax.

$$\text{Year 1} = \frac{\text{€75 million}}{\text{€250 million}} = 0.3 = 30\%$$

$$\text{Year 2} = \frac{\text{€48 million}}{\text{€225 million}} = 0.213 = 21.3\%$$

$$\text{Year 3} = \frac{\text{€88 million}}{\text{€300 million}} = 0.293 = 29.3\%$$

(LOS 35.c)

- . **C** Changes in DTAs and DTLs do not cause statutory and effective rates to differ. Permanent timing differences do cause the rates to differ, as these items affect either taxable income or pretax earnings, but not both. Overseas business operations result in tax being paid to overseas tax authorities, which are unlikely to have the same statutory rate as the firm's domicile country. (LOS 35.c)

READING 36

FINANCIAL REPORTING QUALITY

MODULE 36.1: REPORTING QUALITY

LOS 36.a: Compare financial reporting quality with the quality of reported results (including quality of earnings, cash flow, and balance sheet items).

Financial reporting quality refers to the characteristics of a firm's financial statements. The primary criterion for judging financial reporting quality is adherence to generally accepted accounting principles (GAAP) in the jurisdiction in which the firm operates. However, given that GAAP provide choices of methods, estimates, and specific treatment of many items, compliance with GAAP by itself does not necessarily result in financial reporting of the highest quality.

High quality financial reporting must be *decision useful*. Two characteristics of decision-useful financial reporting are *relevance* and *faithful representation*. Relevance refers to the fact that information presented in the financial statements is useful to users of financial statements in making decisions. Relevant information must also be *material* in that knowledge of it would likely affect the decisions of users of financial statements. Faithful representation encompasses the qualities of *completeness*, *neutrality*, and the absence of errors. We develop the concept of neutrality of financial reports later in this reading.

The **quality of earnings** is, in many respects, a separate issue. The quality of reported earnings (not the quality of earnings reports) can be judged based on the sustainability of the earnings as well as on their level. Sustainability can be evaluated by determining the proportion of reported earnings that can be expected to continue in the future. Increases in reported earnings resulting from changes in exchange rates or by sales of assets that have appreciated over many periods are not typically sustainable, whereas higher profits from increased efficiency or increasing market share would generally be considered sustainable.

One dollar of high-quality earnings is expected to add more value to a company than one dollar of low-quality earnings, based on the criterion of sustainability. The higher probability that high-quality earnings will continue in future periods increases their impact on the value of the firm, calculated as the present value of expected future earnings. At the other end of the sustainability spectrum, a one-time gain of a dollar from favorable currency exchange rate movements is not likely to be repeated and, therefore, has a smaller impact on estimates of a company's value.

The importance of the level of earnings is that reported earnings must be high enough to sustain the company's operations and existence over time, as well as high enough to provide an adequate return to the company's investors. Both of these concerns are important in determining the quality of a company's reported earnings. Sustainability of reported cash flows is also a consideration in determining the quality of reported earnings, as are the value of items reported on the balance sheet. Inadequate accruals for probable liabilities and overstatement of asset values can both decrease the quality of reported earnings and bring sustainability into question.

From our discussion here, we can see that it is quite possible that a firm has high financial reporting quality but a low quality of reported earnings. Reported earnings may be GAAP-compliant and relevant, represent the company's economic activities faithfully, and be decision useful as a result, but have low sustainability or be low enough in amount that the provision of adequate investor returns or the sustainability of the company itself are called into question.

LOS 36.b: Describe a spectrum for assessing financial reporting quality.

Combining both financial reporting quality and the quality of reported earnings, we can categorize the quality of financial reports along a spectrum from best to worst. At the high-quality end of the spectrum, we have financial reports that are compliant with GAAP, decision useful, and report earnings that are sustainable and represent an adequate return on invested capital. At the opposite end of the spectrum are financial reports that are essentially fictitious (fraudulent). When reporting quality is that low, the quality of the reported earnings themselves is impossible to assess. We can identify several levels of quality between these two extremes.

Here is one possible categorization of the quality levels of financial reports, from best to worst:

Reporting is compliant with GAAP and decision useful; earnings are sustainable and adequate.

Reporting is compliant with GAAP and decision useful, but earnings quality is low (earnings are not sustainable or not adequate).

Reporting is compliant with GAAP, but earnings quality is low and reporting choices and estimates are biased.

Reporting is compliant with GAAP, but the amount of earnings is actively managed to increase, decrease, or smooth reported earnings.

Reporting is not compliant with GAAP, although the numbers presented are based on the company's actual economic activities.

Reporting is not compliant and includes numbers that are essentially fictitious or fraudulent.

LOS 36.c: Explain the difference between conservative and aggressive accounting.

Ideally, financial statements should be neutral or unbiased in order to offer the most value to analysts. In general, we describe the choices made within GAAP with respect to reported earnings as **conservative accounting** if they tend to decrease the company's reported earnings and financial position (on the balance sheet) for the current period. We describe choices that increase reported earnings or improve the financial position for the current period as **aggressive accounting**.

Aggressive accounting often results in decreased earnings in future periods, while conservative accounting will tend to increase future period earnings. Both these types of bias are sometimes used by management, for different periods, in an attempt to smooth earnings over time because greater earnings volatility tends to reduce the value of a company's shares. Often **earnings smoothing** is accomplished through adjustment of accrued liabilities that are based on management estimates. During periods of higher-than-expected earnings, management may employ a conservative bias by adjusting an accrued liability upward to reduce reported earnings for that period. This effectively allows deferral of the recognition of these earnings to a future period for which earnings

are less than expected. In such a future period, the accrued liability is adjusted downward to increase reported earnings in that period, perhaps to meet market expectations. Deferral of reported earnings through conservative bias in financial reporting so they can be used opportunistically in a future period is sometimes referred to as putting earnings in the “cookie jar” (presumably to be “enjoyed” later).

Some examples of conservative versus aggressive financial reporting based on management choices and estimates are shown in Figure 36.1.

Figure 36.1: Aggressive and Conservative Accounting

Aggressive	Conservative
Capitalizing current period costs	Expensing current period costs
Longer estimates of the lives of depreciable assets	Shorter estimates of the lives of depreciable assets
Higher estimates of salvage values	Lower estimates of salvage values
Straight-line depreciation	Accelerated depreciation
Delayed recognition of impairments	Early recognition of impairments
Less accrual of reserves for bad debt	More accrual of reserves for bad debt
Smaller valuation allowances on deferred tax assets	Larger valuation allowances on deferred tax assets

Bias can also be present in the way that financial results are presented. A company may present transparent financial statements that help analysts and investors to understand the results and the activities that led to them. Alternatively, a company may provide minimal disclosure in an attempt to emphasize positive developments and obscure information about negative developments.

We should avoid thinking about conservatism in financial reporting as “good” and aggressive reporting as “bad.” Conservative bias can also be considered as a deviation from neutral reporting or faithful representation that reduces the usefulness of financial statements to analysts and investors.

Sometimes GAAP themselves can introduce conservatism by imposing a higher standard of verification for revenue and profit than for expenses and accrual of liabilities. For example:

- Research costs are typically expensed in the period incurred because of the uncertainty about the future benefits to be provided from research activities,

while the associated revenue is not recognized until some future period.

- | Accruals for legal liabilities are recorded when a future payment becomes “probable,” while the standard for recognizing increasing accrued asset value is stricter.
- | Under US GAAP, write-downs of inventory values are required when their future value is likely impaired, but increases in inventory value may not be recorded until the inventory is actually sold.

While conservative bias in financial reporting is not ideal for users of financial statements, it may be beneficial in reducing the probability of future litigation from users claiming they were misled, in reducing current period tax liability (when deductions for tax must also be deducted in the financial statements), and in protecting the interests of those who have less complete information than company management, such as buyers of the company’s debt.

LOS 36.d: Describe motivations that might cause management to issue financial reports that are not high quality and conditions that are conducive to issuing low-quality, or even fraudulent, financial reports.

One important motivation for aggressive accounting choices is to meet or exceed a benchmark number for earnings per share. Specifically, managers may be motivated to report earnings that are greater than:

- | Earnings guidance offered earlier by management.
- | Consensus analyst expectations.
- | Those of the same period in the prior year.

The manager’s motivation here may be career oriented, seeking to enhance her reputation and improve future career opportunities. Because beating certain benchmarks is important to subsequent stock price movements, managers may be motivated by incentive compensation (bonuses) that depends on stock returns. Other possible motivations are to gain credibility with equity market investors or improve the way the company is viewed by its customers and suppliers.

For companies that are highly leveraged and unprofitable, aggressive accounting may be motivated by a desire to avoid violating debt covenants.

When earnings exceed benchmark levels, managers may make conservative accounting choices in ways that allow these earnings to be shown in future

periods, increasing the probability that future period earnings will meet or exceed the relevant benchmark amount.

Three factors that typically exist in cases where management provides low-quality financial reporting are motivation, opportunity, and a rationalization of the behavior. So to the sources of *motivation* previously listed, we can add conditions that increase the *opportunity* to present low-quality financial reports. Circumstances in which low-quality, or even fraudulent, financial reporting is more probable are:

- | The company has weak internal controls.
- | The board of directors provides inadequate oversight.
- | Applicable accounting standards provide a large range of acceptable accounting treatments, provide for inconsequential penalties in the case of accounting fraud, or both.

The third likely element of low-quality financial reporting is *rationalization* by management for less-than-ethical actions. Most people who do something they know is wrong tell themselves a story that seems (at least to them) to justify breaking the rules. Whether the story is “I’ll fix it next period” or “I have to do it to get my bonus and pay for my parents’ care,” the resulting behavior is the same.

LOS 36.e: Describe mechanisms that discipline financial reporting quality and the potential limitations of those mechanisms.

Each country has its own regulatory body responsible for publicly traded securities and the markets in which they trade. For example, in the United States, the regulatory body is the Securities and Exchange Commission (SEC). In the UK, it is the Financial Conduct Authority (FCA). The International Organization of Securities Commissions (IOSCO) coordinates securities regulation on an international basis with over 200 members, such as national securities regulators, stock exchanges, and regional authorities. One such regional authority, the European Securities and Markets Authority (ESMA), coordinates policy among the securities regulators of countries in the European Union.

Securities regulations typically require:

- | A registration process for the issuance of new publicly traded securities.

- | Specific disclosure and reporting requirements, including periodic financial statements and accompanying notes.
- | An independent audit of financial reports.
- | A statement of financial condition (or management commentary) made by management.
- | A signed statement by the person responsible for the preparation of the financial reports.
- | A review process for newly registered securities and periodic reviews after registration.

Enforcement actions by securities regulators may include fines, suspension of participation in issuance and trading of securities, and public disclosure of the results of disciplinary proceedings. Regulators may also pursue criminal prosecution of fraudulent or otherwise illegal activities.

In addition to the audit opinion, a requirement for securities that trade in the United States is that management must include an assessment of the effectiveness of the firm's internal controls.

Note that an unqualified or "clean" audit opinion is not a guarantee that no fraud has occurred but only offers reasonable assurance that the financial reports (prepared under the direction of management) have been "fairly reported" with respect to the applicable GAAP. The auditor is selected and paid by the firm being audited.

Another source of discipline on financial reporting quality is private contracts, such as those with lenders. Such contracts will often specify how financial measures referenced in the loan covenants will be calculated. The counterparties to private contracts with the firm have an incentive to see that the firm produces high-quality financial reports.

LOS 36.f: Describe presentation choices, including non-GAAP measures, that could be used to influence an analyst's opinion.

Firms will sometimes report accounting measures that are not defined or required under GAAP. Such **non-GAAP measures** typically exclude some items in order to make the firm's performance look better than it would using measures defined and required by GAAP. The claim is often made that certain items are excluded because they are one-time or nonoperating costs that will not affect

operating earnings going forward, because the items are non-cash charges, or to “improve comparability with companies that use different accounting methods” for depreciation or restructuring charges.

In the United States, companies that report non-GAAP measures in their financial statements are required to:

- | Display the most comparable GAAP measure with equal prominence.
- | Provide an explanation by management as to why the non-GAAP measure is thought to be useful.
- | Reconcile the differences between the non-GAAP measure and the most comparable GAAP measure.
- | Disclose other purposes for which the firm uses the non-GAAP measure.
- | Include, in any non-GAAP measure, any items that are likely to recur in the future, even those treated as nonrecurring, unusual, or infrequent in the financial statements.

IFRS require that firms using non-IFRS measures in financial reports must:

- | Define and explain the relevance of such non-IFRS measures.
- | Reconcile the differences between the non-IFRS measure and the most comparable IFRS measure.

Overall, the supposition is that firms use non-GAAP measures in an attempt to control the metrics on which they are evaluated and to reduce the focus of analysts and investors on GAAP measures.



MODULE QUIZ 36.1

1. A firm reports net income of \$40 million. The firm’s financial statements disclose in Management’s Discussion and Analysis that \$30 million of net income is attributable to a gain on the sale of assets. Based only on this information, for this period, the firm is *best* described as having high quality of:
 - A. financial reporting only.
 - B. both earnings and financial reporting.
 - C. neither earnings nor financial reporting.
2. Which of the following financial reports are considered to be of the lowest quality? Financial reports that reflect:
 - A. unsustainable earnings.
 - B. biased accounting choices.
 - C. departures from accounting principles.
3. Financial reporting is *most likely* to be decision useful when management’s accounting choices are:

- A. neutral.
 - B. aggressive.
 - C. conservative.
4. Which of the following is *least likely* to be a motivation to overreport earnings?
- A. Reduce tax obligations.
 - B. Meet analyst expectations.
 - C. Remain in compliance with bond covenants.
5. With respect to conditions that may lead to low-quality financial reporting, ineffective internal controls are *best* described as a(n):
- A. motivation.
 - B. opportunity.
 - C. rationalization.
6. A limitation on the effectiveness of auditing in ensuring financial reporting quality is that:
- A. detecting fraud is not the objective of audits.
 - B. public firms are not required to obtain audit opinions.
 - C. auditors may only issue a qualified or unqualified opinion but do not explain why.
7. Under IFRS, a firm that presents a nonstandard financial measure is *least likely* required to:
- A. provide the same measure for at least two prior periods.
 - B. explain the reasons for presenting the nonstandard measure.
 - C. reconcile the nonstandard measure to a comparable standard measure.

MODULE 36.2: ACCOUNTING CHOICES AND ESTIMATES

LOS 36.g: Describe accounting methods (choices and estimates) that could be used to manage earnings, cash flow, and balance sheet items.

Revenue Recognition

One example of how a firm's choices affect the timing of revenue recognition is the choice of where in the shipping process the customer actually takes title to the goods. A firm may choose terms with their customer of **free-on-board (FOB)** at the shipping point (the firm's loading dock) or FOB at the destination (the customer's location). Choosing terms of FOB at the shipping point will mean that revenue is recognized earlier compared to FOB at the destination.

Firms can also manage the timing of revenue recognition by accelerating or delaying the shipment of goods. If additional revenue is required to meet targets, firms can offer discounts or special financing terms to increase orders in the

current period, or ship goods to distributors without receiving an order. Overloading a distribution channel with more goods than would normally be sold during a period is referred to as **channel stuffing**. In periods where high earnings are expected, management may wish to delay recognition of revenue to the next period and hold or delay customer shipments to achieve this.

In a **bill-and-hold transaction**, the customer buys the goods and receives an invoice but requests that the firm keep the goods at their location for a period of time. The use of fictitious bill-and-hold transactions can increase earnings in the current period by recognizing revenue for goods that are actually still in inventory. Revenue for future periods will be decreased as real customer orders for these bill-and-hold items are filled but not recognized in revenue, offsetting the previous overstatement of revenue.

Estimates of Credit Losses

One example of accounting choices that affect financial reports is the estimation of losses from uncollectable customer credit accounts. On the balance sheet, the reserve for uncollectible debt is an offset to accounts receivable. If management determines the probability that accounts receivable will be uncollectible is lower than their current estimate, a decrease in the reserve for uncollectible accounts will increase net receivables reported on the balance sheet, reduce expenses on the income statement, and increase net income. An increase in the allowance for bad debt would have the opposite effect, decreasing net receivables on the balance sheet, increasing expenses, and decreasing net income.

A firm that simply underestimates the percentage of receivables that will be uncollectible will report higher receivables and higher net income as a result. At some point, when actual uncollectible accounts exceed the low estimate, the firm will report an additional expense that will reduce net income and net receivables.

Management can adjust the bad-debt reserve in order to smooth earnings. In periods of high earnings, the allowance for bad debt is increased to reduce reported earnings, in effect storing these earnings for later use. In subsequent periods, if earnings are below benchmark values, the bad-debt reserve can be reduced to meet earnings targets.

Other reserves recorded by a company, such as a reserve for warranty expense, can also be changed to manage reported earnings. A decrease in the estimated warranty expense as a percentage of sales will increase earnings, while an

increase in the reserve for warranty expense will decrease earnings for the period.

Valuation Allowance

Another example of a contra account that can be used to manage earnings is a valuation allowance. Recall that a valuation allowance reduces the carrying value of a deferred tax asset based on the probability it will not be realized. Similar to the effects of an allowance for bad debt, increasing a valuation allowance will decrease the net deferred tax asset on the balance sheet and reduce net income for the period, while a decrease in the valuation allowance will increase the net deferred tax asset and increase net income for the period.

As with the contra account for bad debt, the valuation allowance can be understated to show higher asset values and adjusted over time to smooth earnings.

Depreciation Methods and Estimates

Compared to straight-line depreciation, using an accelerated method of depreciation increases expenses, and decreases net income, in the early years of an asset's life. In the later years of an asset's life, expenses are lower and net income higher when an accelerated depreciation method is used. The carrying value of a depreciable asset on the balance sheet will decrease more rapidly with accelerated depreciation than with straight-line depreciation.

Estimates of the useful life of a depreciable asset and its salvage value upon disposal can also affect net income and the carrying value of the asset. A greater salvage value will slow depreciation so the carrying value of the asset is greater, depreciation expense is less, and net income is higher. A smaller salvage value will have the opposite effects. If the salvage value of an asset is set higher than the actual sale price at the end of the asset's life, a loss on the sale of the asset will decrease net income in the period in which the asset is sold.

Using a longer estimated useful life of a depreciable asset decreases the periodic depreciation expense and increases net income in the early years of the asset's life compared to using a shorter estimated useful life.

Amortization and Impairment

Management choices and estimates regarding amortization of purchased intangible assets are similar to those for depreciation of tangible assets. The intangible asset *goodwill* is not amortized but is subject to a test for impairment. By ignoring or delaying recognition of an impairment charge for goodwill, management can increase earnings in the current period.

Inventory Method

The choice of inventory cost flow methods can have significant effects on both reported earnings and the balance sheet value of inventory. Consider the choice between FIFO and weighted-average inventory costing methods. During periods of rising prices, COGS under the FIFO method will be less than COGS under the weighted-average costing method. Gross profit, gross margin, and earnings will all be greater under the FIFO method than under the weighted-average method as a result. Balance sheet inventory value will be greater under FIFO than under the weighted-average method.

During periods of decreasing prices, the opposite is true; FIFO COGS are greater than weighted-average COGS and FIFO gross profits, gross margin, and earnings less than under the weighted-average method. With decreasing prices, balance sheet inventory will be less under FIFO than under the weighted-average cost method.

In terms of relevance, in an environment of either increasing or decreasing prices, FIFO results in more accurate balance sheet inventory values because inventory value is closer to current replacement cost than under the weighted average cost method. Conversely, COGS are closer to current (replacement) cost under the weighted-average cost method so that gross profit and margin better reflect economic reality. Gross profit under FIFO is distorted in that it includes gains from rising prices (or losses from decreasing prices), so the weighted-average cost method produces “better” information on the income statement. Financial reports that are transparent and provide users with the information needed to understand how the choice of inventory costing method affects income statement and balance sheet values are considered to be higher quality.

Related-Party Transactions

If a public firm does business with a supplier that is private and controlled by management, adjusting the price of goods supplied can shift profits either to or

from the private company to manage the earnings reported by the public company.

Capitalization

Any expense that can be capitalized creates an asset on the balance sheet, and the impact of the expense on net income can be spread over many years.

Consider a firm that has a marketing expense of \$1.5 million and chooses to capitalize this expense and amortize it over three years. In the period in which the expense is incurred, capitalization will reduce the expense on the income statement from \$1.5 million to \$0.5 million, increasing pretax income by \$1 million. At the end of the year, the related balance sheet asset is \$1 million, and an amortization expense of \$0.5 million will be taken (and reduce net income) in each of the following two years. Greater capitalization of research and development costs will shift net income into the current period in the same way.

Capitalization also affects cash flow classifications. If an expense is capitalized, the entire amount is classified as an investing cash outflow so that operating cash flow is increased by that amount.

Other Cash Flow Effects

Management can affect the classification of cash flows through other methods, primarily with the goal of increasing reported cash flow from operations. Taking longer to pay suppliers increases operating cash flows and is referred to as **stretching payables**. Delaying payments that would normally be made near the end of a reporting period until the beginning of the next accounting period will increase operating cash flow in the current period and reduce it in the subsequent period. There is no effect on reported earnings in the current period from stretching payables.

Capitalizing interest expense will decrease cash flow from investing and increase cash flow from operations, along with its effects on the pattern of earnings from depreciating the interest expense over time rather than expensing it all in the current period. More generally, the ability under IFRS to classify interest and dividends paid as either CFO or CFF, and interest and dividends received as either CFO or CFI, gives management an additional way to manage reported operating cash flow.

MODULE 36.3: WARNING SIGNS

LOS 36.h: Describe accounting warning signs and methods for detecting manipulation of information in financial reports.

Following is a list of several warning signs that analysts should look for. The presence of these issues does not indicate fraud or even earnings manipulation, but in each case, the presence of one or more warning signs requires more analysis in order to determine whether there is a real business reason for the item or if earnings manipulation or fraud is driving the decisions and results. Avoiding investment in the company is one alternative when analysts and investors cannot obtain satisfactory answers to the questions raised when multiple warning signs are present.

Revenue Recognition

- | Changes in revenue recognition methods.
- | Use of bill-and-hold transactions.
- | Use of barter transactions.
- | Use of rebate programs that require estimation of the impact of rebates on net revenue.
- | Lack of transparency with regard to how the various components of a customer order are recorded as revenue.
- | Revenue growth out of line with peer companies.
- | Receivables turnover is decreasing over multiple periods.
- | Decreases in total asset turnover, especially when a company is growing through acquisition of other companies.
- | Inclusion of nonoperating items or significant one-time sales in revenue.

Inventories

- | Declining inventory turnover ratio.
- | LIFO liquidations—drawing down inventory levels when LIFO (US GAAP only) inventory costing is used so that COGS reflects the lower costs of items acquired in past periods, which increases current period earnings.

Capitalization Policies

- | Firm capitalizes costs that are not typically capitalized by firms in their industry.

Relationship of Revenue and Cash Flow

- | The ratio of operating cash flow to net income is persistently less than one or declining over time.

Other Warning Signs

- | Depreciation methods, estimated asset lives, or estimates of salvage values are out of line with those of peer companies in the industry.
- | Fourth-quarter earnings show a pattern (either high or low) compared to the seasonality of earnings in the industry or seasonality of revenue for the firm.
- | The firm has significant transactions with related parties (entities controlled by management).
- | Certain expenses are classified as nonrecurring but appear regularly in financial reports.
- | Gross or operating profit margins are noticeably higher than are typical for the industry and peer companies.
- | Management typically provides only minimal financial reporting information and disclosure.
- | Management typically emphasizes non-GAAP earnings measures and uses special or nonrecurring designations aggressively for charges.
- | Growth by purchasing a large number of businesses can provide many opportunities to manipulate asset values and future depreciation and amortization and make comparisons to prior period earnings problematic.

Analysts should consider adjusting prior-period earnings when large restructuring or impairment charges are recognized. Analysts sometimes take such events to be good news because they anticipate better firm performance going forward when poorly performing assets are disposed of. Because the charges represent, to some extent, “corrections” of previously understated expenses and overstated asset values, analysts should consider spreading these costs across prior periods and restating prior earnings to give a more realistic picture of true earnings trends.



1. For the current period, inappropriate capitalization is *most likely* to:
 - A. overstate revenues.
 - B. understate liabilities.
 - C. understate expenses.
2. A potential warning sign that a firm is engaging in channel stuffing is an unusual increase in the firm's:
 - A. receivables turnover.
 - B. days of sales outstanding.
 - C. number of days of payables.

KEY CONCEPTS

LOS 36.a

Financial reporting quality refers to the characteristics of a firm's financial statements. High-quality financial reporting adheres to generally accepted accounting principles (GAAP) and is decision useful in terms of relevance and faithful representation.

Quality of reported results refers to the level and sustainability of a firm's earnings, cash flows, and balance sheet items. High-quality earnings are high enough to provide the firm's investors with an adequate return and are sustainable in future periods.

LOS 36.b

A spectrum for assessing financial reporting quality considers both the quality of a firm's financial statements and the quality of its earnings. One such spectrum, from highest quality to lowest, is the following:

- | Reporting is compliant with GAAP and decision useful; earnings are sustainable and adequate.
- | Reporting is compliant and decision useful, but earnings quality is low.
- | Reporting is compliant, but earnings quality is low and reporting choices and estimates are biased.
- | Reporting is compliant, but earnings are actively managed.
- | Reporting is not compliant, but the numbers presented are based on the company's actual economic activities.
- | Reporting is not compliant and includes numbers that are fictitious or fraudulent.

LOS 36.c

Biased accounting choices that can be made within GAAP include conservative and aggressive accounting. Conservative accounting choices tend to decrease the company's reported earnings and financial position for the current period.

Aggressive accounting choices tend to increase reported earnings or improve the financial position for the current period.

Some managers employ conservative bias during periods when earnings are above target and aggressive bias during poor periods of below-target earnings to artificially smooth earnings.

LOS 36.d

Motivations for firm managers to issue low-quality financial reports may include pressure to meet or exceed earnings targets, career considerations, increasing their compensation, improving perceptions of the firm among customers and suppliers, or meeting the terms of debt covenants.

Conditions that are often present when managers issue low-quality financial reports include motivations, opportunities, and rationalizations. Weak internal controls, inadequate oversight by the board of directors, and wide ranges of acceptable accounting treatments are among the factors that may provide opportunities for low-quality reporting.

LOS 36.e

Mechanisms that help to discipline financial reporting quality include regulation, auditing, and private contracts. Regulators typically require public companies to provide periodic financial statements and notes, including management commentary, and obtain independent audits.

A clean audit opinion offers reasonable assurance that financial statements are free from material errors but does not guarantee the absence of error or fraud. The fact that firms select and pay their auditors may limit the effectiveness of auditing to discipline financial reporting quality.

LOS 36.f

Firms may attempt to influence analysts' valuations by presenting non-GAAP measures, such as earnings that exclude certain nonrecurring items. IFRS requires firms to define and explain the relevance of any non-GAAP measures and reconcile them to the most comparable IFRS measure. Similar requirements apply to US public firms.

LOS 36.g

Accounting choices and estimates that can be used to manage earnings include:

- | Revenue recognition choices such as shipping terms (FOB shipping point versus FOB destination), accelerating shipments (channel stuffing), and bill-and-hold transactions.
- | Estimates of reserves for uncollectible accounts or warranty expenses.
- | Valuation allowances on deferred tax assets.
- | Depreciation methods, estimates of useful lives and salvage values, and recognition of impairments.
- | Inventory cost flow methods.
- | Capitalization of expenses.
- | Related-party transactions.

LOS 36.h

Accounting warning signs that indicate a need for closer analysis may include:

- | Revenue growth out of line with comparable firms, changes in revenue recognition methods, or lack of transparency about revenue recognition.
- | Decreases over time in turnover ratios (receivables, inventory, total asset).
- | Bill-and-hold, barter, or related-party transactions.
- | Net income not supported by operating cash flows.
- | Capitalization decisions, depreciation methods, useful lives, salvage values out of line with comparable firms.
- | Fourth-quarter earnings patterns not caused by seasonality.
- | Frequent appearance of nonrecurring items.
- | Emphasis on non-GAAP measures, minimal information and disclosure in financial reports.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 36.1

- . A Because a large proportion of net income is due to a one-time gain, this period's earnings are likely not sustainable and the firm may be said to have

low quality of earnings for the period. Clear disclosure of this fact in the financial statements suggests high quality of financial reporting. (LOS 36.a)

- . **C** In the spectrum of financial reporting quality, financial reports that depart from generally accepted accounting principles are considered to be of lower quality than those that reflect biased accounting choices. Financial reports that reflect unsustainable earnings, such as one-time gains, can still be of high quality if they state the situation clearly. (LOS 36.b)
- . **A** Financial reporting is most likely to be decision useful when accounting choices are neutral. Either aggressive or conservative accounting choices by management may be viewed as biases. (LOS 36.c)
- . **A** Reducing tax obligations would be a reason to *underreport* earnings. The other choices are motivations to overreport earnings. (LOS 36.d)
- . **B** Ineffective internal controls are a condition that provides an opportunity for low-quality financial reporting. (LOS 36.d)
- . **A** The objective of audits is to provide reasonable assurance that financial statements are presented fairly. A firm that is engaging in accounting fraud may deceive its auditor. Regulators in most countries require publicly traded firms to obtain independent audits of their financial statements. Auditors may issue a qualified opinion noting certain aspects of financial statements that are inconsistent with accounting principles or an adverse opinion if they find that financial statements are materially misstated and do not conform with GAAP. (LOS 36.e)
- . **A** IFRS require a firm that presents a nonstandard financial measure to reconcile that measure to an IFRS measure and explain why the firm believes the nonstandard measure is relevant to users of the financial statements. Presenting the nonstandard measure for prior periods is not a requirement. (LOS 36.f)

Module Quiz 36.2, 36.3

- . **C** Management may make inappropriate capitalization decisions to understate expenses by creating balance sheet assets for items that should instead be recognized as expenses in the current period, increasing net income in the current period. Revenues and liabilities are unlikely to be affected by capitalization decisions. (Module 36.2, LOS 36.g)

- . **B** Channel stuffing, which includes activities such as accelerating deliveries to distributors or sending customers unordered merchandise, would likely increase accounts receivable as a percentage of revenues. This would decrease the receivables turnover ratio and increase days of sales outstanding. Payables would not be affected. (Module 36.3, LOS 36.h)

READING 37

FINANCIAL ANALYSIS TECHNIQUES

INTRODUCTION

This reading presents a “tool box” for an analyst. It would be nice if you can calculate all these ratios, but it is imperative that you understand what firm characteristic each one is measuring—and even more important that you know whether a higher or lower ratio is better in each instance.

Different analysts calculate some ratios differently. It would be helpful if analysts were always careful to distinguish between total liabilities, total interest-bearing debt, long-term debt, and creditor and trade debt, but they do not. Some analysts routinely add deferred tax liabilities to debt or exclude goodwill when calculating assets and equity; others do not. Statistical reporting services almost always disclose how each of the ratios they present was calculated. So do not get too tied up in the details of each ratio, but understand what each one represents and what factors would likely lead to significant changes in a particular ratio.

The DuPont formulas have been with us for a long time and were in the curriculum way back in the 1980s. Decomposing ROE into its components is an important analytic technique, and it should definitely be in your tool box.

MODULE 37.1: INTRODUCTION TO FINANCIAL RATIOS

LOS 37.a: Describe tools and techniques used in financial analysis, including their uses and limitations.

In the first FSA reading, we were introduced to the six steps of the financial statement analysis framework. We now return to look in more detail at Step 3, which is to “Make any appropriate adjustments to the financial statements.

Calculate ratios and perform statistical analysis. Prepare exhibits such as graphs and common-size balance sheets.”

The journey we have taken through the accounting principles, with the focus on areas subject to choice, estimation, and subjectivity, enables us to adjust the financial statements in line with the objectives of our analysis. We can now move on to the calculation of ratios.

Analysts use various tools and techniques to convert financial statement data into formats that facilitate analysis. These include ratio analysis, common-size analysis, graphical analysis, and regression analysis.

Ratio Analysis

Ratios are useful tools for expressing relationships among data that can be used for internal comparisons and comparisons across firms. They are often most useful in identifying questions that need to be answered, rather than answering questions directly. Specifically, ratios can be used to do the following:

- | Project future earnings and cash flow.
- | Evaluate a firm’s flexibility (its ability to grow and meet obligations even when unexpected circumstances arise).
- | Assess management’s performance.
- | Evaluate changes in the firm and industry over time.
- | Compare the firm with industry competitors.

Analysts must also be aware of the limitations of ratios, including the following:

- | Financial ratios are not useful when viewed in isolation. They are only informative when compared to those of other firms, or to the company’s historical performance.
- | Comparisons with other companies are made more difficult by different accounting treatments. This is particularly important when comparing US firms to non-US firms.
- | It is difficult to find comparable industry ratios when analyzing companies that operate in multiple industries.
- | Conclusions cannot be made by calculating a single ratio. All ratios must be viewed relative to one another.

- | Determining the target or comparison value for a ratio is difficult, requiring some range of acceptable values.

Ratios must be evaluated in the context of prior period results, the company's stated strategy, analyst expectations, the current position in the business cycle, and against other firms in the industry.

Definitions of ratios can vary widely among the analytical community. For example, some analysts use all liabilities when measuring leverage, while other analysts only use interest-bearing obligations. The important thing is for each individual analyst to use ratios consistently. Analysts must also understand that reasonable values of ratios can differ among industries.

Common-Size Analysis

Common-size statements normalize balance sheets and income statements and allow the analyst to more easily compare performance across firms, and for a single firm over time:

- | A **vertical common-size balance sheet** expresses all balance sheet accounts as a percentage of total assets.

$$\text{vertical common-size balance sheet ratios} = \frac{\text{balance sheet account}}{\text{total assets}}$$

- | A **vertical common-size income statement** expresses all income statement items as a percentage of sales.

$$\text{vertical common-size income statement ratios} = \frac{\text{income statement account}}{\text{sales}}$$

In addition to comparisons of financial data across firms and time, common-size analysis is appropriate for quickly viewing certain financial ratios. For example, the gross profit margin, operating profit margin, and net profit margin are all clearly indicated within a common-size income statement. Vertical common-size income statement ratios are especially useful for studying trends in costs and profit margins.

The sample common-size statements in Figure 37.1 show balance sheet items as percentages of assets, and income statement items as percentages of sales.

Figure 37.1: Vertical Common-Size Balance Sheet and Income Statement

Balance Sheet, Fiscal Year-End	20X6	20X5	20X4
Assets			
Cash & cash equivalents	0.38%	0.29%	0.37%
Accounts receivable	5.46%	5.61%	6.20%
Inventories	5.92%	5.42%	5.84%
Deferred income taxes	0.89%	0.84%	0.97%
Other current assets	0.41%	0.40%	0.36%
Total current assets	13.06%	12.56%	13.74%
Gross fixed assets	25.31%	23.79%	25.05%
Accumulated depreciation	8.57%	7.46%	6.98%
Net gross fixed assets	16.74%	16.32%	18.06%
Other long-term assets	70.20%	71.12%	68.20%
Total assets	100.00%	100.00%	100.00%
Liabilities			
Accounts payable	3.40%	3.40%	3.79%
Short-term debt	1.00%	2.19%	1.65%
Other current liabilities	8.16%	10.32%	9.14%
Total current liabilities	12.56%	15.91%	14.58%
Long-term debt	18.24%	14.58%	5.18%
Balance Sheet, Fiscal Year-End	20X6	20X5	20X4
Other long-term liabilities	23.96%	27.44%	53.27%
Total liabilities	54.76%	57.92%	73.02%
Preferred equity	0.00%	0.00%	0.00%
Common equity	45.24%	42.08%	26.98%
Total liabilities & equity	100.00%	100.00%	100.00%
Income Statement, fiscal year	20X6	20X5	20X4
Revenues	100.00%	100.00%	100.00%
Cost of goods sold	59.62%	60.09%	60.90%
Gross profit	40.38%	39.91%	39.10%
Selling, general & administrative	16.82%	17.34%	17.84%
Depreciation	2.39%	2.33%	2.18%
Amortization	0.02%	3.29%	2.33%
Other operating expenses	0.58%	0.25%	−0.75%
Operating income	20.57%	16.71%	17.50%
Interest and other debt expense	2.85%	4.92%	2.60%
Income before taxes	17.72%	11.79%	14.90%
Provision for income taxes	6.30%	5.35%	6.17%
Net income	11.42%	6.44%	8.73%

Even a cursory inspection of the income statement in Figure 37.1 can be quite instructive. Beginning at the bottom, we can see that the profitability of the company has increased nicely in 20X6 after falling slightly in 20X5. We can examine the 20X6 income statement values to find the source of this greatly improved profitability. The cost of goods sold seems to be stable, with an improvement (decrease) in 20X6 of only 0.48%. SG&A was down approximately 0.5% as well.

These improvements from (relative) cost reduction, however, only begin to explain the 5% increase in the net profit margin for 20X6. Improvements in two items, amortization and interest and other debt expense, appear to be the most significant factors in the firm's improved profitability in 20X6. Clearly, the analyst must investigate further in both areas to learn whether these improvements represent permanent improvements, or whether these items can be expected to return to previous percentage-of-sales levels in the future.

We can also note that interest expense as a percentage of sales was approximately the same in 20X4 and 20X6. We must investigate the reasons for the higher interest costs in 20X5 to determine whether the current level of 2.85% can be expected to continue into the next period. In addition, more than 3% of the 5% increase in net profit margin in 20X6 is due to a decrease in amortization expense. Because this is a noncash expense, the decrease may have no implications for cash flows looking forward.

This discussion should make clear that common-size analysis doesn't tell an analyst the whole story about this company, but it can certainly point the analyst in the right direction to find out the circumstances that led to the increase in the net profit margin—and to determine the effects, if any, on firm cash flow going forward.

Another way to present financial statement data that is quite useful when analyzing trends over time is a **horizontal common-size balance sheet or income statement**. The divisor here is the first-year values, so they are all standardized to 1.0 by construction. Figure 37.2 illustrates this approach.

Figure 37.2: Horizontal Common-Size Balance Sheet Data

	20X4	20X5	20X6
Inventory	1.0	1.1	1.4
Cash and marketable securities	1.0	1.3	1.2
Long-term debt	1.0	1.6	1.8
PP&E (net of depreciation)	1.0	0.9	0.8

Trends in the values of these items, as well as the relative growth in these items, are readily apparent from a horizontal common-size balance sheet.



PROFESSOR'S NOTE

We have presented data in Figure 37.1 with information for the most recent period on the left, and in Figure 37.2, we have presented the historical values from left to right. Both presentation methods are common, and on the exam, you should pay special attention to which method is used in the data presented for any question.

We can view the values in the common-size financial statements as ratios. Net income is shown on the common-size income statement as net income or revenues, which is the net profit margin, and it tells the analyst the percentage of each dollar of sales that remains for shareholders after all expenses related to the generation of those sales are deducted. One measure of financial leverage, long-term debt to total assets, can be read directly from the vertical common-size financial statements. Specific ratios commonly used in financial analysis and the interpretation of their values are covered in detail in this reading.

Graphical Analysis

Graphs can be used to visually present performance comparisons and composition of financial statement elements over time.

A **stacked column graph** (also called a stacked bar graph) shows the changes in items from year to year in graphical form. Figure 37.3 presents such data for a hypothetical corporation.

Another alternative for graphic presentation of data is a **line graph**. same data as Figure 37.3, but as a line graph. The increase in trade payables and the decrease in cash are evident in either format and would alert the analyst to potential liquidity problems that require further investigation and analysis.

Figure 37.3: Stacked Column (Stacked Bar) Graph

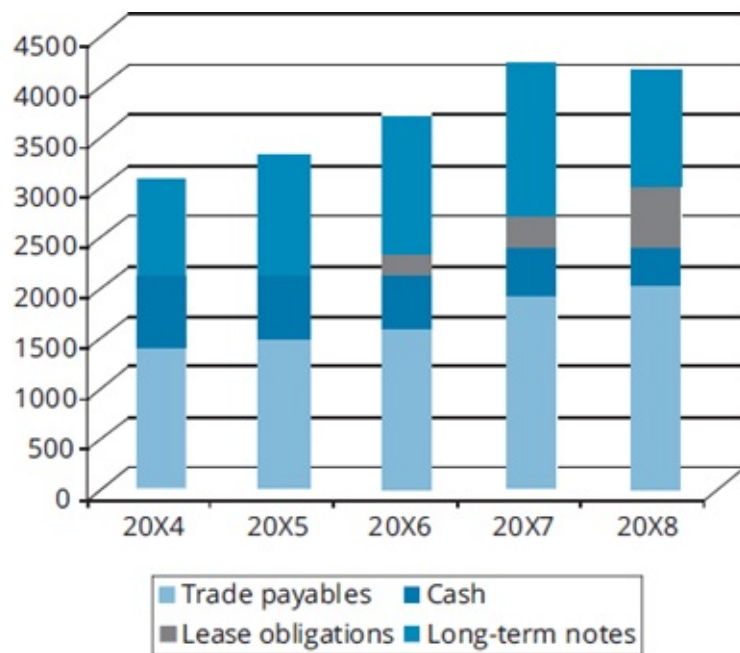
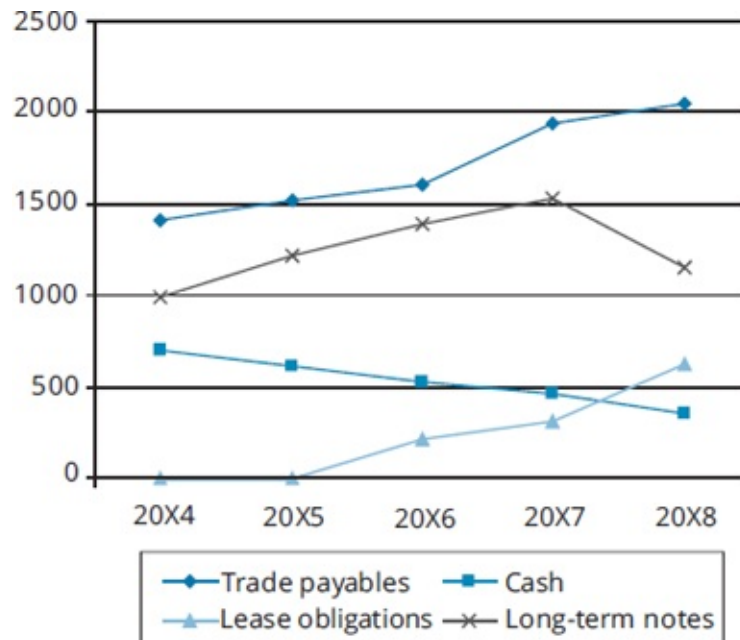


Figure 37.4: Line Graph



Regression Analysis

Regression analysis can be used to identify relationships between variables. The results are often used for forecasting. For example, an analyst might use the relationship between GDP and sales to prepare a sales forecast.

MODULE 37.2: FINANCIAL RATIOS, PART 1

LOS 37.b: Calculate and interpret activity, liquidity, solvency, and profitability ratios.

Financial ratios can be segregated into different classifications by the type of information about the company they provide. Here are such classifications:

- | **Activity ratios.** This category includes several ratios also referred to as asset utilization or turnover ratios (e.g., inventory turnover, receivables turnover, and total assets turnover). They often give indications of how well a firm uses various assets, such as inventory and fixed assets.
- | **Liquidity ratios.** Liquidity, here, refers to the ability to pay short-term obligations as they come due.
- | **Solvency ratios.** Solvency ratios give the analyst information on the firm's financial leverage and ability to meet its longer-term obligations.
- | **Profitability ratios.** Profitability ratios provide information on how well the company generates operating profits and net profits from its sales.

It should be noted that these categories are not mutually exclusive. For example, an activity ratio, such as payables turnover, may also provide information about the liquidity of a company. There is no one standard set of ratios for financial analysis. Different analysts use different ratios and different calculation methods for similar ratios. Some ratios are so commonly used that there is very little variation in how they are defined and calculated. We will note some alternative treatments and alternative terms for single ratios as we detail the commonly used ratios in each category.

Activity Ratios

Activity ratios (also known as asset utilization ratios or operating efficiency ratios) measure how efficiently the firm is managing its assets.

- | A measure of accounts receivable turnover is *receivables turnover*:

$$\text{receivables turnover} = \frac{\text{annual sales}}{\text{average receivables}}$$



PROFESSOR'S NOTE

In most cases, when a ratio compares a balance sheet account (such as receivables) with an income or cash flow item (such as sales), the balance sheet item will be the average of the account instead of simply the end-of-year balance. Averages are calculated by adding the beginning-of-year account value to the end-of-year account value, then dividing the sum by two. For companies operating in seasonal industries, averages of beginning and ending values may not give an accurate view of the average balance over the period. In these cases, an analyst would want to include more data points, throughout the year, when computing averages.

A high receivables turnover ratio could be the result of doing an excellent job of managing credit terms and collections. On the other hand, it might indicate that a company has stringent credit terms, offers a large discount for early payment, or charges high penalties for late payment. A company that has excessively stringent credit terms will lose sales as a result. Insight into why a company has a high receivables turnover rate can be gained from looking at the company's revenue growth compared to peers. Slower growth could indicate that credit terms may be too stringent, while a high receivables turnover—together with revenue growth at or above the peer group average—could indicate superior credit terms and collections management.

- | The inverse of the receivables turnover times 365 is the *average collection period*, or *days of sales outstanding*, which is the average number of days it takes for the company's customers to pay their bills:

$$\text{days of sales outstanding} = \frac{365}{\text{receivables turnover}}$$

- | A measure of a firm's efficiency with respect to its processing and inventory management is *inventory turnover*:

$$\text{inventory turnover} = \frac{\text{cost of goods sold}}{\text{average inventory}}$$



PROFESSOR'S NOTE

Pay careful attention to the numerator in the turnover ratios. For inventory turnover, be sure to use cost of goods sold, not sales.

Inventory turnover that is high may indicate effective management of inventory, but it could also result from holding inventory levels too low so that sales are lost when orders cannot be filled immediately. A low inventory

turnover ratio relative to peers could indicate that some inventory is obsolete and slow-selling. In either case, examining revenue growth relative to peers can provide more insight into whether inventory is well managed or poorly managed.

- | The inverse of the inventory turnover times 365 is the *average inventory processing period*, *number of days of inventory*, or *days of inventory on hand*:

$$\text{days of inventory on hand} = \frac{365}{\text{inventory turnover}}$$

- | A measure of the use of trade credit by the firm is the *payables turnover* ratio:

$$\text{payables turnover} = \frac{\text{cost of goods sold}}{\text{average trade payables}}$$



PROFESSOR'S NOTE

Many analysts calculate payables turnover using purchases in the numerator, rather than COGS. Purchases are not typically a financial statement line item, but we can use the inventory equation to calculate them from the financial statements: Purchases = ending inventory – beginning inventory + cost of goods sold.

- | The inverse of the payables turnover ratio multiplied by 365 is the *payables payment period* or *number of days of payables*, which is the average amount of time it takes the company to pay its bills:

$$\text{number of days of payables} = \frac{365}{\text{payables turnover ratio}}$$



PROFESSOR'S NOTE

We have shown the days calculations for payables, receivables, and inventory based on annual turnover and a 365-day year. If turnover ratios are for a quarter rather than a year, the number of days in the quarter should be divided by the quarterly turnover ratios to get the “days” form of these ratios.

A high payables turnover ratio relative to peers may indicate that a company is not fully taking advantage of supplier credit terms, or that the company is paying suppliers early to take advantage of discounts. A payables turnover rate that is low relative to that of peer companies may indicate that a company is having problems with short-term cash flows—or, alternatively, that a company is simply taking advantage of lenient terms negotiated with

suppliers. As with inventory turnover, examining other ratios (in this case, liquidity ratios) can provide insight into which interpretation of a relatively high or low payables turnover ratio is more likely.

The effectiveness of the firm's use of its total assets to create revenue is measured by its *total asset turnover*:

$$\text{total asset turnover} = \frac{\text{revenue}}{\text{average total assets}}$$

Different types of industries might have considerably different turnover ratios. Manufacturing businesses that are capital intensive might have asset turnover ratios near one, while retail businesses might have turnover ratios near 10. As was the case with the current asset turnover ratios discussed previously, it is desirable for the total asset turnover ratio to be close to the industry norm. Low asset turnover ratios might mean that the company has too much capital tied up in its asset base. A turnover ratio that is too high might imply that the firm has too few assets for potential sales, or that the asset base is outdated.

The use of fixed assets is measured by the *fixed asset turnover* ratio:

$$\text{fixed asset turnover} = \frac{\text{revenue}}{\text{average net fixed assets}}$$

Low fixed asset turnover might mean that the company has too much capital tied up in its asset base or is using the assets it has inefficiently. A turnover ratio that is too high might imply that the firm has obsolete equipment, or at a minimum, that the firm will probably have to incur capital expenditures in the near future to increase capacity to support growing revenues. Because *net* here refers to net of accumulated depreciation, firms with more recently acquired assets will typically have lower fixed asset turnover ratios.

How effectively a company is using its working capital is measured by the *working capital turnover* ratio:

$$\text{working capital turnover} = \frac{\text{revenue}}{\text{average working capital}}$$

Working capital (sometimes called net working capital) is current assets minus current liabilities. The working capital turnover ratio gives us information about the use of working capital in terms of dollars of sales per dollar of working capital. Some firms may have very low working capital if the outstanding payables equal or exceed inventory and receivables. In this

case, the working capital turnover ratio will be very large, may vary significantly from period to period, and is less informative about changes in the firm's operating efficiency.

Liquidity Ratios

Liquidity ratios are employed by analysts to determine the firm's ability to pay its short-term liabilities.

- | The *current ratio* is the best-known measure of liquidity:

$$\text{current ratio} = \frac{\text{current assets}}{\text{current liabilities}}$$

The higher the current ratio, the more likely it is that the company will be able to pay its short-term bills. A current ratio of less than one means that the company has negative working capital and is probably facing a liquidity crisis. Working capital equals current assets minus current liabilities.

- | The *quick ratio* is a more stringent measure of liquidity because it does not include inventories and other assets that might not be very liquid:

$$\text{quick ratio} = \frac{\text{cash} + \text{marketable securities} + \text{receivables}}{\text{current liabilities}}$$

The higher the quick ratio, the more likely it is that the company will be able to pay its short-term bills. Marketable securities are short-term debt instruments, typically liquid and of good credit quality.

- | The most conservative liquidity measure is the *cash ratio*:

$$\text{cash ratio} = \frac{\text{cash} + \text{marketable securities}}{\text{current liabilities}}$$

The higher the cash ratio, the more likely that the company will be able to pay its short-term bills. The current, quick, and cash ratios differ only in the assumed liquidity of the current assets that the analyst projects will be used to pay off current liabilities.

- | The *defensive interval ratio* is another measure of liquidity that indicates the number of days of average cash expenditures the firm could pay with its current liquid assets:

$$\text{defensive interval} = \frac{\text{cash} + \text{marketable securities} + \text{receivables}}{\text{average daily expenditures}}$$

Expenditures, here, include cash expenses for costs of goods, SG&A, and research and development. If these items are taken from the income statement, noncash charges such as depreciation should be added back just as in the preparation of a statement of cash flows by the indirect method.

The *cash conversion cycle (net operating cycle)* is the length of time it takes to turn the firm's cash investment in inventory back into cash, in the form of collections from the sales of that inventory. It measures the time between paying suppliers and receiving cash from customers. The cash conversion cycle is computed from days' sales outstanding, days of inventory on hand, and number of days of payables:

$$\text{cash conversion cycle} = \text{days' sales outstanding} + \text{days of inventory on hand} - \text{number of days of payables}$$

High cash conversion cycles are considered undesirable. A conversion cycle that is too high implies that the company has an excessive amount of capital investment in the sales process. If cash is paid to suppliers before cash is received from customers, the company will require short-term financing to cover the period. For some companies, cash might be received from customers before suppliers are paid, which results in a cash conversion cycle that can be used to finance other parts of the business. Companies in this situation can support a current ratio that is less than one.

MODULE 37.3: FINANCIAL RATIOS, PART 2

Solvency Ratios

Solvency ratios measure a firm's financial leverage and ability to meet its long-term obligations. Solvency ratios include various **debt ratios** that are based on the balance sheet and **coverage ratios** that are based on the income statement.

- | A measure of the firm's use of fixed-cost financing sources is the *debt-to-equity* ratio:

$$\text{debt-to-equity} = \frac{\text{total debt}}{\text{total shareholders' equity}}$$

Increases and decreases in this ratio suggest a greater or lesser reliance on debt as a source of financing.

Total debt is calculated differently by different analysts and different providers of financial information. Here, we will define it as interest-bearing long-term and short-term debt.

Some analysts include non-interest-bearing current liabilities, such as trade payables. Others may deduct cash, cash equivalents, and marketable securities from debt to compute net debt.



PROFESSOR'S NOTE

For the exam, all interest-bearing liabilities should be included in the total debt figure unless otherwise specified. Somewhat surprisingly, the Level I CFA curriculum tells us to exclude leases from total debt calculations, despite the fact that leases are interest bearing. No explanation for excluding them is given. For the exam you should follow this approach when calculating total debt, but be aware that this treatment would result in misleadingly low leverage ratios for industries where using high volumes of leased assets is common, such as airlines.

- | Another way of looking at the usage of debt is the *debt-to-capital* ratio:

$$\text{debt-to-capital} = \frac{\text{total debt}}{\text{total debt} + \text{total shareholders' equity}}$$

Capital equals all short-term and long-term debt plus preferred stock and equity. The ratio shows the proportion of debt in the capital structure of the firm. Increases and decreases in this ratio suggest a greater or lesser reliance on debt as a source of financing.

- | A slightly different way of analyzing debt use is the *debt-to-assets* ratio:

$$\text{debt-to-assets} = \frac{\text{total debt}}{\text{total assets}}$$

Increases and decreases in this ratio suggest a greater or lesser reliance on debt as a source of financing.

- | Another measure that is used as an indicator of a company's use of debt financing is the *financial leverage* ratio (or leverage ratio):

$$\text{financial leverage} = \frac{\text{average total assets}}{\text{average total equity}}$$

Average, here, means the average of the values at the beginning and at the end of the period. A ratio close to 1 indicates that equity is being used to finance assets. If the ratio increases above 1, it is an indication that debt is being used. Greater use of debt financing increases financial leverage and, typically, risk to equity holders and bondholders alike.

- | The remaining risk ratios help determine the firm's ability to repay its debt obligations. The first of these is the *interest coverage ratio*:

$$\text{interest coverage} = \frac{\text{earnings before interest and taxes}}{\text{interest payments}}$$

The lower this ratio, the more likely it is that the firm will have difficulty meeting its debt payments. Because depreciation and amortization are not cash expenses, another ratio that reflects a firm's ability to meet its debt obligations is the debt-to-EBITDA ratio. The ratio indicates how long it would take to repay current total debt from an approximation of operating cash flow:

$$\text{debt-to-EBITDA} = \frac{\text{total debt}}{\text{EBITDA}}$$

- | Another indicator of a company's ability to meet its obligations is the *fixed charge coverage* ratio:

$$\text{fixed charge coverage} = \frac{\text{earnings before interest and taxes} + \text{lease payments}}{\text{interest payments} + \text{lease payments}}$$

Here, lease payments are added back to operating earnings in the numerator. They are also added to interest payments in the denominator. Significant lease obligations will reduce this ratio significantly compared to the interest coverage ratio. Fixed charge coverage is the more meaningful measure for companies that lease a large portion of their assets, such as some airlines. The measure is also useful for US GAAP-based companies with operating leases as currently, unlike finance leases, no interest is recorded in the income statement.



PROFESSOR'S NOTE

With all solvency ratios, the analyst must consider the variability of a firm's cash flows when determining the reasonableness of the ratios. Firms with stable cash flows are usually able to carry more debt.

Profitability Ratios

Profitability ratios measure the overall performance of the firm relative to revenues, assets, equity, and capital.

- | The *net profit margin* is the ratio of net income to revenue:

$$\text{net profit margin} = \frac{\text{net income}}{\text{revenue}}$$

Analysts should be concerned if this ratio is too low. The net profit margin should be based on net income from continuing operations, because analysts should be primarily concerned about future expectations. Below-the-line items, such as discontinued operations, will not affect the company in the future.

Operating profitability ratios look at how good management is at turning their efforts into profits. Operating ratios compare the top of the income statement (sales) to profits. The different ratios are designed to isolate specific costs.

- | The *gross profit margin* is the ratio of gross profit (sales less cost of goods sold) to sales:

$$\text{gross profit margin} = \frac{\text{gross profit}}{\text{revenue}}$$

An analyst should be concerned if this ratio is too low. Gross profit can be increased by raising prices or reducing costs. However, the ability to raise

prices may be limited by competition.

The *operating profit margin* is the ratio of operating profit (gross profit less selling, general, and administrative expenses) to sales. Operating profit is often approximated as earnings before interest and taxes (EBIT):

$$\text{operating profit margin} = \frac{\text{operating income}}{\text{revenue}} \text{ or } \frac{\text{EBIT}}{\text{revenue}}$$

Strictly speaking, EBIT includes some nonoperating items, such as gains on investment. The analyst, as with other ratios with various formulations, must be consistent in his calculation method and know how published ratios are calculated. Analysts should be concerned if this ratio is too low. Some analysts prefer to calculate the operating profit margin by adding back depreciation and any amortization expense to arrive at earnings before interest, taxes, depreciation, and amortization (EBITDA).

Sometimes, profitability is measured using earnings before tax (EBT), which can be calculated by subtracting interest from EBIT or from operating earnings. The *pretax margin* is calculated as follows:

$$\text{pretax margin} = \frac{\text{EBT}}{\text{revenue}}$$

Another set of profitability ratios measures profitability relative to funds invested in the company by common stockholders, preferred stockholders, and suppliers of debt financing. The first of these measures is the return on assets (ROA). Typically, ROA is calculated using net income:

$$\text{ROA} = \frac{\text{net income}}{\text{average total assets}}$$

This measure is a bit misleading, however, because interest is excluded from net income, but total assets include debt as well as equity. Adding interest adjusted for tax back to net income puts the returns to both equity holders and debtholders in the numerator. The interest expense that should be added back is gross interest expense, not net interest expense (which is gross interest expense less interest income). This results in an alternative calculation for ROA:

$$\text{ROA} = \frac{\text{net income} + \text{interest expense} (1 - \text{tax rate})}{\text{average total assets}}$$

A measure of ROA that includes both taxes and interest in the numerator is the *operating ROA*:

$$\text{operating ROA} = \frac{\text{operating income}}{\text{average total assets}} \text{ or } \frac{\text{EBIT}}{\text{average total assets}}$$

Return on invested capital (ROIC), which we described in the Corporate Finance topic area, is the ratio of after-tax operating profit to average long-term capital:

$$\text{return on invested capital} = \frac{\text{after-tax operating profit}}{\text{average long-term capital}}$$

Long-term capital includes long-term debt, preferred equity, and common equity, but excludes working capital. Analysts should be concerned if this ratio is too low.

The *return on equity (ROE)*, sometimes called *return on total equity*, is the ratio of net income to average total equity (including preferred stock):

$$\text{ROE} = \frac{\text{net income}}{\text{average total equity}}$$

Analysts should be concerned if this ratio is too low. Return on equity is often more thoroughly analyzed using the DuPont decomposition, which we describe later in this reading.

A similar ratio to the ROE is the *return on common equity*:

$$\begin{aligned} \text{return on common equity} &= \frac{\text{net income} - \text{preferred dividends}}{\text{average common equity}} \\ &= \frac{\text{net income available to common}}{\text{average common equity}} \end{aligned}$$

This ratio differs from the return on total equity in that it only measures the accounting profits available to, and the capital invested by, common stockholders, instead of common and preferred stockholders. That is why preferred dividends are deducted from net income in the numerator. Analysts should be concerned if this ratio is too low.

EXAMPLE: Calculating ratios

A balance sheet and income statement for Sedgwick Company are shown in the following tables for this year and the previous year.

Using the company information provided, calculate the following ratios for the current year: current ratio, total asset turnover, net profit margin, return on common equity, and total debt to equity.

Sedgwick Company Balance Sheet

	Current Year	Previous Year
Assets		
Cash and marketable securities	\$105	\$95
Receivables	205	195
Inventories	310	290
Total current assets	620	580
Gross property, plant, and equipment	1,800	1,700
Accumulated depreciation	360	340
Net property, plant, and equipment	1,440	1,360
Total assets	\$2,060	\$1,940
Liabilities		
Payables	\$110	\$90
Short-term debt	160	140
Current portion of long-term debt	55	45
Current liabilities	325	275
Long-term debt	610	690
Deferred taxes	105	95
Common stock at par	300	300
Additional paid in capital	400	400
Retained earnings	320	180
Common shareholders' equity	1,020	880
Total liabilities and equity	\$2,060	\$1,940

Sedgwick Company Income Statement

	Current Year
Sales	\$4,000
Cost of goods sold	<u>3,000</u>
Gross profit	1,000
Operating expenses	<u>650</u>
Operating profit	350
Interest expense	<u>50</u>
Earnings before taxes	300
Taxes	<u>100</u>
Net income	<u>200</u>
Common dividends	<u>60</u>

Answer:

- $\text{current ratio} = \frac{\text{current assets}}{\text{current liabilities}}$
 $\text{current ratio} = \frac{620}{325} = 1.9$
- $\text{total asset turnover} = \frac{\text{revenue}}{\text{average assets}}$
 $\text{total asset turnover} = \frac{4,000}{(2,060 + 1,940) / 2} = 2.0$
- $\text{net profit margin} = \frac{\text{net income}}{\text{revenue}}$
 $\text{net profit margin} = \frac{200}{4,000} = 5.0\%$
- $\text{return on common equity} = \frac{\text{net income} - \text{preferred dividends}}{\text{average common equity}}$
 $\text{return on common equity} = \frac{200}{(1,020 + 880) / 2} = 21.1\%$
- $\text{debt-to-equity ratio} = \frac{\text{total debt}}{\text{total equity}}$
 $\text{debt-to-equity ratio} = \frac{610 + 160 + 55}{1,020} = 80.9\%$

Note that preferred equity would be included in the denominator if there were any, and that we have included short-term debt and the current portion of long-term debt in calculating total (interest-bearing) debt.

LOS 37.c: Describe relationships among ratios and evaluate a company using ratio analysis.

EXAMPLE: Relationships among ratios

An analyst calculates the following activity and liquidity ratios for a company over the last three years:

	20X8	20X7	20X6
Current ratio	2.0	1.5	1.2
Quick ratio	0.5	0.8	1.0
Days of inventory	60	50	30
Days' sales outstanding	20	30	40

Determine what the analyst should infer from these ratios taken together.

Answer:

The current ratio has increased over this period, while the quick ratio has decreased. This could result from inventories increasing or from current assets other than inventories decreasing. The increase in days of inventory on hand

suggests increasing inventories explain the opposing trends in the current and quick ratios.

The decrease in days' sales outstanding indicates that the company has been collecting cash from customers sooner than it had been in the past. Taken together, these ratios suggest the company may be accelerating its collections to make up for a drain on cash from poor inventory management.

EXAMPLE: Using ratios to evaluate a company

An analyst is comparing the values of several ratios for the current year to their prior-year values, the current industry average values, and industry average ratios for Sedgwick Company. These selected ratio values are shown in the following table.

	Current Year	Previous Year	Industry Average
Current ratio	1.9	2.1	1.5
Total asset turnover	2.0	2.3	2.4
Net profit margin	5.0%	5.8%	6.5%
Return on common equity	21.1%	24.1%	19.8%
Debt-to-equity	80.9%	99.4%	35.7%

Discuss how these ratios compare with the company's performance last year and with the industry performance.

Answer:

Although the firm's liquidity (as measured by the current ratio) has decreased over the past year, it remains above the industry average.

Total asset turnover has declined over the past year and now appears to be significantly lower than the industry average.

Net profit margin is lower than last year and is much lower than the industry average.

ROE is lower than last year but still higher than the industry average. Given the decline in net profit margin and total asset turnover, it is likely that higher-than-average financial leverage is the reason for this ROE outperformance.

Our supposition about the firm's financial leverage is confirmed by its debt-to-equity ratios. While the current year's ratio is lower than last year's, it is still more than twice the industry average ratio. The significant decrease in the

company's debt-to-equity ratio over the past year suggests the company is trying to get its debt level more in line with the industry average.



MODULE QUIZ 37.1, 37.2, 37.3

1. To study trends in a firm's cost of goods sold (COGS), the analyst should standardize the COGS numbers to a common-sized basis by dividing COGS by:
 - A. assets.
 - B. sales.
 - C. net income.
2. Which of the following is *least likely* a limitation of financial ratios?
 - A. Data on comparable firms are difficult to acquire.
 - B. Determining the target or comparison value for a ratio requires judgment.
 - C. Different accounting treatments require the analyst to adjust the data before comparing ratios.
3. RGB, Inc.'s purchases during the year were \$100,000. The balance sheet shows an average accounts payable balance of \$12,000. RGB's payables payment period is *closest* to:
 - A. 37 days.
 - B. 44 days.
 - C. 52 days.
4. RGB, Inc., has a gross profit of \$45,000 on sales of \$150,000. The balance sheet shows average total assets of \$75,000 with an average inventory balance of \$15,000. RGB's inventory turnover and total asset turnover are *closest* to:

<u>Inventory turnover</u>	<u>Total asset turnover</u>
A. 2.00 times	7.00 times
B. 7.00 times	2.00 times
C. 0.33 times	0.50 times
5. If RGB, Inc., has annual sales of \$100,000, average accounts payable of \$30,000, and average accounts receivable of \$25,000, RGB's receivables turnover and average collection period are *closest* to:

<u>Receivables turnover</u>	<u>Average collection period</u>
A. 2.1 times	174 days
B. 3.3 times	111 days
C. 4.0 times	91 days
6. A company's current ratio is 1.9. If some of the accounts payable are paid off from the cash account, then the:
 - A. numerator would decrease by a greater percentage than the denominator, resulting in a lower current ratio.
 - B. denominator would decrease by a greater percentage than the numerator, resulting in a higher current ratio.
 - C. numerator and denominator would decrease proportionally, leaving the current ratio unchanged.

7. A company's quick ratio is 1.2. If inventory were purchased for cash, then the:
- A. numerator would decrease more than the denominator, resulting in a lower quick ratio.
 - B. denominator would decrease more than the numerator, resulting in a higher quick ratio.
 - C. numerator and denominator would decrease proportionally, leaving the quick ratio unchanged.
8. All other things held constant, which of the following transactions will increase a firm's current ratio if the ratio is greater than 1?
- A. Accounts receivable are collected, and the funds received are deposited in the firm's cash account.
 - B. Fixed assets are purchased from the cash account.
 - C. Accounts payable are paid with funds from the cash account.
9. RGB, Inc.'s receivable turnover is 10 times, the inventory turnover is 5 times, and the payables turnover is 9 times. RGB's cash conversion cycle is *closest* to:
- A. 69 days.
 - B. 104 days.
 - C. 150 days.
10. An analyst who is interested in a company's long-term solvency would *most likely* examine the:
- A. return on total capital.
 - B. defensive interval ratio.
 - C. fixed charge coverage ratio.
11. RGB, Inc.'s income statement shows sales of \$1,000, cost of goods sold of \$400, pre-interest operating expense of \$300, and interest expense of \$100. RGB's interest coverage ratio is *closest* to:
- A. 2 times.
 - B. 3 times.
 - C. 4 times.

MODULE 37.4: DUPONT ANALYSIS

LOS 37.d: Demonstrate the application of DuPont analysis of return on equity and calculate and interpret effects of changes in its components.

The **DuPont system of analysis** is an approach that can be used to analyze return on equity (ROE). It uses basic algebra to break down ROE into a function of different ratios, so an analyst can see the impact of leverage, profit margins, and turnover on shareholder returns. There are two variants of the DuPont system: the original three-part approach and the extended five-part system.

For the original approach, start with ROE defined as follows:

$$\text{ROE} = \frac{\text{net income}}{\text{average equity}}$$

Average or year-end values for equity can be used. Multiplying ROE by (average total assets / average total assets) and rearranging terms produces the following:

$$\text{ROE} = \left(\frac{\text{net income}}{\text{average total assets}} \right) \left(\frac{\text{average total assets}}{\text{average stockholders' equity}} \right)$$

The first term is the ROA, and the second term is a financial leverage ratio that will increase as the use of debt financing increases:

$$\text{ROE} = \text{ROA} \times \text{financial leverage}$$

We can expand this further by multiplying these terms by (revenue / revenue), and rearranging terms:

$$\text{ROE} = \left(\frac{\text{net income}}{\text{revenue}} \right) \left(\frac{\text{revenue}}{\text{average total assets}} \right) \left(\frac{\text{average total assets}}{\text{average stockholders' equity}} \right)$$

The first term becomes net profit margin, the second term is now total asset turnover, and the third term is still financial leverage:

$$\text{ROE} = \text{net profit margin} \times \text{total asset turnover} \times \text{financial leverage}$$



PROFESSOR'S NOTE

The leverage ratio is sometimes called the equity multiplier.

This is the original DuPont equation. It is arguably the most important equation in ratio analysis because it breaks down a very important ratio (ROE) into three key components. If ROE is relatively low, it must be that at least one of the following is true: the company has a poor profit margin, the company has poor asset turnover, or the firm has too little leverage.



PROFESSOR'S NOTE

Often, candidates get confused and think the DuPont method is a way to *calculate* ROE. While you can calculate ROE given the components of either the original or extended DuPont equations, this isn't necessary if you have the financial statements. If you have net income and equity, you can calculate ROE. The DuPont method is a way to *decompose* ROE, to better see what is driving the changes in ROE.

EXAMPLE: Decomposition of ROE with original DuPont

Staret, Inc., has maintained a stable and relatively high ROE of approximately 18% over the last three years. Use traditional DuPont analysis to decompose

this ROE into its three components and comment on trends in company performance.

Staret, Inc., Selected Balance Sheet and Income Statement Items (Millions)

Year	20X3	20X4	20X5
Net income	21.5	22.3	21.9
Revenue	305	350	410
Average equity	119	124	126
Average assets	230	290	350

Answer:

ROE:

$$20X3: 21.5 / 119 = 18.1\%$$

$$20X4: 22.3 / 124 = 18.0\%$$

$$20X5: 21.9 / 126 = 17.4\%$$

DuPont:

$$20X3: 7.0\% \times 1.33 \times 1.93$$

$$20X4: 6.4\% \times 1.21 \times 2.34$$

$$20X5: 5.3\% \times 1.17 \times 2.78$$

Note that values were rounded in these calculations.

While ROE has dropped only slightly, both total asset turnover and net profit margin have declined. The effects of declining net margins and turnover on ROE have been offset by a significant increase in leverage. The analyst should be concerned about the decrease in net margin and determine the combination of pricing pressure and increasing expenses that caused this situation. The company has become more risky due to increased debt financing.

The **extended (5-way) DuPont equation** takes the net profit margin and breaks it down further:

$$ROE = \left(\frac{\text{net income}}{EBT} \right) \left(\frac{EBT}{EBIT} \right) \left(\frac{EBIT}{\text{revenue}} \right) \left(\frac{\text{revenue}}{\text{average assets}} \right) \left(\frac{\text{average assets}}{\text{average equity}} \right)$$

The first term in the 3-part DuPont equation, net profit margin, has been decomposed into three terms:

$\frac{\text{net income}}{\text{EBT}}$ is called the tax burden and is equal to $(1 - \text{tax rate})$

$\frac{\text{EBT}}{\text{EBIT}}$ is called the interest burden

$\frac{\text{EBIT}}{\text{revenue}}$ is called the EBIT margin

We then have the following:

$$\text{ROE} = \left(\frac{\text{tax}}{\text{burden}} \right) \left(\frac{\text{interest}}{\text{burden}} \right) \left(\frac{\text{EBIT}}{\text{margin}} \right) \left(\frac{\text{total asset}}{\text{turnover}} \right) \left(\frac{\text{financial}}{\text{leverage}} \right)$$

An increase in interest expense as proportion of EBIT will increase the interest burden (i.e., decrease the interest burden ratio). Increases in either the tax burden or the interest burden (i.e., decreases in the ratios) will tend to decrease ROE.

EBIT in the second two expressions can be replaced by operating earnings. In this case, we have the operating margin rather than the EBIT margin. The interest burden term would then show the effects of nonoperating income as well as the effect of interest expense.

Note that in general, high profit margins, leverage, and asset turnover will lead to high levels of ROE. However, this version of the formula shows that more leverage *does not always* lead to higher ROE. As leverage rises, so does the interest burden. Hence, the positive effects of leverage can be offset by the higher interest payments that accompany more debt. Note that higher taxes will always lead to lower levels of ROE.

EXAMPLE: Extended DuPont analysis

An analyst has gathered data from two companies in the same industry. Use the extended DuPont analysis to explain the critical factors that account for the differences between the two companies' ROEs.

Selected Income and Balance Sheet Data

	Company A	Company B
Revenues	\$500	\$900
EBIT	35	100
Interest expense	5	0
EBT	30	100
Taxes	<u>10</u>	<u>40</u>
Net income	20	60
Average assets	250	300
Total debt	100	50
Average equity	\$150	\$250

Answer:

return on equity = net income / average equity

Company A: $20 / 150 = 13.3\%$

Company B: $60 / 250 = 24.0\%$

EBIT margin = EBIT / revenue

Company A: $35 / 500 = 7.0\%$

Company B: $100 / 900 = 11.1\%$

total asset turnover = revenue / average assets

Company A: $500 / 250 = 2.0$

Company B: $900 / 300 = 3.0$

interest burden = EBT / EBIT

Company A: $30 / 35 = 85.7\%$

Company B: $100 / 100 = 1$

financial leverage = average assets / average equity

Company A: $250 / 150 = 1.67$

Company B: $300 / 250 = 1.2$

tax burden = net income / EBT

Company A: $20 / 30 = 66.7\%$

Company B: $60 / 100 = 60.0\%$

Company B has a higher tax burden but a lower interest burden (a lower ratio indicates a higher burden). Company B has better EBIT margins and better asset use (perhaps management of inventory, receivables, or payables, or a lower cost basis in its fixed assets due to their age), and less leverage. Company B's higher EBIT margins and asset turnover are the main factors leading to its significantly higher ROE, which it achieves with less leverage than Company A.

MODULE 37.5: INDUSTRY-SPECIFIC FINANCIAL RATIOS

LOS 37.e: Describe the uses of industry-specific ratios used in financial analysis.

As no two industries are the same, the importance of each individual ratio will vary depending on the key performance indicators of an industry. For some industries, analysts calculate ratios to evaluate metrics that might not be relevant for other industries. For example:

- | *Net income per employee* and *sales per employee* are used in the analysis and valuation of service and consulting companies.
- | *Growth in same-store sales* is used in the restaurant and retail industries to indicate growth without the effects of new locations that have been opened. It is a measure of how well the firm is doing at attracting and keeping existing customers. In the case of locations with overlapping markets, decreases in same-store sales growth may indicate that new locations are taking customers from existing ones. *Sales per square foot* is another metric commonly used in the retail industry.
- | In the hotel industry, *average daily rate*, or room revenue divided by number of rooms sold, is a key indicator of profitability. *Occupancy rates*, used to analyze the total number of rooms sold relative to the total number available, are a key activity ratio.
- | For subscription services such as streaming providers, *average revenue per user* is often a key metric.

Banks, insurance companies, and other financial firms carry their own challenges for analysts. Financial firms often have to maintain minimums or

maximums of certain ratios to comply with regulations. Part of the challenge is to understand the commonly used terms and the ratios they represent.

- | *Capital adequacy* typically refers to some monetary measure of a firm's risk, both operational and financial, as a percentage of its equity capital. Regulators monitor capital adequacy ratios to ensure that a financial firm has a buffer to absorb losses. A common measure of capital risk is *value at risk*, which is an estimate of the money size of the loss that a firm will exceed only some specific percentage of the time, over a specific period of time.
- | Banks are subject to minimum *reserve requirements*. Their ratios of various liabilities to their central bank reserves must be above the minimums. The ratio of a bank's liquid assets to certain liabilities is called its *liquid asset requirement*.
- | The performance of financial companies that lend funds is often summarized as the *net interest margin*, which is simply interest income divided by the firm's interest-earning assets.



PROFESSOR'S NOTE

A more thorough discussion of ratios relating to banking and insurance institutions is included at Level II. The focus on financial institutions' capital adequacy intensified due to the 2007–2008 global financial crisis. Contagion is always a risk due to the interconnected nature of banking. Many of the regulations introduced have been directed at reducing this risk. In 2024, the United States and Switzerland experienced events that tested both the effectiveness of the rules and the responsiveness of regulators.

Business Risk

The standard deviation of revenue, standard deviation of operating income, and standard deviation of net income are all indicators of the variation in and the uncertainty about a firm's performance. Because they all depend on the size of the firm to a great extent, analysts employ a size-adjusted measure of variation. The **coefficient of variation** for a variable is its standard deviation divided by its expected value.

Certainly, different industries have different levels of uncertainty about revenues, expenses, taxes, and nonoperating items. Comparing coefficients of variation for a firm across time, or among a firm and its peers, can aid the analyst in assessing

both the relative and absolute degree of risk a firm faces in generating income for its investors.

$$\text{CV sales} = \frac{\text{standard deviation of sales}}{\text{mean sales}}$$

$$\text{CV operating income} = \frac{\text{standard deviation of operating income}}{\text{mean operating income}}$$

$$\text{CV net income} = \frac{\text{standard deviation of net income}}{\text{mean net income}}$$

LOS 37.f: Describe how ratio analysis and other techniques can be used to model and forecast earnings.

Ratio analysis can be used in preparing **pro forma financial statements** that provide estimates of financial statement items for one or more future periods.

A forecast of financial results begins with an estimate of a firm's next-period revenues. If the analyst has no reason to believe that COGS in relation to sales will change for the next period, the COGS percentage from a common-size income statement can be used in constructing a pro forma income statement for the next period based on the estimate of sales.

Similarly, the analyst may believe that certain ratios will remain the same, or change in one direction or the other for the next period. In the absence of any information indicating a change, an analyst may choose to incorporate the operating profit margin from the prior period into a pro forma income statement for the next period. Beginning with an estimate of next-period sales, the estimated operating profit margin can be used to forecast operating profits for the next period.

Rather than point estimates of sales and net and operating margins, the analyst may examine possible changes to create a range of possible values for key financial variables.

Three methods of examining the variability of financial outcomes around point estimates are *sensitivity analysis*, *scenario analysis*, and *simulation*. Sensitivity analysis is based on "what if" questions such as, "What will be the effect on net income if sales increase by 3% rather than the estimated 5%?" Scenario analysis is based on specific scenarios (a specific set of outcomes for key variables) and will also yield a range of values for financial statement items. Simulation is a technique in which probability distributions for key variables are selected, and a

computer is used to generate a distribution of values for outcomes based on a repeated random selection of values for the key variables.



MODULE QUIZ 37.4, 37.5

1. Return on equity using the traditional DuPont formula equals:
 - A. (net profit margin) (interest component) (solvency ratio).
 - B. (net profit margin) (total asset turnover) (tax retention rate).
 - C. (net profit margin) (total asset turnover) (financial leverage multiplier).
2. Which of the following equations *least accurately* represents return on equity?
 - A. (net profit margin)(equity turnover).
 - B. (net profit margin)(total asset turnover)(assets / equity).
 - C. (ROA)(interest burden)(tax retention rate).
3. Which of the following statements about coefficient of variation is *least accurate*?
 - A. A coefficient of variation for an income statement measure represents variation per monetary unit.
 - B. The coefficient of variation is calculated by dividing a mean value by its standard deviation.
 - C. The coefficient of variation is a size-adjusted measure of variation.
4. An analyst who needs to model and forecast a company's earnings for the next three years would be *least likely* to:
 - A. assume that key financial ratios will remain unchanged for the forecast period.
 - B. use common-size financial statements to estimate expenses as a percentage of net income.
 - C. examine the variability of the predicted outcomes by performing a sensitivity or scenario analysis.

KEY CONCEPTS

LOS 37.a

Ratios can be used to project earnings and future cash flow, evaluate a firm's flexibility, assess management's performance, evaluate changes in the firm and industry over time, and compare the firm with industry competitors.

Vertical common-size data are stated as a percentage of sales for income statements, or as a percentage of total assets for balance sheets. Horizontal common-size data present each item as a percentage of its value in a base year.

Ratio analysis has limitations. Ratios are not useful when viewed in isolation and require adjustments when different companies use different accounting treatments. Comparable ratios may be hard to find for companies that operate in

multiple industries. Ratios must be analyzed relative to one another, and determining the range of acceptable values for a ratio can be difficult.

LOS 37.b

Activity ratios indicate how well a firm uses its assets. They include receivables turnover, days of sales outstanding, inventory turnover, days of inventory on hand, payables turnover, payables payment period, and turnover ratios for total assets, fixed assets, and working capital.

Liquidity ratios indicate a firm's ability to meet its short-term obligations. They include the current, quick, and cash ratios; the defensive interval; and the cash conversion cycle.

Solvency ratios indicate a firm's ability to meet its long-term obligations. They include the debt-to-equity, debt-to-capital, debt-to-assets, financial leverage, interest coverage, and fixed charge coverage ratios.

Profitability ratios indicate how well a firm generates operating income and net income. They include net, gross, and operating profit margins; pretax margin; return on assets; operating return on assets; return on total capital; return on total equity; and return on common equity.

LOS 37.c

An analyst should use an appropriate combination of different ratios to evaluate a company over time and relative to comparable companies. The interpretation of an increase in ROE, for example, may be quite different for a firm that has significantly increased its financial leverage compared to one that has maintained or decreased its financial leverage.

LOS 37.d

2-stage decomposition of ROE:

$$\text{ROE} = \left(\frac{\text{net income}}{\text{average total assets}} \right) \left(\frac{\text{average total assets}}{\text{average stockholders' equity}} \right)$$

$$\text{ROE} = \text{ROA} \times \text{leverage}$$

Original DuPont equation (3-stage decomposition):

$$\text{ROE} = \left(\frac{\text{net income}}{\text{sales}} \right) \left(\frac{\text{sales}}{\text{average total assets}} \right) \left(\frac{\text{average total assets}}{\text{average stockholders' equity}} \right)$$

$$\text{ROE} = \text{net profit margin} \times \text{asset turnover} \times \text{leverage}$$

Extended DuPont equation (5-stage decomposition):

$$\text{ROE} = \left(\frac{\text{net income}}{\text{EBT}} \right) \left(\frac{\text{EBT}}{\text{EBIT}} \right) \left(\frac{\text{EBIT}}{\text{revenue}} \right) \left(\frac{\text{revenue}}{\text{total assets}} \right) \left(\frac{\text{total assets}}{\text{total equity}} \right)$$

$$\text{ROE} = \text{tax burden} \times \text{interest burden} \times \text{EBIT margin} \times \text{asset turnover} \times \text{leverage}$$

LOS 37.e

The skill of an analyst involves identifying ratios that relate to the industry being analyzed and the performance, position, and flexibility of the firm, both currently and in the future.

Financial institutions have to comply with capital adequacy directives designed to prevent insolvency and contagion.

LOS 37.f

Ratio analysis, in conjunction with other techniques, can be used to construct pro forma financial statements based on a forecast of sales growth and assumptions about the relation of changes in key income statement and balance sheet items to growth of sales.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 37.1, 37.2, 37.3

- . **B** With a vertical common-size income statement, all income statement accounts are divided by sales. (Module 37.1, LOS 37.a)
- . **A** Company and industry data are widely available from numerous private and public sources. The other statements describe limitations of financial ratios. (Module 37.1, LOS 37.a)
- . **B** payables turnover = (purchases / avg. AP) = 100 / 12 = 8.33
 payables payment period = 365 / 8.33 = 43.8 days
 (Module 37.2, LOS 37.b)
- . **B** total asset turnover = (sales / total assets) = 150 / 75 = 2 times
 inventory turnover = (COGS / avg. inventory) = (150 – 45) / 15 = 7 times
 (Module 37.2, LOS 37.b)
- . **C** receivables turnover = (sales / avg. AR) = 100 / 25 = 4
 average collection period = 365 / 4 = 91.25 days

(Module 37.2, LOS 37.b)

- . **B** Current ratio = current assets / current liabilities. If cash (a current asset) and AP (a current liability) decrease by the same amount and the current ratio is greater than 1, then the numerator decreases less in percentage terms than the denominator, and the current ratio increases. (Module 37.2, LOS 37.b)
- . **A** Quick ratio = (cash + marketable securities + AR) / current liabilities. If cash decreases, the quick ratio will also decrease. The denominator is unchanged. (Module 37.2, LOS 37.b)
- . **C** Current ratio = current assets / current liabilities. If CR is > 1 , then if CA and CL both fall, the overall ratio will increase. (Module 37.2, LOS 37.b)
- . **A** $(365 / 10 + 365 / 5 - 365 / 9) = 69$ days

(Module 37.2, LOS 37.b)

- . **C** Fixed charge coverage is a solvency ratio. Return on total capital is a measure of profitability, and the defensive interval ratio is a liquidity measure. (Module 37.3, LOS 37.b)
- . **B** interest coverage ratio = $\text{EBIT} / \text{interest} = (1,000 - 400 - 300) / 100 = 3$ times

(Module 37.3, LOS 37.b)

Module Quiz 37.4, 37.5

- . **C** This is the correct formula for the three-ratio DuPont model for ROE. (Module 37.4, LOS 37.d)
- . **C** $(\text{ROA})(\text{interest burden})(\text{tax retention rate})$ is not one of the DuPont models for calculating ROE. (Module 37.4, LOS 37.d)
- . **B** Coefficient of variation is computed by dividing standard deviation by mean; therefore, it can be interpreted as risk per monetary unit for the income statement variable considered. Variation in income statement items such as sales, operating profit, and net income is related to the size of a company. By dividing the standard deviation by the mean value of the variable, size is adjusted for, giving risk per unit of the variable. (Module 37.5, LOS 37.e)

- . **B** An earnings forecast model would typically estimate expenses as a percentage of sales. (Module 37.5, LOS 37.f)

READING 38

INTRODUCTION TO FINANCIAL STATEMENT MODELING

MODULE 38.1: FINANCIAL STATEMENT MODELING

LOS 38.a: Demonstrate the development of a sales-based pro forma company model.

A sales-based pro forma company model consists of projected future financial statements, based on an analyst's estimate of a company's future revenues.

- Step 1:* Estimate revenue growth and future revenue, based on market growth and market share, a trend growth rate, or growth relative to GDP.
- Step 2:* Estimate COGS based on a percentage of sales, or on a more detailed method based on business strategy or competitive environment.
- Step 3:* Estimate SG&A as either fixed, growing with revenue, or using some other estimation technique.
- Step 4:* Estimate financing costs using interest rates, debt levels, and the effects of any large anticipated increases or decreases in capital expenditures or anticipated changes in financial structure.
- Step 5:* Estimate income tax expense and cash taxes using historical effective rates and trends, segment information for different tax jurisdictions, and anticipated growth in high- and low-tax segments, taking into account changes in deferred tax items.
- Step 6:* Model the balance sheet based on items that flow from the income statement (working capital accounts).
- Step 7:* Use depreciation and capital expenditures (for maintenance and for growth) to estimate capital expenditures and net PP&E for the balance sheet.
- Step 8:* Use the completed pro forma income statement and balance sheet to construct a pro forma cash flow statement.

LOS 38.b: Explain how behavioral factors affect analyst forecasts and recommend remedial actions for analyst biases.

Like everyone, those in the financial industry are prone to behavioral biases. For analysts, these biases can result in inaccurate forecasts.

Overconfidence bias. This is having too much faith in one's own work. Analysts may underestimate their forecasting errors; hence, they have a narrower confidence interval for their forecasts than warranted. Research has shown that analysts who "go against the grain" (i.e., forecasting what others are not) are more likely to suffer from overconfidence bias. This bias may be mitigated by sharing forecasts and soliciting critique. Analysts should evaluate their past forecasts and learn from their own forecasting errors, which should lead to a widening of their confidence intervals. Using scenario analysis to produce a range of forecasts may help to identify any shortcomings.

Illusion of control bias. This is related to overconfidence, but refers specifically to overestimating what an analyst can control and trying to control things an analyst cannot control. This bias is manifested in two primary ways: seeking "expert" opinions to justify a forecast, and making a model more complex (e.g., by including more independent variables). Overfitted models perform poorly out of sample and can also conceal assumptions that need to be updated based on new information. Illusion of control can be mitigated by focusing only on variables with known explanatory power, and by seeking outside opinions only from those who have a relevant perspective.

Conservatism bias. This is also called *anchoring*, where the analyst makes only small adjustments to their prior forecasts when new information becomes available. Usually, conservatism results in reluctance to incorporate new negative information; however, it could also lead to lags in incorporating positive news. Mitigating this bias requires periodic evaluation of forecasting errors, and using simpler models that are easier to adjust for new or changed assumptions.

Representativeness bias. This bias occurs due to a tendency to rely on known classifications. Sometimes, new information may only be superficially similar to a known classification but may be better viewed from a fresh perspective. One common form of representativeness bias is **base-rate neglect**, where an observation's membership (its base rate, or rate of incidence in a larger population) is neglected in favor of situation or member-specific information. Fixating on a firm's company-specific factors is known as the *inside view*, while viewing the company as a member of a particular industry (focusing on

the base rate) is sometimes known as the *outside view*. Analysts should consider both inside and outside views to generate forecasts.

Confirmation bias. Confirmation bias causes an analyst to seek out (or pay attention to) data that affirms their earlier convictions, and to disregard or underestimate information that calls those opinions into question. For example, an analyst who has a positive view of a particular company may choose to discuss the firm with colleagues who share the same point of view. Two ways to reduce confirmation bias are to keep abreast of research from analysts who have an opposite view, and to seek out the points of view of colleagues who have no emotional investment in the opinion. Analysts should also be aware of their own confirmation bias when they evaluate a company management's representations. Managements tend to portray their companies in a positive way, and analysts who have (or want to have) positive outlooks on companies must be careful not to simply take management's comments at face value.



PROFESSOR'S NOTE

The focus here is on how behavioral biases affect analysts. We will revisit these biases in the context of investors' behavior in the Portfolio Construction topic area.

LOS 38.c: Explain how the competitive position of a company based on a Porter's five forces analysis affects prices and costs.

The competitive environment that a firm operates in and how successful it is in that environment are key in determining the firm's future financial results. There are no formulas for, or clear rules about, how a firm's competitive environment affects its future revenue and costs, but a firm's future competitive success is possibly the most important factor in its future revenue and profitability.

Porter's five forces are a framework analysts commonly use to evaluate a company's competitive position.

Companies have more pricing power when the **threat of substitute products** is low and switching costs are high. They have less pricing power when good substitutes are or may become available, and when it is less costly for customers to switch to those products.

Companies have more pricing power when the **intensity of industry rivalry** is low, and less pricing power when competitive intensity is high. Industry rivalry

tends to be more intense when an industry has many competitors (is less concentrated), when fixed costs and exit barriers are high, when industry growth is slow or negative, and when products are not differentiated significantly.

Pressure on a company's costs may be higher, and its prospects for earnings growth lower, when the **bargaining power of suppliers** is high. If suppliers are few, they may be able to extract a larger portion of any value added.

Companies have less pricing power when the **bargaining power of customers** is higher, especially if a small number of customers are responsible for a large proportion of a firm's sales, or if switching costs are low.

Companies have more pricing power and better prospects for earnings growth when the **threat of new entrants** is low. Significant barriers to entry into an industry make it possible for existing companies to sustain economic profits over time.

LOS 38.d: Explain how to forecast industry and company sales and costs when they are subject to price inflation or deflation.

Input costs can be significant in many industries. The cost of jet fuel in the airline industry, the cost of grains to cereal and baking companies, and the cost of coffee beans to coffee shops are all variable. Changes in these costs can significantly affect earnings.

Companies with commodity-type inputs can hedge their exposure to changes in input prices through derivatives, or more simply through fixed-price contracts for future delivery. Such hedging will reduce the effect of short-term changes in input prices and increase the time until longer-term price changes affect costs and earnings. Companies that are vertically integrated (in effect, their own suppliers) are relatively less exposed to input cost risk.

For a company that neither hedges input price exposure nor is vertically integrated, the issue for an analyst is to determine how rapidly, and to what extent, an increase in costs can be passed on to customers, as well as the expected effect of price increases on sales volume and sales revenue.

An analyst should monitor a company's production costs by product category and geographic location, with a focus on the significant factors that affect input prices, such as weather, government regulation and taxation, tariffs, and the

characteristics of input markets. It may be that a firm can reduce the impact of an increase in an input price by switching to a substitute input; for example, rising oil prices may lead power generation firms to switch from oil to natural gas.

When estimating the effects of an increase in input prices, an analyst must make assumptions about the company's pricing strategy and the effects of price increases on unit sales. When increases in input costs are thought to be temporary, a company may cut other costs (e.g., advertising expenses) to preserve operating margins. This strategy is, however, not appropriate for long-term increases in input costs.

The effects of raising a product's price depend on its elasticity of demand. For most firms, product demand is relatively elastic. With elastic demand, the percentage reduction in unit sales is greater than the percentage increase in price, and a price increase will decrease total sales revenue.

Elasticity of demand is most affected by the availability of substitute products. In a competitive industry, the pricing decisions of other firms in the industry can affect the market shares of its competitors. A company that is the first to increase prices in response to increased costs will experience a greater decrease in unit sales than a company that increases prices after other firms have already done so. A firm may decide to delay increasing prices to gain market share when other firms increase prices in response to increased costs. Firms that are too quick to increase prices will experience declining sales volumes, while firms that are slow to increase prices will experience declining gross margins.

If the money amount of the increase in cost per unit is added to product price and unit sales do not decrease (this is unlikely), the amount of operating profit is unchanged, but gross margins, operating margins, and net margins will decrease.

EXAMPLE: Effect of price inflation on gross profits, gross margins, and operating margins

Alfredo, Inc., sells a specialized network component. The firm's income statement for the past year follows.

Alfredo, Inc., Income Statement for the Year Ended 20X1

Revenues	1,000 @ \$100	\$100,000
COGS	1,000 @ \$40	<u>\$40,000</u>
Gross profit		\$60,000
SG&A		<u>\$30,000</u>
Operating profit		<u>\$30,000</u>

For 20X2, the input costs (COGS) will increase by \$5 per unit.

1. Calculate the gross margin and operating margin for Alfredo, Inc., for 20X1.
2. Calculate the 20X2 gross margin and operating margin assuming the following:
 - a. The entire increase in input cost is passed on to the customers through an equal increase in selling price. The number of units sold is not affected.
 - b. The selling price is increased by 5% and the number of units sold decreases by 5%.
 - c. The selling price is increased by 5% and the number of units sold decreases by 10%.

Answer:

1. gross margin = gross profit / sales = $\$60,000 / \$100,000 = 60\%$
 operating margin = operating profit / sales = $\$30,000 / \$100,000 = 30\%$

2. a. 20X2, given an increase in unit price by \$5 and no change in units sold:

Revenues	1,000 units @ \$105	\$105,000
COGS	1,000 units @ \$45	<u>\$45,000</u>
Gross profit		\$60,000
SG&A		<u>\$30,000</u>
Operating profit		<u>\$30,000</u>
Gross margin		57%
Operating margin		29%

gross margin = gross profit / sales = \$60,000 / \$105,000 = 57%

operating margin = operating profit / sales = \$30,000 / \$105,000 = 29%

- b. 20X2, given an increase in unit price by \$5 and a decrease of 50 in units sold:

Revenues	950 units @ \$105	\$99,750
COGS	950 units @ \$45	<u>\$42,750</u>
Gross profit		\$57,000
SG&A		<u>\$30,000</u>
Operating profit		<u>\$27,000</u>
Gross margin		57%
Operating margin		27%

gross margin = gross profit / sales = \$57,000 / \$99,750 = 57%

operating margin = operating profit / sales = \$27,000 / \$99,750 = 27%

- c. 20X2, given an increase in unit price by \$5 and a decrease of 100 in units sold:

Revenues	900 units @ \$105	\$94,500
COGS	900 units @ \$45	<u>\$40,500</u>
Gross profit		\$54,000
SG&A		<u>\$30,000</u>
Operating profit		<u>\$24,000</u>
Gross margin		57%
Operating margin		25%

gross margin = gross profit / sales = \$54,000 / \$94,500 = 57%

operating margin = operating profit / sales = \$24,000 / \$94,500 = 25%

LOS 38.e: Explain considerations in the choice of an explicit forecast horizon and an analyst's choices in developing projections beyond the short-term forecast horizon.

For a buy-side analyst, the appropriate forecast horizon may simply be the expected holding period for a stock. For example, a portfolio with a 25% annual turnover has an average holding period of four years, so four years may be the most appropriate forecast horizon.

Highly cyclical companies present difficulties when choosing a forecast horizon. The horizon should be long enough that the effects of the current phase of the economic cycle are not driving above-trend or below-trend earnings effects. The forecast horizon should be long enough to include the middle of a business cycle so the analyst's forecast includes a midcycle level of sales and profits.

Normalized earnings are expected midcycle earnings or, alternatively, expected earnings when the current (temporary) effects of events or cyclicalities are no longer affecting earnings.

Events such as acquisitions, mergers, or restructurings should be considered temporary. The forecast horizon should be long enough that the perceived benefits of such events can be realized (or not).

It may also be the case that the forecast horizon is dictated by an analyst's manager.

For earnings projections beyond the short term, one method of forecasting future financial results is to assume that a trend growth rate of revenue over the previous cycle will continue. An analyst can estimate pro forma financial results based on the projection of each future period's revenue.

An analyst will typically value a stock using earnings or some measure of cash flow over a forecast period, along with the stock's terminal value at the end of the forecast horizon. This terminal value is usually estimated using either a relative valuation (i.e., price multiple) approach or a discounted cash flow approach.



PROFESSOR'S NOTE

We will describe these approaches to estimating stock values in the Equities topic area.

When using a multiples approach, an analyst must ensure that the multiple used is consistent with the estimate of the company's growth rate and required rate of return. Using the average P/E ratio for the company over the last 10 years, for example, presupposes that the growth in earnings and required rate of return of the stock will be, on average, the same in the future as it was over the previous 10 years.

When using a discounted cash flow approach to estimate the terminal value, two key inputs are a cash flow or earnings measure and an expected future growth rate. The expected earnings or cash flow should be normalized to a midcycle value that is not affected by temporary events. Because the terminal value is calculated as the present value of a perpetuity, small changes in the estimated (perpetual) growth rate of future earnings or cash flows can have large effects on estimated terminal values—and, hence, the current stock value.

Assuming that growth in future profitability will be the same as average profitability growth in the past may not be justified. A difficult part of an analyst's job is recognizing **inflection points**, those instances when the future will not be like the past. Examples of inflection points include changes in the economic environment or business cycle stage, government regulations, or technology.



MODULE QUIZ 38.1

1. An analyst continues to add independent variables to a forecasting model, even when doing so does not improve its accuracy significantly. The analyst *most likely* exhibits:
 - A. confirmation bias.
 - B. illusion of control bias.
 - C. representativeness bias.
2. Which of Porter's five forces should an analyst focus on when evaluating risks to a company's input costs?
 - A. Threat of new entrants.
 - B. Intensity of industry rivalry.
 - C. Bargaining power of suppliers.
3. When analyzing a cyclical company, the forecast horizon should include:
 - A. a full business cycle.
 - B. the midpoint of a business cycle.
 - C. the next change in phase of a business cycle.

KEY CONCEPTS

LOS 38.a

Developing sales-based pro forma financial statements includes the following steps:

- Step 1:* Estimate revenue growth and future expected revenue.
- Step 2:* Estimate COGS.
- Step 3:* Estimate SG&A.
- Step 4:* Estimate financing costs.
- Step 5:* Estimate income tax expense and cash taxes, taking into account changes in deferred tax items.
- Step 6:* Model the balance sheet based on items that flow from the income statement and estimates for important working capital accounts.
- Step 7:* Use historical depreciation and capital expenditures to estimate future capital expenditures and net PP&E for the balance sheet.
- Step 8:* Use the completed pro forma income statement and balance sheet to construct a pro forma cash flow statement.

LOS 38.b

Behavioral factors may affect analyst's forecasts:

- Overconfidence bias
- Illusion of control bias
- Conservatism bias or anchoring
- Representativeness bias
- Confirmation bias

LOS 38.c

Expectations of a firm's future competitive success are important factors in forecasting future revenue and financial statements. An analyst can evaluate the competitive position of a company based on Porter's five forces:

- Companies have less (more) pricing power when the threat of substitute products is high (low) and switching costs are low (high).
- Companies have less (more) pricing power when the intensity of industry rivalry is high (low).
- Pressure on input costs is higher when the bargaining power of suppliers is high.
- Companies have less pricing power when the bargaining power of customers is high.

Companies have more pricing power when the threat of new entrants is low.

LOS 38.d

Increases in input costs will increase COGS unless the company has hedged the risk of input price increases with derivatives or contracts for future delivery. Vertically integrated companies are likely to be less affected by increasing input costs. The effect on sales of increasing product prices to reflect higher COGS will depend on the elasticity of demand for the products and on the timing and amount of competitors' price increases.

LOS 38.e

For a buy-side analyst, the appropriate forecast horizon to use may be the expected holding period for a stock.

For highly cyclical companies, the forecast horizon should include the middle of a cycle so that the analyst can forecast normalized earnings.

Events such as acquisitions, mergers, or restructurings, should be considered temporary. The forecast horizon should be long enough that the perceived benefits of such events can be realized.

Earnings projections over a forecast period beyond the short term are often based on the historical average growth rate of revenue over the previous economic cycle.

An analyst will typically estimate a terminal value for a stock at the end of the forecast horizon, using either a price multiple or a discounted cash flow approach. Small changes in the estimated growth rate of future profits or cash flows can have large effects on the estimated stock value.

ANSWER KEY FOR MODULE QUIZ

Module Quiz 38.1

- . **B** Making a forecasting model overly complex, even when doing so brings no significant improvement, most likely reflects illusion of control bias. (LOS 38.b)
- . **C** Of Porter's five forces, bargaining power of suppliers has the most direct influence on a company's input costs. The other four forces more directly affect a company's pricing power. (LOS 38.c)

- . **B** The forecast horizon for a cyclical company should include the midpoint of a business cycle, at which the company is expected to produce normalized earnings. (LOS 38.e)

TOPIC QUIZ: FINANCIAL STATEMENT ANALYSIS

You have now finished the Financial Statement Analysis topic section. Please log into your Schweser online dashboard and take the Topic Quiz on this section. The Topic Quiz provides immediate feedback on how effective your study has been for this material. Questions are more exam-like than typical Module Quiz or QBank questions; a score of less than 70% indicates that your study likely needs improvement. These tests are best taken timed; allow 1.5 minutes per question.

READING 39

EQUITY INSTRUMENT FEATURES

MODULE 39.1: PUBLIC AND PRIVATE EQUITY INVESTMENTS

LOS 39.a: Describe basic features and types of equity instruments.

An **equity instrument** is a security that represents ownership of a portion of a corporation. As we discussed in our Corporate Finance reading on organizational forms, the owners of equity instruments (i.e., shareholders) can lose their investment if a company goes bankrupt, but otherwise have no liability for claims against the company. Equity instrument owners have a fractional claim to a company's assets net of any higher-priority claims, such as debts and taxes, and may have voting rights on company policies and decisions.

Two key differences between equity instruments and debt instruments (which we will describe in the Fixed Income topic area) are as follows:

While debt instruments have finite lives (i.e., maturities), equity instruments have indefinite lives. An equity instrument continues to exist unless a corporate event, such as an acquisition by another company or a liquidation of the company, causes it to go out of existence.

Payments of interest and principal to debtholders are contractual obligations of a company. In contrast, payments of **dividends** (distributions of a company's net income to shareholders) are at the company's discretion.

Companies' dividend payment policies typically change over the **company life cycle**. We can think of this life cycle as having the following four phases:

Start-up phase. A start-up company initially has no revenue, and its cash flow is negative as it spends its invested capital to establish its business. At this stage, a company has no earnings to distribute.

Growth phase. Revenue is increasing and cash flow may change from negative to positive, but a company is more likely to reinvest its earnings to grow the business than distribute them to equity holders.

Mature phase. In this phase, a company has stable cash flows and may find fewer avenues to pursue growth, so equity holders prefer that the company distribute its earnings regularly as dividends.

Decline phase. Eventually, a company may reach a phase of decreasing revenues and cash flows, bringing its prospects for future dividend payments into question.

Types of Equity Instruments

Equity instruments differ in their features and legal rights. When a company issues equity securities, it must provide potential investors an **equity prospectus** that discloses these features and rights and describes how the company intends to use the capital it will raise by selling the securities. A prospectus specifies whether the security includes **voting rights** on company matters; **dividend rights** to regularly scheduled or special dividends; **conversion rights** to change the security to another type (e.g., from preferred shares to common shares); **liquidation rights** if the company dissolves and distributes its remaining assets; **split, subdivision, or combination rights** if the company splits, divides, or combines equity shares in the future; and **preemptive rights** that may give the security holder a first opportunity to buy shares the company issues in the future.

Common shares are the most frequently issued form of equity and represent an ownership interest. Common shareholders have a residual claim (after the claims of debtholders, other creditors, and preferred stockholders) on firm assets if the firm is liquidated, and they govern the corporation through voting rights.

Firms are under no obligation to pay dividends on common equity; the firm determines what dividend will be paid periodically. Common stockholders are able to vote for the board of directors, on merger decisions, and on the selection of auditors. If they are unable to attend the annual meeting, shareholders can vote by **proxy**, having someone else vote as they direct them, on their behalf.

Common shares benefit the issuing company by allowing it to raise capital while keeping control of its operations. Common shares benefit investors by allowing them to acquire fractional ownership of a company while investing a small amount relative to a company's size.

Preference shares (or **preferred stock**) have features of both common stock and debt. As with common stock, preferred stock dividends are not a contractual obligation, and the shares usually do not mature. Like debt, preferred shares typically make fixed periodic payments to investors and do not usually have voting rights. Preferred shares have a stated **par value** and pay a percentage dividend based on the par value of the shares. A \$100 par value preferred with an 8% dividend pays a dividend of \$8 per year. If the firm liquidates, preferred shares may be entitled to a predetermined **liquidation preference payout** ahead of any distributions to common shareholders.

Callable preference shares give the firm the right, but not the obligation, to repurchase the shares at a prespecified call price. **Putable preference shares** give the shareholder the right, but not the obligation, to sell the preference shares back to the issuer at a specified price, known as the **put contingency price**.

Cumulative preference shares are usually promised fixed dividends, and any dividends that are not paid must be made up before common shareholders can receive dividends. The dividends of **noncumulative preference shares** do not accumulate over time when they are not paid, but in any given period, preferred dividends must be paid before common shareholders can receive dividends.

Investors in **participating preference shares** receive extra dividends if the firm's profits exceed a predetermined level and may receive a value greater than the par value of the preferred stock if the firm is liquidated. **Nonparticipating preference shares** have a claim equal to par value in the event of liquidation and do not share in firm profits through extra dividends. Smaller and riskier firms whose investors may be concerned about the firm's future often issue participating preferred stock so investors can share in the upside potential of the firm.

Convertible preference shares give the preferred shareholder the right to exchange the shares for common stock at a **conversion ratio** determined when the shares are originally issued. Convertible preferred shares are often used to finance risky early-stage firms. The conversion feature compensates investors for the additional risk they take when investing in such firms and enables the firms to raise capital at a lower cost than issuing common shares.

LOS 39.b: Contrast features of publicly listed versus private equity securities.

The discussion so far has centered on equity that is publicly traded. **Private equity** is usually issued to institutional investors via private placements.

Private equity has several characteristics that distinguish it from public equity. Some of these characteristics are company-specific; some are stock-specific. Company-specific characteristics of private equity include the following:

- | Typically smaller-sized companies
- | Earlier (start-up and growth stages) or later (decline stage) in the company life cycle
- | More limited financial disclosure because private firms are not subject to the reporting requirements that govern publicly traded firms
- | Greater overlap between ownership and management
- | More concentrated ownership

Stock-specific characteristics of private equity include the following:

- | Less liquidity, as no public market exists in which to trade the shares
- | More concentrated control of the company
- | Restrictions on the sale of equity shares by owners

The four main types of private equity investments are venture capital, growth equity, buyout equity, and special situations.

Venture capital refers to the capital provided to firms early in their life cycles to fund their start-up, development, and growth. Investors can be family, friends, wealthy individuals, or private equity funds. Venture capital investments are illiquid and investors often have to commit funds for several years before they can cash out (exit) their investment.

Growth equity refers to capital a firm raises in its growth stage. Raising additional capital at this stage can decrease the concentration of ownership that exists at the venture capital stage.

Buyout equity tends to focus on mature-stage companies, especially when the private equity investors believe a company is underperforming but has value that can be realized by restructuring its capital or operations. A buyout of a publicly traded company that results in the investing group assuming full control is known as a **take-private transaction**. In a **leveraged buyout (LBO)**, investors buy all of a firm's equity using debt financing (leverage). If the buyers are the firm's current management, the LBO is referred to as a

management buyout (MBO). Firms in LBOs usually have cash flow that is adequate to service the issued debt or have undervalued assets that can be sold to pay down the debt over time.

Special situations refer to private equity investment in distressed companies, such as those in the decline stage of the life cycle, or in companies that are restructuring or reorganizing. This may take the form of a **private investment in public equity (PIPE)**, in which a publicly traded firm that needs capital quickly sells private equity to investors.



MODULE QUIZ 39.1

1. The advantage of participating preferred shares versus nonparticipating preferred shares is that participating preferred shares can:
 - A. obtain voting rights.
 - B. receive extra dividends.
 - C. be converted into common stock.
2. Compared to public equity, which of the following is *least likely* to characterize private equity?
 - A. More liquid market.
 - B. Lower reporting costs.
 - C. More concentrated ownership.

KEY CONCEPTS

LOS 39.a

An equity instrument represents ownership of a portion of a corporation.

Common shareholders have a residual claim on firm assets and govern the corporation through voting rights. Common shares have variable dividends, which the firm is under no legal obligation to pay.

Preferred stock typically does not mature, does not have voting rights, and has dividends that are fixed in amount but are not a contractual obligation of the firm.

Cumulative preferred shares require any dividends that were missed in the past to be paid before common shareholders receive any dividends. Participating preferred shares may receive extra dividends if firm profits exceed a prespecified level and may receive proceeds greater than par value if the firm is liquidated. Convertible preferred stock can be converted to common stock at a prespecified conversion ratio.

Callable shares allow the firm the right to repurchase the shares at a prespecified price. Puttable shares give the shareholder the right to sell the shares back to the firm at a prespecified price.

LOS 39.b

Compared to public equity, private equity has less liquidity, fewer financial disclosures, greater overlap between ownership and management, more concentrated ownership, and more restrictions on the sale of shares.

While public equity tends to feature mature-stage companies, private equity investments may occur earlier or later in the company life cycle. Venture capital is provided to firms early in their life cycles. Growth equity refers to capital a firm raises in its growth stage. Buyout equity tends to focus on mature-stage companies. Special situations are investments in distressed, restructuring, or reorganizing firms.

ANSWER KEY FOR MODULE QUIZ

Module Quiz 39.1

- . **B** Participating preferred shares can receive extra dividends if firm profits exceed a prespecified level and a value greater than the par value if the firm is liquidated. (LOS 39.a)
- . **A** Private equity has less liquidity because no public market for it exists. Ownership is typically more concentrated than a publicly traded firm. Reporting costs are lower for private equity because the firm is not subject to the reporting requirements that govern publicly traded firms.(LOS 39.b)

READING 40

EQUITY JURISDICTIONS, CLASSES, AND THE VOTING PROCESS

MODULE 40.1: RIGHTS OF EQUITY SHAREHOLDERS

LOS 40.a: Describe differences in economic and voting rights and other characteristics among different jurisdictions and equity classes.

In our previous reading, we described the economic and voting rights of equity shareholders. These rights may vary if a company has different classes of shares or if an investor owns shares in a foreign company.

Foreign Shares

Many investors exhibit **home asset bias**, overweighting their portfolios with assets in their home countries. One way to counteract home asset bias in a portfolio is by investing in foreign shares.

Direct investing in the securities of foreign companies refers to buying a foreign firm's securities in foreign markets. Some of the obstacles to direct foreign investment include the following:

- | Investors incur extra costs to set up securities accounts in foreign markets.
- | Investors must be familiar with the procedures of each market in which they invest, such as the clearing and settlement of trades.
- | Taxation rules are likely to be different in a foreign market than in an investor's home market.
- | Trading hours may be different than an investor's home market.
- | Reporting requirements of foreign stock exchanges may be different, and required disclosures may be in an unfamiliar foreign language.

- | The investment and return are denominated in a foreign currency, which adds foreign exchange risk and additional transactions costs to an investment.
- | Some countries might regulate or tax the return of capital to an investor's home country.
- | A foreign stock exchange may be illiquid.

Investors can also gain exposure to foreign shares through **indirect investing** vehicles that mitigate some of the obstacles to direct investing. One such method is a **depository receipt (DR)**. DRs represent ownership in a foreign firm and are traded in the markets of other countries in local market currencies.

To create a DR, a domestic bank holds shares deposited by the foreign firm and then issues receipts representing ownership of a specific number of the foreign shares. This **depository bank** acts as a custodian and manages dividends, stock splits, and other events. Although the investor does not have to convert to the foreign currency, the value of a DR is affected by exchange rate changes, alongside factors that affect the value of any stock such as firm fundamentals and economic events. To prevent arbitrage, the price of a DR should equate to the foreign share price at the current exchange rate, but in some situations the arbitrage relationship breaks down.

DRs are often referred to by the markets where they are listed. For example, **American depository receipts (ADRs)** are denominated in US dollars and trade in the United States, and **European depository receipts (EDRs)** trade in Europe and are typically denominated in euros.

Global depository receipts (GDRs) are issued outside the United States and outside the issuer's home country. Although not listed on US exchanges, they are usually denominated in US dollars and can be sold to US institutional investors. GDRs are not subject to the capital flow restrictions imposed by governments, so they offer the firm and the investor greater opportunities for foreign investment. The firm usually chooses to list the GDR in a market where many investors are familiar with the firm.

If the firm is involved with the issue, a DR is a **sponsored DR**; otherwise, it is an **unsponsored DR**. A sponsored DR provides the investor voting rights and is usually subject to greater disclosure requirements. In an unsponsored DR, the depository bank retains the voting rights.

Some large multinational firms encourage share ownership by nondomestic investors through **dual listing**, in which their shares trade on one or more foreign stock exchanges in addition to an exchange in their home market.

Dual-Class Share Structures

A firm may have different classes of common stock (e.g., Class A and Class B shares). Such a **dual-class share structure** often involves some classes of common stock having greater voting rights than others. This can give a specific group of shareholders, such as a company's founders, effective control of the company while holding less than a majority of shares.

Share classes may also be treated differently with respect to economic rights, such as dividends. In some cases, economic rights of different share classes may be in the same proportions as their voting rights. In other cases, one class of common shares may have greater economic rights, while another class has greater voting rights.

Information on the economic and voting rights of different classes of equity shares can be found in the company's filings with securities regulators, such as the Securities and Exchange Commission (SEC) in the United States.

LOS 40.b: Describe the voting process and roles of the issuer's management, board, proxy advisors, asset managers, and asset owners in the process.

As we saw in our Corporate Finance reading on corporate governance, voting rights are how share owners can exercise their authority over a company's actions.

Many of the matters that shareholders vote on routinely, such as the selection of an auditor or candidates for the board of directors, are put forth by a company's management or board. However, share owners themselves can also introduce **shareholder proposals** for actions they believe a company should take. Shareholder proposals are typically nonbinding, but if they attract significant support, they can signal management about issues the shareholders expect them to address. While a company does not have to present all shareholder proposals for a vote, financial market regulations may require management to provide reasons for rejecting one.

Share voting occurs at a company's annual meeting. Shareholders may attend the meeting or vote by proxy, typically with a slightly earlier cutoff date than that for shareholders who attend the meeting.

The company determines which matters need to be presented to shareholders and solicits shareholder proposals. This needs to occur far enough ahead of the annual meeting to allow the company to prepare a **proxy statement** for the annual meeting, which provides the meeting agenda, lists all proposals to be voted on (often with management's or the board's recommendations on voting for or against the proposals), and discloses the requirements and deadlines for voting.

In most cases, equity shares are held by a **custodian** broker or bank in the name of the shares' **beneficial owner**. One of the custodian's responsibilities is distributing proxy statements to shareholders. Another is determining the **shareholder of record** who is eligible to vote at the annual meeting. The **record date** for voting eligibility is often 30 days or more ahead of the meeting. A shareholder who sells the stock after the record date but before the meeting remains eligible to vote the shares.

Investors may choose to delegate their proxy votes to asset managers. As we will see in the Ethical and Professional Standards topic area, asset managers who are CFA Institute members or candidates are required to vote proxies on relevant issues in an informed and responsible manner and should vote in the interests of the shares' beneficial owners, per Standard III(A) Loyalty, Prudence, and Care.

Institutional investors such as mutual funds routinely hold equity shares in many different companies across multiple markets. To exercise their voting rights responsibly, institutional investors often rely on **proxy advisors**, firms that specialize in recommending how to vote proxies based on the most current information and in line with principles of corporate governance.



MODULE QUIZ 40.1

1. Global depository receipts (GDRs) are most often denominated in:
 - A. US dollars and issued in the United States.
 - B. US dollars and issued outside the United States.
 - C. the currency of the country where they trade and issued outside the United States.
2. A company has various securities outstanding. Which of the following *most likely* indicates that the company has a dual-class share structure?
 - A. Common shareholders may vote on company matters, but preferred shareholders may not vote.

- B. Some of a company's bonds are convertible to common shares, while some of its bonds are not convertible.
 - C. Some common shares give the holder two votes on company matters, while other common shares give the holder one vote.
3. An investor hears about a shareholder proposal at XYZ Company that will be voted on at the company's annual meeting the following week. The investor buys 100 shares of XYZ, and the trade settles two days before the meeting. Can this investor vote on the shareholder proposal?
- A. No.
 - B. Yes, because the shareholder is the beneficial owner.
 - C. Yes, because the investor is the shareholder of record.

KEY CONCEPTS

LOS 40.a

Direct investing in the securities of foreign companies refers to buying a foreign firm's securities in foreign markets. Direct investing involves foreign exchange risk and costs of setting up and monitoring investments in markets that may be unfamiliar to the investor.

Depository receipts (DRs) that trade in local market currencies are vehicles for indirect investment in foreign companies. Although the investor does not have to convert to the foreign currency, the value of a DR is affected by exchange rate changes.

A dual-class share structure involves some classes of common stock having greater voting rights or economic rights than others. This structure can give effective control of a company to a specific group of shareholders, such as its founders.

LOS 40.b

A company's management and board of directors propose routine matters for shareholder votes. Shareholders may also make their own proposals, which are typically nonbinding on management but can signal issues that shareholders expect management to address.

Share voting occurs at a company's annual meeting. A proxy statement provides the meeting agenda, lists proposals to be voted on, and discloses the requirements and deadlines for voting.

If shares are held by a custodian, the custodian is responsible for determining the shareholder of record for proxy votes. The record date for eligibility is often 30 days or more ahead of the annual meeting.

Investors can vote proxies themselves or delegate their votes to asset managers.

Proxy advisors are firms that assist institutional investors in managing their proxy votes.

ANSWER KEY FOR MODULE QUIZ

Module Quiz 40.1

- . **B** GDRs are not listed on US exchanges and are most often denominated in US dollars. They are not issued in the United States. (LOS 40.a)
- . **C** A dual-class share structure refers to common shares that have different economic or voting rights. (LOS 40.a)
- . **A** The record date for proxy voting is typically 30 days or more before the company's annual meeting, to allow time for preparing and distributing the proxy statement. (LOS 40.b)

READING 41

EQUITY ISSUANCE AND TRADING

MODULE 41.1: EQUITY MARKETS AND EXCHANGES

LOS 41.a: Describe primary and secondary public equity markets and their functions.

Primary equity markets refer to issuance of new securities to the public. Firms that wish to raise capital by issuing equity securities may use an investment bank to underwrite the sale of new shares to investors. Typically, the sale will take the form of an **initial public offering (IPO)** to investors at large, or a **private placement** directly to qualified investors. Primary markets tend to be most active when share prices are increasing overall.

Some of the circumstances that result in primary equity issuance include the following:

- | An early-stage firm wants to expand beyond venture capital financing to attract new public equity investment.
- | An existing public company spins off a division as a new public company.
- | A firm goes public after being restructured by private owners for a number of years.
- | A government uses the primary equity market to carry out a full or partial **privatization** of an existing state-run entity, transferring its operations to the private sector.

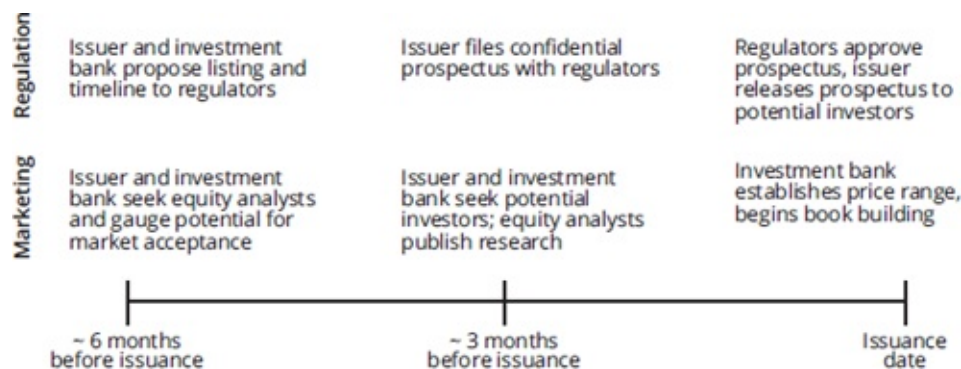


PROFESSOR'S NOTE

We say a company is *publicly owned* or *publicly held* when its equity shares trade in public markets. This is not to be confused with state ownership of business enterprises. Privatizing a state-run enterprise may turn it into a publicly owned corporation.

To carry out an IPO, a corporation must fulfill listing and other regulatory requirements and solicit investors for the new shares. The goals are to introduce the company to equity analysts and investors, gauge market acceptance, determine demand, and establish a price range for the prospective issuance. Figure 41.1 illustrates a typical timeline for an IPO.

Figure 41.1: IPO Timeline



One key IPO requirement is the creation and regulatory review of an **equity prospectus**. Elements of an equity prospectus include a description of the company and its business model; disclosures of key industry and business risks; audited financial statements; terms of the offering (including the number of shares to be issued and the expected price range); and use of issuance proceeds. The prospectus, which is prepared by the issuer and its advisers (including underwriters), must be approved by the relevant market authorities before the issuer releases it. Approval does not imply that the regulator endorses the prospectus or verifies its accuracy.

An equity underwriter assists the firm with raising capital during IPOs or secondary offerings, such as by determining the pricing of the offer, ensuring regulatory compliance, or guaranteeing the purchase of the entire offer.

Equity underwriters and the issuer's owners have an inherent conflict of interest in an IPO. Although underwriters may generate higher fees with a higher offering price, in the long run they may benefit more from placing underpriced shares with institutional investors. Doing so may increase the demand for the shares included in the offering by institutional investors and encourage investors to participate in future IPOs. This could be detrimental to the issuer's owners, who are most concerned with maximizing the value of the business. This conflict may contribute to IPOs often being mispriced such that they see significant gains on the first day of trading.

Less common forms of primary equity issuance include the following:

- | **Direct listing.** In a direct listing, a private company goes public by listing its existing shares on a stock exchange, but without creating any new shares or raising any new capital.
- | **Special purpose acquisition company (SPAC).** A SPAC is a shell company used to acquire an unspecified private company within a specific time frame (e.g., two years). A SPAC raises capital through an IPO and immediately places the net proceeds into a trust account. These funds must either be used within the specified time period to acquire a private company approved by shareholders (a **de-SPAC transaction**), or returned to investors.
- | **Back-door listing.** This involves a private firm acquiring a public company to take advantage of its existing listing.

In contrast to initial offerings, a firm that has previously issued equity securities may issue additional shares. This is referred to as a **seasoned equity offering** and requires a prospectus. These offerings may be dilutive or nondilutive to the ownership proportions of existing shareholders, depending on whether the shares being issued are new or already held by existing owners.

- | A **primary follow-on offering** involves the sale of new shares. This dilutes each existing share's proportional claim to the firm's net income and assets. Firms use this method to reduce debt or raise new equity capital.
- | In a **secondary follow-on offering**, privately held existing shares (typically held by founders, insiders, or private equity funds) are sold for the first time in the public market. This does not change existing shareholders' proportional claim to the firm's net income and assets. A nondilutive offering allows a firm's founders and other private investors to decrease their concentrated ownership position and diversify their holdings.

After shares have been issued, subsequent trades occur in the **secondary equity markets**. Secondary markets provide existing shareholders with liquidity and allow new investors to purchase an interest in a company. Secondary markets are active regardless of the direction of the market.

Observed market prices in the secondary market allow investors, analysts, and other stakeholders to assess periodic share returns, volatilities, and correlations across periods of time, and to gauge a company's performance. For issuers, share price information provides feedback to support decision-making, financing, and compensation. Firms wishing to raise additional equity capital or distribute

capital to shareholders will generally factor secondary market pricing and trading into their decision-making process. Firms that use **restricted stock units (RSUs)** and other share-based compensation use secondary market price changes to evaluate performance.

LOS 41.b: Compare exchange, off-exchange, and over-the-counter equities trading.

An **equity exchange** is a rules-based venue where market participants trade equity securities on a regular basis. Trading rules establish regularly scheduled trading hours and promote liquidity and transparency. An exchange-traded equity market structure consists of **broker-dealers**, financial intermediaries that work with investors to execute trades. Each broker-dealer routes orders through a centralized exchange **order matching system** that is used to pair buy and sell orders for shares.

Broker-dealers act either on behalf of clients as an agent, or as a principal counterparty to a financial transaction with a client. Most often, stock exchanges designate a single dealer, known as **designated market maker**, to ensure continuous trading of each listed security. The designated market maker is responsible for purchasing, holding, or selling shares to promote liquidity and price discovery during normal market hours and times of market imbalance or high volatility.

Securities dealers provide prices at which they will buy and sell shares. The **bid price** is the price at which a dealer will buy a security. The **ask price** or **offer price** is the price at which a dealer will sell a security.

The difference between the bid and ask prices is referred to as the **bid-ask spread** or **bid-offer spread** and is the source of a dealer's compensation. This spread represents the cost of purchasing and selling shares in the current market. The bid-ask spread at any given time is the difference between the lowest ask price and the highest bid price.

A bid-ask spread may be expressed as a percentage of the lowest ask price:

$$\text{bid-ask spread (\%)} = \frac{(\text{lowest ask price} - \text{highest bid price})}{\text{lowest ask price}}$$



PROFESSOR'S NOTE

From an investor's perspective, remember that you always get the side of a bid-ask spread that is *worse for you*. If you are buying a security, you must pay the higher price (the ask). If you are selling, you only get the lower price (the bid).

This “worse for you” rule is especially useful with foreign exchange transactions. As you will see in the Level II curriculum, whether the bid or ask price applies depends on which way around an exchange rate is quoted. Nevertheless, the bid or ask price you get is always the one that gives you *less* of the currency you are acquiring.

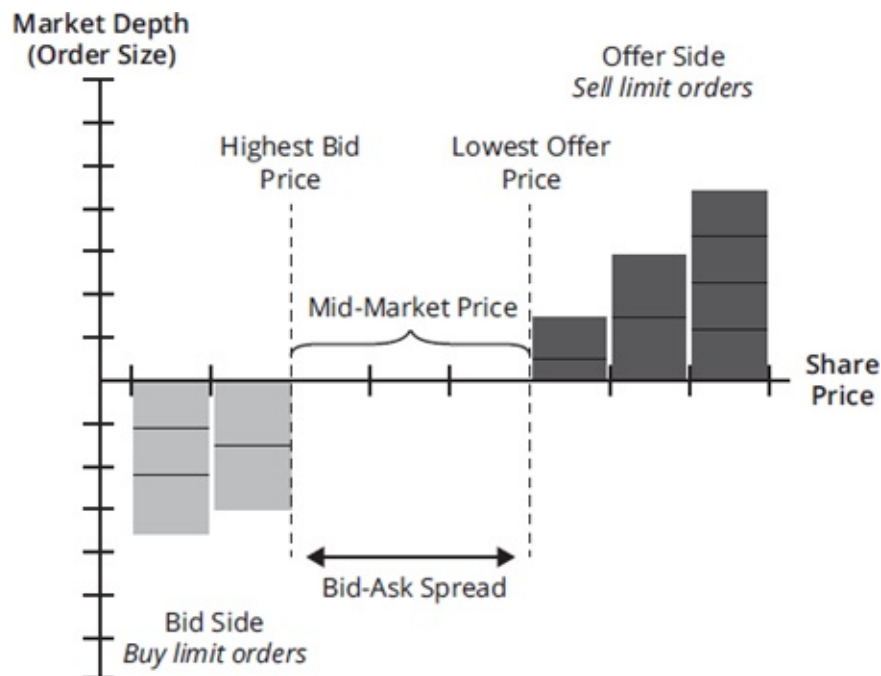
The most common orders, in terms of execution instructions, are market or limit orders. A **market order** instructs the broker to execute the trade immediately at the best available price. A **limit order** places a minimum execution price on sell orders or a maximum execution price on buy orders. For example, a buy order with a limit of \$6 will be executed immediately if the market ask price is \$6 or less, but if the market price is greater than \$6, the order will remain unexecuted unless the market price decreases to \$6 or less.

Market depth is the relative size of orders posted at or near the best available bid and offer prices or the **mid-market price**, the average of the lowest ask and highest bid:

$$\text{mid-market price} = \frac{\text{lowest ask price} + \text{highest bid price}}{2}$$

Figure 41.2 shows market depth as a function of market price, with limit buy orders at low prices and limit sell orders at high prices.

Figure 41.2: Equity Limit Order Book¹



Market participants may track intraday share performance using a **volume-weighted average price (VWAP)**. This measure is derived by weighting the trade price by the volume of shares traded over a specified period:

$$\text{VWAP} = \frac{\sum_{i=1}^N (\text{price}_i \times \text{volume}_i)}{\sum_{i=1}^n \text{volume}_i}$$

Off-exchange trading usually involves either illiquid equities with low trading volumes, or large orders known as **block trades**. Block trades are usually defined as orders of 10,000 shares or more.

Off-exchange trading looks to minimize the **implicit costs of trading**. These are the indirect and unobservable costs resulting from the market impact of trading and arise from the following sources:

- | **Market impact.** This is the effect on transaction prices of executing a large trade immediately.
- | **Delay costs (slippage).** This is the inability to complete a desired trade immediately.
- | **Opportunity costs.** This is unrealized profit that results from a failure to execute a trade quickly.

Over-the-counter (OTC) trading, or **quote-driven trading**, involves a network of broker-dealers who act as market makers, displaying bid and ask prices. The prices they display are firm quotes, meaning they must execute orders received at those prices. Each broker-dealer works with others to route trades by executing offsetting buy or sell orders. OTC markets offer less transparency than exchange-traded markets. Traders may or may not be executing trades at the best available price.

Dealers usually maintain an inventory of securities. Quote-driven markets are sometimes called dealer markets, price-driven markets, or OTC markets. Most securities other than stocks trade in quote-driven markets, and trading often takes place electronically.



MODULE QUIZ 41.1

1. New issues of securities are transactions in the:
 - A. primary market.
 - B. secondary market.
 - C. seasoned market.
2. Trading rules are *least likely* used to:
 - A. promote liquidity.
 - B. provide transparency.
 - C. establish off-hours trading.
3. The price at which a dealer will buy a security is referred to as the:
 - A. bid price.
 - B. ask price.
 - C. mid-market price.

MODULE 41.2: EQUITY INDEXES AND LIQUIDITY MEASURES

LOS 41.c: Describe liquidity measures for a publicly listed security and calculate float and average daily volume.

A firm's **market float** is the total value of the shares that are available to the investing public. This excludes the value of shares held by controlling stockholders because they are unlikely to sell their shares.

$$\text{market float} = \text{total shares outstanding} - \text{restricted shares or shares not available for trading}$$

A stock's **average daily volume** represents its average number of shares traded in a day, measured over a given period:

$$\text{average daily volume} = \frac{\sum_{i=1}^n (\text{shares traded per day})_i}{n}$$

The **turnover ratio** is the ratio of average daily volume to market float and indicates the proportion of outstanding shares that are exchanged on a given trading day:

$$\text{turnover ratio} = \frac{\text{average daily volume}}{\text{market float}}$$

These metrics allow investors to gauge their ability to liquidate positions based on market activity, as the following example illustrates.

EXAMPLE: Liquidating an equity position

Southern Pension Plan owns 600,000 shares of Neptune Corporation. The average daily volume in Neptune over the last month has been 2,250,000 shares. Southern's internal trading guidelines prohibit the fund from buying or selling more than 6% of a stock's average daily trading volume on any given day. Southern's chief investment officer has instructed the plan's trading staff to fully liquidate the plan's holdings in Neptune shares. Calculate the number of days necessary to complete the liquidation.

Answer:

Southern's current position represents $\frac{600,000}{2,250,000} = 26.67\%$ of Neptune's average daily trading volume. Given the plan's 6% constraint, Southern will require $\frac{26.67\%}{6\%} = 4.44$ days. Thus, Southern cannot fully liquidate its Neptune shares in four days, but it may complete the sale within five days.

Market depth and breadth indicate the ability of a market to execute trades without having a significant unfavorable impact on price. Secondary equity market liquidity measures include the following:

- ▮ **Market depth.** This is the relative size of orders posted at or near the best available bid and offer prices.
- ▮ **Market breadth.** This measures the number of orders at different price levels both above and below, but close to, the best bid and offer prices.

- | **Market resiliency.** This is the ability of an issuer's shares to maintain a stable **market-clearing price** at which all buy and sell orders can be executed. In a more resilient market, the stock price will adjust more quickly to a new market-clearing price following a shock.
-

LOS 41.d: Describe types of equity indexes.

A security market index is used to represent the performance of an asset class, security market, or segment of a market. They are usually created as portfolios of individual securities, which are referred to as the **constituent securities** of the index. An index has a numerical value that is calculated from the market prices (actual when available or estimated) of its constituent securities at a point in time.

Investors can use various equity market indexes when evaluating performance and benchmarking investment and investment managers. These equity indexes can be classified as follows:

- | *Broad-based equity indexes* include the largest listed firms adjusted by market float and calculated on a market-capitalization-weighted basis. They may be tiered based on issuer size. A **composite market index** measures a market's overall performance and usually contains more than 90% of a market's total value. For example, the Wilshire 5000 Index contains more than 6,000 equity securities and is, therefore, a good representation of the overall performance of the US equity market.
- | *Sector indexes* measure the returns for an industry sector, such as health care, financial, or consumer goods firms. Investors can use these indexes in cyclical analysis because some sectors perform better than others in various phases of the business cycle. Sector indexes can be for a particular country or global. These indexes are used to evaluate portfolio managers and construct index portfolios.
- | *Investment theme indexes* can be created based on investor trends beyond selecting a certain industry or company size. For example, an index may contain issuers that have been screened to comply with environmental, social, and governance (ESG) principles (e.g., S&P Europe ESG 350 Index).
- | An example of *indexes with alternative weighting methods* is an equally weighted index based on the same companies included in a broad-based index. The goal is to reduce the biases of the greater weighting given to the largest

companies. **Factor-based investing**, which is investment selection based on fundamental factors that may drive performance, has also been a driver of indexes based on alternative weighting (e.g., MSCI Quality Indexes).

Multi-market indexes are typically constructed from the indexes of markets in several countries and used to measure the equity returns of a geographic region (e.g., Latin America indexes), markets based on their stage of economic development (e.g., emerging markets indexes), or the entire world (e.g., MSCI World Index).



MODULE QUIZ 41.2

1. Market float of a stock is *best* described as its:
 - A. total outstanding shares.
 - B. shares that are available to domestic investors.
 - C. outstanding shares, excluding those held by controlling shareholders.
2. The relative size of orders posted at or near the best available bid and offer prices is best described as a market's:
 - A. depth.
 - B. breadth.
 - C. resiliency.
3. A public company has 10.5 million shares outstanding and 2.0 million restricted shares. The firm's market float is:
 - A. 8.5 million shares.
 - B. 10.5 million shares.
 - C. 12.5 million shares.
4. A security market index is *best* described as a:
 - A. value used to adjust nominal security prices for the effects of inflation.
 - B. directory of ticker symbols for the securities listed on a given market.
 - C. group of securities selected to represent the performance of a security market.

KEY CONCEPTS

LOS 41.a

Primary equity markets refer to the issuance of new securities to the public. The IPO process involves fulfilling requirements such as an equity prospectus, and soliciting investors for the new publicly listed shares.

Secondary equity markets are where trades occur after their initial issuance. Observed market prices in the secondary market allow investors to assess returns and gauge company performance.

LOS 41.b

An equity exchange is a rules-based venue where market participants trade equity securities on a regular basis.

Off-exchange trading usually involves block trades (usually defined as an order of 10,000 shares or more) or equities that are illiquid or exhibit low trading volumes.

Over-the-counter (OTC) equity trading involves a network of broker-dealers who buy and sell shares among themselves. OTC markets offer less transparency than exchange-traded markets.

LOS 41.c

A firm's market float is the total value of the shares available to the investing public, excluding shares held by controlling stockholders.

Average daily volume is the average number of issuer shares traded each day over a given period.

The turnover ratio is the ratio of average daily volume to market float and indicates the proportion of outstanding shares exchanged on a given trading day.

Other equity market liquidity measures include market depth, market breadth, and market resiliency.

LOS 41.d

A security market index is used to represent the performance of an asset class, security market, or segment of a market. Equity indexes can be classified as broad based, sector, investment theme, alternatively weighted, and multi-market.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 41.1

- . A The primary market refers to the market for newly issued securities. (LOS 41.a)
- . C Trading rules are used to promote liquidity, provide transparency, and establish regularly scheduled trading hours. (LOS 41.b)
- . A A dealer will buy a security at the bid price or will sell a security at the ask price. (LOS 41.b)

Module Quiz 41.2

- . **C** Market float represents shares available to the investing public and excludes shares held by controlling shareholders. (LOS 41.c)
- . **A** Market depth is the relative size of orders posted at or near the best available bid and offer prices. (LOS 41.c)
- . **A** Market float = total shares outstanding – restricted shares. With 10.5 million shares outstanding and 2.0 million restricted shares, the firm's market float is 8.5 million shares. (LOS 41.c)
- . **C** A security market index is a group of securities (constituent securities) designed to represent the performance of an asset class, security market, or market segment. (LOS 41.d)

¹ Reproduced from the Level I CFA curriculum learning module, "Equity Issuance and Trading," with permission from CFA Institute.

READING 42

SOURCES OF EQUITY RETURNS

MODULE 42.1: DIVIDENDS, STOCK SPLITS, AND SHARE REPURCHASES

LOS 42.a: Describe features and uses of dividends, share repurchases, stock splits, and reverse stock splits.

Cash dividends, as the name implies, are payments made to shareholders in cash, and they are subject to income tax in the period in which the cash flow is received. Payment of dividends is discretionary and decided by a company's board of directors. Mature companies with relatively few growth opportunities are more likely to distribute cash to investors as dividends. Companies with significant growth opportunities typically retain cash for reinvestment and are, therefore, less likely to pay a dividend.

Dividends may be regularly scheduled or one-time special dividends. **Regular cash dividends** occur when a company pays out a portion of profits on a consistent schedule (e.g., quarterly). A long-term record of stable or increasing dividends is widely viewed by investors as a sign of a company's financial stability. Companies often choose to maintain a constant dividend per share over time to avoid adverse signaling.

Special cash dividends (or **extraordinary cash dividends**) are used when favorable circumstances allow the firm to make a one-time cash payment to shareholders, in addition to any regular dividends the firm pays. Many cyclical firms (e.g., automakers) will use a special dividend to share profits with shareholders when times are good but maintain the flexibility to conserve cash when profits are poor.

A company that ceases to operate and has positive net assets may issue a **liquidating dividend**. Unlike other dividends, which are distributions of

earnings, a liquidating dividend may be viewed more as a return of capital to shareholders.

A **share repurchase** is a transaction in which a company buys outstanding shares of its own common stock. Share repurchases are an alternative to cash dividends as a way of distributing cash to shareholders, and they have the same effect on total shareholders' wealth as cash dividends of the same size. However, with a share repurchase, it is up to each shareholder to decide whether to sell back their shares and receive cash.

A company might repurchase shares to support their price or to signal that management believes the shares are undervalued. Companies might repurchase shares to offset an increase in outstanding shares from the exercise of employee stock options, or to increase reported earnings per share by decreasing the number of shares outstanding. Companies that want to preserve cash to invest in growth opportunities when they arise may favor share repurchases over regular cash dividends. In countries that tax capital gains at lower rates than dividends, shareholders may prefer share repurchases to dividend payments.

Stock dividends (a.k.a. **bonus issues** of shares) are dividends paid out in new shares of stock rather than cash. This results in more shares outstanding, but each one is worth less, such that total shareholders' equity remains unchanged. Stock dividends are commonly expressed as a percentage. A 20% stock dividend means every shareholder gets 20% more stock.

Stock splits divide each existing share into multiple shares. As with a stock dividend, this results in more shares, but the price of each share will drop correspondingly to the number of shares created, so there is no change in the owner's wealth or proportional ownership. A stock split that results in more shares is also known as a **forward split**, to distinguish it from a reverse stock split.

Reverse stock splits are the opposite of forward stock splits. After a reverse split, fewer shares are outstanding, but they have a higher price. Because these factors offset one another, shareholder wealth is unchanged. Some stock exchanges require shares to have a minimum price, and a distressed company whose shares are approaching the minimum price may carry out a reverse stock split to increase their price and keep its shares listed.

Splits are expressed as a ratio. For example, in a 3-for-1 stock split, each old share is split into three new shares. In a 1-for-5 reverse stock split, five old

shares become one new share.

Stock splits are currently more common than stock dividends. A company with a high share price may consider a stock split to make their shares more affordable for retail investors, increasing liquidity and broadening the shareholder base.

LOS 42.b: Describe dividend payment chronology.

The dates relevant to dividend payments are shown in Figure 42.1.

Figure 42.1: Dividend Payment Chronology



Here is the breakdown:

- ▮ **Declaration date.** This is the date that the board of directors approves payment of a dividend, specifying the per-share dividend amount, the date shareholders must own the stock to receive the dividend (record date), and the date the dividend payment will be made (payment date).
- ▮ **Ex-dividend date.** This is the first day on which a share purchaser will not receive the next dividend. The ex-dividend date is one or two business days before the record date, depending on the settlement period for stock purchases. If you buy the share on or after the ex-dividend date, you will not receive the dividend.
- ▮ **Record date.** This is the date on which all owners of shares become entitled to receive the dividend payment on their shares. (The record date for dividends is different than the record date for proxy voting that we discussed in an earlier reading.)
- ▮ **Payment date.** This is the date that dividend checks are mailed to, or payment is made electronically to, the shareholders who are entitled to receive the dividend.

On the ex-dividend date, the share price would decrease from the previous day's closing price by the amount of the dividend, in the absence of other market-related and company-specific factors that affect the stock price.

EXAMPLE: Price change on ex-dividend date

Consider shares that are trading at \$25 on the day before the ex-dividend date and will pay a \$1 dividend. On the ex-dividend date, the shares close at \$24.30. A broad market index increased 0.5% on that day. Assuming the company's risk level is the same as that of the overall market, calculate the percentage change in the share price on the ex-dividend date that is attributable to company-specific factors other than the dividend.

Answer:

Other things equal, the \$1 dividend would have resulted in a \$1 decrease in the share price, or in percentage terms:

$$\frac{\$25 - \$1}{\$25} - 1 = -4.0\%$$

The actual percentage change in price was: $\frac{\$24.30}{\$25.00} - 1 = -2.8\%$

Therefore, the percentage change in price that was not caused by the dividend is $-2.8\% - (-4.0\%) = 1.2\%$.

If we assume the company's risk level is the same as that of the overall market, we would expect the stock to have increased 0.5% on the day in the absence of the dividend and any company-specific factors. Therefore, the percentage change we can attribute to company-specific factors is $1.2\% - 0.5\% = 0.7\%$.



MODULE QUIZ 42.1

1. A company is evaluating the likely effects on its share price of declaring a 50% stock dividend or a 3-for-2 stock split. Other things equal, which of these will result in a lower share price?
 - A. A 3-for-2 stock split will result in a lower share price.
 - B. A 50% stock dividend will result in a lower share price.
 - C. Both should have the same effect.
2. The first date on which the purchaser of a stock will not receive a dividend that has been declared is the:
 - A. declaration date.
 - B. ex-dividend date.
 - C. holder-of-record date.

MODULE 42.2: RETURN CALCULATIONS FOR EQUITIES

LOS 42.c: Calculate price and total return for equity securities.

A rate of return on an equity security that is calculated based only on a price increase or decrease (adjusted for stock splits) is referred to as its **price return**. A rate of return that is calculated based on both price changes and income (i.e., cash dividends) is called its **total return**.

EXAMPLE: Price return and total return

An investor buys Stock Q for \$42 and later sells it for \$50. The investor receives a \$2 cash dividend just before selling the stock. Calculate the investor's price return and total return for the holding period.

Answer:

$$\text{Price return} = \frac{\$50 - \$42}{\$42} = 19.05\%$$

$$\text{Total return} = \frac{\$50 - \$42 + \$2}{\$42} = 23.81\%$$



PROFESSOR'S NOTE

This example did not specify the length of the holding period. As we saw in Quantitative Methods, we ordinarily take the additional step of annualizing a holding period return. Read exam questions carefully to be sure whether they are looking for an annualized rate of return or simply a holding period return.

EXAMPLE: Effect of a stock split on equity returns

Continuing the previous example, assume the issuer of Stock Q splits the stock 2-for-1 at the same time it pays the cash dividend. Calculate the investor's price return and total return for the holding period.

Answer:

On a per-share basis, we can adjust the beginning price, the ending price, and the dividend for the stock split:

$$\text{Price return} = \frac{\frac{\$50}{2} - \frac{\$42}{2}}{\frac{\$42}{2}} = 19.05\%$$

$$\text{Total return} = \frac{\frac{\$50}{2} - \frac{\$42}{2} + \frac{\$2}{2}}{\frac{\$42}{2}} = 23.81\%$$

After the stock split, the investor owns two shares that each pay a \$1 cash dividend, so the investor still receives \$2. The stock split does not change the investor's rate of return. This would also be the case for a stock dividend or a reverse stock split.

In the previous examples, we assumed the investor received a cash dividend just before selling the stock. If a cash dividend occurs earlier in the holding period, an investor can earn additional income by investing the cash received.

Reinvested dividends refer to the investment strategy of using cash dividends to buy more shares of the same stock. Alternatively, an investor can use cash dividends to pursue different investments.

If a shareholder receives a cash dividend during the holding period and reinvests it in the same stock, the share price at that time is likely to be different than it was at the beginning of the period. We can state the additional **fractional shares** the investor buys, per share originally owned, as D / P_{PD} , where D is the dividend per share and PPD is the price per share on the dividend payment date. We can express the holding period return with reinvested dividends as follows:

$$r_E = \frac{P_1 \left(1 + \frac{D}{P_{PE}} \right) - P_0}{P_0}$$

where:

P_1 = share price at the end of the period

P_0 = share price at the beginning of the period

P_{PE} = share price on the dividend payment date

D = dividend per share

If the shareholder invests a cash dividend in something other than the stock that paid the dividend, we can account for the return on that investment explicitly, as the following example shows.

EXAMPLE: Return including investment of dividends received

An investor buys Stock Q for \$42 and later sells it for \$50. The investor receives a \$2 cash dividend halfway through the holding period. Calculate the investor's holding period return if the following occur:

1. The shareholder reinvests the dividend in Stock Q at a price of \$46.
2. The shareholder deposits the dividend in a bank account that earns a 3% interest rate over the holding period.

Answer:

1.
$$r_E = \frac{\$50 \left(1 + \frac{\$2}{\$46}\right) - \$42}{\$42} = 24.22\%$$
2. The shareholder will earn 3% interest on the cash dividend, but only over the second half of the holding period, so the value of the dividend plus interest is $\$2 \times (1.03)^{1/2} = \2.03 .
$$r_E = \frac{\$50 - \$42 + \$2.03}{\$42} = 23.88\%$$

Equity Price as a Present Value of Future Cash Flows

We state the total return on a non-dividend-paying stock as follows:

$$r_E = \frac{P_1 - P_0}{P_0}, \text{ or } r_E = \frac{P_1}{P_0} - 1$$

We then can rearrange the latter expression to solve for P_0 , the present value of the share price:

$$1 + r_E = \frac{P_1}{P_0}$$

$$P_0(1 + r_E) = P_1$$

$$P_0 = \frac{P_1}{1 + r_E}$$

Because we do not know at the beginning of the holding period (time = 0) what the share price will be at the end of the holding period (time = 1), it is more appropriate to express this in terms of an *expected* value for the share price:

$$P_0 = \frac{E(P_1)}{1 + r_E}$$

If the stock is expected to pay a dividend at the end of the holding period, we can include the expected value of that dividend in the numerator:

$$P_0 = \frac{E(P_1) + E(D_1)}{1 + r_E}$$

These expressions form the basis of *discounted cash flow models* for valuing equity securities, which we will examine in the next readings. The basic concept is that the value of an equity security today is the present value of its expected future cash flows, discounted at investors' required rate of return on the equity security.



MODULE QUIZ 42.2

1. An investor buys Stock E for \$85. The stock pays a cash dividend of \$3, which the investor reinvests in Stock E for \$84. At the end of the period, the share price of Stock E is \$83. The investor's holding period return is *closest* to:
 - A. -2.4%.
 - B. 0.0%.
 - C. 1.1%.
2. Stock W does not pay a cash dividend but declares and pays a 10% stock dividend. For an investor who owns Stock W, this stock dividend will *most likely*:
 - A. increase the investor's holding period return.
 - B. decrease the investor's holding period return.
 - C. have no effect on the investor's holding period return.

KEY CONCEPTS

LOS 42.a

Regular cash dividends are paid at set intervals. A special or extraordinary cash dividend is a one-time cash payment to shareholders.

Stock dividends are additional shares of stock. Stock splits divide each existing share into multiple shares. In either case, the value of each share will decrease because the total value of outstanding shares is unchanged. The portion of the company owned by each shareholder is also unchanged.

In a reverse stock split, the number of shares owned by each shareholder is decreased, so total shares outstanding are decreased and the value of a single share is increased.

A share repurchase is a purchase by a company of its outstanding shares. Share repurchases are an alternative to cash dividends as a way to distribute cash to shareholders.

LOS 42.b

Dividend payment chronology:

- | The *declaration date* is the date that the board of directors approves payment of the dividend.
- | The *ex-dividend date* is the first day that a share of stock trades without the dividend, one or two business days before the record date. On the ex-dividend date, the value of each share would decrease by the amount of the dividend, in the absence of any other factors that affect the share price.

- ▮ The *record date* is the date on which share owners who will receive the dividend are identified.
- ▮ The *payment date* is the date that the dividend checks are sent to, or payment is transferred to, shareholders.

LOS 42.c

Price return is the rate of return on an equity security based only on its price increase or decrease. Total return is the rate of return based on both price changes and dividends.

$$\text{holding period return with dividend reinvestment} = \frac{P_1 \left(1 + \frac{D}{P_{PE}} \right) - P_0}{P_0}$$

where:

P_1 = share price at the end of the period

P_0 = share price at the beginning of the period

P_{PE} = share price on the dividend payment date

D = dividend per share

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 42.1

- ▮ **C** Both a 50% stock dividend and a 3-for-2 stock split will increase the number of shares by 50%, while neither will affect value of the company. Therefore, the decrease in the share price should be the same in either case. (LOS 42.a)
- ▮ **B** The chronology of a dividend payout is declaration date, ex-dividend date, record date, and payment date. The ex-dividend date is the cutoff date for receiving the dividend: investors who purchase the stock on or after the ex-dividend date will not receive the dividend. (LOS 42.b)

Module Quiz 42.2

$$\text{▮ C } r_E = \frac{\$83 \left(1 + \frac{\$3}{\$84} \right) - \$85}{\$85} = 1.13\%$$

(LOS 42.c)

- ▮ **C** Stock dividends, stock splits, and reverse stock splits have no effect on the rate of return of an equity security. (LOS 42.c)

READING 43

INTRODUCTION TO EQUITY VALUATION

MODULE 43.1: MEASURES AND MODELS OF EQUITY VALUE

LOS 43.a: Contrast price and value for an equity security.

Intrinsic value or **fundamental value** is the rational value investors would place on a stock if they had full knowledge of its characteristics. Unlike its market price, a stock's intrinsic value is not directly observable. Analysts use valuation models to estimate the intrinsic values of stocks.

Individual investors, based on their assumptions about the future, arrive at their own **perceived value** for a stock. Market prices incorporate perceived values of active market participants. If an investor's perceived value is higher than the current market price, the investor considers the stock *undervalued* by the market; if the investor's perceived value is lower than the stock's market price, the investor believes the stock is *overvalued* by the market. If the perceived value equals the market price, the investor views the stock as *fairly valued*. Investors will tend to sell or reduce the portfolio weights of stocks they perceive to be overvalued and buy or increase the portfolio weights of stocks they see as undervalued.

Investors use the following three approaches to determine perceived value:

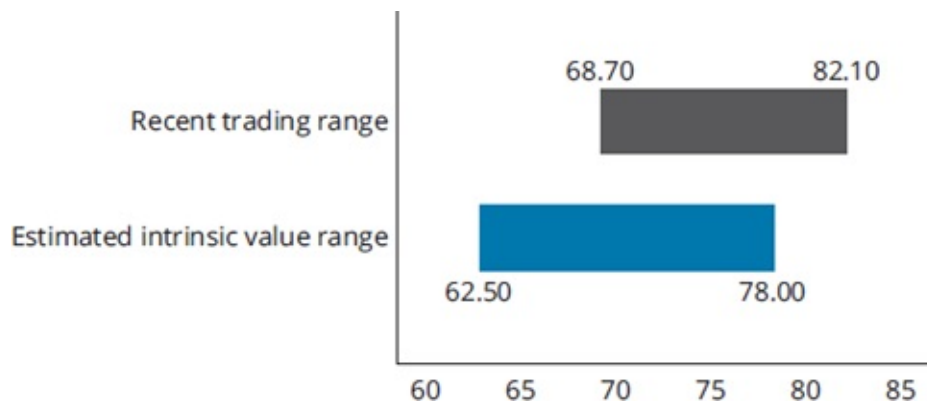
Naïve or story-based valuation is an informal approach that does not link valuation to company finances or fundamentals. Investors often use naïve or story-based valuation for **memestocks** that derive their (baseless) value from social media activity, or **cult companies** that have a dedicated following of the company's founders or products.

Relative valuation relies on comparing a company's fundamentals to a peer group of similar companies. Relative valuation is based on the idea that companies with similar characteristics should trade at similar price multiples.

Absolute or fundamental valuation evaluates a company based on only its own characteristics and fundamentals, without regard to a benchmark. One fundamental approach is to value the stock as the present value of future cash flows available to the investor, discounted at an appropriate risk-adjusted discount rate.

Analysts often estimate a stock's intrinsic value as a range rather than a specific price. A stock's recent low and high prices determine its market price range. Intrinsic value ranges are derived from different estimates of inputs into an analyst's valuation model, or they are based on value estimates from different models. Graphically, we can visualize a side-by-side representation of these ranges in a **floating bar valuation chart**, as shown in Figure 43.1.

Figure 43.1: Floating Bar Valuation for XYZ, Inc.



A **scenario analysis** estimates a base case value for a stock as well as a bullish (more favorable) and a bearish (less favorable) case value. The analyst's expected value for the stock is then a probability-weighted average of these estimated values.

LOS 43.b: Contrast the use of book value of equity, market capitalization, and enterprise value as indicators of intrinsic value.

Book Value of Equity

The **book value of equity** is the value of the firm's assets on the balance sheet minus its liabilities and minority interest. It increases when the firm has positive

net income and retained earnings that flow into shareholders' equity. When management makes decisions that increase income and retained earnings, they increase the book value of equity.

Analysts often evaluate book value of equity on a per-share basis. **Book value per share (BVPS)** is the book value of equity divided by the number of shares outstanding.

Assets of the firm can be tangible (e.g., equipment and inventory) or intangible (e.g., patents, copyrights, and balance sheet goodwill). The book value of tangible property on the balance sheet depends on how recently it was acquired and on the depreciation method chosen for long-lived assets. Intangible property that is internally generated is usually expensed, so it does not appear on the balance sheet. Purchased intangible assets are amortized over their useful lives or subject to impairment testing. Recall from Financial Statement Analysis that goodwill arises when one firm acquires another at a price exceeding the acquired firm's fair value of net identifiable assets.

Differences in accounting methods across companies affect their assets' balance sheet carrying values and, hence, their BVPS.

As a valuation metric, book value is more appropriate for firms with substantial tangible assets. It is less appropriate for early-stage firms whose balance sheet values do not reflect future earnings potential, or for firms with significant intangible assets.

Another approach to estimating book value is to add the components of stockholders' equity, excluding minority or noncontrolling interest:

$$\text{book value} = \text{share capital} + \text{additional paid-in capital} + \text{retained earnings} \\ - \text{treasury stock}$$

EXAMPLE: Book value per share

An analyst has collected the following information from the balance sheet of Apoka Pharmaceuticals:

	20X6 (\$, thousands)
Common stock	1,220
Additional paid-in capital	3,450
Retained earnings	460
Treasury stock	(1,335)
Minority interest	<u>340</u>
Stockholders' equity	4,135

Apoka has 500,000 shares outstanding. Calculate Apoka's book value and book value per share.

Answer:

$$\begin{aligned}\text{Book value} &= \text{stockholders' equity} - \text{minority interest} \\ &= 4,135 - 340 = 3,795\end{aligned}$$

$$\begin{aligned}\text{Or, book value} &= \text{common stock} + \text{additional paid-in capital} \\ &\quad + \text{retained earnings} - \text{treasury stock} \\ &= 1,220 + 3,450 + 460 - 1,335 \\ &= 3,795\end{aligned}$$

$$\text{BVPS} = 3,795,000 / 500,000 = \$7.59$$

A company has a negative book value of equity if its assets are less than its liabilities. This is often the case with firms that are in financial distress or bankruptcy. However, a negative book value can also occur with profitable and valuable companies, such as those in a rapid growth stage or companies that have an aggressive stock buyback program. Repurchases of stock increase treasury stock (which is a negative value within stockholders' equity); thus, they may turn the book value negative.

Market Capitalization

Market capitalization is the total value of a firm's outstanding equity shares based on market prices and reflects investors' expectations about the firm's future performance.

$$\text{market capitalization} = \text{share price} \times \text{number of shares outstanding}$$

While book value is based on balance sheet carrying values (e.g., depreciated cost for tangible long-lived assets), market capitalization reflects the fair value of assets. Further, market capitalization is based on a company's value as a going concern, as opposed to the prospect of selling its assets piecemeal. In other

words, market capitalization includes value attributable to investor expectations about a company's future earnings.

When market capitalization is higher than book value, we may consider the difference to be excess value that the firm is creating through its operations. If market capitalization is lower than book value, an analyst should investigate the shortfall to determine whether it represents an undervalued stock or an uncompetitive company.

A relative valuation metric used by analysts is the **price-to-book ratio (P/B)**:

$$\begin{aligned} \text{P/B} &= \text{market price of stock} / \text{BVPS} \\ &= \text{market capitalization} / \text{book value} \end{aligned}$$

Firms with excess value have a P/B greater than one, and firms with a value shortfall have a P/B less than one.

P/B is used as a metric for valuation relative to a benchmark, such as the average P/B of a peer group. Assuming similar characteristics, a firm's P/B should be close to the benchmark. Absent justifiable reasons, if a stock trades at a P/B significantly higher than the benchmark, the stock would be considered relatively overvalued; and if it trades at a P/B significantly lower than its benchmark, the stock is relatively undervalued.

EXAMPLE: Calculating P/B

Based on the information in the following figure, calculate the P/B for Crisco Systems, Inc., and Soothsayer Corp. as of the end of 20X3.

Data for Crisco Systems, Inc., and Soothsayer Corp.

Company	Book Value of Equity 20X3 (millions of \$)	Shares Outstanding 20X3 (millions)	Price FYE 20X3 (\$)
Crisco Systems, Inc.	\$28,039	7,001	\$17.83
Soothsayer Corp.	\$6,320	5,233	\$12.15

Answer:

Crisco Systems, Inc.:

$$\begin{aligned}\text{book value per share} &= \frac{\text{book value of equity}}{\text{number of shares outstanding}} \\ &= \frac{\$28,039}{7,001} = \$4.01\end{aligned}$$

$$P/B = \frac{\text{market price per share}}{\text{book value per share}} = \frac{\$17.83}{\$4.01} = 4.45 \text{ times}$$

Soothsayer Corp.:

$$\begin{aligned}\text{book value per share} &= \frac{\text{book value of equity}}{\text{number of shares outstanding}} \\ &= \frac{\$6,320}{5,233} = \$1.21\end{aligned}$$

$$P/B = \frac{\text{market price per share}}{\text{book value per share}} = \frac{\$12.15}{\$1.21} = 10.04 \text{ times}$$

Enterprise Value

Enterprise value (EV) measures total company value. EV represents the cost to acquire a firm.

$$EV = \text{market value of common and preferred stock} + \text{net debt}$$

$$EV = \text{market value of common and preferred stock} + \text{market value of debt} - \text{cash and short-term investments}$$

Net debt is the market value of a company's debt minus its cash and short-term investments. Cash and short-term investments are subtracted because an acquirer's cost for a firm would be decreased by the amount of the target's liquid assets. Although an acquirer assumes the firm's debt, it also receives the firm's cash and short-term investments.

EV is appropriate when an analyst wants to compare the values of firms that have significant differences in capital structure. If the companies are located in different countries, an analyst must convert their EVs into a common currency before comparing them.

EBITDA (earnings before interest, taxes, depreciation, and amortization have been subtracted) is commonly used as the denominator for EV multiples (EBIT or revenue can also be used).

EXAMPLE: Calculating EV/EBITDA multiples

Daniel, Inc., is a manufacturer of small refrigerators and other appliances. The following figures are from Daniel's most recent financial statements except for the market value of long-term debt, which has been estimated from financial market data.

Stock price	\$40.00
Shares outstanding	200,000
Market value of long-term debt	\$600,000
Book value of long-term debt	\$900,000
Book value of total debt	\$2,100,000
Cash and marketable securities	\$250,000
EBITDA	\$1,000,000

Calculate the EV/EBITDA multiple.

Answer:

First, we must estimate the market value of the firm's short-term debt and liabilities. To do so, subtract the book value of long-term debt from the book value of total debt: $\$2,100,000 - \$900,000 = \$1,200,000$. This is the book value of the firm's short-term debt. If we assume the company is not in financial distress, the market value of these short-term items should be close to their book value. (As we will see in the Fixed Income topic area, the market values of debt instruments approach their face values as they get close to maturity.)

Add the market value of long-term debt to get the market value of total debt: $\$600,000 + \$1,200,000 = \$1,800,000$.

The market value of equity is the stock price multiplied by the number of shares: $\$40 \times 200,000 = \$8,000,000$.

The EV of the firm is the sum of debt and equity minus cash: $\$1,800,000 + \$8,000,000 - \$250,000 = \$9,550,000$. So, $\text{EV/EBITDA} = \$9,550,000 / \$1,000,000 \approx 9.6$.

If the competitor or industry average EV/EBITDA is above 9.6, Daniel is relatively undervalued. If the competitor or industry average EV/EBITDA is below 9.6, Daniel is relatively overvalued.

LOS 43.c: Describe major categories of equity valuation models and explain their advantages and disadvantages.

Analysts use various models to estimate the value of equities. Usually, an analyst will use more than one model with several different sets of inputs to determine a range of possible stock values. The two basic approaches to valuation are absolute valuation and relative valuation.

Absolute Valuation Approaches

Absolute valuation uses only company-specific information. Two absolute valuation models are present value models and an asset-based approach.

In **present value models** or **discounted cash flow models**, a stock's value is estimated as the present value of cash distributed to shareholders (dividend discount models) or the present value of cash available to shareholders after the firm meets its necessary capital expenditures and working capital expenses (free cash flow to equity models). Alternatively, valuation of the whole firm (value of equity plus value of debt) is calculated as the present value of free cash flow to the firm.

Another discounted cash flow approach is based on **residual income** valuation, where the current book value is added to the present value of future residual income. Residual income is net income less an equity charge that is based on the book value at the beginning of the year.

The discount rate used for dividend, FCFE, and residual income models is the required return on equity, while the discount rate for FCFF models is the weighted average cost of capital.



PROFESSOR'S NOTE

In our next readings, we will examine all these models in detail.

While present value models are based on sound fundamental theory, the value estimates are highly sensitive to the estimates of their inputs. For mature firms with stable and predictable dividend policies, using dividend discount models makes sense. FCFE models are appropriate for non-dividend-paying firms with stable capital structures. When a firm has a negative FCFE or when the capital structure is not stable (e.g., high reliance on debt or significant stock repurchases), FCFF models are more appropriate. Residual income models are useful for firms that do not pay dividends or have negative free cash flow.

Advantages of discounted cash flow models are as follows:

- | They are based on the fundamental concept of discounted present value and are well grounded in finance theory.
- | They allow analysts to change assumptions, such as revenue growth and margins, to evaluate their impact on valuation.

Disadvantages are as follows:

- | Inputs must be estimated.
- | Value estimates are highly sensitive to input values.

The second type of absolute valuation approach is an **asset-based model**. In an asset-based model, an analyst estimates the intrinsic value of a company's common stock as the company's assets minus its liabilities and preferred stock. Analysts typically adjust the book values of the assets and liabilities to their fair values.

Asset-based models are suitable for companies with assets whose market values are readily available. For example, with resource-rich firms (e.g., mining companies with known proven deposits), the value of the firm's assets can be determined readily. Some firms may have real estate holdings with values that exceed their cash flow generating potential, and an asset-based valuation would need to incorporate that information.

Advantages of asset-based models are as follows:

- | They can provide floor values for a stock or a company.
- | They are useful for valuing divisions that are being sold or liquidated.
- | They are most reliable when the firm has primarily tangible short-term assets, assets with ready market values, or when the firm is being liquidated.
- | They are increasingly useful for valuing public firms that report fair values of assets.

Disadvantages are as follows:

- | They do not value the firm as a going concern, so they ignore margins, the efficient use of assets, and the competitive landscape.
- | They ignore effects of taxes.

Relative Valuation Approaches

Two basic types of **relative valuation models** (or **multiplier models**) can be used to estimate intrinsic values. The first type values a company from the

perspective of equity owners. The ratio of stock price to fundamentals such as earnings, sales, book value, or cash flow per share is used to determine whether a stock is fairly valued. For example, many analysts use the **price-to-earnings ratio (P/E)**. A **trailing P/E ratio** uses past EPS, while a **leading P/E ratio** uses forecast EPS for the next year. Analysts routinely adjust companies' reported earnings to exclude nonrecurring or extraordinary items. If EPS is negative or very low, this multiple loses its usefulness.

EXAMPLE: Method of comparables

MK Technologies shares are selling for \$50. Earnings for the last 12 months were \$2 per share. The average trailing P/E for firms in MK's industry is 32 times. Determine whether MK is overvalued or undervalued using the method of comparables.

Answer:

$$\text{MK's trailing P/E} = \frac{\$50}{\$2} = 25 \text{ times}$$

MK is relatively undervalued because its observed trailing P/E (25 times) is less than the industry average trailing P/E (32 times).

The second type of relative valuation model, which measures value from the perspectives of both equity and debt investors, is based on the ratio of EV to either EBITDA, EBIT, or revenue. As with other comparables-based models, values of an EV multiple that differ significantly from their benchmark may suggest overvaluation or undervaluation, unless they are justified by firm-specific attributes.



PROFESSOR'S NOTE

We will examine both of these comparables-based approaches to valuation in our next readings.

Analysts often use multiple methods to estimate value, which may result in inconsistencies in valuations. A weighted average can be used to arrive at the value estimate, with higher weights for models in which the analyst has greater confidence in input estimates. Analysts should also examine the consistency of assumptions used in various approaches to reconcile significant valuation differences.

MODULE QUIZ 43.1



1. Alcott Sidney is a tech influencer with a loyal fan base. Sidney started SMX Company, and its first product is a browser with integrated social media. Fans have purchased stock in SMX, and it is currently trading at a P/E multiple three times those of comparable companies. SMX stock is *most accurately* described as:
 - A. undervalued.
 - B. a meme stock.
 - C. a cult stock.
2. Alpha Semiconductors is a growing company that uses large amounts of debt to repurchase its stock. Alpha's dividend payout ratio varies from year to year. Which measure of future cash flows would be *most appropriate* to use in present value models to value Alpha stock?
 - A. Free cash flow to equity.
 - B. Free cash flow to the firm.
 - C. Dividends.
3. Beta Consumer Products Limited has a dominant market position in several consumer products. Beta has a stable revenue stream and pays 80% of its earnings in dividends. Which measure of future cash flows is *most appropriate* to use in present value models to value Beta stock?
 - A. Free cash flow to equity.
 - B. Free cash flow to the firm.
 - C. Dividends.
4. Spills, Inc., has 250,000 shares outstanding. Spills stock trades at a price of €15, while its book value per share is €16. Based only on this information, it is *most appropriate* to conclude that Spills has:
 - A. a value shortfall.
 - B. excess value.
 - C. an undervalued stock.
5. Valley Company has announced that it will cease operating and liquidate within a year. An analyst who wants to determine the value of Valley's equity shares should *most appropriately* use a(n):
 - A. asset-based model.
 - B. relative valuation model.
 - C. discounted cash flow model.

KEY CONCEPTS

LOS 43.a

Intrinsic value is the rational value investors would place on a stock if they had full knowledge of the company's characteristics. Perceived value varies based on the assumptions used by different investors and can be based on naïve or story-based valuation, relative valuation, or absolute valuation.

LOS 43.b

Book value of equity is the value of a firm's balance sheet assets minus its liabilities and minority interest. Book value per share is book value divided by the number of outstanding shares.

Market capitalization is the current market price of a firm's stock multiplied by the number of outstanding shares. When market capitalization exceeds (falls short of) book value, the firm is generating excess value (a value shortfall) attributable to its ability to generate future cash flows.

Enterprise value (EV) is the market value of common and preferred stock, plus the market value of debt, minus cash and short-term investments. The ratio of EV to earnings before interest, taxes, depreciation, and amortization (EBITDA) is frequently used for relative valuation.

LOS 43.c

Absolute valuation models include present value (discounted cash flow) models based on future dividends, free cash flow, or residual income; and asset-based models based on the fair value of a company's assets minus its liabilities and preferred stock.

Relative valuation models based on multiples such as price-to-earnings, price-to-book value, and EV-to-EBITDA are used to value a company relative to its peers.

ANSWER KEY FOR MODULE QUIZ

Module Quiz 43.1

- . **C** SMX stock is best characterized as a cult stock because its valuation results from a loyal following of the company's founder. (LOS 43.a)
- . **B** Because the company's capital structure is volatile (using debt to repurchase stock), FCFF would be preferred over FCFE, and dividend-based present value models are not appropriate for companies with low or volatile dividend payments. (LOS 43.b)
- . **C** Dividend discount models are appropriate for companies with stable and predictable dividend payments. (LOS 43.b)
- . **A** Companies with a lower share price than their book value per share are considered to have a value shortfall. An analyst should investigate the

reason for a value shortfall before concluding that the stock is undervalued. (LOS 43.b)

- . **A** Asset-based models can be applied to firms that are expected to cease operating. Other valuation techniques are more appropriate for evaluating going concerns. (LOS 43.c)

READING 44

DISCOUNTED CASH FLOW (DCF) AND GROWTH MODELS

MODULE 44.1: CASH FLOW METRICS

LOS 44.a: Contrast the inputs used in dividend discount, free cash flow to equity (FCFE), free cash flow to the firm (FCFF), and residual income models, and explain the process for using present value models to value equity.

Present value (PV) models calculate the value of equity as the discounted value of future cash flows. Unlike fixed-coupon bonds, common equity does not pay fixed dividends and does not have a finite life. Hence, we need to make some assumptions about estimated cash flows, $E(CF_t)$, during a forecast horizon and estimate a **terminal value**, $E(TV_n)$, at the end of that horizon. The value today is then determined as the PV of the series of cash flows and the terminal value.

$$\text{value} = \sum_{t=1}^n \frac{E(CF_t)}{(1+r)^t} + \frac{E(TV_n)}{(1+r)^n}$$

Metrics used to value equity with present value models include dividends, free cash flow to equity, free cash flow to the firm, EBITDA, and residual income. (While EBITDA and residual income are not cash flows, analysts can use them in much the same way in these models.)

Dividends

Dividend discount models (DDMs) define the future cash flows as the dividends that shareholders expect to receive. The primary advantage of using dividends as the definition of cash flow is that it is theoretically justified, because dividends are the actual cash that equity investors receive.

DDMs assume a firm's earnings are used to pay dividends or retained to support a growth in future earnings. If a firm uses debt (i.e., changes capital structure) to finance payment of dividends or repurchases its stock, this assumption is violated, and a DDM is not an appropriate choice of valuation model.

DDMs are suitable for firms with positive and stable or growing earnings that consistently pay a high proportion of earnings as dividends, particularly if they have a stable capital structure and little change in number of shares outstanding (i.e., no significant stock issuance or buybacks). Under such conditions, future dividends can be forecast with confidence. DDMs are also useful for industries where a high dividend payout is legally mandated, such as real estate investment trusts or regulated utilities.

The primary disadvantage of DDMs is that they are difficult to implement for non-dividend-paying companies. A second disadvantage of measuring cash flow with dividends is that it takes the perspective of an investor who owns a minority stake in the firm and cannot control its dividend policy.

Free Cash Flow to Equity

Free cash flow to equity (FCFE) is often used in DCF models instead of dividends because it represents the potential amount of cash that *could* be paid out to common shareholders. That is, FCFE reflects the firm's capacity to pay dividends. FCFE is also useful for firms that do not currently pay dividends. FCFE valuation takes the perspective of a controlling shareholder (one that can decide how much of the free cash flow is to be paid in dividends).

FCFE is defined as the cash remaining after a firm meets its debt obligations and provides for capital expenditures and working capital investment necessary to maintain its existing assets and fund new assets needed to support the firm's growth. In other words, it is the cash available to the firm's equity holders after a firm meets all its other obligations. FCFE for a period is often calculated as:

$$\text{FCFE} = \text{net income} + \text{depreciation} - \text{increase in working capital (WCInv)} - \text{capital expenditure (CapEx)} + \text{net borrowing}$$

The first three terms in the formula combine to approximate cash flow from operations (CFO), so we may also calculate FCFE as:

$$\text{FCFE} = \text{CFO} - \text{FCInv} + \text{net borrowing}$$

Net borrowing is the increase in debt during the period (i.e., principal borrowed minus principal repaid) and represents the portion of capital expenditures and working capital investment that comes from borrowed funds.

From the statement of cash flows, we can also derive FCFE as:

$$\text{FCFE} = \Delta \text{cash} + \text{dividends} - \Delta \text{equity}$$

where:

$$\Delta \text{equity} = \text{share issuance} - \text{share repurchases}$$

We can interpret this as saying a company can use its FCFE to increase cash, pay dividends, or buy back stock.

Analysts often assume that a company follows a target capital structure, maintaining a constant debt-to-assets ratio (DR). Consider that depreciation during the year accounts for part of capital expenditures. Hence, we can define net investment as:

$$\text{net investment} = \text{CapEx} - \text{depreciation} + \text{WCInv}$$

We can then restate FCFE as:

$$\text{FCFE} = \text{net income} - \text{net investment} + \text{net borrowing}$$

To keep the DR constant, the amount of borrowing needed to fund the company's net investment is:

$$\text{net borrowing} = \text{DR} \times \text{net investment}$$

Substituting this into our restated formula for FCFE:

$$\text{FCFE} = \text{net income} - [\text{net investment} + (\text{DR} \times \text{net investment})]$$

Therefore:

$$\text{FCFE} = \text{net income} - (1 - \text{DR}) (\text{net investment})$$

Free Cash Flow to the Firm

FCFE is less useful as a cash flow metric for valuation when a firm has high capital expenditures, which may result in negative FCFE, or when its capital structure is volatile, making it difficult to estimate the firm's future borrowing. In such cases, we can use **free cash flow to the firm (FCFF)**. FCFF is defined as the cash flow generated by the firm's operations, before paying interest, that exceeds the capital investment required to sustain the firm's current productive capacity.

$$\text{FCFF} = \text{CFO} - \text{CapEx} + \text{interest} (1 - \text{tax rate})$$

We add back only the after-tax interest because interest expense reduces the firm's tax expense.

FCFF is a pre-leverage cash flow that is available to distribute to the firm's debtholders and shareholders. It can also be related to the firm's FCFE as:

$$\text{FCFF} = \text{FCFE} + \text{interest} (1 - \text{tax rate}) - \text{net borrowing}$$

For a firm with no debt, $\text{FCFE} = \text{FCFF}$. Unlike dividends and FCFE, which are discounted at the required rate of return on equity, FCFF should be discounted at the firm's weighted average cost of capital.

EBITDA

Earnings before interest, taxes, depreciation, and amortization (EBITDA) is another pre-leverage earnings metric, but is different from FCFF due to the following:

- | EBITDA is before tax.
- | EBITDA does not capture tax savings on depreciation.
- | EBITDA does not account for capital expenditures or working capital investment.

We can state these differences as follows:

$$\text{FCFF} = [\text{EBITDA} \times (1 - \text{tax rate})] + (\text{depreciation} \times \text{tax rate}) - \text{FCInv} - \text{WCInv}$$

Some analysts adjust EBITDA for non-recurring or non-operating items, but these adjustments are subjective. Many firms report **adjusted EBITDA** based on their own estimates of these items, but the accounting standards do not recognize this as an earnings measure.

Residual Income

When FCFE and FCFF are negative, it can be difficult to predict when they are likely to turn positive. In these cases, we can instead use **residual income** as a substitute for future cash flows. Residual income is the firm's earnings after subtracting the cost of equity capital from reported net income. In other words, the value of the firm is linked to its ability to earn a return above its cost of capital. We can view residual income as a measure of the **economic profit** a firm

generates. In a valuation model, residual income is discounted at the required rate of return on equity.

On a per-share basis:

$$\text{residual income per share at time } t = \text{EPS}_t - r \times \text{BVPS}_{t-1}$$

A residual income model uses book values and assumes **clean surplus accounting**. This means that the company's book value of equity is affected only by the income statement (retained earnings) and by cash flows to or from shareholders (dividends, stock issuances, or buybacks). Any change in the book value of equity that bypasses the income statement, such as foreign currency translation gains or losses, is inconsistent with clean surplus accounting and requires an analyst to adjust the reported book value.

Process for Using Present Value Models

In summary, the equity valuation process has the following steps:

- Step 1:* Determine which cash flow metric to use.
- Step 2:* Forecast the metric over the appropriate time horizon. This varies based on the company's characteristics.
- Step 3:* Choose the required rate of return that corresponds with the cash flow metric being used.
- Step 4:* Calculate a terminal value that represents the company's value beyond the forecast horizon.
- Step 5:* Calculate the PV of the cash flows and terminal value, and add these to determine the value of the stock.

We discuss these steps in our next module.



MODULE QUIZ 44.1

1. Which of the following is *most accurate* regarding using EBITDA as a proxy for FCFF?
 - A. EBITDA accounts for tax benefits of depreciation and interest expense.
 - B. EBITDA accounts for investment in fixed capital and working capital.
 - C. EBITDA is a pretax measure, while FCFF is an after-tax measure.
2. A firm reports EBITDA per share of \$2.05. On a per-share basis, the firm's depreciation is \$0.22, interest expense is \$0.15, working capital investment is \$0.38, and capital expenditures are \$0.76. If the firm's tax rate is 30%, what is the firm's FCFF per share?
 - A. \$0.36.

- B. \$0.51.
- C. \$0.66.

MODULE 44.2: VALUATION MODELS

LOS 44.b: Calculate and interpret the intrinsic value of an equity security based on present value models that assume constant cash flow growth or multistage cash flow growth.

Firms go through a life cycle of start-up, growth, maturity, and decline. While not all firms follow the same pattern, many firms exhibit similarities:

- | During the start-up phase, earnings tend to be negative and decline initially. Start-up firms tend to have negative earnings, and many of them fail. Accordingly, cash flow-based models are unsuitable for valuing them.
- | During the growth phase, earnings turn positive and start growing rapidly. Analysts estimate the time horizon of a firm's growth phase to help them forecast their chosen cash flow metric.
- | During the mature phase, earnings growth tends to moderate to a long-run rate consistent with the overall growth rate of the economy. A constant growth assumption is most plausible when a firm enters the mature phase. Multinational firms operate in many countries, so they can grow at a rate higher than the growth rate of their domestic country.
- | Firms with obsolete products and technology enter a decline phase.

Constant Growth Rate Model

Companies that are (1) mature, (2) have stable leverage that is close to the industry average, and (3) have a high dividend payout ratio are the best candidates for valuation by assuming a constant growth rate.

We also use a constant growth rate to estimate a terminal value at the end of the forecast horizon, or when the growth rate stabilizes from a high-growth period.

Assuming a constant growth rate in dividends and earnings, we can estimate value today ($t = 0$) as:

$$\text{value}_t = \frac{E(CF_{t+1})}{(r - g)}$$

where:

g = constant growth rate

r = required rate of return

The assumed constant growth rate cannot be higher than the required rate of return and should be less than the growth rate of the economy.

EXAMPLE: Calculating value with the constant growth model

DownUnder Financial recently paid a dividend of 1.80 Australian dollars. An analyst has examined the financial statements and historical dividend policy of DownUnder and expects that the firm's dividend rate will grow at a constant rate of 3.5% indefinitely. The analyst also determines that the required rate of return on equity is 10%. Calculate the current value of DownUnder's shares using these assumptions.

Answer:

Using the constant growth model to estimate share value:

$$\text{value} = \frac{D_1}{(r - g)} = \frac{D_0(1 + g)}{(r - g)} = \frac{1.80(1.035)}{(0.10 - 0.035)} = 28.66$$



PROFESSOR'S NOTE

Pay close attention to whether a question gives you a past dividend or an expected dividend. The constant growth model requires the expected dividend in the *next* period. This example says the company “recently paid” a dividend of 1.80. This tells us it was a *past* dividend that we need to grow by one period to get the expected dividend. If, instead, it had said the company “is expected to pay” a dividend of a given amount, we would not have needed to grow it.

If we determine that a constant growth rate model is appropriate for a company, we can calculate the **implied growth rate** priced by the market by rearranging the constant growth rate formula:

$$g = r - \frac{E(CF_1)}{\text{price}}$$

EXAMPLE: Implied growth rate

DownUnder Financial is expected to have FCFE of AUD 2.40 per share, and its required rate of return on equity is 10%. Assuming a market price of AUD 40.00 per share, calculate the constant growth rate for DownUnder that its share price implies.

Answer:

$$g = r - \frac{E(CF_1)}{\text{price}} = 0.10 - \left(\frac{2.40}{40} \right) = 0.10 - 0.06 = 0.04, \text{ or } 4\%.$$

Two-Stage Growth Models

There are two types of two-stage models:

During the initial n periods, the growth rate is g_s , after which it drops to a perpetual constant growth rate g_L . Typically, $g_s > g_L$.

During the initial n periods, the growth rate varies. This requires us to forecast the cash flows for those periods individually. Individual cash flows are estimated for n periods, and the terminal value is estimated based on the constant growth rate. This approach is best carried out using a spreadsheet.

For the first scenario, value is determined as the sum of the PVs of cash flows during the short-run growth period, plus the PV of a terminal value for the long-run constant growth period:

$$\text{value} = \sum_{t=1}^n \frac{E(CF_0)(1 + g_s)^t}{(1 + r)^t} + \frac{TV_n}{(1 + r)^n}$$

where:

$$TV_n = \frac{CF_{(n+1)}}{(r - g_L)}$$

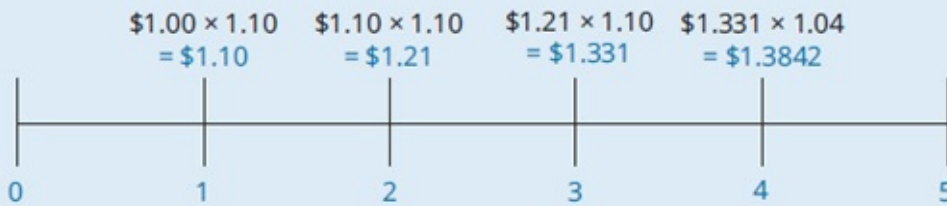
EXAMPLE: Calculating value with a two-stage DDM

Sea Island Recreation currently pays a dividend of \$1.00. An analyst forecasts growth of 10% for the next three years, and 4% growth thereafter. The required return is 12%. Calculate the current value per share.

Answer:

The forecast dividends are shown in the following figure.

Dividend Cash Flows

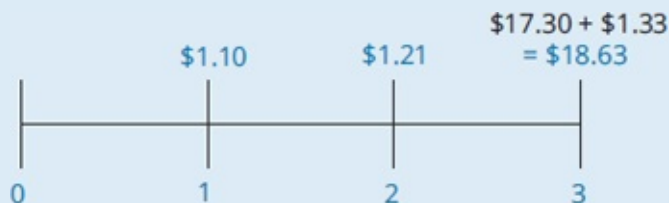


Constant growth at 4% begins after the third year, and we can employ the DDM to determine the value of the stock at time $t = 3$. Accordingly:

$$V_3 = \frac{D_3 \times (1 + g)}{r - g} = \frac{D_4}{r - g} = \frac{\$1.3842}{0.12 - 0.04} = \$17.30$$

Now we know the dividends in Years 1, 2, and 3, the terminal value in Year 3, and the discount rate. The cash flows that we need to solve the problem are shown in the following figure.

Dividend and Terminal Value Cash Flows



The value of the stock today is the present value of these cash flows:

$$\frac{\$1.10}{1.12} + \frac{\$1.21}{1.12^2} + \frac{\$1.33 + \$17.30}{1.12^3} = \$15.21$$

Or, using the cash flow functions on a financial calculator: $CF_0 = 0$; $C_01 = 1.10$; $C_02 = 1.21$; $C_03 = 18.63$; $I/Y = 12$; $CPT \rightarrow NPV = 15.21$

EXAMPLE: A firm with no current dividend

A firm currently pays no dividend but is expected to pay a dividend at the end of Year 4. Year 4 earnings per share are expected to be \$1.64, and the firm will maintain a dividend payout ratio of 50%. Assuming a constant growth rate of 5% and a required rate of return of 10%, estimate the current value of this stock.

Answer:

The first step is to find the value of the stock at the end of Year 3. V_3 is the PV of dividends from Year 4 onward. Remember, the constant growth model gives us a stock's value one period *before* the first dividend is paid.

Calculate D_4 , the estimate of the dividend that will be paid at the end of Year 4:

$$D_4 = (\text{dividend payout ratio})(\text{earnings in Year 4}) = (0.5)(1.64) = \$0.82$$

Apply the constant growth model to estimate V_3 :

$$V_3 = \frac{D_4}{(r_e - g_c)} = \frac{\$0.82}{(0.10 - 0.05)} = \$16.40$$

Finally, calculate the current value, V_0 :

$$V_0 = \frac{16.40}{1.1^3} = \$12.32$$

Estimating the Constant Growth Rate

To estimate the growth rate, an analyst can use two methods:

Use the historical growth in dividends for the firm. This approach may not be appropriate for firms that have changes in capital structure, margins, competitive position, or technology.

Estimate the sustainable growth rate using fundamental factors.

The **sustainable growth rate** of dividends is the rate at which equity, earnings, and dividends can continue to grow indefinitely, assuming that ROE is constant, the dividend payout ratio is constant, and no new equity is sold:

$$SGR_{Div} = (1 - \text{dividend payout ratio}) \times ROE$$

or

$$SGR_{Div} = \left(1 - \frac{\text{dividend}}{\text{net income}}\right) \times \left(\frac{\text{net income}}{\text{beginning book value}}\right)$$



PROFESSOR'S NOTE

This method of estimating the sustainable growth rate is sometimes stated as $RR \times ROE$, where RR is the earnings retention ratio, which equals one minus the dividend payout ratio.

EXAMPLE: Sustainable growth rate

Green, Inc., is expected to pay dividends equal to 25% of earnings. Green's ROE is 21%. Calculate and interpret its sustainable growth rate.

Answer:

$$g = (1 - 0.25) \times 21\% = 15.75\%$$

With long-run economic growth typically in the single digits, it is unlikely that a firm can sustain 15.75% growth. The analyst should examine the industry growth rate and the firm's historical growth rate to determine whether this estimate is reasonable.

Similarly, the sustainable growth rate for FCFE is:

$$SGR_{FCFE} = \left(1 - \frac{FCFE}{\text{net income}}\right) \left(\frac{\text{net income}}{\text{beginning book value}}\right)$$

To estimate a sustainable growth rate for FCFF, we use net operating profit after taxes (NOPAT), a metric for income available to all capital providers, instead of net income:

$$SGR_{FCFF} = \left(1 - \frac{FCFF}{NOPAT}\right) \left(\frac{NOPAT}{\text{beginning book value of total capital}}\right)$$

where:

$NOPAT = \text{operating income} (1 - \text{tax rate})$

The estimated sustainable growth rate is unlikely to be the same for all three metrics, even though in theory it should be. To reconcile the differences, analysts should evaluate the assumptions used.

Weighted Average Cost of Capital

The discount rate used to calculate the PV must be consistent with the cash flow metric used. For dividends, FCFE, and residual income, the discount rate is the required rate of return on equity. For discounting FCFF, the discount rate is the **weighted average cost of capital (WACC)**, which is the weight of debt in the company's capital structure times its after-tax cost of debt, plus the weight of equity times the cost of equity. Weights are based on market values. Because a firm's debt is less risky (i.e., it has a higher priority of claims) than its equity, the cost of debt is lower than the cost of equity.

$$WACC = \left(\frac{\text{MV of debt}}{\text{MV of debt} + \text{MV of equity}}\right) r_d (1 - \text{tax rate}) + \left(\frac{\text{MV of equity}}{\text{MV of debt} + \text{MV of equity}}\right) r_e$$

where:

r_d = cost of debt

r_e = cost of equity

The PV of future FCFF is the value of the firm. Subtracting the market value of debt provides the value of equity.

EXAMPLE: FCFF valuation

Tykes Toys has a required rate of return on equity of 15%, a WACC of 12%, and a marginal tax rate of 40%. Tyke's FCFF over the next three years (in millions of dollars) is expected to be 209, 259, and 312, while its NOPAT is expected to be 300, 345, and 390. After three years, free cash flow is expected to grow at a sustainable constant rate. Book value is expected to be 1,950 at the end of the second year. What is the value of the firm?

Answer:

Estimate the sustainable growth rate at which the firm will grow after three years.

$$SGR_{FCFF} = \left(1 - \frac{FCFF}{NOPAT}\right) \left(\frac{NOPAT}{\text{beginning book value}}\right)$$

Because this is at the end of the third year, we need FCFF and NOPAT for Year 3 and book value at the end of Year 2.

$$SGR = \left(1 - \frac{312}{390}\right) \left(\frac{390}{1,950}\right) = (1 - 0.80)(0.20) = 0.04$$

Next, use the constant growth model to determine the value for Year 3, using WACC as the discount rate.

$$V_3 = \frac{312(1.04)}{(0.12 - 0.04)} = 4,056$$

Finally, discount the FCFF cash flows to their present value.

$$\text{Total PV} = \frac{209}{(1.12)} + \frac{259}{(1.12)^2} + \frac{(312 + 4,056)}{(1.12)^3} = \$3,502 \text{ million}$$



MODULE QUIZ 44.2

1. CFC Regs, Inc., just paid a dividend of \$2.00 per share. The required return is 13%, and the expected constant growth rate is 6%. The value of the stock today is *closest* to:
A. \$28.60.
B. \$30.30.
C. \$35.30.
2. Forever, Inc., will pay a \$4 dividend in the next period, and expects dividends to grow by 3% per year thereafter. If the discount rate for the company is 9%, the current value of one share of the company's common stock is *closest* to:

- A. \$66.67.
 - B. \$68.67.
 - C. \$133.33.
3. Arena Distributors currently pays no dividends. The company recently reported earnings of \$1.50 per share, which an analyst expects to grow at a 15% rate for the next four years. Beginning in Year 5, the analyst expects Arena to distribute 20% of its earnings in the form of dividends, and to have a constant growth rate of 5%. Using a required rate of return of 12%, the value of Arena shares today is *closest* to:
- A. \$5.00.
 - B. \$7.86.
 - C. \$39.30.

MODULE 44.3: MODEL LIMITATIONS AND PREFERRED STOCK VALUATION

LOS 44.c: Explain the shortcomings of constant and multistage cash flow growth assumptions in present value models.

The valuation estimates derived from PV models are only as good as the quality of inputs into the models. The choice and amount of the cash flow metric, the time horizon, and the discount rate are all subject to estimation errors.

While dividend discount models are appropriate for mature firms with high and stable payouts, they ignore payouts in the form of stock buybacks. Also, using past growth in dividends as indicative of future growth would be inappropriate for firms facing a deteriorating competitive environment.

Analysts need to evaluate a firm's industry position, its competitive advantages, changes in technology, and the overall macroeconomic environment to reinforce their estimated input values. Valuation is affected significantly by the estimated discount rate. The costs of both debt and equity rely on the risk-free rate in the economy, which fluctuates.

LOS 44.d: Calculate the intrinsic value of a non-callable, non-convertible preferred stock and describe how contingency features affect intrinsic value.

Preferred stock has priority of claims over common equity for dividend payments and in liquidation. Accordingly, the required rate of return on preferred stock is less than the required return on common equity. Preferred stock pays a fixed dividend (as a percentage of face value) and may have features such as

cumulative or participating dividends and contingency provisions such as being callable, putable, or convertible. First, we will examine preferred stock without such features or provisions.

Finite-Maturity Preferred Stock

For a preferred stock with a finite maturity N , the value of the stock is given by the present value of preferred dividends (D_p) and the face value (FV):

$$\text{finite-maturity preferred stock value} = \sum_{i=1}^N \frac{D_p}{(1 + r_p)^i} + \frac{FV}{(1 + r_p)^N}$$



PROFESSOR'S NOTE

“Preferred stock with a finite maturity” is effectively a bond. On an issuer’s balance sheet, such securities are typically classified as debt liabilities rather than equity.

Perpetual Preferred Stock

A perpetual preferred stock value is determined by setting the growth rate equal to zero in the constant growth model, which reduces the formula to the following:

$$\text{perpetual preferred stock value} = \frac{D_p}{r_p}$$

EXAMPLE: Preferred stock valuation

A company’s \$100 par preferred stock pays a \$5 annual dividend and has a required return of 8%. Calculate the value of the preferred stock.

Answer:

$$\text{Value of the preferred stock} = \frac{D_p}{r_p} = \frac{\$5.00}{0.08} = \$62.50$$

Contingency features in preferred stock refer to **embedded options** that give the issuer or the investor a right, but not an obligation. Options have a non-negative value to the party that holds the right to exercise them. Therefore, the value of an embedded option is subtracted from (if the right to exercise is held by the issuer) or added to (if the right to exercise is held by the investor) the intrinsic value of a preferred stock.

With a **callable preferred stock**, the *issuer* has the right to redeem the preferred shares for a predetermined fixed price. Because the right to call is held by the issuer, the value of a callable preferred is lower than the value of a comparable non-callable preferred. Issuers usually offer higher dividends on callable preferred stock.

With a **putable preferred stock**, the *investor* has the right to sell the preferred shares back to the issuer for a predetermined price. Because the investor has the right to exercise, the embedded put option makes the shares more valuable than otherwise identical option-free preferred stock. Dividends on putable preferred stock are typically lower than for option-free preferred stock.

A **convertible preferred stock** gives the *investor* the right to convert the preferred shares into common shares of the issuer at a predetermined **conversion ratio** after a fixed period. A convertible preferred has both debt-like (fixed dividends, priority of claims) and equity-like (conversion feature) characteristics. As with putable preferred stock, the fact that the investor holds the right to exercise makes convertible preferred shares more valuable than otherwise identical option-free preferred shares.



MODULE QUIZ 44.3

1. A perpetual preferred stock with a par value of \$100 pays an annual dividend of \$6. If the required rate of return on these preferred shares is 8%, their price is *closest* to:
 - A. \$75.
 - B. \$98.
 - C. \$133.
2. For preferred shares that are otherwise identical except for their contingency features, a company's:
 - A. callable preferred stock will have a higher required rate of return than its putable preferred stock.
 - B. putable preferred stock will have a higher required rate of return than its callable preferred stock.
 - C. convertible preferred stock will have a higher required rate of return than its option-free preferred stock.

KEY CONCEPTS

LOS 44.a

Dividend discount models are suitable for valuing equity of firms with positive and stable or growing earnings that consistently pay a high proportion of

earnings as dividends.

Free cash flow to equity discount models are suitable for firms that do not pay dividends and have a stable capital structure. Free cash flow to the firm can be used when free cash flow to equity is negative, or when the capital structure is not stable.

Dividend discount models take a minority shareholder perspective, while free cash flow models take a controlling shareholder perspective.

Residual income models can be used when free cash flow is negative, as long as book value is consistent with clean surplus accounting.

Steps in discounted cash flow valuation are the following:

Step 1: Determine which cash flow metric to use.

Step 2: Forecast the metric over the appropriate time horizon.

Step 3: Choose the required rate of return that corresponds with the cash flow metric being used.

Step 4: Calculate a terminal value that represents the company's value beyond the forecast horizon.

Step 5: Calculate the present value of the cash flows and terminal value.

LOS 44.b

Assuming a constant growth rate, we can value a stock as:

$$\text{value}_t = \frac{E(\text{CF}_{t+1})}{(r - g)}$$

The sustainable growth rate of dividends can be estimated as $(1 - \text{dividend payout ratio}) \times \text{return on equity}$.

With a multistage model, we can forecast cash flows over a time horizon, after which we assume their growth rate becomes constant. The growth rate can be a historical average or based on fundamental factors.

While dividends, FCFE, and residual income are discounted at the required rate of return on equity, FCFF should be discounted at the firm's weighted average cost of capital.

LOS 44.c

Valuation estimates derived from discounted cash flow models are only as good as the quality of the inputs into the model. The choice and amount of the cash flow metric, the time horizon, and the discount rate are all subject to estimation errors.

LOS 44.d

The value of a perpetual preferred stock is given by:

$$\text{value} = \frac{D_p}{r_p}$$

Callable preferred stock has a lower value than otherwise identical option-free preferred stock because the issuer holds the option to call the shares.

Putable or convertible preferred stock has a higher value than otherwise identical option-free preferred stock because the investor holds the option to put or convert the shares.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 44.1

- . C EBITDA is a pretax measure. EBITDA does not account for tax benefits of interest and depreciation, nor does it account for investment in fixed capital and working capital. (LOS 44.a)
- . A $FCFF = [EBITDA \times (1 - \text{tax rate})] + (\text{dep.} \times \text{tax rate}) - FCInv - WCInv$
 $= 2.05(1 - 0.30) + (0.22 \times 0.30) - 0.76 - 0.38 = \$0.36.$
(LOS 44.a)

Module Quiz 44.2

- . B $V_0 = \frac{D_1}{r - g}$

Because the dividend was “just paid,” the \$2.00 represents the dividend in the current period, D_0 . To get D_1 we must grow the dividend by one period at the given growth rate:

$$D_1 = \$2.00(1.06) = \$2.12$$

$$V_0 = \frac{\$2.12}{0.13 - 0.06} = \$30.29$$

(LOS 44.b)

- . A The growth rate is 3% and we are given the next period's dividend (D_1).
Therefore:

$$\text{stock value} = \frac{D_1}{(r - g)} = \frac{\$4.00}{[0.09 - 0.03]} = \$66.67.$$

(LOS 44.b)

- . A First, forecast the earnings in Year 5, then calculate the dividends in Year 5 as 20% of Year 5 earnings. Applying the constant growth model to the Year 5 dividend gives us an estimate of the terminal value in Year 4. The terminal value discounted back four years is the current value of the stock:

$E_4 = \$1.50 \times (1.15)^4 = \2.62 (earnings grow at 15% per year for four years).

$E_5 = \$2.62 \times 1.05 = \2.75 (earnings grow at 5% starting in the fifth year).

$D_5 = \$2.75 \times 0.20 = \0.55 (payout ratio is 20%)

$$V_4 = \frac{D_5}{(r - g)} = \frac{0.55}{(0.12 - 0.05)} = 7.86$$

Because there are no dividends for the first four years, the value today is PV of the terminal value at the end of Year 4:

$$V_0 = \frac{V_4}{(1 + r)^4} = \frac{7.86}{(1.12)^4} = \$5$$

(LOS 44.b)

Module Quiz 44.3

- . A $\frac{\$6}{0.08} = \75

(LOS 44.d)

- . A An option granted to the issuer will increase the required rate of return on preferred stock. Because the issuer of callable preferred holds the right to exercise, while the investor holds the right to exercise put and conversion options, callable preferred stock will have a higher required rate of return than otherwise identical putable, convertible, or option-free preferred stock.
(LOS 44.d)

READING 45

RELATIVE VALUE EQUITY VALUATION APPROACHES

MODULE 45.1: PRICE MULTIPLES AND EQUITY VALUATION

LOS 45.a: Explain the process for using multiples to value equity based on the method of comparables and the method of forecasted fundamentals.

Relative valuation models compare the price or enterprise value of a security to some metric of value. **Price multiples** divide the current market price of a stock by its earnings per share, free cash flow to equity per share, or book value per share. **Enterprise value (EV) multiples** divide the EV by EBITDA, free cash flow to the firm, or revenue.

A multiple that uses a value from the past period in its denominator is called a **trailing multiple**. One that uses a forecast value for the next period is called a **leading multiple** or **forward multiple**. A **justified multiple** is an analyst's estimate of the multiple that a stock trading at its fair value should exhibit.

An analyst must choose the most appropriate multiple for any given valuation. For example, when a company's earnings are negative, its price-to-earnings ratio is meaningless.

To determine whether a stock is fairly valued at its current price, analysts compare the company's multiples to a benchmark. When a price multiple is higher than its benchmark value, the stock is considered overvalued; when a price multiple is lower than its benchmark, the stock is seen as undervalued.

There are two approaches to defining a benchmark for a company: (1) the method of comparables and (2) the method of forecasted fundamentals.

Method of Comparables

The **method of comparables** values a stock based on the average or median of peer companies' price multiples. The economic rationale for the method of comparables is the "law of one price," which implies that two similar assets should be priced at similar multiples (e.g., P/E ratios).

The similar companies chosen as a benchmark are called a **peer group**. A valid peer group must be truly comparable in terms of the included companies' competitive positions, risk, and earnings growth rates. Variations in accounting methods, nonrecurring items, and timing differences (due to fiscal year-end reporting periods differing among companies) can make comparisons difficult.

While the average multiple of a firm's peer group is commonly used as the benchmark or justified multiple, if the distribution of the multiple is skewed, it makes more sense to use the median multiple. Analysts may also remove outliers from the peer group before calculating the mean.

Analysts must choose an appropriate multiple to use in a comparables-based valuation. For example, a dividend-based multiple is inappropriate when the peer group includes non-dividend-paying firms.

Method of Forecasted Fundamentals

The **method of forecasted fundamentals** determines the benchmark value based on a discounted cash flow model. For example, we might start with a constant growth model:

$$P_0 = \frac{D_1}{r - g}$$

If we divide both sides by expected earnings per share for the next period (E_1), we arrive at a **justified leading P/E ratio**:

$$\frac{P_0}{E_1} = \frac{\frac{D_1}{E_1}}{(r - g)}$$

The leading P/E ratio indicates that the multiple is positively related to the earnings growth rate and negatively related to the required rate of return (i.e., the risk of the company).

We can also create a justified price-to-book value multiple based on this model. First, we divide both sides by the firm's book value of equity per share:

$$\frac{P_0}{BVPS_0} = \frac{\frac{D_1}{BVPS_0}}{r - g}$$

We can refine this multiple further by stating the expected dividend in the next period (D_1) as the product of the firm's expected earnings per share and its dividend payout ratio:

$$\frac{P_0}{BVPS_0} = \frac{\frac{(E_1 \times \text{dividend payout ratio})}{BVPS_0}}{r - g}$$

Then, because $\frac{EPS}{BVPS}$ is the firm's return on equity (ROE), we can restate this relationship:

$$\frac{P_0}{BVPS_0} = \frac{ROE \times \text{dividend payout ratio}}{r - g}$$

Next, recall that the relationship between ROE and the payout ratio gives us an estimate of the firm's sustainable growth rate (g). We can rearrange this relationship:

$$g = ROE \times (1 - \text{dividend payout ratio})$$

$$g = ROE - (ROE \times \text{dividend payout ratio})$$

$$ROE \times \text{dividend payout ratio} = ROE - g$$

Substituting this into the price-to-book-value relationship, we arrive at the **justified P/B ratio**:

$$\frac{P_0}{BVPS_0} = \frac{ROE - g}{r - g}$$

This relationship implies that the multiple depends on the difference between a company's actual ROE and its required rate of return on equity (or cost of equity). A higher spread of ROE over r leads to a higher P/B multiple.

Reconciling the Methods

The method of comparables and method of forecasted fundamentals may yield conflicting results that need reconciliation. For example, a firm's justified P/E ratio might be higher than the average of its peers, but that is acceptable when the firm's earnings growth rate is higher than those of its peers.

We can reconcile these approaches in two ways:

Modify the peer value based on the fundamental variable that causes the difference, then compare the modified multiples. This approach only considers one fundamental variable at a time.

The **price/earnings-to-growth ratio (PEG)** is commonly used to standardize the P/E ratio based on firms' expected growth rates:

$$\text{PEG ratio} = \frac{P/E}{g}$$

The growth rate g in the PEG ratio is expressed as a percentage, rather than a decimal.

The PEG ratio controls for growth rate differences across firms. If a firm has a higher PEG ratio relative to peers, its P/E is higher than justified by its growth rate, so its stock is overvalued.

EXAMPLE: Calculating and using the PEG ratio

Med-Ready, Inc., has a trailing P/E of 28.75 and a five-year consensus growth rate forecast of 14.5%. The median PEG ratio for a group of companies comparable in risk to Med-Ready is 2.34. Calculate the firm's PEG ratio and explain whether the stock appears to be correctly valued, overvalued, or undervalued.

Answer:

The firm's PEG is $\frac{28.75}{14.5} = 1.98$. Given the comparable group median PEG ratio of 2.34, Med-Ready's stock may be undervalued.

- A regression approach produces an estimate of the multiple after controlling for one or more fundamental variables. The fundamental variables are the independent (or X) variables, and the observed multiple is the dependent (or Y) variable. Peer group data are used to estimate the regression coefficients.

Consider a model using the P/B ratio as the dependent variable, and ROE and growth rates as independent variables:

$$P/B_i = b_0 + b_1 \times ROE_i + b_2 \times g + e_i$$

Using the peer group as the sample, estimates of b_0 , b_1 , and b_2 are generated. These estimates are then used to calculate the justified P/B ratio of the subject

company.



PROFESSOR'S NOTE

Multiple regression is addressed in the Level II curriculum. The Level I exam will not require you to perform or analyze a multiple regression, but it might ask you to interpret the results of one, such as what we show in the following example.

EXAMPLE: Calculating justified P/B ratio

Alpha, Inc.'s, ROE is 14%, its required rate of return is 8%, and its expected growth rate is 4%. Rob Elwell uses a two-factor regression model using ROE and g to analyze the P/B ratio of a peer group of 10 companies.

The estimated regression intercept is 1.42, the estimated coefficient for ROE is 6.78, and the estimated coefficient for the growth rate is 12.79.

Calculate and interpret a justified P/B ratio for Alpha, using both the method of forecasted fundamentals and the regression method.

Answer:

1. Forecasted fundamentals:

$$\begin{aligned}\text{Justified P/B ratio} &= \frac{(\text{ROE} - g)}{(r - g)} \\ &= \frac{(0.14 - 0.04)}{(0.08 - 0.04)} \\ &= 2.50\end{aligned}$$

2. Regression method:

$$\text{Justified P/B ratio} = 1.42 + 6.78 \times 0.14 + 12.79 \times 0.04 = 2.88$$

Based on forecasted fundamentals, the stock should trade at 2.50 times book value per share. Based on the regression model, the stock should trade at 2.88 times book value per share.

Price multiples can also be used to determine a terminal value to use in present value models. Analysts can estimate a justified price multiple at the end of the forecast horizon, or when the growth rate stabilizes. Using the forecast fundamental variables, an analyst can calculate a justified future price multiple and use it to estimate the terminal value.

For example, a terminal value based on a justified P/E ratio is:

terminal value in year n = justified leading P/E ratio
× forecast earnings in year $(n + 1)$



MODULE QUIZ 45.1

1. Sudesh Mukherjee calculates the P/B ratio of Allied Pharma and compares it to the average of a peer group. Mukherjee notices that Allied's ROE, retention rate, and risk differ from those of its peer group companies. To control for these differences, Mukherjee should *most appropriately* use:
 - A. PEG ratios.
 - B. a regression approach.
 - C. the method of comparables.
2. A company's trailing P/E is 12.50, while its expected growth rate is 5%. Its PEG ratio is *closest* to:
 - A. 2.39.
 - B. 2.50.
 - C. 2.62.
3. Limitations of the comparables method in relative valuation *least likely* include differences among peer companies':
 - A. earnings.
 - B. risk levels.
 - C. accounting methods.

MODULE 45.2: PRICE AND ENTERPRISE VALUE MULTIPLES

LOS 45.b: Calculate and interpret price and enterprise value multiples, and describe their appropriate uses.

A firm that earns a return above its required rate of return would benefit from retaining earnings and investing in those growth opportunities, rather than paying out dividends. Viewed this way, the value of its stock represents the present value of its current dividend as a perpetuity (i.e., assuming no growth) plus its **present value of growth opportunities (PVGO)**:

$$P_0 = \frac{E_1}{r} + \text{PVGO}$$

This means the value of a firm's equity has two components:

The value of its assets in place $\left(\frac{E_1}{r}\right)$, which is the present value of a perpetual cash flow of E_1 .

The present value of its future investment opportunities (PVGO).

EXAMPLE: Calculating PVGO

Reliable, Inc.'s, shares trade at 60.00 with expected earnings of 5.00 per share and a required return of 10%. Calculate Reliable's present value of growth opportunities.

Answer:

$$P_0 = \frac{E_1}{r} + \text{PVGO}$$

$$60.00 = \frac{5.00}{0.10} + \text{PVGO}$$

$$\text{PVGO} = 60.00 - \frac{5.00}{0.10} = 10.00$$

Earnings yield is the reciprocal of the P/E ratio. A company's **forward earnings yield** is the next period's expected earnings divided by its current share price, or $\frac{E_1}{P_0}$.

We can rearrange the PVGO to state it in terms of forward earnings yield, as follows:

$$\frac{E_1}{P_0} = r \left(1 - \frac{\text{PVGO}}{P_0} \right)$$

Or:

$$\frac{\text{PVGO}}{P_0} = 1 - \frac{\text{forward earnings yield}}{r}$$

EXAMPLE: Ratio of PVGO to price

Using the data from the previous example, calculate the ratio of PVGO to the current price.

Answer:

$$\frac{PVGO}{P_0} = \frac{10.00}{60.00} = 16.67\%$$

$$\text{Or, forward earnings yield} = \frac{5.00}{60.00} = 8.33\%$$

$$\begin{aligned} \frac{PVGO}{P_0} &= 1 - \frac{\text{forward earnings yield}}{r} \\ &= 1 - \left(\frac{0.0833}{0.10} \right) = 0.1667, \text{ or } 16.67\% \end{aligned}$$

Some firms have a negative PVGO. This suggests that their managers are investing in negative NPV projects and destroying value. Those firms should distribute earnings as dividends instead of retaining them.

Core and Normalized Earnings

The P/E ratio is meaningless when earnings are negative, and it can be unrepresentative of a stock's value when earnings include substantial nonrecurring items. Analysts typically try to estimate a company's **core earnings** (also called **persistent earnings** or **continuing earnings**). Core earnings exclude nonrecurring components such as gains and losses from asset sales, asset write-downs, provisions for future losses, changes in accounting estimates, or income or losses from discontinued operations.

Earnings may also exhibit cyclicity, changing systematically over a business cycle. Analysts adjust P/E ratios for cyclicity by estimating **normalized earnings**, which estimate EPS in the middle of a business cycle. The following two methods are used to normalize earnings:

Under the method of historical average EPS, normalized EPS is estimated as the average EPS over the most recent business cycle.

Under the average ROE method, normalized EPS is estimated as average ROE over the most recent business cycle times the current book value per share. Using BVPS reflects the effect of changes in firm size more accurately than does the method of historical average EPS.

EPS also needs to be adjusted to reflect the impact of **dilution**, changes in the number of shares outstanding that result from exercise of employee stock options, warrants, and convertible bonds.

EXAMPLE: Calculating normalized earnings

Using the data in the following figure, calculate normalized earnings for Magnolia Enterprises, using both the method of historical average EPS and the method of average ROE.

Data for Magnolia Enterprises

Year	20X1	20X2	20X3	20X4
EPS	\$4.20	\$3.75	\$4.75	\$4.30
BVPS	\$26.02	\$27.78	\$29.25	\$32.29
ROE	14.0%	12.0%	16.0%	14.0%

Answer:

Average EPS approach:

$$\text{normalized earnings} = \frac{(4.20 + 3.75 + 4.75 + 4.30)}{4} = \$4.25$$

Average ROE approach:

$$\text{average ROE} = \frac{(0.14 + 0.12 + 0.16 + 0.14)}{4} = 0.14 = 14.00\%$$

$$\text{normalized earnings} = \text{average ROE} \times \text{BVPS}_{20X4} = 0.14 \times \$32.29 = \$4.52$$

Income Statement Measures

When earnings are negative or when differences in accounting methods make earnings non-comparable across firms, analysts may use the **price-to-sales (P/S) multiple**. The P/S multiple is suitable when comparing firms with similar revenue recognition practices and for mature and cyclical firms.

We can calculate a justified P/S ratio $\left(\frac{P_0}{S_0}\right)$ by multiplying a firm's trailing P/E ratio by its profit margin $\left(\frac{E_0}{S_0}\right)$:

$$\frac{P_0}{S_0} = \frac{P_0}{E_0} \times \frac{E_0}{S_0} = \frac{(\text{dividend payout ratio})(1 + g)(\text{profit margin})}{(r - g)}$$

EXAMPLE: Calculating justified P/S ratio

A stock has a dividend payout ratio of 60%, an ROE of 8.3%, earnings per share of \$4.25, sales per share of \$218.75, and an expected growth rate of 5%. Shareholders require a return of 10% on their investment. Calculate a justified P/S multiple based on these fundamentals.

Answer:

$$\begin{aligned}\text{Profit margin} &= \frac{\text{EPS}}{\text{sales per share}} \\ &= \frac{\$4.25}{\$218.75} = 0.0194\end{aligned}$$

$$\frac{P_0}{S_0} = \frac{(\text{dividend payout ratio})(1 + g)(\text{profit margin})}{(r - g)}$$

$$\frac{P_0}{S_0} = \frac{(0.6)(1 + 0.05)(0.0194)}{(0.10 - 0.05)} = 0.244$$

Sales revenue is available to both providers of debt and equity capital, while price is an equity-only metric. For this reason, the P/S ratio is appropriate for firms with little to no leverage. For more leveraged firms, it may be more appropriate to use enterprise value in the numerator.

Other income statement measures include EBIT and EBITDA, which are available to both shareholders and debtholders. Therefore, they are commonly used in ratios with **enterprise value (EV)**.

$$\text{EV} = \text{market value of common and preferred stock} + \text{market value of debt} - \text{cash and short-term investments}$$

Alternatively, analysts may use **firm value**, which is the market value of a firm's equity and debt (the first two terms in EV).

Analysts also use free cash flow to the firm (FCFF) in EV ratios. Assuming a constant growth rate:

$$\text{EV} = \frac{\text{FCFF}_1}{\text{WACC} - g} = \frac{\text{EBITDA}(1 - t) + \text{depreciation}(t) - \text{CapEx} - \text{WCInv}}{\text{WACC} - g}$$

Dividing both sides by EBITDA gives us the following expression for the **EV/EBITDA multiple**:

$$\frac{\text{EV}}{\text{EBITDA}} = \frac{1 - t}{\text{WACC} - g} + \frac{\frac{\text{depreciation}(t)}{\text{EBITDA}}}{\text{WACC} - g} - \frac{\frac{\text{CapEx}}{\text{EBITDA}}}{\text{WACC} - g} - \frac{\frac{\text{WCInv}}{\text{EBITDA}}}{\text{WACC} - g}$$

EXAMPLE: EV/EBITDA multiple

Accel Industries has \$100 million of EBITDA, \$50 million of depreciation, \$60 million of capital expenditures, and \$10 million of working capital

investment., Accel's WACC is 7%, its tax rate is 20%, and its growth rate is 2%. Calculate Accel's justified EV/EBITDA multiple.

Answer:

$$\text{FCFF} = 100(1 - 0.20) + (50 \times 0.20) - 60 - 10 = 20 \text{ million}$$

$$\begin{aligned} \text{EV} &= \frac{\text{FCFF}_1}{\text{WACC} - g} \\ &= \frac{20}{(0.07 - 0.02)} = 400 \text{ million} \end{aligned}$$

$$\text{EV/EBITDA} = \frac{400}{100} = 4x$$

The EV/EBITDA ratio is often used for capital-intensive industries where depreciation and amortization are large. EV/EBITDA is positively related to the growth rate and negatively related to WACC.

Balance Sheet Measures

Book value of equity is commonly used in the P/B ratio. The P/B ratio is useful for comparing firms with mostly tangible assets and similar accounting practices, leverage, and asset profiles. For some industries, such as banks, analysts use **price-to-tangible book value**, which excludes intangibles such as goodwill from book value.

When peer firms have differences in financial leverage, or when book value is negative, analysts may prefer to use the **value-to-book multiple**, which is firm value divided by the book value of capital (i.e., book value of equity plus book value of debt):

$$\text{value-to-book} = \frac{\text{market value of equity} + \text{market value of debt}}{\text{book value of equity} + \text{book value of debt}}$$

The **enterprise value-to-book ratio** uses EV in the numerator and removes cash from the denominator:

$$\text{EV-to-book} = \frac{\text{market value of equity} + \text{market value of debt} - \text{cash}}{\text{book value of equity} + \text{book value of debt} - \text{cash}}$$

We can also state an EV-to-book ratio in terms of the factors that drive it. Recall the constant growth formula for EV:

$$\text{EV} = \frac{\text{FCFF}_1}{\text{WACC} - g} = \frac{\text{EBITDA}(1 - t) + \text{depreciation}(t) - \text{CapEx} - \text{WCInv}}{\text{WACC} - g}$$

Depreciation, capital expenditures, and working capital investment together make up a firm's reinvestment, the amount of its EBIT that is not available to debtholders and equity holders. A firm's **reinvestment rate** is its reinvestment as a proportion of EBIT. We can state free cash flow to the firm in terms of the reinvestment rate:

$$\text{FCFF} = \text{EBIT}(1 - t) \times (1 - \text{reinvestment rate})$$

We can define a firm's **return on capital (ROC)** as the ratio of its after-tax EBIT to its book value of capital:

$$\text{ROC} = \frac{\text{EBIT}(1 - t)}{\text{book value of capital}}$$

Using these definitions for ROC and the reinvestment rate (along with some algebra), we can express a ratio of EV-to-book value as follows:

$$\frac{\text{EV}}{\text{BV of capital}} = \frac{\text{ROC}(1 - \text{reinvestment rate})}{\text{WACC} - g}$$

EXAMPLE: EV-to-book value

Accel Industries has an EV / BV of capital ratio of 2.1x. Accel's earnings are growing at 3% and its WACC is 8%. Accel's return on capital is 12% and its reinvestment rate is 30%. Determine whether Accel is fairly valued, overvalued, or undervalued.

Answer:

$$\begin{aligned} \text{Justified EV / BV of capital} &= \frac{\text{ROC}(1 - \text{reinvestment rate})}{\text{WACC} - g} \\ &= \frac{0.12(1 - 0.3)}{(0.08 - 0.03)} = 1.68x \end{aligned}$$

Because Accel's current EV / BV of capital of 2.1 is greater than 1.68x, Accel is overvalued.

Cash Flow Statement Measures

The **price-to-dividend multiple** is the ratio of a stock price to its dividends per share. The **dividend yield** is the reciprocal of price-to-dividend multiple. A **trailing dividend yield** is based on the dividend paid in the past year. A **leading dividend yield** uses the expected dividend in the next year.

We can expand the dividend yield to express it as a function of the dividend payout ratio and the firm's earnings yield:

$$\frac{D_0}{P_0} = \frac{D_0}{E_0} \times \frac{E_0}{P_0}$$

Another way to express the dividend yield starts with the constant growth model:

$$P_0 = \frac{D_0(1+g)}{(r-g)}$$

Dividing both sides by D_0 , we get:

$$\frac{P_0}{D_0} = \frac{(1+g)}{(r-g)}$$

Inverting both sides of the equation gives us this expression for a justified dividend yield:

$$\frac{D_0}{P_0} = \frac{(r-g)}{(1+g)}$$

Unlike price multiples, with yield measures such as dividend yield or earnings yield, a higher value than justified indicates an undervalued stock; a lower value indicates an overvalued stock. As with multiples, yield measures are compared to peers after controlling for differences in fundamental factors such as growth rates and required ROE.

EXAMPLE: Justified dividend yield

Accel's dividend payout ratio is 60%, while its justified trailing P/E multiple is 12. Betax's required rate of return on equity is 10%, while its earnings have a constant growth rate of 4%. Calculate Accel and Betax's justified dividend yield.

Answer:

Accel's justified earnings yield is the reciprocal of its justified P/E, or 1/12.

$$\begin{aligned}\text{Justified dividend yield} &= \text{dividend payout ratio} \times \text{justified earnings yield} \\ &= 0.60 \times 1/12 = 0.05, \text{ or } 5\%.\end{aligned}$$

$$\begin{aligned}\text{Betax's justified dividend yield} &= \frac{(r-g)}{(1+g)} \\ &= \frac{(0.10 - 0.04)}{(1.04)} = 0.0577 = 5.77\%.\end{aligned}$$

While other cash flow statement items like CFO are also used in multiples, FCFE is more appropriate because it more accurately reflects the cash available to shareholders. Similarly, FCFF is appropriate to use for EV multiples because it reflects cash available to both shareholders and debt holders.

Justified P/FCFE and EV/FCFF can be derived from their respective constant growth models (after dividing both sides by the respective cash flow metric):

$$\frac{P_0}{\text{FCFE}_1} = \frac{1}{r - g}$$

$$\frac{\text{EV}}{\text{FCFF}_1} = \frac{1}{\text{WACC} - g}$$

One concern with either free cash flow-based multiple is that these measures are sensitive to fluctuations in capital expenditures and working capital investment, which tend to be cyclical.

Analysts use unique price or EV multiples for some industries, based on measures specific to those industries. For example, mining and oil firms may have multiples based on production or developed reserves. A problem with this approach is that these multiples are not comparable to other industries, which can potentially lead to an entire industry becoming mispriced.



MODULE QUIZ 45.2

1. Allied Pharma's stock currently trades at \$35 and has a forward earnings yield of 7.5%. If Allied's required rate of ROE is 10%, the proportion of its current price represented by growth opportunities is *closest* to:
A. 25%.
B. 50%.
C. 75%.
2. The average ROE for Lever, Inc., over the last business cycle was 32%. Lever's earnings per share for the coming year are expected to be \$5.00. Lever's dividend payout ratio is 30%, and its current book value per share is \$14. Shares are trading at \$54. Lever's normalized earnings per share are *closest* to:
A. \$4.48.
B. \$5.00.
C. \$5.26.
3. Lawler, Inc.'s, EBITDA is forecast to be \$8 million. An analyst forecasts Lawler's capital expenditures to be \$4 million, its working capital investment to be \$2 million, and its depreciation to be \$4.2 million. The analyst estimates that Lawler's WACC is 9%, its EBITDA growth rate is 4%, and its tax rate is 25%. Lawler's justified EV/EBITDA multiple is *closest* to:
A. 2.63x.

- B. 3.51x.
- C. 4.20x.

MODULE 45.3: PEER GROUPS AND MULTIPLES-BASED VALUATION

LOS 45.c: Determine a potential peer group for multiples-based equity valuation based on a company's industry classification or specific drivers of expected return.

Comparables-based valuation uses mean or median multiples of a peer group as benchmarks. Peer groups can be determined based on (1) companies with the same business activity as the subject company (an industry-based peer group) or (2) companies with same statistical **factor exposures** to risk and return as the subject company.

An industry-based peer group consists of firms with similar competitive forces, supply chain dynamics, and technology. Analysts should still take care when identifying peer group members because firms may have differences in their fundamental factors (e.g., leverage) or exposure to multiple industries and markets. Peers should be similar in terms of cyclicalities, life cycle stage, and market structure. The idea is that similar firms should generate similar cash flows.

Issues analysts may encounter while using an industry-based peer group include the following:

- Unique firms with no (or few) similar companies
- High variation among life cycle stages of the firms in an industry
- Firms with exposure to different tax or regulatory jurisdictions
- Multi-industry companies

Statistical factor-based peer groups comprise companies with the same or similar risk-return factor exposures. While the CAPM is a single-factor model that measures risk premium based on a company's systematic risk (beta), multifactor models (which we will discuss in a later reading) are more commonly used in identifying a factor-based peer group. A disadvantage of this approach is that statistical relationships based on past data might not persist into the future, which can result in models being misspecified.

LOS 45.d: Describe the use of multiples based on past, current, and projected future values in estimating equity value.

One concern with comparables-based valuation is that it fails to identify overvalued or undervalued stocks when their peer companies are also overvalued or undervalued. Another concern is that the multiples based on current values (e.g., of earnings) may be influenced by accounting choices, nonrecurring items, or cyclicalities.

Alternatively, comparables may be based on two elements:

Historical values of the subject company's multiples. This allows analysts to recognize changes in the multiples over time and link them to changes in the company's fundamentals.

Historical averages of peer group multiples. This allows analysts to compare the company's multiples to its peers over time.

As mentioned earlier, multiples based on projected future values can be used to estimate terminal values for present value models. In such a case, the forecast multiple is applied to a forecast metric, such as earnings per share.

EXAMPLE: Estimating a terminal value using forecast multiples

Level Partners is expected to have earnings in 10 years of \$12 per share, a dividend payout ratio of 30%, and a required return of 11%. At that time, the dividend growth rate is expected to fall to 4% in perpetuity. Estimate the terminal value at the end of 10 years.

Answer:

$$\begin{aligned}\text{Forecast justified trailing P/E} &= \frac{P_{10}}{E_{10}} = \frac{(\text{dividend payout ratio})(1 + g)}{(r - g)} \\ &= \frac{(0.3)(1 + 0.04)}{(0.11 - 0.04)} = 4.46x\end{aligned}$$

$$\begin{aligned}\text{Terminal value in Year 10} &= E_{10} \times \text{justified trailing P/E} = 12 \times 4.46 \\ &= \$53.52\end{aligned}$$



MODULE QUIZ 45.3

1. Which of the following is *least likely* an issue with comparables valuation based on industry-based peer groups?
A. Unique firms.

- B. Multi-industry companies.
 - C. Similar life cycle stages of firms in an industry.
2. An analyst observes that Tiller Company's P/E ratio is less than those of its industry peers, and concludes that Tiller's stock is undervalued. The analyst is *most likely* to be mistaken about Tiller's valuation if the industry as a whole is:
- A. overvalued.
 - B. undervalued.
 - C. fairly valued.

KEY CONCEPTS

LOS 45.a

The method of comparables values a stock based on the average or median price multiple of its peer companies. The method of forecasted fundamentals estimates the benchmark multiple based on a discounted cash flow model.

The PEG ratio controls for growth rate differences between comparable firms:

$$\text{PEG ratio} = \frac{\text{P/E multiple}}{\text{growth rate}}$$

A regression approach can control for differences in multiple fundamentals. Regression coefficients are estimated from a peer group sample.

LOS 45.b

A stock price theoretically represents the present value of a perpetuity of its current dividend plus the present value of the company's growth opportunities (PVGO):

$$P_0 = \frac{E_1}{r} + \text{PVGO}$$

Core earnings exclude nonrecurring components. Normalized earnings are estimated for the middle of the business cycle.

Normal EPS is calculated as the average EPS over the most recent business cycle, or the average ROE over the cycle applied to the most recent book value per share. Average ROE accounts for changes in firm size over time.

Compared to the price-to-earnings (P/E) multiple, a price-to-sales (P/S) multiple is more suitable for mature, cyclical, and negative-earnings firms.

$$\text{justified P/S} = \frac{(\text{dividend payout ratio})(1 + g)(\text{profit margin})}{(r - g)}$$

An EV/EBITDA multiple can be used for firms with different capital structures, and for capital-intensive industries with significant depreciation and amortization.

$$\text{justified price-to-book ratio (P/B)} = \frac{(\text{ROE} - g)}{(r - g)}$$

$$\begin{aligned} \text{value-to-book ratio} &= \frac{\text{EV}}{\text{book value of equity} + \text{book value of debt}} \\ &= \frac{\text{ROC}(1 - \text{reinvestment rate})}{\text{WACC} - g} \end{aligned}$$

$$\text{justified dividend yield} = \frac{r - g}{1 + g}$$

LOS 45.c

Peer groups can be identified as companies in the same industry (engaging in the same principal activity), or as companies with the same statistical factor exposures as the subject company.

Issues with industry-based peer group include unique firms, multi-industry firms, variation in life cycle stages of companies in an industry, and firms in different tax or regulatory jurisdictions.

LOS 45.d

Concerns for comparables-based valuation include that it can fail when the comparables are mispriced, and that multiples are based on accounting data.

In addition to comparing multiples to peer companies, analysts can compare multiples to their historical values for the subject company or to historical peer averages.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 45.1

- . **B** A regression approach can control for multiple factors. The PEG ratio only controls for growth rate differences. The method of comparables is not appropriate when peer companies are not comparable. (LOS 45.a)
- . **B** $\text{PEG ratio} = \frac{\text{P/E}}{g} = \frac{12.50}{5} = 2.50$. (LOS 45.a)

- A Earnings differences are controlled by calculating the P/E ratio. Differences in accounting methods, growth rates, and risk can make a comparables-based approach inappropriate. (LOS 45.a)

Module Quiz 45.2

• A
$$\frac{PVGO}{P_0} = 1 - \frac{\frac{E_1}{P_0}}{r}$$
$$= 1 - \left(\frac{0.075}{0.10} \right) = 0.25, \text{ or } 25\%.$$

(LOS 45.b)

- A Only the average ROE and the book value per share are relevant for calculating normalized earnings:

$$\begin{aligned} \text{Normalized earnings} &= \text{average ROE} \times \text{BVPS} \\ &= 0.32 \times \$14 = \$4.48 \text{ per share.} \end{aligned}$$

(LOS 45.b)

• A
$$\begin{aligned} FCFF_1 &= EBITDA(1 - t) + \text{depreciation}(t) - \text{CapEx} - \text{WCInv} \\ &= 8(1 - 0.25) + (4.2 \times 0.25) - 4 - 2 \\ &= \$1.05 \text{ million} \end{aligned}$$

$$\begin{aligned} EV &= \frac{FCFF_1}{(WACC - g)} \\ &= \frac{1.05}{(0.09 - 0.04)} \\ &= \$21 \text{ million} \end{aligned}$$

$$\frac{EV}{EBITDA} = \frac{21}{8} = 2.625x$$

(LOS 45.b)

Module Quiz 45.3

- C Firms in an industry may be at *different* life cycle stages, which can make them less relevant for comparables-based valuation. Issues with industry-based peer group include unique firms and multi-industry firms. (LOS 45.c)
- A If the industry peer group as a whole is overvalued, a firm that has a lower price multiple may have a fairly valued stock, or one that is less overvalued than its industry. (LOS 45.d)

READING 46

FINANCIAL STATEMENT FORECASTING IN EQUITY VALUATION

MODULE 46.1: FINANCIAL STATEMENT FORECAST MODELS

LOS 46.a: Explain the rationale, construction, and uses of financial statement forecast models in equity valuation.

The present value models we have seen in earlier readings value equity using a constant growth rate or multiple growth rates over a forecast horizon.

Disaggregated valuation models use forecast financial statements, allowing the analyst more flexibility and granularity. Analysts can incorporate their industry projections, competitive forecasts, and opinions about individual company attributes into these forecasts.

A disaggregated valuation model has four components: forecasting a company's revenues, expenses, assets, and financing.

Revenue Modeling

Revenue forecasts can be based on a historical approach, a bottom-up approach, or a top-down approach:

- | A *historical approach* relies on a company's past growth rates in revenue as estimates of its future growth rate. We can also estimate revenue growth rates of a company's individual business segments separately, because public companies operating in multiple lines of business must disclose revenue and other financial data for economically significant operating segments. A historical approach is useful for mature firms that have sufficiently long

historical data and stable business models, so that we can reasonably expect their past trends to continue.

- | A *bottom-up approach* incorporates expectations about a company's specific business units. For example, airlines use available seat kilometers (ASK) as a capacity metric. Analysts combine their capacity utilization forecasts, the airlines' planned changes in capacity, and an expected price per ASK to produce a bottom-up revenue forecast. Another example of a bottom-up approach is forecasting for retail or restaurant chains, where a growth rate in same-store sales is combined with data on planned new store openings.
- | A *top-down approach* begins with expectations about a macroeconomic variable, often the expected growth rate of nominal GDP. For example, revenue projections are derived from an estimate of GDP growth and an expected relationship between GDP growth and company sales. In the "market growth and market share" top-down approach, we begin with an estimate of industry sales (market growth), then estimate company revenue as a percentage of industry sales (market share). For example, if the industry revenue growth forecast is 3% and company market share is expected to increase from 12% to 14%, the forecast growth in revenue is $3\% + \left(\frac{14}{12} - 1\right) = 19.67\%$.

Expense or Profit Modeling

Expense forecasts can be based on a historical approach, itemizing economically significant expenses, or a direct profit margin approach.

- | A *historical approach* links individual expenses to revenues based on past data. A regression model can capture the fixed and variable components of expenses:

$$\Delta \text{Expense} = b_0 + b_1 \times \Delta \text{revenue}$$

The intercept b_0 represents the fixed component and the slope coefficient b_1 represents the variable component. This is done for each expense, as shown in the following example.

EXAMPLE: Regression model for expense forecasting

Shiv Dugal is analyzing Allied Pharma and has modeled each element of Allied's expenses using a separate regression, with the change in the expense as the dependent variable and the change in revenue as the independent variable. Regression results for the changes in cost of goods sold (COGS) and

selling, general, and administrative (SG&A) expenses are shown in the following tables:

ΔCOGS	Coefficient
Intercept	0.01
b_1	0.90

$\Delta \text{SG\&A}$	Coefficient
Intercept	0.02
b_1	0.64

Estimate the growth rate in COGS and SG&A expense if revenues are forecast to increase by 5%.

Answer:

1. $\Delta \text{COGS} = 0.01 + (0.90 \times 0.05) = 0.055$, or 5.5%.
2. $\Delta \text{SG\&A} = 0.02 + (0.64 \times 0.05) = 0.052$, or 5.2%.

Because regression models require a minimum sample size to be reliable, this historical approach can only be used for established companies with sufficient historical data.

- | *Itemizing expenses* allows analysts to incorporate new information or assumptions in their forecasts. For example, the terms of a new labor agreement should be reflected in forecasts of wage and salary expenses. Analysts can use company annual reports to identify significant line-item expenses for which they need to fine-tune their estimates.
- | A *direct profit margin approach* assumes a profit margin explicitly, without disaggregating individual expenses. For some industries, COGS is an overwhelmingly large component of total cost and can be reasonably expected to vary with revenues. An analyst can fine-tune the estimated gross margin based on a solid understanding of industry and company dynamics. Profit margin estimates are difficult to forecast in practice, especially if companies have one-off charges such as restructuring or impairment. Some expenses, such as interest, tend not to vary directly with revenues.

EXAMPLE: Direct profit margin forecast

An analyst estimates that Allied's current revenue of \$116,230,000 will grow 5% next year, and that its gross margin will increase from 14% to 15%. Estimate the COGS for Allied for the next year.

Answer:

$$\begin{aligned}\text{Revenue} &= 116,230,000 \times 1.05 = 122,041,500 \\ - \text{gross margin} &= 122,041,000 \times 0.15 = \underline{18,306,225} \\ &= \text{COGS} \qquad \qquad \qquad 103,735,275\end{aligned}$$

Asset Modeling

Disaggregated models allow analysts to estimate different levels of assets to support the forecast revenues. A constant growth rate model assumes assets grow at the same rate as revenues. In other words, it assumes total asset turnover remains constant. Alternatively, disaggregated analysis of individual asset categories can be used to forecast assets.

A regression model similar to the one we described for expenses can also be used for assets, although this approach is limited by the nature of capital expenditures (fixed assets must be put in place before they can start generating revenues). Only assets or liabilities that have cash flow implications should be considered. For example, changes in goodwill due to impairments do not have cash flow implications, so they should be left out of a model.

Financing Modeling

The final step in financial statement forecasting is to determine the sources and amount of funding a company needs.

$$\text{financing needs} = \Delta \text{operating assets} - \Delta \text{non-debt liabilities} - \Delta \text{retained earnings}$$

Operating assets can be viewed in this context as all assets other than cash. Non-debt liabilities are non-interest-bearing liabilities such as accounts payable and accrued expenses.

When the quantity ($\Delta \text{operating assets} - \Delta \text{non-debt liabilities}$) is greater than zero, the company has funding needs. These can be funded with existing cash, by retaining earnings, by borrowing, or by issuing additional stock.

When the quantity ($\Delta \text{operating assets} - \Delta \text{non-debt liabilities}$) is less than zero, the company has a surplus that can be used to pay dividends, repurchase shares, pay down debt, or increase cash.

Analysts should ensure that their assumptions are reasonable about how a company will meet its funding needs or use its surplus. For example, the company's forecast debt ratio should be within its normal range. Similarly, an analyst should not assume new stock issuance or repurchases unless a company's past actions support this. Cash balances should not be assumed to be either too high or below an acceptable, company-specific minimum.

EXAMPLE: Financing sources

Select information for Allied Pharmaceuticals for 20X1 and 20X2 (forecast) is shown in the following table:

	20X1	20X2(f)
Operating assets	\$148,760	\$160,400
Non-debt liabilities	\$74,150	\$77,890
Net income	\$5,000	\$7,800

If Allied continues to pay no dividends, and assuming no change in cash, calculate the new financing needed for 20X2.

Answer:

$$\begin{aligned}\text{Financing needed} &= \Delta \text{operating assets} - \Delta \text{non-debt liabilities} \\ &\quad - \Delta \text{retained earnings} \\ &= (160,400 - 148,760) - (77,890 - 74,150) - 7,800 = \$100\end{aligned}$$



MODULE QUIZ 46.1

1. Goodlife operates a chain of restaurants. An analyst expects that Goodlife's same-restaurant sales will increase by 5%, and Goodlife will open three new restaurants. What type of revenue model is the analyst using?
 - A. Top-down approach.
 - B. Bottom-up approach.
 - C. Historical approach.
2. An analyst using a historical approach to forecast a company's expenses is *most likely* to:
 - A. use a regression model to estimate the fixed and variable components of the company's expenses.
 - B. anticipate that the company's COGS will remain constant as a percentage of revenues.
 - C. project an above-trend increase in the company's SG&A expenses to reflect the company's planned new advertising campaign.

MODULE 46.2: MODELS BASED ON COMPANY CHARACTERISTICS

LOS 46.b: Evaluate the construction of an equity valuation model based on characteristics of the subject company.

Equity valuation models may vary with a company's life cycle stage.

Start-Up Stage Modeling

Because a typical start-up firm has negative earnings and no debt, the value of equity equals the value of the firm. Start-ups derive their value from their growth potential and are typically valued as a multiple of anticipated future sales.

Start-ups need **venture capital (VC)** from specialist private equity firms called VC firms. Because of the high risk of investing in a start-up, VC firms demand a high return on investment. This return metric is expressed as a multiple of original investment, over the period of the investment:

$$\text{ROI multiple} = (1 + r)^n$$

where:

n = number of years to exit of VC investment

r = required rate of return

A start-up firm's intrinsic value can be expressed in terms of its projected revenue and price-to-sales ratio (P/S):

$$IV_t = \frac{\text{expected revenue} \left(\frac{\text{price}}{\text{sales}} \right)}{\text{ROI multiple}}$$

The value of the firm immediately after the VC investment at $t = 0$ is the **post-money value**. **Pre-money value** is the firm's value before the VC investment.

$$\text{pre-money value} = \text{post-money value} - \text{VC investment}$$

The VC investor's fractional equity ownership (f) is:

$$f = \frac{\text{VC investment}}{\text{post-money value}}$$

EXAMPLE: Valuation of a start-up

Xeta Networks is a start-up that needs a VC investment of \$2 million to bring its product to market. Over the VC investment horizon of five years, revenues are expected to reach \$10 million, and the firm is expected to be valued at a P/S

ratio of 4x. Based on the risk of the firm, the VC investor wants an ROI multiple of 8x. Calculate the pre-money value of Xeta Networks and the VC investor's fractional ownership.

Answer:

$$\text{post-money value} = \frac{\text{expected sales} \left(\frac{\text{price}}{\text{sales}} \right)}{\text{ROI multiple}}$$

$$\frac{\$10 \text{ million} (4)}{8} = \$5 \text{ million}$$

$$\begin{aligned} \text{pre-money value} &= \text{post-money value} - \text{VC investment} \\ &= \$5 \text{ million} - \$2 \text{ million} = \$3 \text{ million} \end{aligned}$$

$$\text{fractional ownership} = \frac{\text{VC investment}}{\text{post-money value}} = \frac{\$2 \text{ million}}{\$5 \text{ million}} = 40\%$$

Growth Stage Modeling

Firms in the growth phase tend to have some historical financial data, use cash flows to fund growth objectives, and can be valued based on their profit margins and sustainable growth rates. Growth firms derive most of their value from their terminal value, as shown in the following example.

EXAMPLE: Valuation of a growth firm

Xeta Networks was able to achieve revenues of \$10 million. Over the next five years, all earnings will be reinvested in the business. Revenues are expected to grow at 30% per year, with COGS expected to be 40% of revenue and SG&A expenses expected to be 20% of revenue.

Assuming a 30% tax rate, calculate the value of Xeta's equity, assuming a P/E multiple of 12x after five years and a required rate of return of 10%.

Answer:

$$\text{Revenues after five years} = \$10 \text{ million} \times (1.30)^5 = \$37,129,300$$

$$\text{Earnings before tax \%} = 1 - \text{COGS\%} - \text{SG\&A\%} = 1 - 40\% - 20\% = 40\%$$

$$\text{Net income \%} = \text{EBT\%}(1 - t) = 40\%(1 - 0.30) = 28\%$$

$$\text{Net income} = 0.28 \times \$37,129,300 = \$10,396,204$$

$$\begin{aligned}
 \text{Terminal value in five years} &= \text{net income} \times \text{P/E multiple} \\
 &= 10,396,204 \times 12 = \$124,754,448 \\
 \text{Value}_{t=0} &= \frac{124,754,448}{1.10^5} = \$77,462,697
 \end{aligned}$$

Mature-Stage and Decline-Stage Modeling

Mature-stage firms tend to have stable growth in earnings and cash flow and ample historical financial data. A constant growth model can be used to value these companies' equity.

Decline-stage firms have declining profit margins and are primarily concerned about returning capital to investors. Firms may have difficulty in meeting their obligations and may be in financial distress.

Analysts adjust decline-stage firms' historical data to reflect any key income statement and balance sheet items that are affected by deterioration in the business. Due to greater risk faced by both debt and equity investors, discount rates are higher for decline-stage firms.

A going-concern assumption may not be valid if bankruptcy becomes likely. A probability-weighted average value of a decline-stage firm's equity can be calculated as follows:

$$\text{value} = \text{probability of going concern} \times \text{going-concern value} + \text{probability of distress} \times \text{distress sale value}$$

EXAMPLE: Valuation of a firm in decline

After 10 years, Xeta Networks is facing significant competition, and its margins have narrowed. An analyst estimates a 30% probability of bankruptcy that would cause the firm's assets to be surrendered to creditors. Xeta's equity is valued at \$12 million on a going-concern basis. Calculate the value of Xeta's equity given its probability of distress.

Answer:

In the event of bankruptcy, equity value = 0 (firm will be owned by creditors)

$$\begin{aligned}
 \text{value} &= \text{probability of going concern} \times \text{going-concern value} \\
 &\quad + \text{probability of distress} \times \text{distress sale value} \\
 &= (0.70 \times \$12 \text{ million}) + (0.30 \times \$0) = \$8.4 \text{ million}
 \end{aligned}$$

LOS 46.c: Estimate and interpret the value of an equity security based on the outputs of a financial statement forecast model.

**PROFESSOR'S NOTE**

This LOS requires candidates to apply techniques learned in prior modules based on a provided output. A multiple-choice question on the Level I exam would most likely present a small subset of an example like the one that follows.

EXAMPLE: Alpine Semiconductors

Beatrice Simpson is analyzing Alpine Semiconductors, a computer chip maker. The following table shows the firm's income statement and balance sheet for 20X1. Alpine's tax rate is 20%.

Table 1: Alpine Semiconductors Income Statement (Millions)

	20X1
Revenue	1250
COGS	812.5
Gross Profit	437.5
SG&A	188
D&A	78
EBIT	171.5
Interest Expense	12
EBT	159.5
Taxes	31.9
Net Income	127.6
Cash & ST Inv.	420
Operating Assets	1150
Total Assets	1570
—	—
Non-Debt Liabilities	72
Total Debt	150
Total Liabilities	222
Equity	1348
Total Liabilities & Equity	1570
Notes:	
Net Investment	68

Simpson calculates Alpine's WACC as 8%. She believes a constant growth assumption is not valid for the next three years and estimates individual growth rates, as shown in the following table.

Table 2: Growth Estimates

	Growth Rates		
	20X2	20X3	20X4
Revenue	5%	3%	2%
COGS	4%	3%	1%
SG&A	3%	2%	2%
D&A	3%	3%	1%
Interest Expense	5%	2%	1%
Net Investment	5%	2%	2%

Simpson prepares the following forecasts for 20X2–20X4.

Table 3: Forecasts for 20X2–20X4

	20X2	20X3	20X4
Revenue	1312.50	1351.88	1378.91
COGS	845.00	870.35	879.05
Gross Profit	467.50	481.53	499.86
SG&A	193.64	197.51	201.46
D&A	80.34	82.75	83.58
EBIT	193.52	201.26	214.82
Interest Expense	12.60	12.85	12.98
EBT	180.92	188.41	201.84
Taxes	36.18	37.68	40.37
Net Income	144.74	150.73	161.47
Notes:			
Net Investment	71.40	72.83	74.28

1. Based on the assumptions she has used, does Simpson expect Alpine's profit margins to increase, decrease, or remain stable over the next three years?
2. Does Simpson assume Alpine will meet its funding shortfall with retained earnings, by reducing its cash balances, or by borrowing?
3. What is the sustainable growth rate under Simpson's assumptions, based on FCF?

Answers:

1. Simpson's assumptions are shown in Table 2 and indicate that revenues are expected to grow faster than expenses. No calculations are needed to conclude that Simpson expects Alpine's profit margins to increase.
2. Table 2 shows a positive growth rate in interest expense, which indicates a higher borrowing level in the future. This is consistent with Alpine funding at least part of its shortfall using debt.
3. From Table 1: $\text{NOPAT} = \text{EBIT} (1 - t) = 171.50 (1 - 0.20) = 137.20$

$$\text{FCFF} = \text{NOPAT} - \text{net investment} = 137.20 - 68 = 69.20$$

$$\text{Book value of capital} = \text{book value of debt} + \text{book value of equity} = 150 + 1,348 = 1,498$$

Using the model for estimating a sustainable growth rate based on free cash flow to the firm, which we introduced in our earlier reading on Discounted Cash Flow and Growth Models:

$$\begin{aligned} \text{SGR}_{\text{FCFF}} &= \left(1 - \frac{\text{FCFF}}{\text{NOPAT}}\right) \left(\frac{\text{NOPAT}}{\text{beginning book value of total capital}}\right) \\ &= \left(1 - \frac{69.20}{137.20}\right) \left(\frac{137.20}{1,498}\right) = 0.0454 = 4.54\% \end{aligned}$$



MODULE QUIZ 46.2

1. Cytus Pharma needs a VC investment of \$30 million. Its VC investor's investment horizon is six years, when the company's annual revenues are forecast to be \$120 million and its value is expected to reflect a P/S ratio of 5x. The VC investor has an ROI multiple of 6x. The fractional ownership demanded by the VC investor is *closest* to:
 - A. 5%.
 - B. 30%.
 - C. 50%.
2. Beta Industrials has \$10 million par value of debt coming due at the end of the year. Beta's auditors have issued a qualified audit report indicating doubts about the firm's going-concern assumption. Beta's current annual sales are \$15 million. An analyst estimates that the firm's enterprise value-to-sales ratio is 2x as a going concern, with a 50% probability, or a liquidation enterprise value-to-sales ratio of 0.8x. The analyst's equity valuation of Beta is *closest* to:
 - A. \$2 million.
 - B. \$11 million.
 - C. \$21 million.

KEY CONCEPTS

Disaggregated valuation models allow an analyst to incorporate industry projections, the competitive environment, and company attributes into their forecasts.

Revenue forecasts can be based on historical growth rates, a bottom-up approach, or a top-down approach.

Expense forecasts can be based on a historical approach, itemizing significant expenses, or a direct profit margin approach.

Asset models can assume constant growth or forecast individual assets based on a regression model.

$$\text{financing needs} = \Delta \text{operating assets} - \Delta \text{non-debt liabilities} - \Delta \text{retained earnings}$$

LOS 46.b

Appropriate equity valuation models vary by a company's life cycle stage

Start-ups are typically valued as a multiple of anticipated future sales.

$$\text{ROI multiple} = (1 + r)^n$$

$$\text{post-money value} = \frac{\text{expected revenue} \left(\frac{\text{price}}{\text{sales}} \right)}{\text{ROI multiple}}$$

$$\text{venture capital investor's fractional equity ownership} = \frac{\text{VC investment}}{\text{post-money value}}$$

Growth firms derive most of their value from their terminal value. Mature-stage firms tend to be valued using a constant growth model.

Decline-phase firms may be in financial distress. Their valuation is a probability-weighted average of their values as a going concern and in liquidation.

LOS 46.c

Given selected outputs from a financial statement forecast model, candidates should be prepared to value an equity security using an appropriate valuation method.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 46.1

- . **B** Combining a forecast growth rate in same-store sales with planned new store opening data is an example of a bottom-up approach to revenue modeling. (LOS 46.a)
- . **A** Using a regression model to estimate the fixed and variable components of expense items is consistent with a historical approach to expense modeling. Considering information that affects a specific cost item is an example of itemizing expenses. Analyzing the relationship between revenues and COGS is consistent with a direct profit margin approach. (LOS 46.a)

Module Quiz 46.2

- . **B** $120 \text{ million} \times 5 = \600 million

$$\text{Post-money value} = \frac{\text{expected sales} \left(\frac{\text{price}}{\text{sales}} \right)}{\text{ROI multiple}}$$

$$= \frac{120 \text{ million} (5)}{6} = 100 \text{ million}$$

$$\text{Fractional ownership} = \frac{\text{VC investment}}{\text{post-money value}} = \frac{30}{100} = 30\%.$$

(LOS 46.b)

- . **B** Enterprise value as a going concern = $\$15 \text{ million} \times 2 = \30 million .

Equity value as a going concern = enterprise value as a going concern – value of debt = $\$30 \text{ million} - \$10 \text{ million} = \$20 \text{ million}$.

Enterprise value in liquidation = $\$15 \text{ million} \times 0.8 = \12 million .

Equity value in liquidation = $\$12 \text{ million} - \$10 \text{ million} = \$2 \text{ million}$.

Probability-weighted equity value = $0.50 \times \$20 \text{ million} + 0.50 \times \$2 \text{ million} = \$11 \text{ million}$.

(LOS 46.b)

READING 47

INDUSTRY AND COMPETITIVE ANALYSIS

MODULE 47.1: INDUSTRY ANALYSIS FRAMEWORK

LOS 47.a: Describe the process of industry and competitive analysis.

Analysts conduct industry and competitive analysis to assess the environment in which a company operates. An industry, in this context, is a group of firms that offers comparable products or services. By evaluating the drivers of an industry's size and profits, an analyst can clearly explain the rationale behind the inputs in valuation models and the narrative that shapes and supports these inputs.



PROFESSOR'S NOTE

As we will see in the Ethical and Professional Standards topic area, being able to explain these things helps an analyst demonstrate that an investment recommendation has a reasonable basis.

Conducting industry and competitive analysis involves five steps:

Step 1: Define the industry in terms of its scope, which is often subjective.

This requires identifying the factors that define an industry (e.g., product similarity or geographical region) and deciding how to treat companies that operate in multiple industries. Classification systems produced by third parties provide standardized guidance.

Step 2: Survey the industry in terms of its size, growth rate, profitability, and composition.

Step 3: Analyze the industry structure using a framework such as Porter's five forces to determine the degree of competitive rivalry and key profitability

drivers.

Step 4: Examine external influences using a framework such as PESTLE: political, economic, social, technological, legal, and environmental influences.

Step 5: Analyze companies' competitive strategies in terms of how each company fits within the industry and what competitive advantages each company has.

LOS 47.b: Describe industry classification methods and compare methods by which companies can be grouped.

The first step in industry and competitive analysis is to define the company's industry. Industries are composed of firms with common competitive forces and value drivers. Once a company has been classified into an industry, an analyst can search for peers against which to compare the company.

Commercial or Third-Party Classification Systems

One way to determine how to classify a company is to consult a commercial, or third-party, classification service. These global organizations independently assess the products and services offered by companies and classify those that are similar, according to hierarchical taxonomies. The lowest tier, where the company sits, is a single group; higher tiers are aggregate groups. Widely used classification systems include:

- | *Global Industry Classification Standard (GICS)*. This was developed by S&P Dow Jones Indices and MSCI for public companies. It classifies by sector, industry groups, industries, and subindustries.
- | *Industry Classification Benchmark (ICB)*. This was developed by FTSE Russell for public companies. It classifies by industries, supersectors, sectors, and subsectors.
- | *The Refinitiv Business Classification (TRBC)*. This was developed by Refinitiv for public companies, private companies, not-for-profits, and government entities. It classifies by economic sectors, business sectors, industry groups, industries, and activities.

These third-party providers review and update their classifications annually at a minimum, with some adding new companies to their offerings more frequently.

In addition to these global organizations, regional systems that provide classifications include the North American Industry Classification System (NAICS) and the European Nomenclature of Economic Activities (NACE).

Data providers such as Bloomberg, FactSet, and S&P Capital IQ, as well as stock exchanges such as the Nasdaq, either use these industry classification systems or construct their own similar taxonomies.

The primary advantage of using a commercial classification system is its consistency. However, these systems have limitations. The classifications may be too broad or too narrow and often require some judgment.

- | Industry groupings can include companies with different business models or substitute products. For example, McDonald's Corporation (MCD) and Yum! Brands, Inc. (YUM) are classified under the GICS Restaurant subindustry, but McDonald's owns a significant franchisee real estate portfolio, which some analysts believe should be valued separately for better comparison.
- | Companies might operate in multiple industries. Some classification systems with strict taxonomies classify companies based on the product or service that generates the most revenue. Not all systems follow the same guidelines, which can cause problems when attempting to make comparisons. For example, some systems assign Amazon to the "consumer discretionary retailer" industry despite its significant growth in cloud computing.
- | Industries may have different constituents due to geographical differences. These include health care or insurance companies, which cannot operate on a global basis. These industries may be less easy to compare using the taxonomies of the major third-party providers. Regulations, cultural behavior, or shipping costs may also make global taxonomies impractical.
- | Groupings shift or evolve over time. Groupings that shift or evolve over time to reflect natural business changes may undergo reclassification. For example, in 2023, GICS moved two companies, Visa and Mastercard, to the "transaction and payments processing services" classification in the financial sector (from the data processing industry within the information technology sector). Other examples of evolution include industries with new entrants, or delistings resulting from acquisitions and bankruptcies. To account for these types of changes, analysts may need to restate historical data to ensure that the current groupings are comparable.

Classification by Geography

Classification systems can be based on a company's country of domicile. Countries are, in turn, categorized by level of industrialization, such as developed, emerging, or frontier economies. The country used for classification is most often where a company is headquartered or listed on an exchange, even if it does most of its business elsewhere. For example, Toyota is considered a Japanese company, but it generates most of its revenues in North America. Analysts should be aware that a company's geographic classification might not reflect its revenue profile or the location of its assets.

Classification by Business Cycle Sensitivity

Depending on the degree to which economic fluctuations affect their revenue and profitability, companies can be classified as defensive, cyclical, or growth.

A **defensive company** produces goods and services with relatively stable demand over the business cycle. Defensive sectors include health care, utilities, telecommunications, and consumer staples. Those sectors produce items required for day-to-day living, the demand for which is not affected significantly by fluctuations in macroeconomic activity.

A **cyclical company** is one whose earnings are highly dependent on the stage of the business cycle. Such companies have high earnings volatility and typically high operating leverage. Their products often have long lifespans, such as materials or equipment sold to other producers; or high-unit-value items whose purchase can be delayed until the economy improves, such as automobiles or home appliances. The commodities that serve as inputs for these products also vary with demand. Fluctuating revenue and high fixed costs can result in volatile operating profits. Examples of cyclical industries include basic materials and processing, consumer discretionary, energy, financial services, and industrial and producer durables.



PROFESSOR'S NOTE

The curriculum learning module omits telecommunications from defensive companies and energy and producer durables from cyclical companies. We include them here for completeness.

A **growth company** typically demonstrates strong revenue growth through all cycles. These companies may or may not be profitable, depending on which

phase of the company life cycle characterizes them. When evaluating growth companies, analysts should try to determine how long growth will remain above trend and, as growth plateaus, what the profit levels are likely to be in the industry. Growth companies are often in industries such as emerging technology or biotechnology.

When an analyst covers companies in countries that vary with respect to their positions in the business cycle, combining business cycle sensitivity with geography to classify companies may be appropriate.

LOS 47.c: Determine an industry's size, growth characteristics, profitability, and market share trends.

The second step in industry and competitive analysis is to survey the industry to gauge its size, historical rate of growth, growth drivers, profitability, and key constituents and their market share trends.

Industry Size

Industry size is the total annual sales of the product or service under consideration. This is not always the same as total annual sales of all the companies in the industry. For companies that operate in multiple product lines, only the relevant portion of their total annual sales should be included in industry size.

Figure 47.1 describes two approaches to estimating industry size: top down and bottom up.

Figure 47.1: Approaches to Estimating Industry Size

Method	Definition	Advantages	Key Limitations
Top down	■ Estimates total industry revenue using data from government agencies or third-party providers	■ Best for industries with many private companies or small businesses that do not report earnings publicly	■ Data may lack granularity or specific detail required for niche analysis
Bottom up	■ Sums the individual revenues of every company producing goods and services within the industry	■ Provides a direct, precise measure when large, publicly traded companies dominate the industry	■ May be hard to isolate revenue for companies with multiple product lines ■ Inconsistent reporting segments make comparisons tough ■ Poor fit for industries with many private or unincorporated firms

Industry Growth Characteristics

Industry growth should be evaluated against broad measures of economic activity to determine an industry's life cycle stage, and over the business cycle to determine its sensitivity to fluctuations in the economy.

Growth industries typically expand faster than the general economy no matter what the stage of the business cycle because they have not yet reached all of their total addressable market. Analysts must judge the persistence of high growth rates and estimate when they will peak. This is more problematic when an industry's growth is based on an emerging technology.

Mature industries have little or no growth potential remaining in their markets. Given that the growth rate of all industries combined should equal the rate of economic growth, maturing industries should grow near or below the rate of the economy. Analysts should focus on any disruptors, such as significant changes in the intensity of competition among industry participants, as well as any signs that the industry might be declining overall.

Business cycle sensitivity is an additional consideration. Industries exposed to interest rate movements or those involving discretionary sales or capital goods experience more revenue volatility over a business cycle.

Structural changes, such as those that occurred during the COVID-19 pandemic, may also affect industry demand, and they need to be factored into any analysis.

Industry Profitability

Industry profitability is commonly measured using **operating profit margin** or **return on invested capital (ROIC)**:

$$\text{operating profit margin} = \frac{\text{operating income}}{\text{total revenue}}$$

$$\text{ROIC} = \frac{\text{NOPAT}_t}{\text{average invested capital}}$$

where:

NOPAT_t = net operating profit after taxes

average invested capital = the average of long-term liabilities plus equity over a given reporting period

ROIC measures value creation across all investments and can be compared to a company's weighted average cost of capital (WACC). When $\text{ROIC} > \text{WACC}$, the company is providing a positive return.

Unfortunately, it is often impractical to estimate ROIC for private companies that do not have publicly available financial statements. Instead, analysts may use the returns of publicly traded companies and assume they are similar for private companies, or they may use data provided by a government or a third-party provider.

Market Share

Another key component of the industry survey is to determine **market share** for each company within the industry. Over a specified period, market share is calculated as follows:

$$\text{market share} = \frac{\text{company sales}}{\text{industry size}}$$

Because industry size is difficult to measure, market share should be estimated as a range instead of a specific point.

The trend in a company's market share over time helps determine whether customers view its products favorably. By tracking market share changes over time, an analyst can also detect increased competition from new entrants to the market. Conversely, industry consolidation may increase a company's market share, and an analyst should determine whether a company is increasing its market share net of acquisitions.

Unless an industry is segmented or has a high degree of product differentiation,

greater industry concentration indicates less competition, and less industry concentration suggests more competition. An analyst can gauge industry concentration using the **Herfindahl-Hirschman Index (HHI)**. The HHI is the sum of the squares of the market shares of all industry participants.

$$HHI = \sum_{i=1}^n (\text{market share}_i)^2$$

For example, a market consisting of five companies, with shares of 35%, 25%, 20%, 10%, and 10%, would have an HHI of $35^2 + 25^2 + 20^2 + 10^2 + 10^2 = 2,450$.

The maximum HHI is 100^2 or 10,000, which would indicate that a company is a monopoly. In some countries, regulators use HHI to determine which industries require closer scrutiny. An HHI greater than 1,800 is often interpreted to indicate high concentration.



MODULE QUIZ 47.1

1. Industry classification systems from commercial index providers typically classify firms by:
 - A. statistical methods.
 - B. products and services.
 - C. business cycle sensitivity.
2. Firms and industries are *most appropriately* classified as cyclical or noncyclical based on:
 - A. their stock price fluctuations relative to the market.
 - B. the sensitivity of their earnings to the business cycle.
 - C. the volatility of their earnings relative to their competitors.
3. A pet food company earned \$500 million in revenue in the current year. Based on an estimated market share of 10%, what is the market size for pet food?
 - A. \$0.5 billion.
 - B. \$4.5 billion.
 - C. \$5.0 billion.

MODULE 47.2: PORTER'S FIVE FORCES AND PESTLE ANALYSIS

LOS 47.d: Analyze an industry's structure and external influences using Porter's Five Forces and the PESTLE framework.

The third step in industry and competitive analysis is to examine the industry's structure and the competitive rivalry within it.

The analytical framework developed by Michael Porter¹ describes industry competitive forces that determine long-run profitability. If some or all the forces he described are strong, industry return on capital is unlikely to exceed the cost of capital. Conversely, if the forces are less intense, industry profitability should be high.

To illustrate the concept of competitive advantage Porter discussed, Warren Buffet coined the phrase **economic moat**. Like the water that protected castles from invaders, an economic moat shields company profits from competitors. The longer a company can maintain its competitive advantage, the wider its economic moat, and the longer its profit will remain above its capital costs. Some examples of factors that contribute to an economic moat are production cost advantages, the ability to operate at an efficient scale, ownership of proprietary intangible assets, network effects from using a company's products, and high switching costs for customers.

In an industry with few competitors, such as an oligopoly, all companies have wide moats. In more competitive industries, fewer companies have wide moats, while most generate returns below their cost of capital.

To evaluate competitive rivalry within an industry, analysts can compare **Porter's five forces** with the factors that determine the width of a company's economic moat:

Threat of new entrants. Both the number of new entrants over time and their industry impact are important to consider when assessing market share. To address the threat posed by new rivals, a company may rely on the following courses of action:

- Partner with or acquire new rivals to mitigate their impact.
- Lobby for government regulation that restricts entry into an industry, or for government subsidies for domestic companies.
- Support restrictive licensing requirements so that the cost of complying with government regulation is a barrier to entry.
- File for patents that grant their holders exclusive rights to their products or technologies for numerous years.
- Develop intangible assets, such as customer loyalty.
- Increase retention by emphasizing the high costs of switching, which include the price, opportunity cost, transition risk, and missed opportunity to receive

upgrades from the existing provider.

- Leverage network effects to solidify market dominance as user adoption grows and enhances product value.
- Pursue cost advantages. These may coincide with other sources of competitive advantage such as scale, brand loyalty, proprietary technology, or economies of scope (when the cost of producing a combination of goods and services is less than producing each separately). Mature industries with inelastic demand for commoditized products must operate at an efficient scale. The fact that new entrants would potentially push returns in these industries below the cost of capital acts as a barrier to entry.

Threat of substitutes. Goods and services that address the same needs as an existing product limit the profit potential of an industry by increasing the elasticity of demand. These substitute products, in turn, limit the prices firms can charge. Substitutes may come from seemingly different industries and shift consumer norms, much like videoconferencing software has changed the landscape for workers and become a substitute for physical office space.

Bargaining power of suppliers. Suppliers' ability to raise prices or limit supply influences industry profitability. Suppliers are more powerful if just a few of them exist, if their products are scarce or customized, or if switching costs are high. For example, Microsoft is one of the few suppliers of operating system software, so it has pricing power. Supplier bargaining power may also be exerted through proprietary technology licenses, intellectual property, or specialized and highly skilled labor, including labor that requires licensing or is well organized. By contrast, supplier bargaining power is weak when there are many small-scale providers of commoditized goods that have ready substitutes.

Bargaining power of customers. Buyers' ability to bargain for lower prices or higher quality influences industry profitability. For example, bargaining by governments and ever-larger health care providers have put downward pressure even on patented drugs. Customers tend to have greater power when they are a large and concentrated base, when products are more commodity-like, or when switching costs are low. In these cases, customers may influence payment terms, pricing, quality standards, and delivery times. In response, companies may reward customer loyalty, bundle services, or differentiate their products. Companies looking to mitigate the effects of external bargaining power may pursue **vertical integration**, expanding their operations to an earlier stage of production (upstream or backward integration) or a later stage (downstream or

forward integration) to control more of the delivery process. Companies can also participate in joint ventures or exclusive supplier agreements, or they can eliminate intermediaries.

Rivalry among existing competitors. Rivalry may manifest as price competition, product differentiation, market share changes, or changes in industry growth or profitability. Slow growth leads to competition as firms fight for market share, and high fixed costs lead to price decreases as firms try to operate at full capacity.

EXAMPLE: Porter's five forces analysis for the passenger airline industry²

Threat of New Entrants: Low to Moderate

Potential entrants in the passenger airline industry face significant capital investment requirements, with industry incumbents benefiting from economies of scale. Such regulatory hurdles as strict aviation regulations, licensing requirements, and even barriers to exit in some markets serve as deterrents to new industry entrants.

While established global airlines benefit from extensive networks and frequent flyer programs, in some regions or markets, low-cost carriers (LCCs) have a lower entry barrier due to flexible leasing options, limited schedules, and the use of secondary airports. As a result, the threat of entry is low for major global carriers but moderate for LCCs in many regional and underserved markets.

Threat of Substitutes: Moderate

While driving and high-speed rail travel represent substitutes for short-haul passenger air travel, long-haul flights have few alternatives. Business-related travel remains affected by remote work trends and virtual meetings, as described in an earlier example. However, few alternatives to air transportation exist for leisure travel and tourism. As a result, substitutes for passenger air travel are a moderate threat and are more pronounced for shorter trips and business-related travel, while longer flights and leisure travel are less susceptible to substitution.

Bargaining Power of Suppliers: High

Passenger airline companies are dependent on aircraft manufacturers, fuel deliveries, and unionized employees as key suppliers. The global dominance of jet aircraft manufacturers Boeing and Airbus leave airlines with little

bargaining power. As a major expense item for passenger airlines, oil prices are subject to significant volatility, which affects operating margins. Airline pilots, engineers, and maintenance workers are often well organized; this affords them the ability to negotiate wage and benefit increases, driving up airline operating expenses. Airlines are heavily dependent on these suppliers, which have strong bargaining power due to their critical role in operations and the lack of substitutes.

Bargaining Power of Customers: High

Passenger airline customers are very sensitive to price differences, which are highly transparent given the widespread practice of online travel booking. While airlines use such strategies as branded credit cards, frequent flyer programs, premium seating, and travel lounge access, customer loyalty is limited, given low switching costs—particularly for short-haul trips and popular destinations.

Demand for non-essential air travel is also sensitive to the business cycle. Business travelers may face constraints due to restrictive travel policies, limit travel in economic downturns, and may forgo travel altogether in favor of remote meetings as described earlier. Leisure travelers may delay discretionary holiday travel or substitute ground travel for flights when seeking to limit expenses. As a result, customer bargaining power in the passenger airline industry is considered high.

Rivalry Among Existing Competitors: High

A high degree of industry rivalry results from numerous global and regional airlines and LCCs competing on price, routes, and service quality while facing high operating expenses and suppliers with significant bargaining power.

Regulatory constraints limit the degree of industry consolidation, while limitations on reducing capacity during economic downturns leads to greater price competition. As illustrated in an earlier example, the airline industry faces cyclical demand and returns below the cost of capital over the cycle, intensifying the rivalry among competitors.

PESTLE Analysis

The fourth step in industry and competitive analysis is to examine the industry's external influences. The **PESTLE framework** identifies external influences as political, economic, social, technological, legal, and environmental factors.

Political influences have an important and widespread effect on businesses through various channels, including taxes, subsidies, and reimbursements. Influences may be indirect (e.g., changes in monetary policy, regulation, and geopolitical factors) or direct (e.g., a government purchase of an industry's products or services). Sectors that are notably exposed to political influences include energy, health care, and defense.

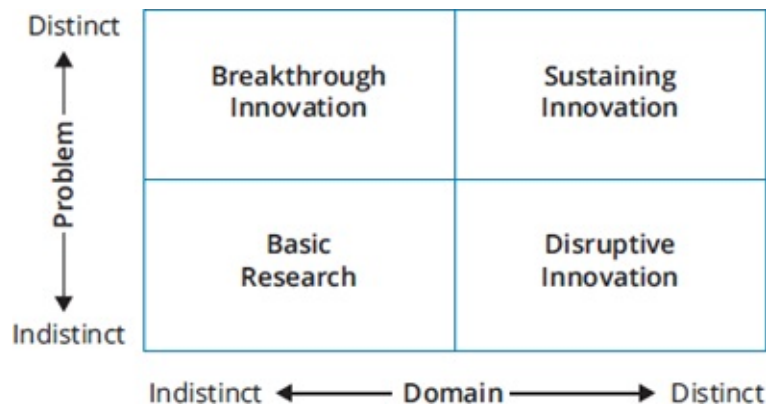
Economic influences can be cyclical trends in economic output or structural changes, such as an economy's competitiveness and stage of development. For example, capital investment in a developing country may increase labor productivity and lift its growth rate above those of developed countries. In conducting industry and competitive analysis, an industry's expected performance should align with economic forecasts after adjusting for its stage in the business cycle. Exchange rates also play a role in industry growth as they affect the costs and competitiveness of goods in other markets. A depreciating currency boosts exports, while a strengthening currency makes imports cheaper for domestic buyers.

Social influences relate to trends in how people work, play, spend their money, and conduct their lives. These factors are particularly important for industries that sell directly to individuals. At times, these trends evolve with economic change, such as a shift away from low-cost goods to brand-name products as income levels rise. For more affluent consumers, quality, brand image, and features such as flexibility, convenience, or health consciousness become increasingly important. Some companies are bowing to pressures to use socially responsible procurement of raw materials, and to ensure that production processes are ethical and observe human rights.

Technological influences can change an industry dramatically through the introduction of new or improved products or services that can make existing ones redundant. Technology change is key to total factor productivity within an industry and in the overall economy.

To assess the impact of technology advancement on industry competition, innovations can be classified according to their domain (i.e., the products, market segments, or business processes they affect) and the problems they are meant to solve. An **innovation matrix**, as shown in Figure 47.2, categorizes innovations into four categories: basic research, breakthrough innovation, sustaining innovation, and disruptive innovation.

Figure 47.2: Innovation Matrix³



- **Basic research.** This is *indistinct* in terms of its *domain* and the *problem*, and it deepens the understanding of scientific principles without a specific commercial objective.
- **Breakthrough innovation.** This has an *indistinct domain* and a *distinct problem*, and it introduces a new product, service, or business model that has significant impact across industries. An example is the rechargeable lithium battery used in smartphones, computers, and cars.
- **Sustaining innovation.** This focuses on *distinct problems* within *distinct domains*, such as feature enhancements that improve functionality.
- **Disruptive innovation.** This focuses on *indistinct problems* within *distinct domains*; it creates new markets or redefines existing markets. An example is the introduction of streaming services that have surpassed cable subscriptions. When markets are disrupted, companies can either change their business models or adapt more slowly and lose market share.

Legal influences consider changes in laws and regulations that present both business risks and opportunities. Consider the tobacco industry, which has faced declining revenues due to laws implemented in many jurisdictions over the past few decades, such as prohibiting smoking in public places and requiring extensive disclosures on cigarette packages. In contrast, consider the relatively new cannabis industry that has developed in jurisdictions that have legalized its sale and use.

Environmental influences, such as climate change and concerns about environmental sustainability of some industries, are often connected with political, social, and legal influences. The energy industry is often challenged to address environmental concerns. Many in the industry are grappling with the

energy transition, which is a government-mandated structural shift to reduce the emission of greenhouse gases through policy, taxes, and subsidies.

Analyzing individual companies within an industry, the fifth step in industry and competitive analysis, will be discussed in the next reading.



MODULE QUIZ 47.2

1. Two of the five competitive forces in the Porter framework are:
 - A. threat of entry and barriers to exit.
 - B. power of suppliers and threat of substitutes.
 - C. rivalry among competitors and power of regulators.
2. Assessing the impact on an industry of the economic outlook is *best* described as an example of:
 - A. surveying the industry.
 - B. analyzing the industry structure.
 - C. examining external influences.

KEY CONCEPTS

LOS 47.a

Industry and competitive analysis involves five steps:

Step 1: Define the industry.

Step 2: Survey the industry's size, growth, profitability, and market share trends.

Step 3: Analyze the industry's structure.

Step 4: Analyze external influences on the industry.

Step 5: Determine industry participants' competitive strategies.

LOS 47.b

Three commercial classification systems and their hierarchical structures (four or five tiers) are as follows:

GICS: sector, industry groups, industries, subindustries

ICB: industries, supersectors, sectors, subsectors

TRBC: economic sectors, business sectors, industry groups, industries, activities

Limitations of classification systems include inappropriate groupings with companies with different business models or substitute products; inaccurate classifications of companies selling multiple products; the inability to deal with products and services offered in local or national markets only; and groupings that change over time.

Other ways to group companies include geography, business cycle sensitivity, or a combination of these two approaches.

LOS 47.c

Industry size is the total annual sales of the product, which is not always the same as total annual sales of all the companies in the industry.

A top-down approach to evaluating industry size uses data from a government agency or third-party provider. A bottom-up approach involves adding the revenue from all the companies that produce the goods or services in question.

Growth industries expand faster than the general economy. Mature industries have growth rates that are in line with the general economy or decreasing.

Industry profitability is commonly measured as follows:

$$\text{operating profit margin} = \frac{\text{operating income}}{\text{total revenue}}$$

$$\text{ROIC} = \frac{\text{NOPAT}_t}{\text{average invested capital}}$$

Market share is a company's annual revenues as a percentage of the industry total sales.

Industry concentration is often expressed numerically using the Herfindahl-Hirschman Index (HHI), the sum of the squares of the market shares of all industry participants.

LOS 47.d

Porter's five forces that determine industry competition are as follows:

- Threat of new entrants

- Threat of substitutes

- Bargaining power of customers

- Bargaining power of suppliers

- Rivalry among existing competitors

PESTLE analysis consists of six elements:

- Political influences
- Economic influences
- Social influences
- Technology influences
- Legal influences
- Environmental influences

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 47.1

- . **B** Commercial classification systems classify firms according to the product or service they produce. (LOS 47.b)
- . **B** For industry analysis, cyclical firms and industries are those with earnings that are highly dependent on the business cycle, while defensive firms and industries are those with earnings that are relatively less sensitive to the business cycle. (LOS 47.c)
- . **C** $\text{Market share} = \frac{\text{revenue}}{\text{market size}}$. Here, $0.10 = \frac{\$500 \text{ million}}{\text{market size}}$.
So, $\text{market size} = \frac{\$500 \text{ million}}{0.10} = \5 billion . (LOS 47.c)

Module Quiz 47.2

- . **B** Porter's five forces are rivalry among existing competitors, threat of entry, threat of substitutes, bargaining power of buyers, and bargaining power of suppliers. (LOS 47.d)
- . **C** The economic outlook is an external influence on an industry. The PESTLE framework, which includes political, economic, social, technology, legal, and environmental influences, is used to evaluate the external environment. (LOS 47.d)

¹ Michael Porter, "The Five Competitive Forces That Shape Strategy," *Harvard Business Review*, Volume 86, No. 1: pp. 78–93.

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READING 48

COMPANY ANALYSIS: PAST, PRESENT, AND FUTURE

MODULE 48.1: CORPORATE STRATEGY AND INDUSTRY POSITION

LOS 48.a: Explain a company's competitive strategy and position within an industry.

The fifth and final step of industry and competitive analysis, which we described in our previous reading, is to evaluate a company's competitive strategy, including its competitive advantage, its position within its industry, and its resources and capabilities. This involves identifying the company's drivers of value creation, then analyzing its financial statements to evaluate its performance.

A company's **business model** describes how it defines and creates value in terms of who it serves, what it sells, how it prioritizes resources, and how it defends itself against competitors. A company's **competitive strategy** is a critical component, and every company has one, whether intentionally or not.

- | **Intentional competitive strategy.** Proactive and driven by clear, measurable goals, an intentional strategy is shared with internal staff and external investors. The company monitors its performance and refines its business model through repeated cycles of execution and evaluation. Having an intentional competitive strategy increases the likelihood of long-term value creation.
- | **Unintentional competitive strategy.** This is a reactive approach. Managers may prioritize their individual agendas and tend toward maintaining the status quo or simply following industry norms. With little clear direction and no

formal process for improvement, an unintentional competitive strategy decreases the likelihood of long-term value creation.

Effective competitive strategies are most easily recognized in hindsight, after they produce consistent positive economic profits. To assess competitive strategy on a forward-looking basis, an analyst should evaluate the following:

- | Whether the strategy can respond appropriately to threats within the industry, as identified using Porter's five forces
- | Whether the strategy is neutral to, or benefits from, relevant external influences presented in the PESTLE analysis
- | How well the company can execute its strategy given its resources and stage of development, as well as that of its industry

Competitive Strategies for Each Stage of the Company Life Cycle

A company's life cycle stage affects its strategy and positioning, as we summarize in Figure 48.1.

Figure 48.1: Company Life Cycle Stages and Strategy Implications

Life Cycle Stage	Revenue and Cash Flow	Primary Strategic Focus	Key Challenges
Start-up	■ Little or no revenue ■ Negative cash flow	■ Innovative product and business model development; partnership formation; capital raising for research	■ Few assets; lack of operating history; vulnerable to setbacks
Growth	■ Strong revenue growth ■ Cash flow moves from negative to positive, but remains below revenue due to reinvestment	■ Growth through sales innovation; identifying demand drivers	■ Operational constraints; regulatory hurdles; competitive challenges
Mature and established	■ Revenue growth slows, then plateaus ■ Cash flow rises more slowly or flattens	■ Efficiency; defending market share; product evolution	■ Saturation; risk of “stuck-in-the-middle” competitive strategy
Decline	■ Revenue falls or grows less than nominal GDP or competitors ■ Cash flow flattens, then declines	■ Cost management; restructuring; reinvention; downsizing or exiting	■ Lower demand; new substitutes; excess industry capacity; technological obsolescence

Evaluating Competitive Strategies for Companies in the Start-up, Growth, and Decline Stages

To evaluate start-up or growth companies entering established markets, analysts focus on gauging the total market potential and the company’s ability to achieve above-trend growth. This is done by calculating the total addressable market, serviceable market, and obtainable market for each of the company’s goods or services.

- ▮ **Total addressable market (TAM).** This is the largest potential market for these goods and services.
- ▮ **Serviceable market.** This is the portion of the TAM relevant to the company, given its focus or geographical footprint.
- ▮ **Obtainable market.** This is the portion of the serviceable market the company can realistically hope to capture, given its current scale and strategy.

Analysts must stay attuned to market innovation and changes in consumer demand that can redefine the competitive landscape. Shifts in consumer preferences or breakthroughs in technology are forces that may enable a company to achieve above-trend growth in the start-up or growth stages. Alternatively, these same forces may push a company into the decline stage if it loses market share and must refocus on cost efficiencies and downsizing.

EXAMPLE: Estimating market potential

Volt-Vantage is a fictional specialized manufacturer of ultra-fast charging (UFC) stations for heavy-duty electric trucks. Having launched in the European Union (EU) in 20X0, they have seen rapid growth. Analysts use total charging gigawatt hours (GWh) as the primary measure of market capacity for this industry.

Market Context

In the logistics industry, *interoperability standards* act similarly to *cabotage*, or trade restrictions imposed by a country to restrict foreign operators from carrying freight or cargo within their borders. Chargers designed for the CCS2 standard (common in Europe) cannot serve trucks using the Megawatt Charging System (MCS) standard without expensive hardware retrofits. Currently, the European heavy-duty EV market is split between these standards.

Total Estimated Energy Demand for Heavy-Duty EVs in Europe (GWh)

Fleet Operator	20X1 Actual GWh	% of Total	20X8 Projected GWh	% of Total
Total, European market	12,500	100%	45,000	100%
Eurofreight Logistics	3,750	30%	11,250	25%
Nordic Haulage	1,250	10%	9,000	20%
Volt-Vantage (Current)	625	5%	6,750	15%
Alpine Express	2,500	20%	4,500	10%
Other operators	4,375	35%	13,500	30%

Market Segmentation Details

1. *TAM*. This is the total energy demand for all heavy-duty electric trucks in Europe (45,000 GWh).
2. *Serviceable market constraints*. Due to interoperability standards, Volt-Vantage chargers only work with the CCS2 standard, which represents 65%

of the total European GWh demand. Further, they only have permits to operate in the EU. Excluding non-EU markets like the UK and Switzerland further reduces their reachable energy market to 80% of the CCS2 segment.

3. *Obtainable market constraints*. Volt-Vantage's production is capped by their factory output. They can only manufacture and maintain enough stations to service 7,000 GWh of demand.

TAM, Serviceable Market, and Obtainable Market

1. *TAM* reflects the total revenue or capacity available in the entire industry category. In this case, total European energy demand for heavy-duty EVs is 45,000 GWh.
2. *Serviceable market* is the TAM adjusted for geographic and technical reach. In this case, TAM (45,000) $\times$ interoperability fit (65%) $\times$ geographic permit reach (80%) = 23,400 GWh.
3. *Obtainable market* is the 7,000 GWh that Volt-Vantage can manufacture and maintain currently, as there is no mention of a planned expansion.

Evaluating Strategies for Mature-Stage Companies

To compare established companies with operating track records, Professor Michael Porter identified three types of competitive strategies: a cost leadership (low-cost) strategy, a product or service differentiation strategy, and a focus strategy. According to Porter, a firm must choose one of these to compete effectively and avoid being “stuck in the middle”—an undesirable outcome where the company has no clear competitive advantage.

In a **cost leadership strategy**, a company seeks to have the lowest production costs in its industry, offer the lowest prices, and generate enough volume to make a superior return. This strategy tends to be used in mature industries or industries with commoditized products. It can be used defensively to protect market share, or offensively to gain market share. A company focused on cost leadership should have customers that focus on costs, rather than differentiation, and it should gear its managerial incentives toward improving operating efficiency through strategies such as scaling production, optimizing utilization, or achieving vertical integration.

In a **differentiation strategy**, the company's products and services should be distinctive in terms of type, quality, or positioning. To succeed, a differentiation

strategy (e.g., filing a patent or increasing customer engagement) must provide an economic moat that shields the company from competition, and it must be difficult for competitors to copy.

A **focus strategy** refers to targeting a niche market across products and services, regions, or stages of product development. Executing a focus strategy can include aspects of both cost leadership and differentiation.

Figure 48.2 summarizes the three generic corporate strategies.

Figure 48.2: Generic Competitive Strategies¹

	Cost Leadership	Differentiation	Focus
Forms	■ Economies of scale ■ Access to materials or supply chain ■ Cost controls ■ Aggressive pricing ■ Low cost distribution ■ Economies of scope	■ Quality/features ■ Advertising/brand ■ Service/distribution ■ Intellectual property ■ Customer experience ■ Premium pricing ■ Product or service may be bundled with other services	■ Elements of both cost leadership and differentiation focused on specific group or product
Five forces impact	■ Threat of new entrants ■ Customer bargaining power ■ Industry rivalry	■ Threat of new entrants ■ Threat of substitutes ■ Customer bargaining power ■ Industry rivalry	■ Threat of new entrants ■ Threat of substitutes ■ Customer bargaining power
Industry applicability	■ Capital intensity ■ Price-conscious customers ■ Lack of product differentiation ■ Limited industry innovation	■ Less price-conscious customers ■ Customers value product uniqueness ■ Industry innovation with changing product features	■ Competitors face challenges in servicing specific customers, regions, or delivering specific products
Risks	■ Cost inflation, lack of discipline ■ Technological change affecting costs or market share ■ Customer desire to elevate product quality (premiumization)	■ Imitation ■ Increased buyer sophistication ■ Pricing premium becomes excessive ■ Exclusivity prevents market share growth	■ Large competitors outcompete on price ■ Narrowing of differences among groups ■ Increased buyer sophistication



MODULE QUIZ 48.1

- Which of the following *best* describes a growth company?
 - Product innovation and negative cash flow.
 - Product innovation and operational constraints.
 - Innovative sales practices and negative cash flow.
- Which of the following *best* describes a cost leadership strategy?
 - Volume sold can range from modest to high.
 - Managerial incentives promote operational efficiency.

- C. Success depends heavily on investments in customer service and proprietary distribution channels.

MODULE 48.2: REVENUE, PROFITABILITY, AND CAPITAL

LOS 48.b: Evaluate a company's revenue drivers, operating profitability, and pricing power.

Revenue Drivers

To link a company's performance to its business model, competitive strategy, market position, and operating environment, an analyst must review the company's past and current financial statements.

Financial statement analysis most commonly begins with revenues from the income statement. The focus is on **revenue drivers**, which are the components of a bottom-up or top-down analysis. In a bottom-up analysis, revenue is broken down into specific drivers such as price and volume, business segments, or geography. In a top-down analysis, macroeconomic variables such as market share, economic growth, or industry performance serve as drivers of revenue.

Best practice is for an analyst to use both approaches when evaluating revenue. Key considerations include the following:

- | The level of, and changes in, drivers (e.g., recurring vs. episodic revenue)
- | The evolution of drivers over time (e.g., changes to the product mix and the impact on the future share of the company's obtainable market)
- | Future changes relative to other industry participants

Operating Profitability

Assessing profitability is the next step in linking strategy to performance. Three key metrics to examine are EBITDA, EBIT, and net income:

- . $\text{EBITDA} = \text{revenue} - \text{cost of goods sold} - \text{operating expenses}$
- . $\text{EBIT} = \text{EBITDA} - \text{depreciation and amortization}$
- . $\text{Net income} = \text{EBIT} - \text{interest expense} - \text{taxes}$



PROFESSOR'S NOTE

The formula presented here for EBITDA is the approximation presented in the Level I CFA curriculum. Given that depreciation and amortization are operating expenses, the full equation would read $\text{EBITDA} = \text{revenue} - \text{cost of goods sold} - \text{operating expenses excluding depreciation and amortization}$.

Dividing each of the metrics just listed by revenue obtains its corresponding margin: EBITDA margin, EBIT margin (or operating margin), and net margin. We summarize the characteristics of each metric in Figure 48.3.

Figure 48.3: Characteristics of Profitability Metrics

Metric	Definition	Strengths	Weaknesses
Net income and net margin	■ Total earnings after all expenses and taxes	■ Most comprehensive; includes all operating, financing, and investing activities and taxes ■ Useful for companies in the growth, mature, or decline life cycle stages	■ Highly volatile; difficult to compare across companies for a specified period; difficult to assess a company over time ■ Can be distorted by changes in capital structure, expenditures, or nonrecurring items ■ Less useful for start-ups or highly cyclical industries
Operating income (EBIT) and operating margin	■ Earnings before interest, taxes, and nonrecurring items	■ Increases comparability across companies for a specified period, and for the same company over time by removing financing and tax effects	■ Can be distorted by depreciation/amortization policies (CapEx) ■ Example: companies investing in production facilities show lower operating margins than those that underinvest
EBITDA and EBITDA margin	■ EBIT plus depreciation and amortization	■ Allows comparison of firms with different investment policies, leverage, and taxes	■ Not a metric recognized by GAAP/IFRS; ignores the cost of replacing assets; adjustments may not be standard

Rather than relying solely on profitability ratios, an analyst may also evaluate the components of a company's cost structure over multiple periods or benchmark its cost drivers against peer groups.

Pricing Power, Market Structure, and Price Elasticity

Pricing power is a company's ability to adjust the economic terms of its offering (primarily price) to its advantage without hurting sales. This is a function of three interrelated factors: price elasticity, market structure, and a company's competitive position in its industry.

Price elasticity measures how much demand changes when price changes, making it a fundamental gauge of pricing power.

- | Inelastic demand implies high pricing power. Consumers are insensitive to price changes, such as for necessities (e.g., health care).
- | Elastic demand implies little pricing power. Consumers are sensitive to price changes, such as for luxuries (e.g., leisure travel).

Increased pricing power may come from a higher degree of product differentiation, higher switching costs, strong brand loyalty, high barriers to entry, and few substitutes. Relative market share may have more impact on pricing power than absolute market share, with a greater relative share implying more pricing power.

A company's industry also affects its ability to set prices. In general, as competition increases, pricing power decreases. In markets that are less competitive, such as monopolies, oligopolies, or those with monopolistic competition, companies have more pricing power.

In highly competitive markets, companies have less pricing power and are said to be **price takers**. Over the long run, competition drives prices to equal marginal costs, and returns on capital to equal the cost of capital. An exception is a low-cost producer (e.g., Saudi Aramco in petroleum production) that can maintain margins even when prices are dictated by the global market.

Operating Leverage

A critical factor in the stability of a company's profit margins is its **operating leverage**, or the proportion of fixed costs to variable costs. In terms that express how cost structure affects performance, operating profit can be defined as follows:

$$\text{operating profit} = [Q \times (P - VC)] - FC$$

where:

Q = number of units sold

P = price per unit

VC = variable costs per unit, those that change with the level of output
(e.g., materials, direct labor, inventory costs)

FC = fixed costs (stated as a total rather than on a per-unit basis), those that
do not change within a specific range of output (e.g., rent, management
salaries, insurance)

Operating leverage results from fixed operating costs. The larger the proportion of costs that are fixed, the higher the operating leverage, and the more rapidly operating profits will increase with a given increase in quantity sold (and the faster they will decrease with a given decrease in quantity).

Regression analysis is a useful tool in determining how sensitive expenses are to revenue changes. Less sensitive expenses are likely to be fixed costs, while the more sensitive expenses are variable costs. When using historical data in this way, an analyst assumes that the company's cost structure has remained constant.

LOS 48.c: Evaluate a company's working capital, capital investments, and capital structure.

To link performance to a company's business model, analysts also need to consider key aspects of the balance sheet, including long-term fixed assets, working capital, and incremental financing.

Analyzing free cash flow to equity (FCFE) involves using both the income statement and balance sheet to determine how much cash remains after the company has paid its operating expenses and made the necessary investments.

$$\text{FCFE} = \text{net income} + \text{depreciation and amortization} - \Delta \text{ working capital} - \Delta \text{ LT assets} + \Delta \text{ debt}$$

Each positive term is a source of cash, while each negative term is a use of cash.

$$\Delta \text{ working capital} = \Delta \text{ current assets} - \Delta \text{ current liabilities}$$

$$\text{net debt issued } (\Delta \text{ debt}) = \text{additional borrowing} - \text{debt repaid}$$

Levels and composition of FCFE vary among companies and among industries. They also vary over the stages of the company life cycle, with FCFE often negative for start-ups and growth companies that must spend heavily to achieve scale and do not have stable enough cash flows to support debt issuance. Mature

firms typically have positive FCFE, with fewer capital investments and more stable debt capacity.

Working Capital Management

To evaluate a company's performance, analysts assess **working capital management**, how the company optimizes its use of current assets and current liabilities to run its business efficiently. Recall that working capital equals current assets minus current liabilities.

To assess working capital management, analysts look at the length of a company's **operating cycle**, or how efficiently the firm converts raw materials into cash (the time from receiving materials to collecting cash). The operating cycle is measured by efficiency ratios, including days sales outstanding, days payables outstanding, inventory on hand, and receivables turnover.

To narrow the focus further, analysts also consider the **cash conversion cycle** (time from paying for materials to collecting cash) as follows:

$$\text{cash conversion cycle} = \text{days of inventory on hand} + \text{days sales outstanding} - \text{days payable outstanding}$$



PROFESSOR'S NOTE

At times, the Level I CFA curriculum references the more general term *working capital management* when discussing the cash conversion cycle. Understand that these terms are like nesting dolls. Working capital management is the big picture. The operating cycle narrows the focus to the production timeline, and the cash conversion cycle zooms in on the cash gap.

The longer the cash conversion cycle, the greater the need to finance working capital. A shorter cash conversion cycle improves a company's market position and its returns. A negative cash conversion cycle indicates that operations are managed to provide a source of funding for the company.



PROFESSOR'S NOTE

In cash conversion cycle examples, the Level I CFA curriculum highlights that industries may handle inventory differently. One example refers to *program accounting*, which is used to measure inventory under GAAP for specific long-term programs or projects (e.g., heavy equipment manufacturing). Program accounting takes overall project profitability into account, rather than simply focusing on unit costs as

one would in the retail industry, for example. Program accounting is a term that provides context, but its calculation is not required for the exam.

Figure 48.4 shows examples of how companies in selected industries align working capital management with their competitive strategies.

Figure 48.4: Working Capital Management Examples from Different Industries

Industry	Strategy/Focus	Performance Examples
Retail	High turnover; bargaining power with suppliers	Costco outperforms Walmart by maintaining a shorter cash conversion cycle via fewer items, fewer brands per item, and high volume
Manufacturing	Capital-intensive, complex production which can amplify the impact of production challenges	Boeing's cash conversion cycle tripled due to safety groundings, leading to massive inventory backlogs
Services and software	Minimal physical assets	Low working capital needs; often focused solely on receivables and payables
Beverages and brewers	Using supplier credit to fund the business	AB InBev uses high days payable outstanding (paying suppliers slowly) to create negative working capital, effectively a free source of funding



PROFESSOR'S NOTE

Companies in the beverages and brewers industry and subindustry do not always operate with negative working capital, but AB InBev used it effectively. It works best when revenue is growing and suppliers are healthy.

Capital Investments

When a company acquires an asset for long-term use, it typically capitalizes the purchase, recognizing the expense over time through depreciation and amortization. Analysts monitor **net capital expenditure** to assess a company's productive capacity and future earnings growth.

$$\text{net capital expenditure} = \text{total capital expenditure} - \text{depreciation and amortization}$$

Net capital expenditure illustrates how a company's business model affects its investment in long-term fixed assets. For example, software firms with licensing

fees and other intellectual property have a high portion of intangible assets that are amortized over their useful lives. On the other hand, manufacturing firms looking to increase capacity must purchase tangible assets such as machinery, which are depreciated.

Analysts also evaluate the **return on invested capital (ROIC)** to measure a company's relative performance and assess whether it has created economic value from investors' capital. ROIC measures value creation across investments. When ROIC exceeds a company's weighted average cost of capital, the company has created value for its investors.

$$\text{ROIC} = \frac{\text{NOPAT}}{\text{average invested capital}}$$

where:

NOPAT = net operating profit after taxes

invested capital = total long-term liabilities + total equity

$$\text{average invested capital} = \frac{(\text{investment capital}_{\text{beginning of period}} + \text{investment capital}_{\text{end of period}})}{2}$$



PROFESSOR'S NOTE

The Level I CFA curriculum approximates the investment capital calculation. The complete version is invested capital = total debt + total equity – cash and equivalents. In this context, the approximation is sufficient.

Capital Structure

When evaluating a company, an analyst should consider its reliance on debt financing as well as its access to capital. A company bases its capital structure decisions on the nature of its operations, its business model, its stage in the company life cycle, and its asset conversion cycle.

Early-stage and growth firms with negative cash flow usually finance their operations with equity capital. Mature companies with stable cash flows use more debt to finance their operations, and analysts should assess the relative mix. Industries that require more long-term capital investment, such as manufacturing or utilities, tend to have more long-term debt financing, while service-based or retail industries tend to use shorter-term debt.

When evaluating debt, analysts should consider a company's loans outstanding, the type and maturity of its bonds, and its ability to take on additional debt. To assess how a company's debt affects its overall financial performance, analysts look to the income statement to calculate the **degree of financial leverage (DFL)**, which indicates how sensitive net income is to changes in operating income.

$$DFL = \frac{\% \Delta \text{ net income}}{\% \Delta \text{ operating income}}$$

DFL increases when a company adds interest expense by borrowing. Using financial leverage can increase profits, but it can also compound losses and increase the likelihood of financial distress when earnings weaken. As a company evolves its strategy (e.g., by completing an acquisition, restructuring operations, or committing additional capital), analysts should prepare disaggregated financial models to confirm that cash will be available to service and repay debt. Other ratios to monitor include the debt ratio $\left(\frac{\text{debt}}{\text{assets}}\right)$ and interest coverage ratio $\left(\frac{\text{EBITDA}}{\text{interest expense}}\right)$.



MODULE QUIZ 48.2

1. An analyst notices that a company's EBITDA margin is expanding, while its net margin is contracting. Which of the following internal developments would *most likely* explain this divergence?
 - A. Improved manufacturing efficiency.
 - B. A significant shift from equity to debt financing.
 - C. An increase in the proportion of variable costs to fixed costs.
2. Jayco, Inc., sells 10,000 units at a price of \$5 per unit. Jayco's fixed costs are \$8,000, interest expense is \$2,000, variable costs are \$3 per unit, and EBIT is \$12,000. Disregarding taxes, the degree of financial leverage (DFL) is *closest* to:
 - A. 1.00.
 - B. 1.20.
 - C. 2.00.

KEY CONCEPTS

LOS 48.a

An analyst evaluates how a company creates value relative to its peers by evaluating its competitive strategy, which includes its competitive advantage, its position within its industry, and its resources and capabilities.

The strategy a company pursues is influenced by its stage in its life cycle:

- | A start-up has little to no revenue, negative cash flow, and innovative product and business model development. It may have few assets and little to no operating history.
- | A growth company has robust revenue growth, cash flow that may be negative or positive, and an innovative sales strategy that optimizes demand drivers. It may face potential operational constraints, regulatory hurdles, and competitive challenges.
- | A mature or established company sees revenue growth slow, then plateau, while cash flow increases more slowly or flattens. It may lean into operational efficiency or product evolution to defend market share and avoid being “stuck in the middle” without a differentiating strategy as its market reaches saturation.
- | A company in decline (due to lower demand, new substitutes, excess industry capacity, or technological obsolescence) has revenue that is below trend or falling, and cash flow that flattens or declines. It may focus on cost management, restructuring, reinvention, downsizing, or exiting.

In a cost leadership strategy, the firm seeks to have the lowest costs of production in its industry, offer the lowest prices, and generate enough volume to make a superior return.

In a differentiation strategy, the firm’s products and services should be distinctive in terms of type, quality, or delivery.

In a focus strategy, a company targets a niche market. The strategy may include elements of cost leadership and differentiation.

LOS 48.b

Revenue drivers can be analyzed bottom up (based on financial statements) or top down (based on economic and industry factors).

Operating profitability links strategy to performance and can be assessed via the following metrics:

- | EBITDA = revenue – cost of goods sold – operating expenses. It is used to compare firms with different investment policies, leverage, and taxes, but it ignores the cost of replacing assets.
- | EBIT = EBITDA – depreciation and amortization. It can be distorted by capital investment policies, but it can generally be used to compare companies within an industry over a distinct period.

- | Net income = EBIT – interest expense – taxes. It is useful for companies in the life cycle stages of growth, mature, or decline, but it is less useful for start-ups or cyclical industries. It can be difficult to compare companies using this metric, or to evaluate a company over time because net income can be distorted by changes in capital structure.

Pricing power is a function of market structure and a company's competitive position:

- | Companies in highly competitive markets have low pricing power.
- | The higher the elasticity of demand for a company's offering, the lower the pricing power.
- | Pricing power for companies in less competitive markets may result from greater product differentiation, lack of good substitutes, high barriers to entry, high customer loyalty, and high switching costs.

LOS 48.c

FCFE = net income + depreciation and amortization – Δ working capital – Δ LT assets + Δ debt.

Working capital = current assets – current liabilities.

Efficiency is measured by the cash conversion cycle:

cash conversion cycle: days inventory + days sales outstanding – days payable outstanding

A longer conversion cycle indicates greater need for external financing.

Net capital expenditures indicate if a company is maintaining or expanding productive capacity.

Return on invested capital (ROIC) measures whether a company creates value (ROIC > WACC).

Capital structure, or the mix of debt and equity, is evaluated using the degree of financial leverage (DFL) to assess risks.

$DFL = \% \Delta \text{ net income} / \% \Delta \text{ operating income}$

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 48.1

- . **C** The growth stage of a company's life cycle is characterized by growth through sales innovation and identifying demand drivers. Cash flow may be negative or positive, as this is the stage where cash flow moves into positive territory. Companies in this stage may face operational constraints, regulatory hurdles, and competitive challenges. Product innovation is more typically seen in the start-up stage. (LOS 48.a)
- . **B** Companies that use a cost leadership strategy should have managerial incentives to create efficient operations. In a cost leadership strategy, the firm seeks to generate a high enough sales volume to make a superior return. Investments in customer service and proprietary distribution channels are key elements of a differentiation strategy. (LOS 48.a)

Module Quiz 48.2

- . **B** The company has most likely made a significant shift from equity to debt financing. EBITDA is calculated before interest and taxes. Increasing the level of debt increases interest expense, which reduces net income (and net margin) but leaves EBITDA (and EBITDA margin) unaffected. Improving manufacturing efficiency should improve both EBITDA and net margin. Increasing the ratio of variable to fixed costs would generally affect both metrics in the same direction. (LOS 48.b)
- . **B** Jayco's net income is $\$12,000 - \$2,000 = \$10,000$. If its operating income increases by 1% to $1.01(\$12,000) = \$12,120$, its net income will increase to $\$12,120 - \$2,000 = \$10,120$, which is an increase of $\$120 / \$10,000 = 0.012$, or 1.2%. Therefore, Jayco's DFL is $1.20\% / 1\% = 1.20$. (LOS 48.c)

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READING 49

EQUITY ANALYST RESEARCH REPORTS

MODULE 49.1: EQUITY ANALYST RESEARCH REPORTS

LOS 49.a: Describe the elements that should be covered in a detailed company research report.

A **company research report** includes an analyst's valuation and investment recommendations, based on analysis of the current and forecast company performance and position.

An initial research report for external distribution (said to be **initiating coverage** of a company) is likely to be thorough, followed by subsequent updates that are less thorough. These updates may be produced after widely anticipated events, such as mergers or earnings releases.

The basic structure and key items typically included in an initial company research report, with the intention of introducing the company to investors, are as follows:

- | *Front page.* While presented first, the front page should be written last. The front page should show the analyst's buy/hold/sell recommendation and target buying or selling prices. It should include an investment thesis that summarizes the arguments supporting the analyst's recommendation. Also on the front page should be the subject company's name, ticker symbol, primary sector and industry, market capitalization and float, key recent developments, and a summary of recent and forecast financial data, such as earnings. All of these will then be covered in further detail in the remainder of the research report.
- | *Analysis of the business model.* This section introduces the company to the reader. It will cover how the firm generates value, including revenue drivers

(e.g., which products or services are offered, pricing strategies, major customers) and expense drivers (e.g., major suppliers, distribution strategies).

| *Industry overview and competitive positioning.* The analyst producing the report may have industry expertise, but he should not assume that the reader does, so introducing the industry is important. Many firms choose not to explicitly name their competitors, so it is up to the analyst to establish which companies should be considered as peers. Industry analysis should incorporate quantitative comparisons, such as market capitalization, P/E multiples, earnings growth, and return figures—as well as qualitative analysis using models such as PESTLE and Porter’s five forces. Full analysis of these models does not necessarily need to be included in the research report; the analyst should apply discretion in including only the pertinent information. For example, the technological element of PESTLE analysis will be crucial in industries like telecommunications, but less important in others, such as basic materials.

| *Financial analysis and valuation.* The analyst will conduct thorough financial analysis of the company, but the research report does not necessarily need an exhaustive commentary on the financial statements. It should include only those measures and financial ratios (general or industry specific) that are relevant for supporting the analyst’s price target. Applying a critical eye to which information to include is increasingly important in an age where AI can produce seemingly endless amounts of financial analysis at the press of a button.

Using financial statements (including footnotes and disclosures) and other information, the analyst will use both actual past performance and forecast performance to produce a price target. Forecast performance will be based on company-specific factors as well as the wider industry and economy, focusing on any specific short-term or long-term catalysts the analyst identifies as relevant. The analyst will likely use various valuation techniques to calculate the intrinsic value of a company’s equity shares, then use it to identify a price target. The target can be an absolute monetary amount or stated against a benchmark, such as an index, industry average, or the analyst’s universe of companies. The price target may be a point estimate or a range and typically covers a period of one to two years.

| *Investment risks.* This section provides further detail on key risks the analyst identified in the section on valuation. These risks may be company specific,

industry specific, or related to the wider economy. Some may be related to the quantitative analysis used in the valuation models, such as sensitivity to the analyst's assumptions.

- | *Environmental, social, and governance factors.* The content of this section may vary depending on the industry or countries in which the company operates, but it may be crucial in industries such as air travel or data storage, where environmental concerns have increasing importance.
- | *Recommendation.* The research report must reach a conclusion and recommend a trading action for the stock. *Buy, sell, or hold* is the most common terminology. Some firms use *overweight, underweight, and neutral* with the same meanings. The analyst will consider all information gathered when reaching their conclusion.

No single approach can determine where a hold recommendation turns into a buy or sell. However, a rough guide is to assign a buy recommendation if a stock is expected to appreciate at least 10%-15% on an absolute basis or relative to a benchmark; a sell recommendation if the stock is expected to decline at least 10%-15%; or a hold recommendation for anything in between.

Key items typically included in a subsequent company research report are as follows:

- | Front matter (similar to what was described previously, but more condensed)
- | Recommendation, including rationales for changes
- | Analysis of new information (e.g., updated financial statements, variance analysis of expected vs. actual results)
- | Changes to the initial valuation, with supporting rationales
- | Changes in risks since the initial or most recent report

Regardless of the type of research report produced, maintaining the highest level of ethical and analytical standards is paramount. A well-written, timely report should distinguish clearly between facts and opinions, disclose any potential conflicts of interest the analyst or firm may have, and provide adequate information about risks an investor may face.

LOS 49.b: Compare sell-side and buy-side equity analyst reports and their conclusions.

The main differences between a **sell-side research report** and a **buy-side research report** are the entity that creates the report and the purpose of the report.



PROFESSOR'S NOTE

A common misconception is that sell-side reports give sell recommendations, and buy-side reports give buy recommendations. This is not the case. *Sell-side* and *buy-side* refer to the producers of the report.

Sell-Side Research Reports

Sell-side research reports are typically produced by investment banks and are created for their clients. These research reports can be used to support relationships between the bank and its clients (e.g., in issuing or underwriting debt and equity securities, increasing volumes for broker or dealer trading, or advisory work).

Typically, the analyst producing the reports will have expertise in a specific industry and focus on selected companies within that industry. However, unless the research is directly sold, creating sell-side research reports incurs costs without bringing in any revenue. Therefore, a sell-side firm needs to limit the universe of companies its research analysts cover. Within an industry, a research analyst must consider how many companies are feasible to cover, and which ones have desirable characteristics for coverage.

High-quality ongoing coverage of a stock can benefit both the stock issuer and the investment bank. Research coverage may lend credibility to a stock and increase its trading volume, which helps to bring in commissions from trades for the investment bank.

Investment banks have potential conflicts of interest if they provide other services to a company, such as underwriting. The relevant teams face regulatory constraints to ensure that research remains objective, and that research analysts do not have access to any material nonpublic information that teams elsewhere in the organization may hold. Regulations may also limit issuing research reports close to an IPO or secondary equity offering, or may require research (if sold) to be paid for by asset managers separately from any trade execution costs. While it is common for sell-side research reports to give buy or hold recommendations (or switch between the two), analysts may be reluctant to issue sell

recommendations if they perceive a risk that doing so will harm the firm's relationship with the company in question.

While some sell-side research reports may present views contrary to the market consensus, this requires a high degree of certainty on the part of the analyst, so it is more typical for sell-side recommendations to be in line with the consensus. Because platforms such as Bloomberg often show select information from analysts such as forecast profit margins, it is straightforward to obtain information about the market consensus.

Buy-Side Research Reports

The buy side refers primarily to firms that invest in securities. Unlike sell-side reports, buy-side research is typically confidential and is used internally by a firm's investment committee to support investment decisions or monitor existing positions. Buy-side analysts write their own reports, but they will often use publicly available sell-side reports to enhance their knowledge of a particular company. Using these sell-side reports does not necessarily mean the buy-side analyst will reach the same investment recommendation.

Without the pressure of considering potential trading volumes and potential commissions, buy-side research generally has more scope to focus on smaller stocks, should the analyst see fit.

Buy-side firms engage in **stock screening** to narrow the universe of equities to which they will devote research resources. Increasingly, they use finance software, AI, and platforms such as Bloomberg to assist with stock screening. The initial level of screening determines the universe of stocks (e.g., a specific index). Next, the firm applies quantitative criteria, such as growth rates and profit margins, alongside any other metrics that fit the desired investment strategy. The firm refines these criteria until the appropriate number of stocks remain, which the firm's research analysts can cover with appropriate depth. Equity analysts should, however, be aware that any arbitrary criterion set (e.g., a specific profit margin) may result in an appealing stock being omitted.

One branch of buy-side research is **activist short selling**. This is typically carried out by small equity analyst teams and focuses on stocks the analysts believe are likely to decline significantly in value. The analysts look to uncover misinformation in the market that is inflating the market price, or undiscovered fraud which, once made public, will reduce the stock price significantly. In

contrast with most buy-side research, which remains internal and confidential, activist short sellers often circulate their research publicly after they have released it to their limited partner investors.

Sell-side research reports have the following characteristics:

- | *Objective.* The objective is to provide information and recommendations to clients.
- | *Level of analysis and detail.* Sell-side analysts cover a wide range of industries and stocks for a diverse audience, but they tend to use a standardized format.
- | *Target audience.* External investors and traders are the main target audience for sell-side research.
- | *Valuation methods.* Methods include discounted cash flow, multiples, and other common models, focusing on future earnings.
- | *Regulatory constraints.* Constraints include regulations dealing with conflicts of interest due to additional relationships with covered companies and investment banking clients.

Buy-side research reports have the following characteristics:

- | *Objective.* The objective is to support internal decision-making.
- | *Level of analysis and detail.* Reports are customized and in depth, with a specific focus on stocks or industries.
- | *Target audience.* The target audiences are internal investment teams and portfolio managers.
- | *Valuation methods.* Analysts might use models created internally at their firms in addition to traditional valuation methods.
- | *Regulatory constraints.* These reports are mostly unregulated, as the information is typically not made public.

LOS 49.c: Explain why value estimates from the same modeling approach may differ based on analyst assumptions regarding revenue, profitability, investment, and financing.

Valuation models require several estimates or forecasts, which can lead to differing outcomes for different analysts. One analyst may estimate an intrinsic value that is above the current market value and provide a buy recommendation, while another analyst may value the same stock below the current market price

and provide a sell recommendation. This is not a weakness of the valuation process; differing valuations encourage price discovery and efficient markets. Differing price targets can also be due to other factors, such as location. A domestic analyst may value a stock differently than a foreign analyst who might see a particular firm as a good opportunity to gain exposure to a new market. Stocks for which different analysts produce widely dispersed valuations are known as **hard-to-value stocks**.

To understand what assumptions have been used, if not explicitly stated, one can assume that an analyst's price target is the intrinsic value estimated using a discounted cash flow model, then solve for one of the underlying assumptions of that model, such as long-term growth.

EXAMPLE: Deriving an analyst's growth rate assumption

An analyst is initiating coverage of a new firm and states a price target of \$46.50 per share. The firm has 10 million shares outstanding. Assuming the analyst forecasts FCFE for next year of \$22 million, FCFE for the year after of \$24 million, and a required return of 8%, calculate the long-term growth rate implied by the analyst's valuation.

Answer:

$FCFE_1$ should be discounted back one year to find its present value, and then $FCFE_2$ treated as a delayed perpetuity, as follows:

$$(\$46.50 \times 10 \text{ million}) = \frac{\text{FCFE}_1}{(1+r)^1} + \frac{\left(\frac{\text{FCFE}_2}{r-g}\right)}{(1+r)^1}$$

Using the inputs just listed, we can solve for g :

$$(\$46.48 \times 10 \text{ million}) = \frac{\$22 \text{ million}}{1.08} + \frac{\left(\frac{\$24 \text{ million}}{0.08 - g}\right)}{1.08}$$

$$\$464.8 \text{ million} = \$20.37 \text{ million} + \frac{\left(\frac{\$24 \text{ million}}{0.08 - g}\right)}{1.08}$$

$$\$444.43 \text{ million} = \frac{\left(\frac{\$24 \text{ million}}{0.08 - g}\right)}{1.08}$$

$$\$479.98 \text{ million} = \frac{\$24 \text{ million}}{0.08 - g}$$

$$0.08 - g = \frac{\$24 \text{ million}}{\$479.98 \text{ million}}$$

$$g = 0.08 - \frac{\$24 \text{ million}}{\$479.98 \text{ million}}$$

$$g = 0.08 - 0.05 = 0.03 = 3\%$$

Some key drivers for the range of valuations analysts may produce are estimated revenue growth and its effect on profitability, as well as their assumptions regarding terminal value and a company's investment and financing requirements.

Revenue and Profitability

Greater revenue growth is typically seen as an indicator of a higher-valued stock, going hand-in-hand with higher free cash flow. However, looking at revenue growth in isolation can be misleading. An analyst must consider the source of changes in a company's revenue and cash flow, such as changes in its cost structure. The following example illustrates the different impacts of revenue and cost changes on profit margins.

EXAMPLE: Evaluating the impact of cost structure assumptions

Deck Health, Inc., has released its most recent earnings, showing revenue for 20X6 at \$10 million and operating profit of \$4 million, giving an operating profit margin of 40%. Three analysts update their forecasts for the next two

years. They all expect revenue to grow by 3% annually due to higher sales volume, but they have differing opinions about how costs will change.

Analyst 1 assumes that operating costs will rise in line with revenue at 3% for the next two years.

Analyst 2 assumes that costs will rise by 4% due to necessary operating changes in the production processes, such as hiring additional staff.

Analyst 3 assumes that Deck Health will be able to achieve some economies of scale by producing higher volumes, and that costs will only rise by 2% each year.

Evaluate the impact the different assumptions have on Deck Health's forecast operating profit margin.

Answer:

Analyst 1:

	20X7	20X8
Revenue	\$10,300,000	\$10,609,000
Operating costs	\$6,180,000	\$6,365,400
Operating profit	\$4,120,000	\$4,243,600
Operating profit margin	40%	40%

Analyst 2:

	20X7	20X8
Revenue	\$10,300,000	\$10,609,000
Operating costs	\$6,240,000	\$6,489,600
Operating profit	\$4,060,000	\$4,119,400
Operating profit margin	39.4%	38.8%

Analyst 3:

	20X7	20X8
Revenue	\$10,300,000	\$10,609,000
Operating costs	\$6,120,000	\$6,242,400
Operating profit	\$4,180,000	\$4,366,600
Operating profit margin	40.6%	41.2%

Despite all three analysts forecasting the same revenue for 20X7 and 20X8, the operating profit margin varies due to the different cost assumptions. Analyst 1, who predicts costs to grow proportionally to revenue, has forecast a steady 40% operating profit margin. Analyst 2 has forecast a worsening operating

profit margin due to higher labor costs, while Analyst 3 believes economies of scale will result in an increasing operating profit margin.

Capital Investment, Terminal Value, and Financing Requirements

Analysts must consider a company's investment needs when forecasting cash flow. If growth in investment matches growth in net income, no overall change in free cash flow results. If investment grows at a slower rate than income, the forecast free cash flow will increase. While this may seem positive, an analyst must ask whether growth in free cash flow is sustainable without significant capital investment, and should be skeptical of long-term revenue or profitability growth projections that are not backed up by investment in the capital needs of the business. On the other hand, if a company is increasing investment and this is not resulting in income growth, forecast free cash flow will decline, and the company will produce a lower return on capital in the long run.

In most valuation models such as FCFE, a substantial portion of the value of the business comes from the terminal value. An analyst should consider carefully how long a forecast horizon can be forecast accurately and what assumptions, such as growth rate, are built into the terminal value, as these can have a material impact on the valuation produced.

While financing choices have some impact on valuation, they are less important than factors that result from operating and strategic decisions. One should be wary of a valuation that places a large emphasis on financing decisions.



MODULE QUIZ 49.1

1. Which of the following statements regarding company research reports subsequent to initiating coverage is *most accurate*?
 - A. The report updates the recommendation in light of new information about the company.
 - B. The primary audience is those who are not already knowledgeable about the company or security.
 - C. The report provides an industry overview and information on competitive positioning and ESG considerations.
2. An analyst expects a stock to increase in value by 9%, while its benchmark index is expected to grow by 6%. Which of the following recommendations from the analyst is *least likely*?
 - A. Buy.
 - B. Sell.
 - C. Hold.

3. Which of the following equity analyst teams is *most likely* to create a research report on a small, newly formed company that shows great potential?
 - A. Buy-side equity analysts.
 - B. Sell-side equity analysts.
 - C. Activist short sellers.
4. An analyst is creating a forecast for a new company. Using the prior year's results as a starting point, she estimates that for the next three years, revenue and operating costs will grow at the same percentage per year, with a lower growth rate in capital investment. All else equal, what does her forecast imply about the company's operating profit margin over the three years? The analyst is forecasting that the operating profit margin will:
 - A. increase.
 - B. decrease.
 - C. not change.

KEY CONCEPTS

LOS 49.a

Key items typically included in an initial company research report are as follows:

- | Front page
- | Analysis of the business model
- | Industry overview and competitive positioning
- | Financial analysis and valuation
- | Investment risks
- | Environmental, social, and governance factors
- | Recommendation

Key items typically included in a subsequent company research report are as follows:

- | Front matter
- | Recommendation, including rationales for changes
- | Analysis of new information
- | Changes to the initial valuation, with supporting rationales
- | Changes in risks since the initial or most recent report

LOS 49.b

Sell-side research reports are typically produced by investment banks and are created for the investment bank's clients. These research reports can be used to support relationships between the bank and the clients.

Sell-side research reports have the following characteristics:

- | *Objective.* The objective is to provide information and recommendations to clients.
- | *Level of analysis and detail.* Sell-side analysts cover a wide range of industries and stocks for a diverse audience, but they tend to use a standardized format.
- | *Target audience.* External investors and traders are the main target audience for sell-side research.
- | *Valuation methods.* Methods include discounted cash flow, multiples, and other common models, focusing on future earnings.
- | *Regulatory constraints.* Constraints include regulations dealing with conflicts of interest due to additional relationships with covered companies and investment banking clients.

Buy-side research is typically confidential, used internally by the investment committee to support investment decisions or monitor existing positions.

One branch of buy-side research is activist short selling. The analysts look to uncover misinformation in the market that is inflating the market price, or undiscovered fraud that, once made public, will decrease the stock price.

Buy-side research reports have the following characteristics:

- | *Objective.* The objective is to support internal decision-making.
- | *Level of analysis and detail.* Reports are customized and in depth, with a specific focus on stocks and/or industries.
- | *Target audience.* The target audiences are internal investment teams and portfolio managers.
- | *Valuation methods.* Analysts might use models created internally at their firms in addition to traditional valuation methods.
- | *Regulatory constraints.* These reports are mostly unregulated, as the information is typically not made public.

Valuation models require several estimates or forecasts that can lead to differing valuations of the same stock. The main drivers of differences are assumptions around revenue and profitability growth, terminal value, and capital investment requirements. Financing choices may have some impact on valuation, but they tend not to be as significant.

ANSWER KEY FOR MODULE QUIZ

Module Quiz 49.1

- . **A** Subsequent research reports update the recommendation and rationale from an initial research report. The primary audience for an *initial* company report is an audience who is not already knowledgeable about the issuer or security. (LOS 49.a)
- . **B** It is unlikely based on the information given that the analyst would issue a sell recommendation. Although the stock is not expected to appreciate by the guideline 10%-15%, this is only a guideline, and it is expected to outperform the benchmark index. It is possible, therefore, that the analyst may issue a buy recommendation, or a hold. (LOS 49.a)
- . **A** Buy-side analysts are more able to focus on smaller stocks. Sell-side analyst research reports are typically a cost center, so factors such as potential trading volumes need to be considered when deciding which companies to provide research on, generally leading to a focus on larger stocks. Buy-side analysts do not have these restrictions. Activist short sellers focus their research on stocks that are expected to decline significantly in value. (LOS 49.b)
- . **C** Revenue growing at the same percentage per year as operating costs will result in a steady operating profit margin. Capital investment will not have an impact on operating profit margin, but it is a use of cash when calculating FCFE. A lower growth rate of capital investment compared to operating profit implies increasing FCFE, although the analyst should question whether the profit growth is sustainable with low capital investment. (LOS 49.c)

READING 50

THE CAPITAL ASSET PRICING MODEL, MARKET MODEL, AND OTHER FACTOR-BASED EQUITY MODELS

MODULE 50.1: ESTIMATING THE COST OF EQUITY WITH THE MARKET MODEL AND CAPM

LOS 50.a: Explain the use of the CAPM and the market model to estimate a company's required return on equity.

The discount rate or required rate of return is a key input into present value and other models for equity valuation. The required rate of return should incorporate the risk of the investment.

Recall that the **capital asset pricing model (CAPM)** relates a company's required rate of return on equity to its systematic risk, or beta. The required rate of return depends on the risk-free rate, the company's beta, and the **market risk premium**, which is the difference between the expected return on the market portfolio and the risk-free rate.

$$E(R_i) = R_f + \beta_i (E(R_m) - R_f)$$

where:

$E(R_i)$ = expected return on equity security i .

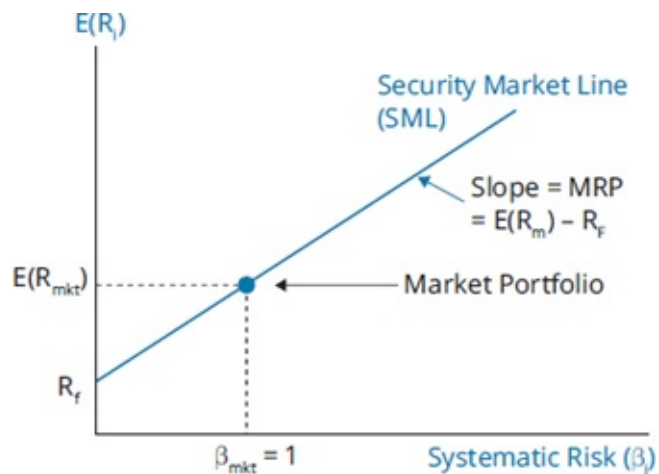
$E(R_m)$ = expected return on the market

R_f = risk-free rate of return

$$\beta_i = \frac{\sigma_{i,m}}{\sigma_m^2} = \frac{\sigma_i}{\sigma_m} \rho_{i,m}$$

This equation can be described graphically using the **security market line (SML)**, as shown in Figure 50.1.

Figure 50.1: The Capital Asset Pricing Model



The y-intercept of the SML is the risk-free rate, and its slope is the market risk premium, $E(R_m) - R_f$. The SML is not static; both the risk-free rate and expectations about returns change over time.

EXAMPLE: CAPM and cost of equity

Stellar's historical stock returns have a standard deviation of 22% and a correlation of 0.80 with market returns. The risk-free rate is 4%. The expected market rate of return is 12% with a standard deviation of 11%. Calculate the expected rate of return on Stellar stock.

Answer:

$$\begin{aligned}\beta_{\text{Stellar}} &= \frac{\sigma_{\text{Stellar}}}{\sigma_{\text{market}}} \rho_{\text{Stellar,market}} \\ &= \frac{22\%}{11\%} \times 0.8 = 1.6 \\ E(R_{\text{Stellar}}) &= 4\% + 1.6(12\% - 4\%) = 16.8\%\end{aligned}$$

Estimating Beta

Beta estimates for a company can be obtained from data providers such as Bloomberg. These are typically based on a linear regression using periodic (e.g., daily, monthly) returns data on the specific stock, and on a market proxy (e.g., S&P 500 Index) over a lookback period (e.g., the past five years).

Rearranging the CAPM, we can relate the excess return on a stock with the excess return on the market:

$$E(R_i) - R_f = \beta_i[E(R_m) - R_f]$$

A linear regression model using actual past data is fit to estimate the values of α and β in the following model:

$$(R_{i,t} - R_{f,t}) = \alpha + \beta_i(R_{m,t} - R_{f,t}) + \varepsilon_t$$



PROFESSOR'S NOTE

Recall that we examined this technique for estimating beta in our Quantitative Methods reading on linear regression.

A similar regression using the **market model** uses actual (rather than excess) returns and provides similar estimates of beta:

$$R_{i,t} = \alpha + \beta_i R_{m,t} + \varepsilon_t$$

The standard error of beta captures the level of statistical noise in its estimation. The higher the standard error of beta, the lower the reliability of the estimate.

The R^2 of a regression indicates its explanatory power. In this model, the square root of R^2 is the correlation of returns between the stock and the market index.

Market model betas exhibit **mean reversion** toward one. That is, estimated beta values higher or lower than one will tend to drift toward one over time. While beta is estimated using past data, its purpose is to estimate the cost of equity to discount *future* cash flows.

We can adjust an estimated (or raw) beta by calculating an **adjusted beta** as a weighted average, giving a weight of ω to the raw beta and a weight of $(1 - \omega)$ to 1:

$$\text{adjusted beta} = \omega (\text{raw beta}) + (1 - \omega)$$

An adjusted beta will be closer to one than the estimated beta, which makes it useful to account for the expected mean reversion of beta. Data sources that provide adjusted betas often use two-thirds as the weight for raw beta.

EXAMPLE: Adjusted beta

A market model estimate of beta for Allied Pharma is 1.30. Calculate the adjusted beta with $\omega = 2/3$.

Answer:

$$\text{adjusted beta} = \omega(\text{raw beta}) + (1 - \omega)$$

$$= (2/3) \times 1.30 + 1/3 = 1.20$$

Because the raw beta of 1.30 is greater than 1, the adjusted beta is lower.

The lookback sampling period used in estimating beta is important. Although a large sample size reduces the standard error of beta, if the lookback period is too far into the past, the data may reflect conditions that were significantly different from what they are likely to be in the future. If so, the estimated beta will inaccurately describe the future systematic risk of the company.

To address statistical noise in estimating beta, analysts may choose to estimate beta for an equally weighted portfolio of peer companies. Because the beta of an individual company captures both its business and financial risk, we may consider it a **levered beta**. While estimating beta for a portfolio of similar companies ensures that we capture the industry's business risk, we must account for differences in leverage between the subject company and its peers.

We first convert the beta of the comparables to an **unlevered beta**:

$$\beta_U = \frac{\beta_{L, \text{comparables}}}{\left[1 + (1 - t_{\text{comparables}}) \frac{D}{MVE}\right]}$$

where:

β_U = unlevered beta of the comparables portfolio

$\beta_{L, \text{comparables}}$ = levered beta of the comparables portfolio

$t_{\text{comparables}}$ = average tax rate of the comparables portfolio

$\frac{D}{MVE}$ = average ratio of debt to market value of equity of the comparables portfolio

Then, to estimate the levered beta of the subject company, we adjust the unlevered beta of the comparables portfolio by re-levering it using the subject company's leverage:

$$\beta_{L, \text{firm}} = \beta_U \left[1 + (1 - t_{\text{firm}}) \frac{D}{MVE_{\text{firm}}}\right]$$

where:

$\beta_{L, \text{firm}}$ = levered beta of subject company

$\frac{D}{MVE_{\text{firm}}}$ = subject company's ratio of debt to market value of equity

t_{firm} = subject company's tax rate

Industry-based beta estimation is useful for private companies, which lack market data. It is also useful for firms with significant changes in their capital

structures, such that their past data do not capture the changes in their risk profiles.

EXAMPLE: Industry beta

Allied Pharma has a debt to market value of equity ratio of 1.0. The pharmaceutical industry's average beta is 1.12, and its average debt to market value of equity ratio is 0.50. Allied's tax rate of 25% is the same as the industry average. Calculate Allied's comparables-based beta.

Answer:

$$\begin{aligned}\beta_U &= \frac{\beta_{L, \text{comparables}}}{\left[1 + (1 - t) \frac{D}{MVE}\right]} \\ &= \frac{1.12}{\left[1 + (1 - 0.25)(0.50)\right]} = 0.8145 \\ \beta_{L, \text{Allied}} &= \beta_U \left[1 + (1 - t) \frac{D}{MVE_{\text{Allied}}}\right] \\ &= 0.8145[1 + (1 - 0.25)(1.0)] = 1.4255\end{aligned}$$

Allied's levered beta is higher than the industry average beta of 1.12 because Allied's debt to market value of equity ratio of 1.0 is higher than the industry average ratio of 0.50.

Estimating the Market Risk Premium

Analysts can estimate the market risk premium using a historical approach or a forward-looking approach.

Historical estimation is based on the average market risk premium over a lookback period. This average can be an arithmetic mean or a geometric mean. Because variations in returns decrease the effect of compounding, the geometric mean is less than the arithmetic mean.

The risk-free rate used can be a short-term rate (e.g., 1-year T-bill rate) or a long-term rate (e.g., 10-year T-bond rate). Because short-term rates are normally lower, using a short-term risk-free rate results in a higher estimate of the market risk premium. For equity valuation models, which are typically long term, it is more appropriate to use long-term risk-free rates to estimate the market risk premium.

Historical market risk premium estimates assume that past average returns are an unbiased estimate of future returns.

Forward-looking estimates are derived from current market data; they are implied in current prices. One approach is based on rearranging the constant growth model for valuation of a market index:

$$E(R_M) = \frac{D_{1,M}}{P_{0,M}} + g_M$$

This relationship simply states that the market's required rate of return is equal to its current dividend yield plus an assumed constant growth rate.

Subtracting the risk-free rate gives us an **implied market risk premium**:

$$MRP_{\text{Implied}} = [E(R_M) - R_f] = \frac{D_{1,M}}{P_{0,M}} + g_M - R_f$$

Market Risk Premium in Emerging Markets

To value stocks in developing markets, an analyst should add an additional risk premium, the **country risk premium**, to the market risk premium. A country risk premium is denominated in a reserve currency (e.g., USD, EUR) and captures the incremental risk unique to investing in developing countries.

$$MRP_{\text{developed}} = [E(R_m) - R_f]$$

$$CRP = \text{sovereign bond yield spread} \times \frac{\sigma_E}{\sigma_D}$$

where:

$$\text{sovereign bond yield spread} = \frac{\text{yield on reserve-currency-denominated emerging market sovereign debt} - \text{yield on reserve market sovereign debt}}{\text{yield on reserve market sovereign debt}}$$

$$\sigma_E = \text{annualized standard deviation emerging market equity returns}$$

$$\sigma_D = \text{annualized standard deviation of emerging market debt returns}$$

Including a country risk premium in the CAPM results in the following estimate for the required return on an emerging market stock:

$$E(R_i) = R_f + \beta_i (MRP + CRP)$$

EXAMPLE: Calculating required rate of return in an emerging market

Renee Felipe is analyzing Elipse Industrials, an Egyptian company. Egyptian EUR-denominated sovereign bonds trade at a spread of 3.5% over the EUR

benchmark rate of 2.0%, which Felipe uses as the risk-free rate. Felipe estimates Elipse's beta as 1.5, that the market risk premium for the EUR zone is 4.0%, and that Egyptian equity and debt markets have historical standard deviations of 18% and 12%, respectively.

Calculate the required rate of return on Elipse's stock using the CAPM with a country risk premium.

Answer:

$$\begin{aligned}\text{CRP}_{\text{Egypt}} &= \text{sovereign bond yield spread} \times \frac{\sigma_{\text{equity}}}{\sigma_{\text{debt}}} \\ &= 0.035 \left(\frac{0.18}{0.12} \right) = 0.0525 \\ E(R_{\text{Elipse}}) &= R_f + \beta_{\text{Elipse}}(\text{MRP} + \text{CRP}) \\ &= 0.02 + 1.5(0.04 + 0.0525) = 15.88\%\end{aligned}$$

Issues in Estimating the Market Risk Premium

Historical estimates of the market risk premium can be biased by the historical sampling period from which they are estimated. Estimates based on boom periods, when equity returns tend to exceed the risk-free rate by more than average, may overestimate the expected future market risk premium. Using this higher estimate to discount future cash flows would result in lower valuations.

By contrast, using high current market prices to produce a forward-looking estimate of the market risk premium results in a lower estimate. In other words, estimates of the market risk premium may conflict between a historical approach and a forward-looking approach. Analysts should revisit the assumptions they used in arriving at a market risk premium and fine-tune their estimates.



MODULE QUIZ 50.1

1. The dividend yield of a market index is estimated at 2.3%. The risk-free rate is 2%, and the index is projected to grow at 5% per year. A forward-looking estimate of the market risk premium is *closest* to:
 - A. 5.3%.
 - B. 7.3%.
 - C. 9.3%.
2. Steve Samuel is estimating the market risk premium using a historical approach based on the past six years of weekly data. Samuel observes that this period was characterized by a booming equity market with abnormally high equity returns. Based on this, Samuel should *most appropriately* conclude that his estimated market risk premium is too:
 - A. high, and valuations using that estimate would be biased downward.

- B. low, and valuations using that estimate would be biased downward.
 - C. high, and valuations using that estimate would be biased upward.
3. The slope of the security market line (SML) is *best* described as:
- A. beta.
 - B. the risk-free rate.
 - C. the market risk premium.
4. To account for additional risk of investing in an emerging market company, analysts using the capital asset pricing model (CAPM) should *most appropriately* add a country risk premium to the:
- A. risk-free rate.
 - B. market risk premium.
 - C. required rate of return.

MODULE 50.2: ARBITRAGE PRICING THEORY

LOS 50.b: Describe the arbitrage pricing theory and the importance of multi-factor models used in equity investing.

Academic studies have shown that a single-factor model such as CAPM, which prices only market risk, is inadequate for explaining return variances across stocks. This suggests additional risk factors are needed.

Arbitrage pricing theory (APT) is a framework for estimating equities' expected returns using a **multifactor model**. We can state a generalized multifactor model that includes k risk factors:

$$E(r_i) = R_f + \beta_{1i}f_1 + \beta_{2i}f_2 + \dots + \beta_{ki}f_k$$

where:

$E(r_i)$ = expected return of stock i

R_f = risk-free rate of return

β_{ki} = beta of stock i to k th risk factor

f_k = risk premium for k th risk factor

APT is a statistical factor model. It does not specify what the factors are, or the optimal number of them. Researchers have tried to use various macroeconomic and company-specific risk factors to identify the ones that affect stock returns.

One example is a four-factor model that uses size, value, and momentum risk factors in addition to market risk. The size, value, and momentum risk factors are each captured by the return on a zero-cost, long-short portfolio (i.e., one that

has equal overall values in long positions and short positions). This model is specified as follows:

$$E(r_i) = R_f + \beta_{i,\text{market}} \times E(R_m - R_f) + \beta_{i,\text{size}} \times E(\text{SMB}) + \beta_{i,\text{value}} \times E(\text{HML}) + \beta_{i,\text{momentum}} \times E(\text{UMD})$$

where:

$E(\text{SMB})$ = expected “small minus big” return of a zero-cost portfolio that is long small-cap stocks and short large-cap stocks

$E(\text{HML})$ = expected “high-value minus low-value” return of a zero-cost portfolio that is long value stocks and short growth stocks

$E(\text{UMD})$ = expected “up minus down” return of a zero-cost portfolio that is long recent winners and short recent losers

A stock with a positive beta for size would indicate that the stock is a small-cap stock, and it is expected to generate a positive incremental return for the time periods when small caps outperform large caps.

The betas are estimated with a multiple regression using historical data. Unlike the beta for the market return, for which a value of one indicates average risk, we would expect factor betas to average zero (that is, they are often negative).

EXAMPLE: Four-factor model

Peter Simyar has collected factor betas for three companies, as shown in the following table. Assume the risk-free rate is 2%.

Four-Factor Model Betas

Stocks	$\beta_{i,\text{Market}}$	$\beta_{i,\text{Size}}$	$\beta_{i,\text{Value}}$	$\beta_{i,\text{Momentum}}$
AirFlow	1.20	0.50	0.46	-0.08
Boon	0.80	-0.60	-0.48	0.03
Casper	1.50	-0.30	-1.17	-0.27

Based on the data given in the table:

1. Which stock is a small-cap stock?
2. Which stock is expected to benefit from recent upswing in its price?
3. Which stock has the greatest exposure to the overall market?
4. Which stock is a value stock?

Answer:

1. AirFlow has a positive beta for size, indicating that it is a small-cap stock.
2. Boon has a positive momentum beta, indicating that it will benefit from recent increases in its price.
3. Casper has the highest market beta, indicating that it has the highest exposure to overall market.
4. AirFlow has a positive beta for value, indicating that it is a value stock.



MODULE QUIZ 50.2

1. A stock has a market, size, value and momentum betas of 1.20, 0.44, -0.62, and 0.12, respectively. This stock is *most appropriately* characterized as a:
 - A. small-cap value stock.
 - B. small-cap growth stock.
 - C. large-cap momentum stock.
2. A stock has market, size, value, and momentum betas of 1.20, 0.44, -0.62, and 0.12, respectively. The respective risk premiums are 5%, 11%, 2%, and 8%, respectively. If the risk-free rate is 3%, the expected return of this stock is *closest* to:
 - A. 10.5%.
 - B. 11.1%.
 - C. 13.6%.

KEY CONCEPTS

LOS 50.a

The capital asset pricing model (CAPM) is a single-factor model that prices only the systematic risk of a company:

$$E(R_i) = R_F + \beta_i [E(R_m) - R_f]$$

The market model uses historical stock and market index return data to estimate beta:

$$R_{i,t} = \alpha + \beta_i R_{m,t} + \varepsilon_t$$

An adjusted beta accounts for the tendency of equity betas to mean-revert toward one.

Beta based on comparables is unlevered using the comparables' debt to market value of equity:

$$\beta_U = \frac{\beta_{L, \text{comparables}}}{\left[1 + (1 - t_{\text{comparables}}) \frac{D}{MVE}\right]}$$

The unlevered beta is then levered using the subject company's debt to market value of equity:

$$\beta_{L, \text{firm}} = \beta_U \left[1 + (1 - t_{\text{firm}}) \frac{D}{MVE_{\text{firm}}}\right]$$

A historical estimate of the market risk premium can be the arithmetic mean or the geometric mean of a market index's excess return over the risk-free rate.

A forward-looking estimate of the market risk premium is based on the constant growth model:

$$MRP_{\text{implied}} = [E(R_M) - R_f] = \frac{D_{1,M}}{P_{0,M}} + g_M - R_f$$

For equities in emerging markets, a country risk premium (CRP) is added to the market risk premium:

$$CRP = \text{sovereign bond yield spread} \times \frac{\sigma_{\text{emerging market}}}{\sigma_{\text{developed market}}}$$

LOS 50.b

Arbitrage pricing theory (APT) provides a framework for estimating the expected return of an equity using a multifactor model, but it does not specify the number of factors or what the factors should represent.

A four-factor model includes size (small cap vs. large cap), style (value vs. growth), and momentum (recent winners vs. recent losers) factors in addition to the market risk factor.

ANSWER KEY FOR MODULE QUIZZES

Module Quiz 50.1

- . A $R_M = \text{dividend yield} + \text{growth rate} = 2.3\% + 5\% = 7.3\%$. So, $MRP = R_M - R_f = 7.3\% - 2\% = 5.3\%$. (LOS 50.a)
- . A High market returns in the past period sampled result in a high value of the market risk premium ($R_M - R_f$). A higher estimate of the market risk premium increases the cost of equity used in discounted cash flow models.

Because this causes future cash flows to be discounted at a higher rate, the valuations obtained will be biased downward. (LOS 50.a)

- . **C** The slope of the SML is the market risk premium. (LOS 50.a)
- . **B** To estimate the required rate of return of a nondeveloped market company, an analyst should add a country risk premium to the market risk premium. (LOS 50.a)

Module Quiz 50.2

- . **B** Because the beta for size is positive and the beta for value is negative, the stock is a small-cap growth stock. (LOS 50.b)
- . **C** $E(R) = 3\% + (1.20 \times 5.00\%) + (0.44 \times 11\%) + (-0.62 \times 2\%) + (0.12 \times 8\%) = 13.56\%$. (LOS 50.b)

TOPIC QUIZ: EQUITIES

You have now finished the Equities topic section. Please log into your Schweser online dashboard and take the Topic Quiz on this section. The Topic Quiz provides immediate feedback on how effective your study has been for this material. Questions are more exam-like than typical Module Quiz or QBank questions; a score of less than 70% indicates that your study likely needs improvement. These tests are best taken timed; allow 1.5 minutes per question.

FORMULAS

Financial Statement Analysis

$$\text{cash-flow-to-revenue ratio} = \frac{\text{CFO}}{\text{net revenue}}$$

$$\text{cash-return-on-assets ratio} = \frac{\text{CFO}}{\text{average total assets}}$$

$$\text{cash-return-on-equity ratio} = \frac{\text{CFO}}{\text{average total equity}}$$

$$\text{cash-to-income ratio} = \frac{\text{CFO}}{\text{operating income}}$$

$$\text{cash flow per share} = \frac{\text{CFO} - \text{preferred dividends}}{\text{weighted average number of common shares}}$$

$$\text{debt coverage} = \frac{\text{CFO}}{\text{total debt}}$$

$$\text{interest coverage} = \frac{\text{CFO} + \text{interest paid} + \text{taxes paid}}{\text{interest paid}}$$

$$\text{reinvestment} = \frac{\text{CFO}}{\text{cash paid for long-term assets}}$$

$$\text{debt payment} = \frac{\text{CFO}}{\text{cash long-term debt repayment}}$$

$$\text{dividend payment} = \frac{\text{CFO}}{\text{dividends paid}}$$

$$\text{investing and financing} = \frac{\text{CFO}}{\text{cash outflows from investing and financing activities}}$$

$$\text{average age} = \frac{\text{accumulated depreciation}}{\text{annual depreciation expense}}$$

$$\text{total useful life} = \frac{\text{historical cost (gross cost)}}{\text{annual depreciation expense}}$$

$$\text{remaining useful life} = \frac{\text{ending net PP\&E}}{\text{annual depreciation expense}}$$

$$\text{tax expense} = \text{tax payable} + \Delta\text{DTL} - \Delta\text{DTA}$$

$$\text{effective tax rate} = \frac{\text{income tax expense}}{\text{pretax income}}$$

$$\text{cash tax rate} = \frac{\text{tax paid (cash)}}{\text{pretax income}}$$

$$\text{vertical common-size balance sheet ratios} = \frac{\text{balance sheet account}}{\text{total assets}}$$

$$\text{vertical common-size income statement ratios} = \frac{\text{income statement account}}{\text{sales}}$$

$$\text{receivables turnover} = \frac{\text{annual sales}}{\text{average receivables}}$$

$$\text{days of sales outstanding} = \frac{365}{\text{receivables turnover}}$$

$$\text{inventory turnover} = \frac{\text{cost of goods sold}}{\text{average inventory}}$$

$$\text{days of inventory on hand} = \frac{365}{\text{inventory turnover}}$$

$$\text{payables turnover} = \frac{\text{cost of goods sold}}{\text{average trade payables}}$$

$$\text{number of days of payables} = \frac{365}{\text{payables turnover ratio}}$$

$$\text{total asset turnover} = \frac{\text{revenue}}{\text{average total assets}}$$

$$\text{fixed asset turnover} = \frac{\text{revenue}}{\text{average net fixed assets}}$$

$$\text{working capital turnover} = \frac{\text{revenue}}{\text{average working capital}}$$

$$\text{current ratio} = \frac{\text{current assets}}{\text{current liabilities}}$$

$$\text{quick ratio} = \frac{\text{cash} + \text{marketable securities} + \text{receivables}}{\text{current liabilities}}$$

$$\text{cash ratio} = \frac{\text{cash} + \text{marketable securities}}{\text{current liabilities}}$$

$$\text{defensive interval} = \frac{\text{cash} + \text{marketable securities} + \text{receivables}}{\text{average daily expenditures}}$$

$$\text{cash conversion cycle} = \text{days' sales outstanding} + \text{days of inventory on hand} - \text{number of days of payables}$$

$$\text{debt-to-equity} = \frac{\text{total debt}}{\text{total shareholders' equity}}$$

$$\text{debt-to-capital} = \frac{\text{total debt}}{\text{total debt} + \text{total shareholders' equity}}$$

$$\text{debt-to-assets} = \frac{\text{total debt}}{\text{total assets}}$$

$$\text{financial leverage} = \frac{\text{average total assets}}{\text{average total equity}}$$

$$\text{interest coverage} = \frac{\text{earnings before interest and taxes}}{\text{interest payments}}$$

$$\text{debt-to-EBITDA} = \frac{\text{total debt}}{\text{EBITDA}}$$

$$\text{fixed charge coverage} = \frac{\text{earnings before interest and taxes} + \text{lease payments}}{\text{interest payments} + \text{lease payments}}$$

$$\text{net profit margin} = \frac{\text{net income}}{\text{revenue}}$$

$$\text{gross profit margin} = \frac{\text{gross profit}}{\text{revenue}}$$

$$\text{operating profit margin} = \frac{\text{operating income}}{\text{revenue}} \text{ or } \frac{\text{EBIT}}{\text{revenue}}$$

$$\text{pretax margin} = \frac{\text{EBT}}{\text{revenue}}$$

$$\text{return on assets} = \frac{\text{net income}}{\text{average total assets}}$$

$$\text{operating return on assets} = \frac{\text{operating income}}{\text{average total assets}} \text{ or } \frac{\text{EBIT}}{\text{average total assets}}$$

$$\text{return on invested capital} = \frac{\text{after-tax operating profit}}{\text{average long-term capital}}$$

$$\text{return on equity} = \frac{\text{net income}}{\text{average total equity}}$$

$$\begin{aligned} \text{return on common equity} &= \frac{\text{net income} - \text{preferred dividends}}{\text{average common equity}} \\ &= \frac{\text{net income available to common}}{\text{average common equity}} \end{aligned}$$

Original DuPont decomposition:

$$\text{ROE} = \left(\frac{\text{net income}}{\text{revenue}} \right) \left(\frac{\text{revenue}}{\text{average total assets}} \right) \left(\frac{\text{average total assets}}{\text{average stockholders' equity}} \right)$$

Extended DuPont decomposition:

$$\text{ROE} = \left(\frac{\text{net income}}{\text{EBT}} \right) \left(\frac{\text{EBT}}{\text{EBIT}} \right) \left(\frac{\text{EBIT}}{\text{revenue}} \right) \left(\frac{\text{revenue}}{\text{average assets}} \right) \left(\frac{\text{average assets}}{\text{average equity}} \right)$$

Coefficients of variation:

$$\text{CV sales} = \frac{\text{standard deviation of sales}}{\text{mean sales}}$$

$$\text{CV operating income} = \frac{\text{standard deviation of operating income}}{\text{mean operating income}}$$

$$\text{CV net income} = \frac{\text{standard deviation of net income}}{\text{mean net income}}$$

Equities

$$\text{bid-ask spread (\%)} = \frac{(\text{lowest ask price} - \text{highest bid price})}{\text{lowest ask price}}$$

$$\text{volume-weighted average price} = \frac{\sum_{i=1}^N (\text{price}_i \times \text{volume}_i)}{\sum_{i=1}^n \text{volume}_i}$$

$$\text{average daily volume} = \frac{\sum_{i=1}^n (\text{shares traded per day})_i}{n}$$

$$\text{turnover ratio} = \frac{\text{average daily volume}}{\text{market float}}$$

Holding period return with reinvested dividends:

$$r_E = \frac{P_1 \left(1 + \frac{D}{P_{PE}} \right) - P_0}{P_0}$$

where:

P_1 = share price at the end of the period

P_0 = share price at the beginning of the period

P_{PE} = share price on the dividend payment date

D = dividend per share

Free cash flow to equity (FCFE):

$$\text{FCFE} = \text{net income} + \text{depreciation} - \text{increase in working capital (WCInv)} - \text{capital expenditure (CapEx)} + \text{net borrowing}$$

$$\text{FCFE} = \text{CFO} - \text{FCInv} + \text{net borrowing}$$

$$\text{FCFE} = \Delta \text{cash} + \text{dividends} - \Delta \text{equity}$$

where:

$$\Delta \text{equity} = \text{share issuance} - \text{share repurchases}$$

$$\text{FCFE} = \text{net income} - (1 - \text{debt-to-assets ratio}) (\text{net investment})$$

Free cash flow to the firm (FCFF):

$$\text{FCFF} = \text{CFO} - \text{CapEx} + \text{interest} (1 - \text{tax rate})$$

$$\text{FCFF} = \text{FCFE} + \text{interest} (1 - \text{tax rate}) - \text{net borrowing}$$

$$\text{FCFF} = [\text{EBITDA} \times (1 - \text{tax rate})] + (\text{depreciation} \times \text{tax rate}) - \text{FCInv} - \text{WCInv}$$

Residual income per share at time $t = \text{EPS}_t - r \times \text{BVPS}_{t-1}$

$$\text{preferred stock value} = \frac{D_p}{r_p}$$

Constant growth dividend discount model (Gordon growth model):

$$V_0 = \frac{D_1}{r_e - g_c}$$

$$\text{Implied constant growth rate} = r - \frac{E(CF_1)}{\text{price}}$$

Multistage dividend discount model:

$$\text{value} = \frac{D_1}{(1 + k_e)} + \frac{D_2}{(1 + k_e)^2} + \dots + \frac{D_n}{(1 + k_e)^n} + \frac{P_n}{(1 + k_e)^n}$$

where:

$$P_n = \frac{D_{n+1}}{r_e - g_c}$$

g_c = constant growth rate of dividends

n = number of periods of supernormal growth

Sustainable growth rates:

$$SGR_{Div} = (1 - \text{dividend payout ratio}) \times ROE$$

$$SGR_{Div} = \left(1 - \frac{\text{dividend}}{\text{net income}}\right) \times \left(\frac{\text{net income}}{\text{beginning book value}}\right)$$

$$SGR_{FCFE} = \left(1 - \frac{FCFE}{\text{net income}}\right) \left(\frac{\text{net income}}{\text{beginning book value}}\right)$$

$$SGR_{FCFF} = \left(1 - \frac{FCFF}{NOPAT}\right) \left(\frac{NOPAT}{\text{beginning book value of total capital}}\right)$$

where:

$NOPAT$ = operating income $(1 - \text{tax rate})$

Weighted average cost of capital (WACC):

$$\begin{aligned} WACC = & \left(\frac{\text{MV of debt}}{\text{MV of debt} + \text{MV of equity}}\right) r_d (1 - \text{tax rate}) \\ & + \left(\frac{\text{MV of equity}}{\text{MV of debt} + \text{MV of equity}}\right) r_e \end{aligned}$$

where:

r_d = cost of debt

r_e = cost of equity

$$\text{PEG ratio} = \frac{P/E}{g}$$

Justified P/E ratio:

$$\frac{P_0}{E_1} = \frac{D_1/E_1}{r_e - g}$$

Enterprise value = market value of common and preferred stock + market value of debt – cash and short-term investments

Present value of growth opportunities = $P_0 - \frac{E_1}{r}$

Justified dividend yield:

$$\frac{D_0}{P_0} = \frac{(r - g)}{(1 + g)}$$

Justified price-to-FCFE ratio:

$$\frac{P_0}{FCFE_1} = \frac{1}{r - g}$$

Justified enterprise value-to-FCFF ratio:

$$\frac{EV}{FCFF_1} = \frac{1}{WACC - g}$$

financing needs = Δ operating assets – Δ non-debt liabilities – Δ retained earnings

ROI multiple = $(1 + r)^n$

where:

n = number of years to exit of venture capital investment

r = required rate of return

Start-up firm's intrinsic value = $\frac{\text{expected revenue} \left(\frac{\text{price}}{\text{sales}} \right)}{\text{ROI multiple}}$

Return on invested capital = $\frac{NOPAT_t}{\text{average invested capital}}$

where:

$NOPAT_t$ = net operating profit after taxes

average invested capital = the average of long-term liabilities plus equity over a given reporting period

operating profit = $[Q \times (P - VC)] - FC$

where:

Q = number of units sold

P = price per unit

VC = variable costs per unit, those that change with the level of output (e.g., materials, direct labor, inventory costs)

FC = fixed costs (stated as a total rather than on a per-unit basis), those that do not change within a specific range of output (e.g., rent, management salaries, insurance)

Capital Asset Pricing Model (CAPM):

$$E(R_i) = R_f + \beta_i (E(R_m) - R_f)$$

where:

$E(R_i)$ = expected return on equity security i .

$E(R_m)$ = expected return on the market

R_f = risk-free rate of return

$$\beta_i = \frac{\sigma_{i,m}}{\sigma_m^2} = \frac{\sigma_i}{\sigma_m} \rho_{i,m}$$

adjusted beta = ω (raw beta) + $(1 - \omega)$

INDEX

A

absolute valuation, 289
accounting profit, 168
accounts receivable, 13
acquisition method, 127
activist short selling, 378
activity ratios, 213
adjusted beta, 389
adjusted EBITDA, 298
adverse opinion, 6
aggressive accounting, 191
American depository receipt (ADR), 256
amortization, 20
antidilutive securities, 34
arbitrage pricing theory (APT), 393
ask price, 264
asset-based model, 290
audit, 6
available-for-sale securities, 53
average age, 136
average daily volume, 267
average depreciable life, 136

B

back-door listing, 263
balance sheet ratios, 58
bargaining power of customers, 239
bargaining power of suppliers, 239

- base-rate neglect, 238
- basic EPS, 32
- basic research, 353
- beneficial owner, 257
- bid-ask spread, 264
- bid-offer spread, 264
- bid price, 264
- bill-and-hold transaction, 196
- block trades, 265
- bonus issues, 274
- book value of equity, 284
- book value per share (BVPS), 284
- bottom-up analysis, 364
- breakthrough innovation, 353
- broker-dealers, 263
- business model, 359
- business risk, 230
- business segment (operating segment), 4
- buyout equity, 252
- buy-side research report, 377

C

- callable preference shares, 250
- callable preferred stock, 306
- capital asset pricing model (CAPM), 387
- capitalization, 18, 20, 199
- carrying value, 168
- cash conversion cycle, 216, 367
- cash dividends, 273
- cash flow from financing activities (CFF), 78, 86, 96
- cash flow from investing activities (CFI), 78
- cash flow from operating activities (CFO), 68, 96
- cash flow per share, 101

cash flow statement, 65
cash flow-to-revenue ratio, 100
cash return-on-assets ratio, 101
cash return-on-equity ratio, 101
cash tax rate, 179
cash-to-income ratio, 101
change in accounting estimate, 30
change in accounting policy, 30
channel stuffing, 196
clean surplus accounting, 298
coefficient of variation, 230
combination rights, 250
common shares, 250
common-size balance sheet, 57, 209
common-size income statement, 40, 209
common-size statements, 209
company life cycle, 249
company research report, 375
competitive strategy, 359
complex capital structure, 32
composite market index, 268
confirmation bias, 239
conservatism bias, 238
conservative accounting, 191
constituent securities, 268
contingency features, 306
continuing earnings, 319
contract, 14
contributed capital, 82
conversion ratio, 251, 307
conversion rights, 250
convertible preference shares, 251
convertible preferred stock, 307
core earnings, 319

cost leadership (low-cost) strategy, 362
cost leadership strategy, 362
country risk premium, 391
coverage ratios, 101, 217
cult companies, 283
cumulative preference shares, 251
current ratio, 216
custodian, 257
cyclical company, 345

D

days of inventory on hand, 214
days of sales outstanding, 214
debt coverage ratio, 101
debt payment ratio, 101
debt ratios, 217
declaration date, 275
decline phase, 250
defensive company, 345
defensive interval, 216
deferred compensation, 152
deferred tax asset (DTA), 168, 170
deferred tax liability (DTL), 55, 168, 169
defined benefit plan, 153
defined contribution plan, 153
degree of financial leverage (DFL), 370
delay costs, 265
depletion, 20
depository bank, 256
depository receipt (DR), 256
depreciation, 20, 219
derecognition, 131
derivative instruments, 52

- designated market maker, 264
- de-SPAC transaction, 263
- differentiation strategy, 362
- diluted EPS, 34
- diluted EPS equation, 35
- dilution, 319
- dilutive securities, 34
- direct financing leases, 149
- direct investing, 255
- direct listing, 262
- direct method, 68
- disaggregated valuation model, 331
- disclaimer of opinion, 6
- discontinued operation, 29
- discounted cash flow model, 289
- disruptive innovation, 353
- dividend discount model (DDM), 295
- dividend payment ratio, 101
- dividend rights, 250
- dividends, 249
- dividends, cash flow classification, 78, 85
- dividend yield, 322
- dual-class share structure, 256
- dual listing, 256
- DuPont system, 225

E

- earnings before interest, taxes, depreciation, and amortization (EBITDA), 219, 297
- earnings guidance, 8
- earnings per share (EPS), 32
- earnings smoothing, 191
- earnings yield, 318

- economic goodwill, 52
- economic moat, 349, 362
- economic profit, 298
- effective tax rate, 41, 171, 179
- effective tax rate reconciliation, 178
- embedded options, 306
- employee stock options, 156
- enterprise value (EV), 288, 320
- enterprise value (EV) multiples, 313
- enterprise value (EV)-to-book ratio, 321
- environmental influences, 354
- equity exchange, 263
- equity instrument, 249
- equity prospectus, 250, 262
- European depository receipt (EDR), 256
- EV/EBITDA multiple, 321
- ex-dividend date, 275
- expense, 20
- extended (5-way) DuPont equation, 227
- extraordinary cash dividends, 273

F

- factor-based investing, 268
- factor exposures, 324
- fair value, 155
- finance lease, 144
- Financial Accounting Standards Board (FASB), 2
- financial assets, 52
- financial instruments, 52
- financial leverage ratio, 218
- financial reporting, 2
- financial reporting quality, 189
- financial statement analysis, 2

- financial statement analysis framework, 1
- financial statement notes, 3
- firm value, 320
- first-in, first-out (FIFO), 20
- fixed asset turnover, 215
- fixed charge coverage ratio, 218
- floating bar valuation chart, 284
- focus strategy, 362
- forecast horizon, 243
- Form 8-K, 4
- Form 10-K, 4
- Form 10-Q, 4
- Form 144, 4
- Form DEF-14A, 4
- Form S-1, 4
- Forms 3, 4, and 5, 4
- forward earnings yield, 318
- forward multiple, 313
- forward split, 274
- fractional shares, 277
- free cash flow, 97
- free cash flow to equity (FCFE), 99, 296
- free cash flow to the firm (FCFF), 98, 297
- free-on-board (FOB), 196
- fundamental value, 283
- funded status, 153

G

- geographic segment, 5
- global depository receipts (GDRs), 256
- goodwill, 51, 126, 127
- grant date, 155
- gross investment, 136

- gross profit margin, 41, 219
- growth company, 346
- growth equity, 252
- growth industries, 347
- growth in same-store sales, 229
- growth phase, 250

H

- hard-to-value stocks, 379
- held-to-maturity securities, 52
- Herfindahl-Hirschman Index (HHI), 348
- home asset bias, 255
- horizontal common-size balance sheet or income statement, 211

I

- identifiable intangible asset, 126
- identifiable intangible assets, 49
- illusion of control bias, 238
- impairment, 129
- implicit costs of trading, 265
- implied growth rate, 300
- implied market risk premium, 391
- income tax expense, 168
- income tax paid, 168
- indirect investing, 256
- inflection points, 244
- initial public offering (IPO), 261
- initiating coverage, 375
- innovation matrix, 353
- intangible assets, 49, 125
- intensity of industry rivalry, 239

intentional competitive strategy, 359
interest coverage ratio, 101, 218
internal controls, 6
International Accounting Standards Board (IASB), 2
International Organization of Securities Commissions (IOSCO), 2, 193
intrinsic value, 283
inventories, 20
inventory disclosures, 114
investing and financing ratio, 102
investment in working capital, 73

J

justified leading P/E ratio, 314
justified multiple, 313
justified P/B ratio, 315

L

last-in, first-out (LIFO), 20
leading dividend yield, 322
leading multiple, 313
leading P/E ratio, 291
lease liability, 144
lease receivable, 148
lessee, 143
lessor, 143
leveraged buyout (LBO), 252
levered beta, 390
LIFO liquidation, 111
limit order, 264
line graph, 211
liquid asset requirement, 230

- liquidating dividend, 273
- liquidation preference payout, 250
- liquidation rights, 250
- liquidity ratios, 59, 213, 216
- long-term financial liabilities, 55
- lower of cost or market, 106

M

- management buyout (MBO), 252
- management commentary, 5
- management discussion and analysis (MD&A), 5
- market breadth, 267
- market capitalization, 286
- market-clearing price, 267
- market depth, 264, 267
- market float, 266
- market impact, 265
- market model, 389
- market order, 264
- market resiliency, 267
- market risk premium, 387
- market share, 348
- mark-to-market accounting, 52
- matching principle, 18
- mature industries, 347
- mature phase, 250
- mean reversion, 389
- measurement date, 29
- meme stock, 283
- method of comparables, 314
- method of forecasted fundamentals, 314
- mid-market price, 264
- multifactor model, 393

multiplier models, 291

N

net borrowing, 296
net capital expenditure, 369
net debt, 288
net income per employee, 229
net interest expense or income (defined benefit plan), 154
net interest margin, 230
net investment in the lease, 149
net pension asset, 153
net pension liability, 153
net profit margin, 41, 218
net realizable value, 105
net realizable value (NRV), 105
noncumulative preference shares, 251
non-GAAP measures, 194
nonparticipating preference shares, 251
normalized earnings, 243, 319
number of days in inventory, 214
number of days of payables, 214

O

obtainable market, 361
offer price, 264
off-exchange trading, 265
operating cycle, 367
operating lease, 144
operating leverage, 366
operating profit margin, 41, 219, 347
operating return on assets, 220

opportunity costs, 265
order matching system, 263
overconfidence bias, 238
over-the-counter (OTC) trading, 266

P

participating preference shares, 251
par value, 250
payables turnover, 214
payment date, 275
peer group, 314
pension, 153
perceived value, 283
performance obligation, 14
performance ratios, 100
performance shares, 156
period costs, 18
permanent difference, 168, 171
persistent earnings, 319
PESTLE analysis, 352
PESTLE framework, 352
phantom stock, 157
phaseout period, 29
Porter's five forces, 239, 349
post-money value, 336
potentially dilutive securities, 32
preemptive rights, 250
preference shares, 250
preferred stock, 250
pre-money value, 336
present value model, 289
present value of the growth opportunities (PVGO), 317
pretax margin, 42, 219

- price/earnings-to-growth ratio (PEG), 315
- price elasticity, 366
- price multiples, 313
- price return, 276
- price takers, 366
- price-to-book ratio (P/B), 286
- price-to-dividend multiple, 322
- price-to-earnings ratio (P/E), 291
- price-to-sales (P/S) multiple, 320
- price-to-tangible book value, 321
- pricing power, 365
- primary equity markets, 261
- primary follow-on offering, 263
- prior-period adjustment, 30
- private equity, 251
- private investment in public equity (PIPE), 252
- private placement, 261
- privatization, 261
- product or service differentiation strategy, 362
- profitability ratios, 213, 218
- pro forma financial statements, 237
- prospective application, 30
- proxy, 250
- proxy advisor, 257
- proxy statement, 257
- proxy statements, 3
- putable preference shares, 250
- putable preferred stock, 307
- put contingency price, 250

Q

- qualified opinion, 6
- quality of earnings, 189

quick ratio, 216
quote-driven trading, 266

R

record date (for dividends), 275
record date (for proxy voting), 257
recoverability test, 129
recoverable amount, 129
regular cash dividends, 273
regulatory authorities, 2
reinvested dividends, 277
reinvestment rate, 322
reinvestment ratio, 101
related-party transactions, 198
relative valuation models, 291, 313
remaining useful life, 136
remeasurements, 154
representativeness bias, 238
research and development costs, 27, 126
research costs, 27, 126
reserve requirements, 230
residual income, 289, 298
restricted stock units, 156
restricted stock units (RSUs), 263
retrospective application, 30
return on assets (ROA), 219
return on capital (ROC), 322
return on common equity, 220
return on equity (ROE), 220
return on invested capital (ROIC), 347, 369
revenue drivers, 364
reverse stock splits, 274
right-of-use asset (ROU asset), 144

S

- sales per employee, 229
- sales per square foot, 230
- sales-type lease, 149
- scenario analysis, 231, 284
- seasoned equity offering, 263
- secondary equity markets, 263
- secondary follow-on offering, 263
- sector, 344
- Securities and Exchange Commission, 2, 193
- security market line (SML), 388
- sell-side research report, 377
- sensitivity analysis, 231
- serviceable market, 361
- service cost, 154
- shareholder of record, 257
- shareholder proposal, 257
- share repurchase, 274
- simple capital structure, 32
- simulation, 231
- slippage, 265
- software development costs, 27, 126
- solvency ratios, 59, 213, 217
- special cash dividends, 273
- special purpose acquisition company (SPAC), 263
- special situations, 252
- spinoff, 132
- split rights, 250
- sponsored DR, 256
- stacked column graph, 211
- standard auditor's opinion, 6
- standard-setting bodies, 2
- start-up phase, 249

- statutory tax rate, 171, 179
- stock-based appreciation rights (SARs), 157
- stock dividends, 32, 274
- stock grants, 155
- stock screening, 378
- stock splits, 32, 274
- stretching payables, 199
- subdivision rights, 250
- sustainable growth rate, 303
- sustaining innovation, 353

T

- take-private transaction, 252
- taxable income, 167
- tax base, 168
- taxes payable, 167
- tax loss carryforward, 168
- tax return terminology, 167
- temporary difference, 168
- terminal value, 295
- threat of new entrants, 239
- threat of substitutes, 239
- top-down analysis, 364
- total addressable market (TAM), 361
- total asset turnover, 215
- total return, 276
- total useful life, 136
- trading securities, 52
- trailing dividend yield, 322
- trailing multiple, 313
- trailing P/E ratio, 291
- transaction price, 14
- treasury stock method, 38

turnover ratio, 267

U

unearned revenue, 13

unidentifiable intangible asset, 126

unidentifiable intangible assets, 49

unintentional competitive strategy, 359

unlevered beta, 390

unqualified opinion, 6

unsponsored DR, 256

unusual or infrequent items, 29

V

valuation allowance, 168, 177, 197

value-at-risk, 230

value in use, 129

value-to-book multiple, 321

venture capital, 251

venture capital (VC), 335

vertical common-size balance sheet, 209

vertical common-size income statement, 209

vertical integration, 350

vesting period, 155

volume-weighted average price (VWAP), 265

voting rights, 250

W

weighted average cost method, 20

weighted average cost of capital (WACC), 304

weighted average number of common shares, 32

working capital management, 367

working capital turnover, 215